Haojue

DK160

SERVICE MANUAL

99500H2M3R0E010

FOREWORD

This manual contains an introductory description on the DK160 and procedures for its inspection service and overhaul of its main components. Other information considered as generally known is not included.
Read the GENERAL INFORMATION section to familiarize yourself with the motorcycle and its maintenance.
Use this section as well as other sections to use as a guide for proper inspection and service.
This manual will help you know the motorcycle better so that you can assure your customers of fast and reliable service.

* This manual has been prepared on the basis of the latest specifications at the time of publication. If modifications have been made since then, differences may exist between the content of this manual and the actual motorcycle.
* Illustrations in this manual are used to show the basic principles of operation and work procedures. They may not represent the actual motorcycle exactly in detail.
* This manual is written for persons who have enough knowledge, skills and tools, including special tools, for servicing Haojue motorcycles. If you do not have the proper knowledge and tools, ask your authorized Haojue motorcycle dealer to help you.

⚠ WARNING

Improper repair may result in injury to the mechanic and may render the motorcycle unsafe for the rider and passenger.

⚠ CAUTION

Inexperienced mechanics or mechanics without proper tools, not to able to perform the services described in this manual properly.

HAOJUE HOLDING.,LTD
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(The 1st Edition)

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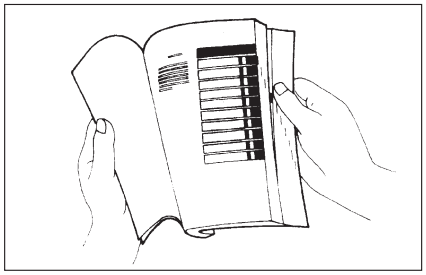
HOW TO USE THIS MANUAL
TO LOCATE WHAT YOU ARE LOOKING FOR:

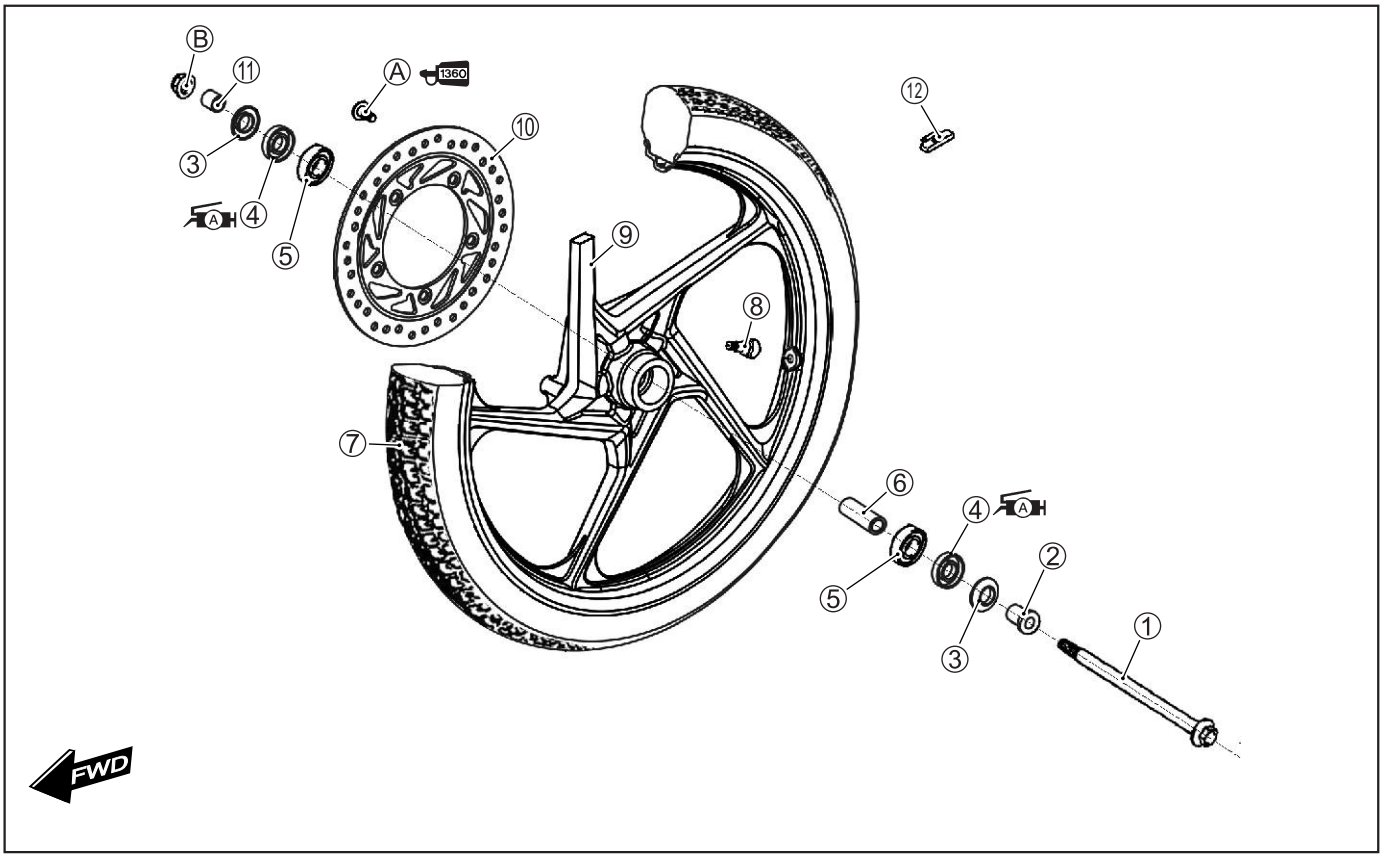
1.The text of this manual is divided into sections.
2.The section titles are listed in the GROUP INDEX.
3. Holding the manual as shown as at the right will allow you to find the first page of the section easily.
4.The contents are listed on the first page of each section to help you find the item and page you need.



COMPONENT PARTS AND WORK TO BE DONE

Under the name of each system or unit, is its exploded view. Work instruction and tightening torque, lubricating points and locking points, are provided.

EX: Front wheel



①	Front axle	⑧	Valve
②	Sleeve	⑨	Front wheel rim
③	Dust cover	⑩	Front brake disc
④	Oil seal	⑪	Spacer
⑤	Roller bearing	⑫	Balancer
⑥	Spacer	Ⓐ	Brake disc bolt
⑦	Front tire	Ⓑ	Front axle

ITEM	N•m
Ⓐ	18-28
Ⓑ	36-52

SYMBOL

The following symbols are instructions and necessary information for maintenance.

SYMBOL	DEFINITION	SYMBOL	DEFINITION
	Torque control required. Data beside it indicates specified torque.		Apply THREAD LOCK “1360” 99000-32130
	Apply oil. Use engine oil unless otherwise specified.		Apply or use the absorber oil
	Apply SUPER GREASE “A” 99000-25010		Apply or use the brake fluid
	Apply MOLY PASTE 99000-25140		Measure in voltage
	Apply sealant “1215” 99000-31110		Measure in resistance
	Apply sealant “1207B” 99000-31140		Measure in continuity test range
	Apply THREAD LOCK “1342” 99000-32050		Use special tool
	Apply THREAD LOCK “1303” 99000-32030		The data of maintenance

WIRE COLOR

B ..... Black	Gr..... Grown	R ..... Red
Bl ..... Blue	Lbl..... Light blue	V ..... Violet
Br ..... Brown	Lg..... Light green	W ..... White
Dg..... Deep green	O ..... Orange	Y ..... Yellow
G ..... Green	P ..... Pink	

B/Bl ..... Black with Blue tracer
B/Br ..... Black with Brown tracer
B/G ..... Black with Green tracer
B/R ..... Black with Red tracer
B/W ..... Black with White tracer
B/Y ..... Black with Yellow tracer
Bl/W..... Blue with White tracer
Bl/Y ..... Black with Yellow tracer
Br/Y ..... Brown with Yellow tracer
G/Y ..... Green with Yellow tracer
O/B ..... Orange with Black tracer
O/Bl ..... Orange with Blue tracer
O/W ..... Orange with White tracer
O/Y ..... Orange with Yellow tracer
R/B ..... Red with Black tracer
R/Y ..... Red with Yellow tracer
W/B ..... White with Black tracer
W/R ..... White with Red tracer
Y/G ..... Yellow with Green tracer
Y/R ..... Yellow with Red tracer
Y/W ..... Yellow with White tracer

GENERAL INFORMATION

1

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WARNING/CAUTION/NOTE

Please read this manual and follow its instructions carefully. To emphasize special information, the symbol and the words WARNING,CAUTION and NOTE have special meanings. Pay special attention to the messages highlighted by these signal words.

⚠ WARNING

Indicates a potential hazard that could result in death or injury.

⚠ CAUTION

Indicates a potential hazard that could result in motorcycle damage.

NOTE:
Indicates special information to make maintenance easier or instructions clearer.

Please note, however, that the warnings and the cautions contained in this manual cannot possibly cover all potential hazards relating to the servicing, or lack of servicing of the motorcycle. In addition to the WARNING and CAUTION stated, you must use good judgment and basic mechanical safety principles. If you are unsure about how to perform a particular service operation, ask a more experienced technician for advice.

GENERAL PRECAUTION

⚠ WARNING

- Proper service and repair procedures are important for the safety of the service technician and the safety and reliability of the motorcycle.
- When 2 or more persons work together, pay attention to safety of each other.
- When it is necessary to run the engine indoors, make sure that exhaust gas is forced outdoor.
- When working with the toxic or flammable materials, make sure that the area you work in is well-ventilated and that you follow all of the material manufacturer's instructions.
- Never use gasoline as a cleaning solvent.
- To avoid getting burned, do not touch the engine, engine oil or exhaust system during or for a while after engine operation.
- After servicing fuel, oil, exhaust or brake system, check all lines and fittings related to the system for leakage.

⚠ CAUTION

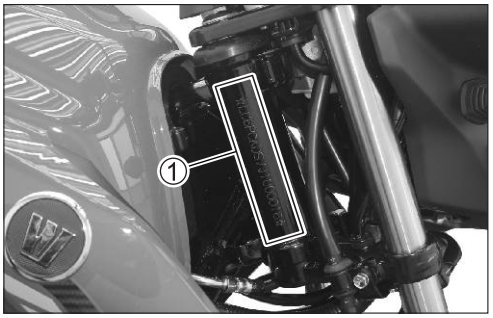
- If parts replacement is necessary, replace the parts with SUZUKI Genuine Parts or their equivalent.
- When removing parts that are to be reused, keep them arranged in an orderly manner so that they may be reinstalled in the proper order and orientation.
- Be sure to use special tools when instructed.
- Make sure that all parts used in reassembly are clean. Lubricate them when specified.Use the specified lubricant, bonds or sealant.
- When removing the battery, disconnect the negative cable first and then the positive cable. When reconnecting the battery, connect the positive cable first and then the negative cable, and replace the terminal cover on the positive terminal.
- When performing service to electrical parts, if the service procedures not require use of battery power, disconnect the negative battery cable.
- When tightening the cylinder head, case bolts and nuts, tighten the larger sizes first. Always tighten the bolts and nuts from the inside working out, in a crisscross manner.
- Whenever you remove oil seals, gaskets, packing, O-rings, locking washers, self-locking nuts, cotter pins, circlips and certain other parts as specified, be sure to replace them with new ones. Also, before installing these new parts, be sure to remove any left over material from the mating surfaces.
- Never reuse a circlip. When installing a new circlip, take care not to expand the end gap larger than required to slip the circlip over the shaft. After installing a circlip, always ensure that it is completely seated in its groove and securely fitted.
- Use a torque wrench to tighten fasteners to the specified torque. Wipe off grease and oil if a thread is smeared with them.
- After reassembling, check parts for tightness and proper operation.

NOTE:

- *To protect the environment, do not unlawfully dispose of used motor oil and all other fluids, batteries and tires.*
- *To protect the Earth's natural resources, properly dispose of used motorcycle and parts.*

SERIAL NUMBER LOCATIONS

The frame serial number or V.I.N. (Vehicle Identification Number) ① is stamped on the steering head pipe. The engine serial number ② is located on the right side of the crankcase. These numbers are required especially for registering the motorcycle and ordering spare parts.



FUEL AND OIL RECOMMENDATIONS

FUEL

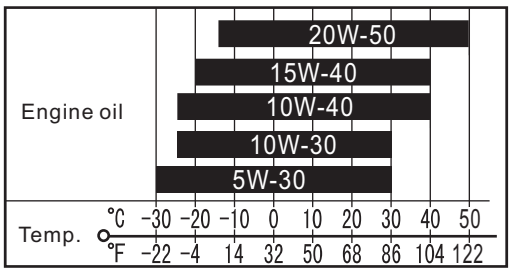
Use unleaded gasoline of grade 89 octane (Research Method) or higher. Ethanol gasoline E10.

⚠ CAUTION

Use non-unleaded gasoline and low quality engine oil will shorten the spark plug and muffler accelerant life. Dirty gasoline will clog the fuel hose. Do not overfill the gasoline.

ENGINE OIL

Use a premium quality 4-stroke motor oil to ensure longer service life of your motorcycle. Use only oils which are rated SL of the API service classification.
The recommended viscosity is SAE 10W-40. In winter of northen cold area 5W-30 should be used, If an SAE 10W-40 or 5W-30 motor oil is not available, select an alternative according to the right chart.



BRAKE FLUID

BF **Specification and classification: DOT 3 or DOT4**

BREAK-IN PROCEDURE

The correct "BREAK-IN" extends the engine life and maximums the engine performance. The general rules are as follows:

- The throttle must not be opened to the full during the new motorcycle's break-in period. Throttle opening should be limited to less than 3/4 of its maximum, while violent acceleration should be avoided.
- Do not maintain constant engine gear and speed for an extended time period during any portion of the break-in. Try to vary the throttle position. During the period of break-in, increase the throttle opening properly to enable break-in fully.
- When engine is running at constant low speed (low load), parts will wear rapidly leads to bad brake-in.

SPECIFICATIONS

DIMENSIONS AND CURB WEIGHT

Overall length	2010 mm
Overall width	765 mm
Overall height	1100mm
Wheelbase	1285 mm
Ground clearance	195 mm
Curb weight	135 kg

ENGINE

Type	Single cylinder, vertical, air-cooled, four stroke
Number of cylinders	1
Bore	60 mm
Stroke	57.4 mm
Piston displacement	162 cc
Compression ratio	9.65:1
Fuel supply	Fuel injection
Air cleaner	Oiled paper filter
Starter system	Electric startup
Lubrication system	Wet sump

TRANSMISSION

Clutch	Wet multi-plate type
Transmission	5-speed constant mesh
Primary reduction ratio	3.091
Final reduction ratio	2.933
Gear ratios, Low	3.077
2nd	1.875
3rd	1.409
4th	1.120
Top	0.938
Drive chain	428HS 132 links

ELECTRIC

Ignition type	Electronic ignition
Spark plug	CPR8EA-9
Battery	12V 7Ah
Fuse	10 A/15A/20A
Headlamp	12V 35W/35W
Position light	12V 5W
Tail light / brake light	12V 5W/21W
Turn signal light	12V 10W
License light	12V5W

CHASSIS

Front suspension	Telescopic, coil spring, oil damped
Rear suspension	Swing arm type, oil dampened, 5-way adjustable
Steering angle	42°
Caster	26.5°
Trail	98 mm
Turning diameter	4200 mm
Front brake	Disc brake/Brake lever
Rear brake	Drum brake/Brake pedal/CBS
Front tire size	80/100- 18 47P
Rear tire size	100/80-18 59P
Dynamic wheel balance	8 g
Front absorber stroke	110 mm

CAPACITIES

Fuel tank	16.5 L
Engine oil, oil change	1 100 ml
Overhaul	1 200 ml
Front absorber oil (single)	152 ml

These specifications are subject to change without notice.

PERIODIC MAINTENANCE

2

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PERIODIC MAINTENANCE

The chart below lists the recommended intervals for all the required periodic service work necessary to keep the motorcycle operating at peak performance and economy.

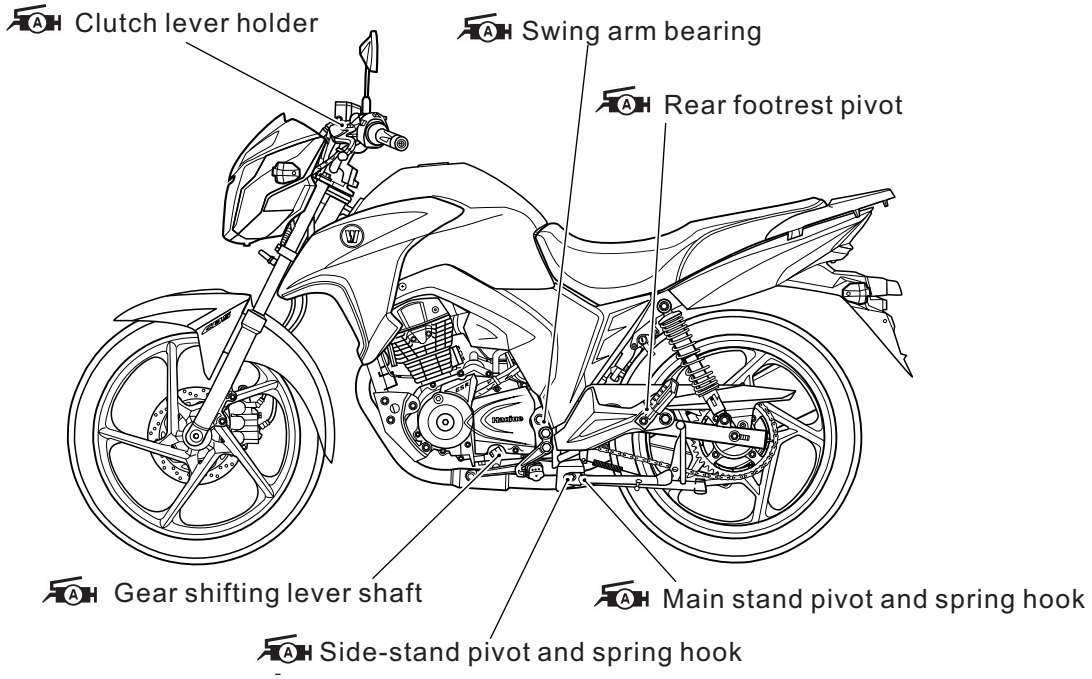
NOTE:
More frequent servicing may be performed on motorcycles that are used under severe conditions.

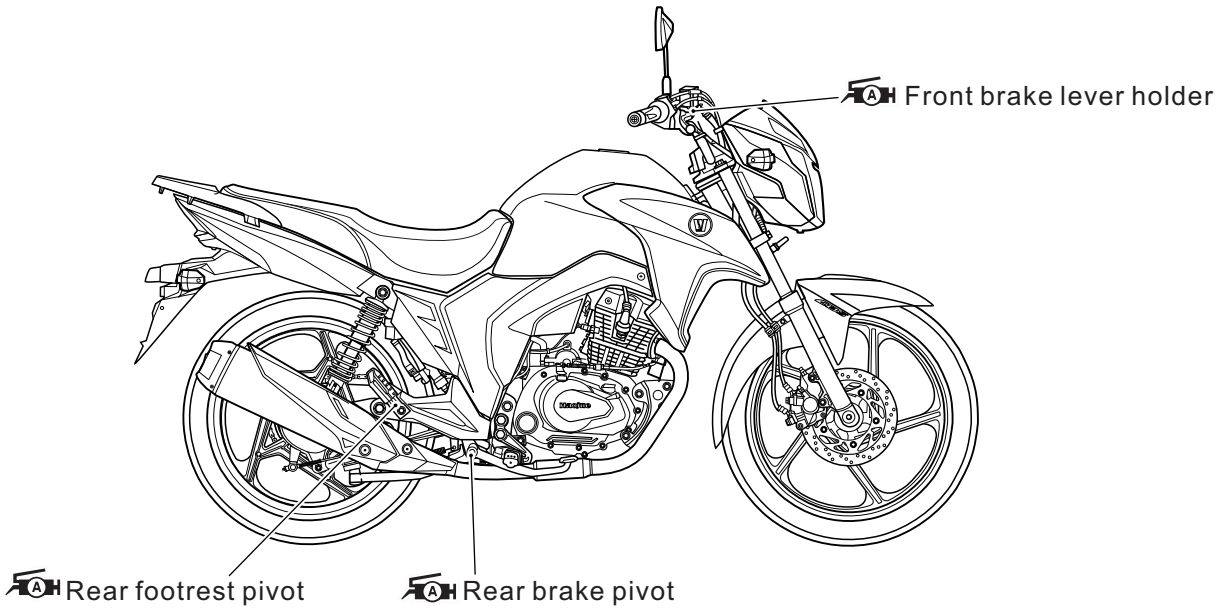
PERIODIC MAINTENANCE CHART

Item \ Interval		Km	Initial 1 000	Every 3 000	Every 6 000	Page
		Month	3	6	12	
Battery / Fuse			Inspect	Inspect	———	2-3
Air filter			———	Clean	———	2-4
			Clean every 18 months or every 12000 km			
Exhaust pipe bolts and muffler bolts			Inspect	Inspect	———	2-5
Valve clearance & cylinder head nuts			Inspect	Inspect	———	2-5
Spark plug			Inspect	Inspect	———	2-6
			Replace every 10 000 km			
Engine oil			Clean	Clean	———	2-7
Oil sump filter			Clean every 18 months or 12 000 km			2-7
Engine oil centrifugal filter			Clean every 18 months or 12 000 km			2-7
Clutch cable play			Inspect	Inspect	———	2-8
Throttle body			———	Clean	———	2-8
Throttle cable			Inspect	Inspect	———	2-8
Fuel hose			Inspect	Inspect	———	2-9
			Replace every 10 years			
Evaporative emission control system			Inspect	Inspect	———	2-9
Drive chain/sprocket			Inspect, clean & lubricate for every 1,000 km			2-10
Brakes	Brakes		Inspect	Inspect	———	2-12
	Brake hose		Inspect	Inspect	———	
			Replace every 4 years			
	Brake fluid		Inspect every 3 000 km, replace every 2 years			
Tire			Inspect	Inspect	———	2-15
Dynamic wheel balance			After tire replacement or repair			2-15
Steering			Inspect	Inspect	———	2-15
Front and rear shock absorbers			———	Inspect	———	2-16
Lights and signals			Inspect	Inspect	———	2-16
Chassis bolts and nuts			Inspect	Inspect	———	2-17

LUBRICATION POINTS

Proper lubrication is important for smooth operation and long life of each working part of the motorcycle.
Major lubrication points are indicated below.





⚠ WARNING

Be careful not to apply too much grease to the brake cam shafts. If grease gets on the linings, brake slippage will result.

⚠ CAUTION

Wipe out the grease, oil, rust, or dust before lubricating any parts.
Apply anti-rust solvent to the part which is exposed easy to get rusted when the motorcycle is used in raining day or moist environment.

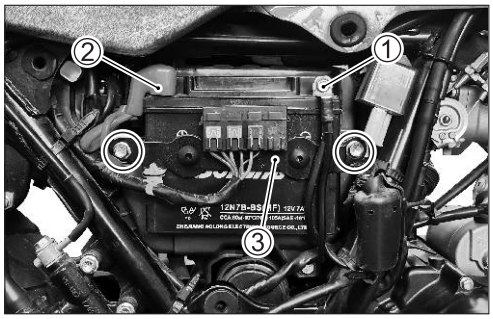
MAINTENANCE AND TUNE-UP PROCEDURES

This section describes the service procedures for each section of periodic maintenance.

BATTERY/FUSE

Inspect initial 1 000 km and every 4 000 km

BATTERY/FUSE



The battery is fixed inside the left side cover. This battery is a sealed type battery, and requires no maintenance. However, charging is required once the voltage falls below 12.4V.

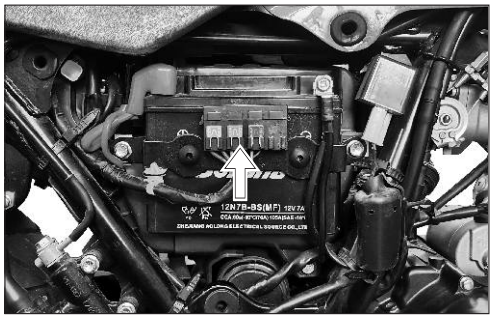
- Turn off the ignition switch.
- Remove the right cover. (Refer to page 6-1)
- Remove the ⊖ lead① first, then remove ⊕ lead②.
- Remove the battery holder③.
- Take out the battery.

Install the battery in the reverse order of the disassembly.

⚠ CAUTION

When install the battery lead wires, fix the ⊕ lead first and the ⊖ lead last.

FUSE



- The fuses are located on the battery holder. If engine is shut down suddenly during the driving or circuit is open, please check the fuse.

⚠ WARNING

Do not use a fuse of a different specification , or it will incur negative effect to electric, fire and power loss of engine which is very dangerous.

⚠ CAUTION

Do not use aluminum, iron wire or any other substitute for the fuse. If the fuse blows frequently, it indicates that the electric system may has problem. Always investigate the cause, correct it and then replace the fuse.

AIR FILTER

Clean initial 1 000 km and every 2 000 - 3 000 km

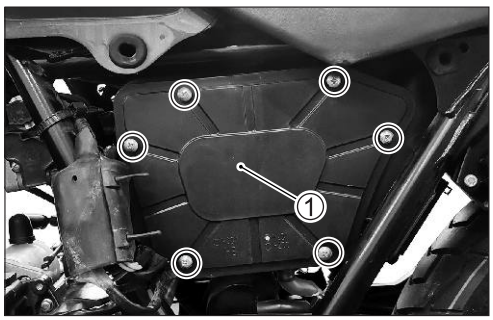
If the air filter is clogged with dust, intake resistance will be increased with a resultant decrease in output and an increase in fuel consumption.

⚠ CAUTION

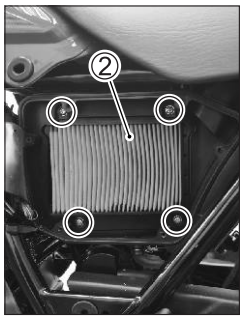
If driving under dusty condition, clean the air filter element more frequently.

Check and clean the filter in the following manner.

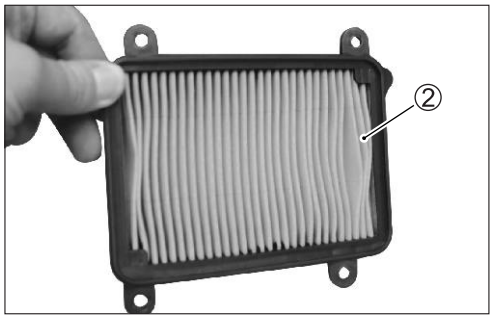
- Remove the left cover. (Refer to page 6-1)
- Loosen the air filter covet screws, take off the air filter cover ①.



- Remove the paper filter ② and fire-proof filter ③, replace the filter as necessary.
- Clean the air filter case and cover.



- Check to see if the paper filter is clogged with dust, replace the paper filter as necessary.



Reinstall the air cleaner in the reverse order of removal. Make sure the air filter is fitted and sealed property at specified position.

⚠ CAUTION

If driving under dusty conditions, clean the air cleaner element more frequently. The surest way to accelerate engine wear is to operate the engine without the element or to use a torn element. Make sure that the air cleaner is in good condition at all times. The life of the engine depends largely on this component.

DRAIN PLUG

Remove the air cleaner drain plug and drain water and oil at the periodic maintenance interval.

⚠ CAUTION

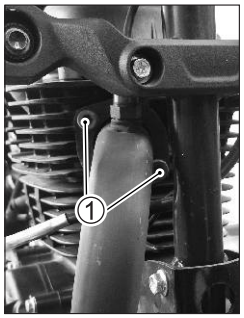
After cleaning, make sure the drain plug is fitted properly.

EXHAUST PIPE AND MUFFLER BOLTS

Inspect initial 1 000km and every 3 000km

- Tighten the exhaust pipe/muffler bolts to the specified torque.

Exhaust pipe bolt ①: 11-15 N•m
Muffler bolt ②: 35-45 N•m



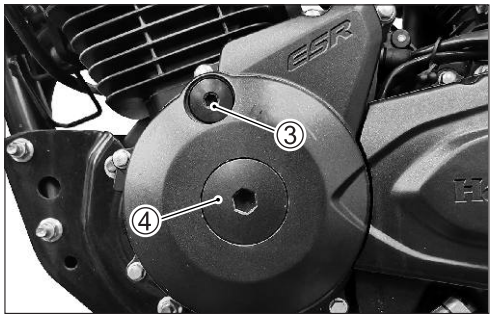
VALVE CLEARANCE & CYLINDER HEAD NUTS

Inspect initial 1 000km and every 3 000km

At the distances indicated above, check and adjust the clearance to the following specification. The procedure for adjusting the valve clearance is as follows:

- Remove the left and right covers. (Refer to page 6-1)
- Remove the spark plug cap ①and the cylinder head shell ②.

- Remove the valve timing inspection plug ③and generator cover cap ④.

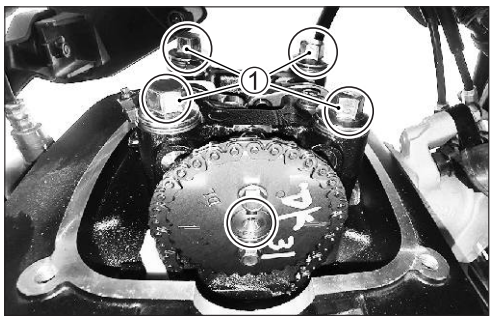


- Rotate the generator rotor counter clockwise until the white line on the rotor is aligned with center (mark line) of hole on the crankcase to set the piston at TDC of the compression stroke.

***NOTE*:**
The valve clearance is to be checked when the engine is cold.

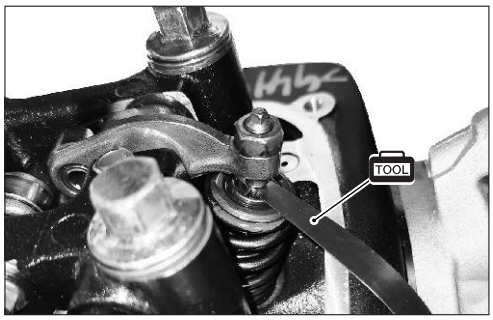
- The cam sprocket carved line should parallel with the cylinder head top surface and the notch of camshaft should face upward.
- Tighten the cylinder head cover nuts to ① the specified torque when engine is cool.

Cylinder head nut: 25-35 N•m



- Inspect the intake and exhaust valve clearance with the special tool

09900-20803: Thickness gauge

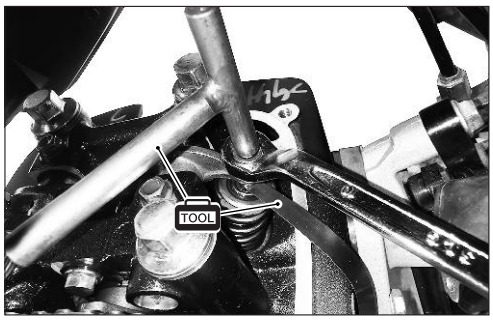


Valve clearance

Valve clearance (When cold): IN: 0.04-0.08mm
EX: 0.09-0.13mm

- If the valve clearance is out of specification, bring it into the specified range by using the special tool.

09900-20803: thickness gauge
11F14-019: T valve adjust wrench

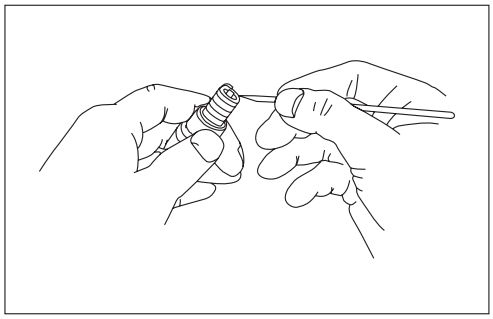


- Reassemble the removed parts in the reverse order of removal.

SPARK PLUG

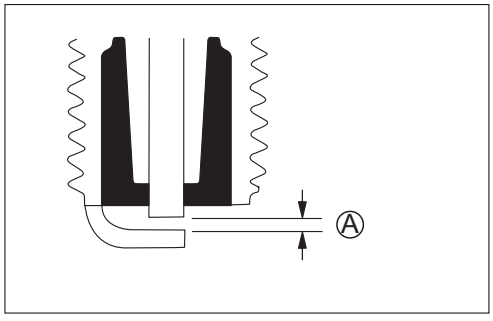
Inspect Initial 1 000 km and every 3 000 km.
Replace it every 10 000 km.

Remove the carbon deposits with a wire of pin and adjust the spark plug gap Ⓐ to 0.8-0.9 mm with a thickness gauge. Make sure to observe the appearance of the plug. If the standard plug is apt to get wet, a hotter plug should be used. If the standard plug is apt to overheat (porcelain is whitish in appearance), replace with a cooler one.



Spark plug gap Ⓐ: 0.8-0.9 mm

09900-20803: thickness gauge



TYPE	NGK
Hot type	CPR7EA
Standard	CPR8EA
Cold type	CPR9EA

ENGINE OIL

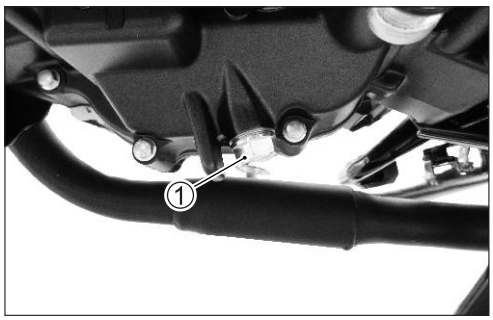
Change initial 1 000 km and every 3 000 km

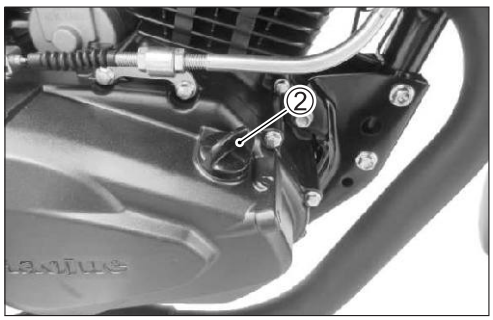
The oil should be changed when the engine is warm. The procedure is as follows:

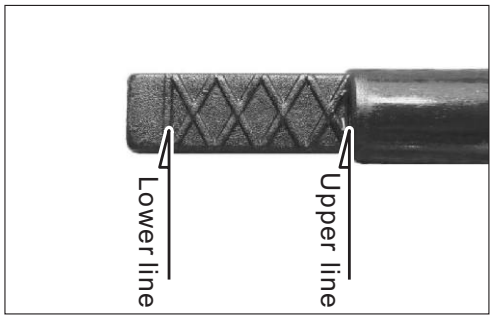
- Support the motorcycle by the main stand.
- Drain the oil by removing the drain plug ① and oil dipstick②.
- Tighten the drain plug with a new washer, pour new engine oil through the oil filler. The engine will hold about 1 100 ml of oil (1 200 ml for overhaul). Use of SL in API with an viscosity of SAW 10W-40.

DATA Required amount of engine oil:
Oil change: 1 100 ml
Overhaul: 1 200 ml

- Start up the engine and allow it to run for several seconds at idling speed.
- Shut down the engine and wait about 1 minutes.
- Remove the oil dipstick, wipe the tip of the oil dipstick with clean cloth.
- Keep the motorcycle on level ground and hold it vertically.
- Insert the oil dipstick without tightening.
- Remove the oil dipstick and inspect the oil level. The oil level should be between the lower line and upper line of the oil dipstick. If the oil level is below lower line, fill with oil to the upper line. If oil level exceeds upper line, drain some oil until below upper line.



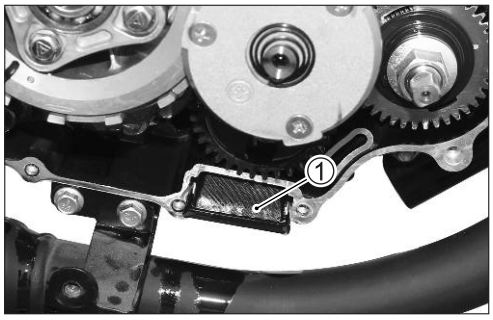


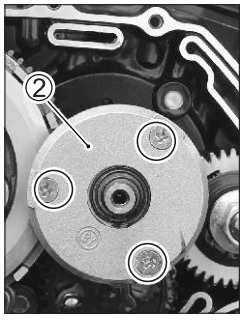


OIL SUMP FILTER AND OIL CENTRIFUGAL FILTER

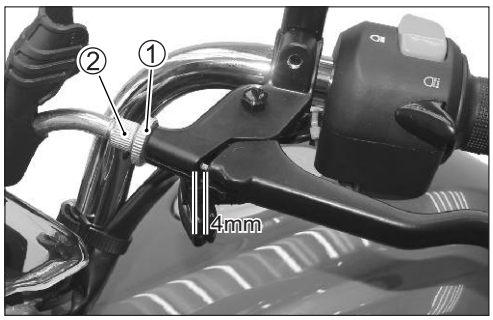
Clean every 18 months or 12,000 km

- Drain the engine oil.
- Remove the clutch cover. (Refer to page 3-21)
- Remove the oil sump filter ①.
- Check and clean oil sump filter, if it is damaged, replace it with a new one.
- After cleaning, reinstall the oil sump filter.
- Remove the oil centrifugal filter cover ②.
- Clean the oil centrifugal filter rotor and cover with a clean cloth.
- Reinstall the oil centrifugal filter cover after cleaning.
- Install the clutch cover.(Refer to page 3-25)
- Pour new engine oil and check the engine oil level.





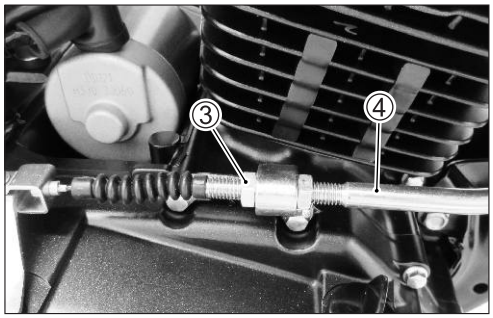
CLUTCH CABLE PLAY



Inspect initial 1 000 km and every 3 000 km.

Clutch play should be 4 mm as measured at the clutch lever holder before the clutch begins to disengage. If the play in the clutch is incorrect, adjust it in the following way:

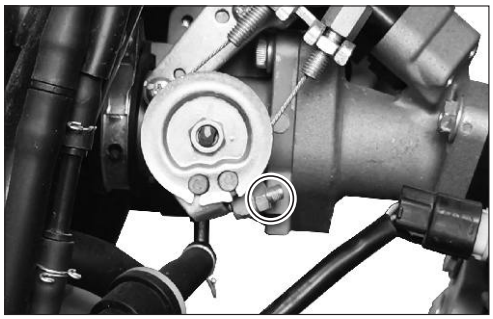
- Remove the jacket, loosen the lock nut ①and turn the adjuster ② on the clutch lever holder all the way in.
- Loosen clutch cable adjuster lock nut ③.
- Turn the clutch cable adjuster ④ in or out to acquire the specified play.
- Use the adjuster ② for minor adjustment.
- Tighten lock nut ① and ③ while holding the adjuster in position.



⚠ CAUTION

Excessive clutch cable play may cause clutch dragging and clutch and gear shifting mechanism wear and damage.

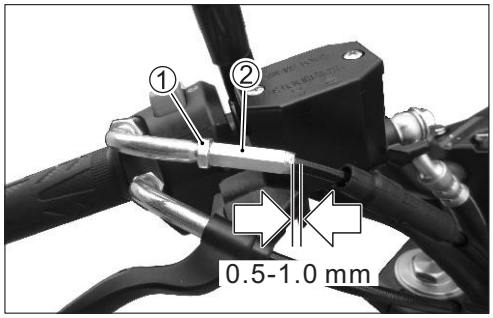
THROTTLE BODY



Clean every 6 000 km.

- The throttle position screw is set precisely at factory. Do not make adjustment to it. Check the idle speed of motorcycle, if the idle speed is not stable, clean as necessary. (Refer to page 5-8)

THROTTLE CABLE PLAY



Inspect initial 1 000 km and every 3 000 km

Adjust the throttle cable play as follows.

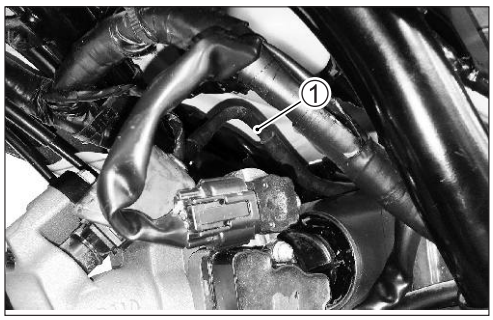
- Loosen the adjuster lock nut ①.
- Turn the adjuster ② so that the throttle cable play has 0.5 -1.0 mm play.
- Tighten the lock nut after adjustment.

⚠ WARNING

After the adjustment is completed, check that handlebar movement does not raise the engine idle speed and that the throttle grip returns smoothly and automatically.

FUEL HOSE

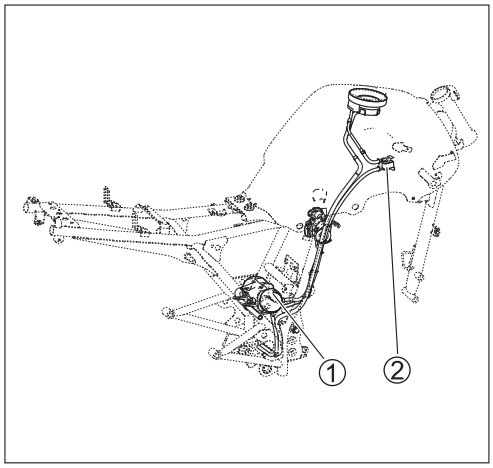
Inspect initial 1 000 km and every 3 000 km, replace every 10 years.



Check the fuel hose ① for leakage, cracks, wear and damage. If any defects are found, replace the fuel hose with a new one.

EVAPORATION EMISSION CONTROL SYSTEM

Inspect every 3 000 km



This motorcycle is equipped with evaporation emisssion control system. Inspect the following items periodically.

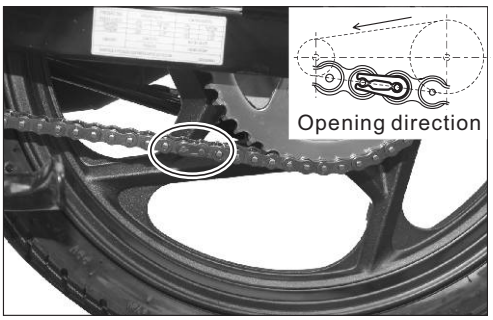
- Inspect the connection of each hose.
- Inspect each hose and EVAP canister ①bodies for damage, if any defects are found, replace the hose and canister with a new one.
- Check each hose, tilt stop valve ②and EVAP canister to see whether they are clogged, clean or replace the clogged part as necessary.

⚠ WARNING

If fuel evaporation emission control system is required to be inspected and servicing, we strongly recommend you do this job in a qualified service shop.

DRIVE CHAIN/SPROCKET

Inspect, clean & lubricate for every 1,000 km



Inspection

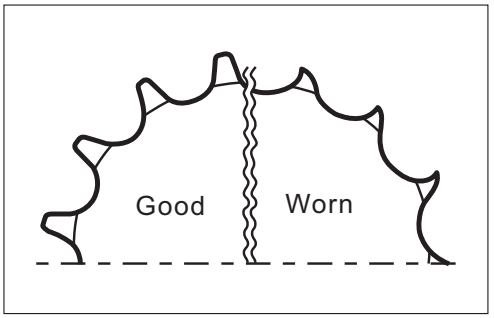
When inspecting the drive chain, look for the following:

1. Loose pins 2. Damage rollers
3. Rusted links 4. Twisted or seized links
5. Excessive wear

If any defects are found, the drive chain must be placed.

Damage to the drive chain means that the sprocket may also be damaged. Inspect the sprocket for the following:

1.Excessively worn teeth
2.Broken or damaged teeth
3.Loose sprocket mounting nuts

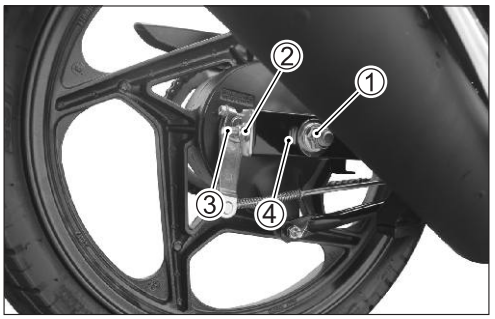


Cleaning and Lubrication

After throughly washing the drive chain with detergent liquid and allowing it to dry, lubricate the drive chain with chain lube or new engine oil.

Drive Chain Adjustment

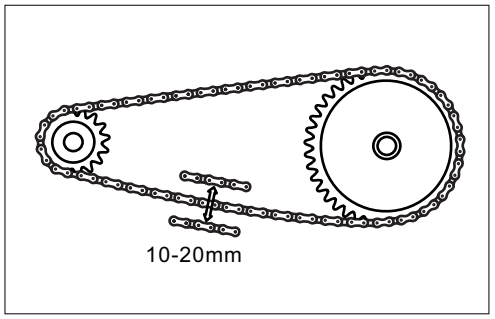
Adjust the drive chain slack to the proper specification.
The drive chain may require more frequent adjustment than periodic maintenance schedule depending upon your riding conditions.



⚠ WARNING

Too much chain slack can cause the chain to come off the sprockets, resulting in an accident or serious damage to the motorcycle. Inspect and adjust the drive chain slack before each use.

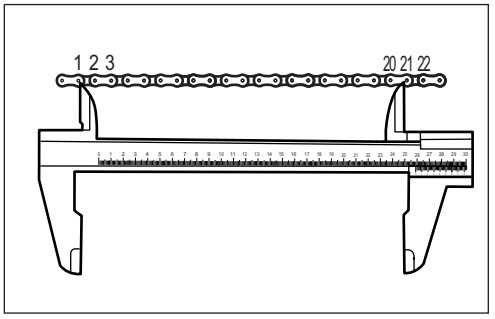
- Support the motorcycle with the main stand.
- Loosen rear axle nut①.
- Loosen or tighten both chain adjuster nuts ② till there is 10-20mm of slack at the middle of the chain between engine and rear sprocket. The mark④ on the chain adjuster③ must be at the same position on the scale to ensure that front and rear wheels are correctly aligned.
- After adjusting the drive chain, tighten rear wheel axle nut to the specified torque.
- Check the rear brake pedal height. (Refer to page 2-14)



Rear axle nut: 50-80 N•m

Drive chain pitch length

Count out 21 pins of the chain and measure the distance between. If the distance exceeds 259.4 mm, the chain must be replaced.



DATA **Drive chain 20 pitch length limit: 259.4 mm**

BRAKES

Brakes:
Inspect initial 1 000 km and every 3 000 km
Brake hose:
Inspect initial 1 000 km and every 3 000 km
Replace the brake hose every 4 years.
Brake fluid:
Inspect every 3 000km,
Replace the brake fluid every 2 years.

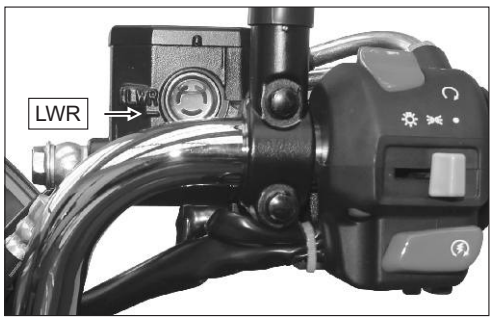
BRAKE FLUID LEVEL

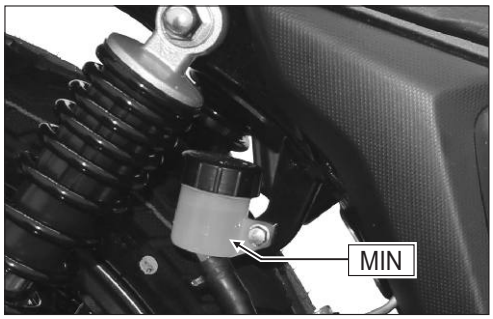
- Support the motorcycle body on the center stand, and place the handlebars straight.
- Check the brake fluid level by observing the “LWR” or “MIN” line on the brake fluid reservoir.
- When the level is below the “LWR” or “MIN” line, replenish with brake fluid that meets the following specification.

DOT3 or DOT4

⚠ WARNING

Brake fluid is harmful or fatal if swallowed, and harmful if it comes in contact with skin or eyes. If brake fluid is swallowed, immediately contact a poison control center or a physician. If brake fluid gets in eyes, flush eyes with water and seek medical attention.
The brake system of this motorcycle is filled with a glycol-based brake fluid.
Do not use or mix different types of fluid.
Do not use any brake fluid taken from old, used or unsealed containers.
Never re-use brake fluid left over from the last servicing or stored for a long period.
Do not wash the master cylinder with high pressure water.

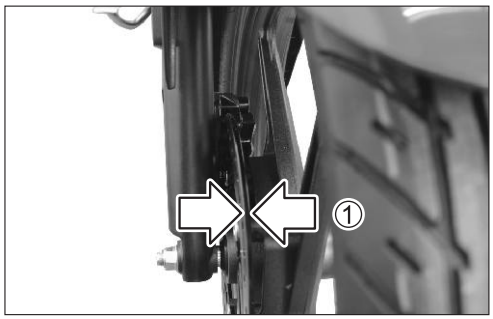




BRAKE DISC AND BRAKE PADS

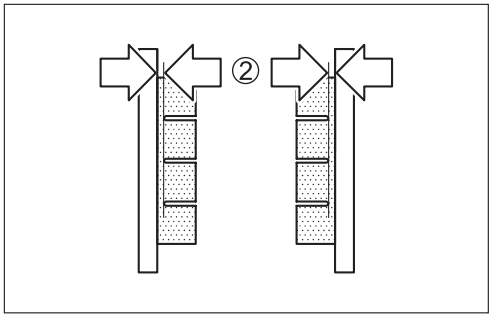
FRONT BRAKE

Check the thickness① of front brake disc, replace the front brake disc with new one if the thickness is less than 3.5mm. The extent of brake pad wear can be checked by observing the grooved limit line ② on the pad. When the wear exceeds the grooved limit line, replace the pads with new ones.



⚠ CAUTION

Replace the brake pad with a set otherwise brake performance will be adversely affected.

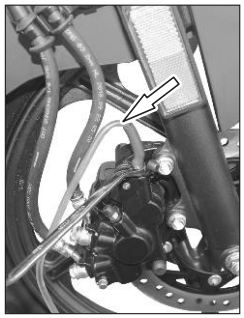


AIR BLEEDING THE BRAKE FLUID CIRCUIT

Air trapped in the fluid circuit acts like a cushion to absorb a large proportion of the pressure developed by the master cylinder and thus interferes with the full braking performance of the caliper brake. The presence of air is indicated by "sponginess" of the brake lever and also by lack of braking force. Considering the danger to which such trapped air exposes the machine and rider, it is essential that, after remounting the brake and restoring the brake system to the normal condition, the brake fluid circuit be purged of air in the following manner:

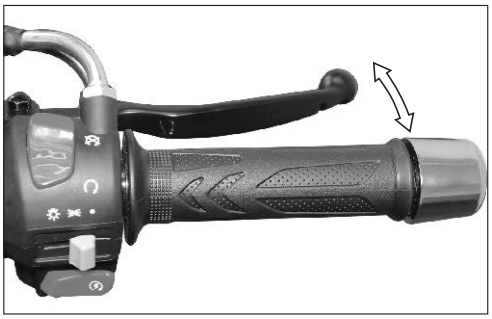
⚠ CAUTION

Spilled brake fluid can damage painted surfaces and plastic parts.
Be careful not to spill any fluid when filling the brake fluid reservoir. Wipe spilled fluid up immediately.



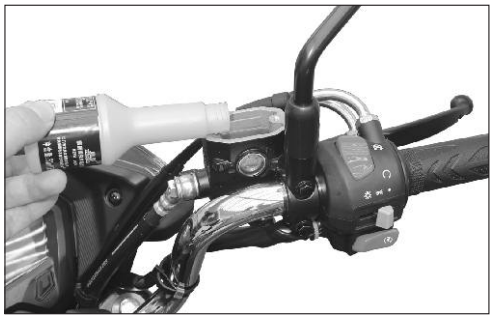
FRONT BRAKE

- Fill up the master cylinder reservoir to the upper line.
- Attach a hose to the air bleeder valve and insert the free end of the hose into a receptacle.
- Squeeze and release the brake lever several times in rapid succession and squeeze the lever fully without releasing it.
- Loosen the air bleeder valve by turning it a quarter of a turn so that the brake fluid runs into the receptacle. This will remove the tension of the brake lever causing it to touch the handlebar grip. Then, close the air bleeder valve, pump and squeeze the lever, and open the valve.
  Repeat this process until fluid flowing into the receptacle contains no air bubbles.
- While bleeding the brake system, replenish the brake fluid in the reservoir as necessary. Make sure that there is always some fluid visible in the reservoir.



- Close the air bleeder valve and disconnect the pipe.
- Tighten the bleeder valve to the specified torque.

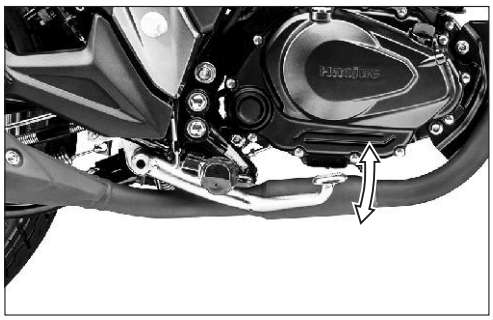
Bleeder valve: 7-10 N•m



- Fill the reservoir to the upper level line, install the reservoir cap.

CBS SYSTEM

- Fill the rear master cylinder reservoir to the upper level.
- Attach a hose to the front brake caliper rear air bleeder valve and insert the free end of the hose into a receptacle.



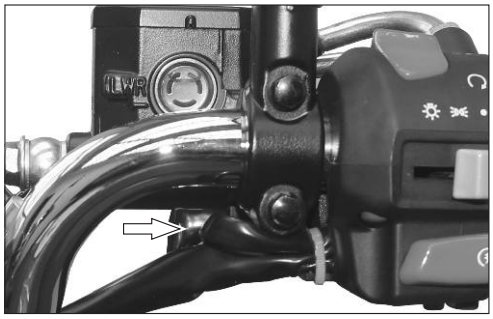
- Depress and release the brake pedal several times in rapid succession and depress the pedal fully without releasing it.
- Loosen the air bleeder valve by turning it a quarter of a turn so that the brake fluid runs into the receptacle. Then, close the air bleeder valve, release and depress the pedal, and open the valve. Repeat this process until fluid flowing into the receptacle contains no air bubbles.
- Replenish the brake fluid reservoir as necessary while bleeding the brake system. Make sure that there is always some fluid visible in the reservoir.

- Close the air bleeder valve and disconnect the pipe.
- Tighten the bleeder valve to the specified torque.

Bleeder valve: 7-10 N•m

- Fill the reservoir between “MAX” and “MIN” level line, then install the reservoir cap.

FRONT BRAKE LIGHT SWITCH



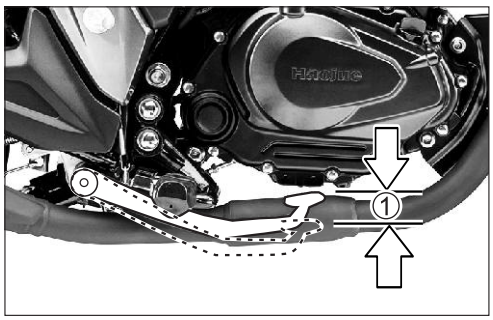
Squeeze the front brake lever to inspect the front brake light switch. If the brake light switch works abnormally, it may be inspected, adjusted or replaced.

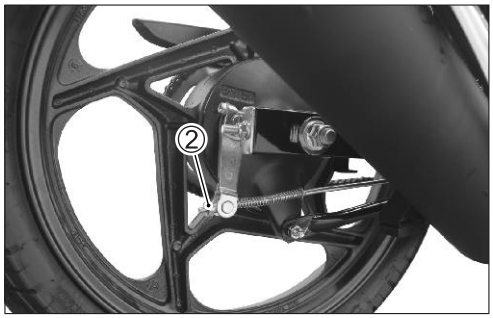
REAR BRAKE

REAR BRAKE PEDAL FREE TRAVEL

Rear brake pedal travel ① is the distance from the free position to the position that the brake starts taking effect. Turn the adjuster nut ② to adjust the rear brake pedal free travel to 20-30 mm.

DATA **Rear brake pedal free travel ①: 20-30 mm**

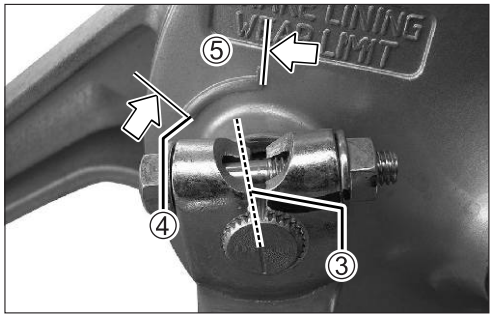




BRAKE LINING WEAR LIMIT

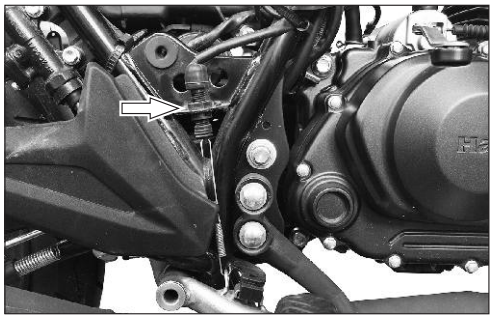
This motorcycle is equipped with brake lining wear limit indicators on front brake. To check wear of the break lining, follow the steps below.

- Check if the brake system is properly adjusted first.
- While operating the brake, check to see the extension line ③ from the index mark ④ is within the range ⑤ on the brake panel as shown in the illustration at right.
  If the index mark is outside the range, the brake shoe assembly should be replaced to ensure safe operation.



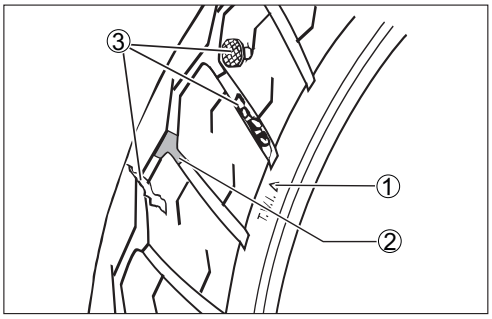
REAR BRAKE LIGHT SWITCH

Step on the rear brake pedal to inspect the brake light switch. If the brake light switch works abnormally, it may be inspected, adjusted or replaced.

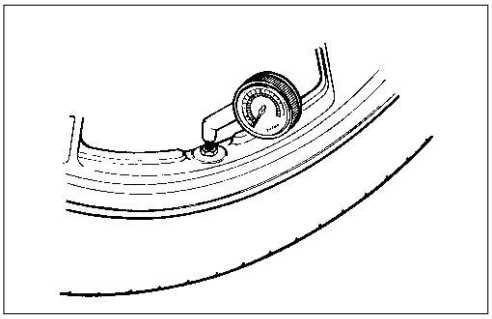


TIRES

Inspect initial 1 000 km and every 3 000 km



Check the surface and tire pressure during the maintenance.
There are serial marks ①"T.W.I." (tire wear indicator) in the edge of the tire. Check the block ② of T.W.I. in the tread near the mark. If the tire wears off to reach the block, the tire should be replaced.
Check the damage ③(punctures or fractures) on the tire surface visually. As surface damage may impede driving stability, such tires should be replaced.



COLD INFLATION TIRE PRESSURE	SOLO RIDING		DUAL RIDING	
	kPa	kgf/cm^2	kPa	kgf/cm^2
FRONT	175	1.75	175	1.75
REAR	225	2.25	225	2.25

⚠ CAUTION

Proper tire pressure and condition affect vehicle performance. Please check the tire pressure and surface condition periodically.。

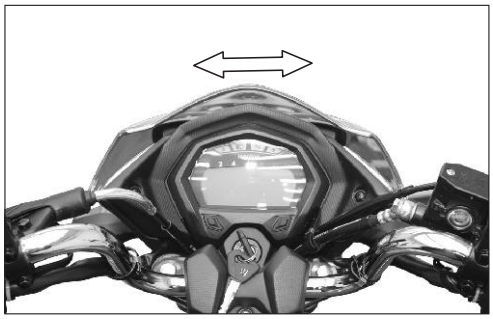
DYNAMIC WHEEL BALANCE

Process the dynamic wheel balance after tire replacement or repair.

DATA **Dynamic wheel balance: ≤ 8 g**

STEERING

Inspect Initial 1 000 Km and every 3 000 Km



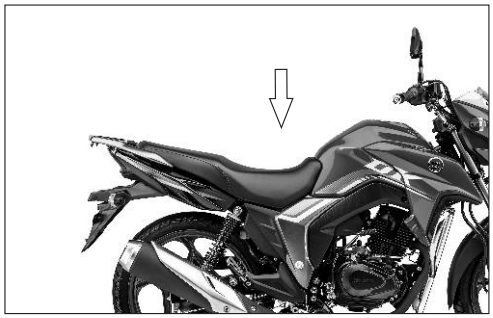
Steering stem bearings should be adjusted properly for smooth turning of the handlebars and safe running.
Steering which is too stiff prevents smooth movement of handlebars.
Steering which is too loose will cause vibration and damage to the steering bearing. Check to see that there is no play in the front fork attachment.
If the play is found, perform the steering bearing adjustment as described in page 5-17 of this manual.

FRONT AND REAR SHOCK ABSORBERS

Inspect every 6 000 km.

FRONT SHOCK ABSORBER

- Grasp the front brake lever, squeeze the front shock absorber to check if it moves smoothly.
- Check for leaks or damage.



REAR SHOCK ABSORBER

- Depress the rear carrier, move the rear shock absorber up and down four or five times to check for the noise and smooth movement.
- Inspect the rear absorber spring for damage, deformation or oil leakage.

LIGHTING AND SIGNAL

Inspect initial 1 000 km and every 3 000 km

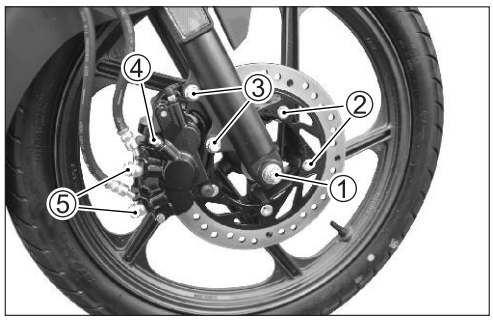
Inspect the headlight, left and right turn signal lights, tail light/brake light and combination meter signal light, replace the light with new one if there is something unusual. (Refer to page 6-7)

CHASSIS AND ENGINE MOUNTING BOLTS AND NUTS

Inspect Initial 1 000 km and every 3 000 km

The nuts and bolts listed are important parts, and they must be in good condition for safety. They must be re-tightened, as necessary, to the specified torque with a torque wrench.

NO.	ITEM		N•m	Kg•m
1	Front axle nut		36-52	3.6-5.2
2	Brake disc mounting bolt		18-28	1.8-2.8
3	Brake caliper mounting bolt		18-28	1.8-2.8
4	Air bleeder valve		7-10	0.7-1.0
5	Brake hose mounting bolt		20-25	2.0-2.5
6	Master cylinder mounting bolt		8-12	0.8-1.2
7	Handlebar clamp upper bolt (whole handlebar)		12-20	1.2-2.0
8	Upper bracket clamp bolt		35-55	3.5-5.5
9	Steering stem lock bolt		25-35	2.5-3.5
10	Rear shock absorber mounting bolt/nut		22-35	2.2-3.5
11	Rear axle nut		50-80	5.0-8.0
12	Rear brake cam nut		8-12	0.8-1.2
13	Rear link nut		10-16	1.0-1.6
14	Rear swing arm shaft nut		50-80	5.0-8.0
15	Rear sprocket nut		30-37	3.0-3.7
16	Engine hanging plate mounting nut		22-33	2.2-3.3
17	Engine mounting bolts	Front	33-39	3.3-3.9
18		Rear	50-60	5.0-6.0
19	Exhaust pipe bolt		11-15	1.1-1.5
20	Muffler mounting bolt		20-26	2.0-2.6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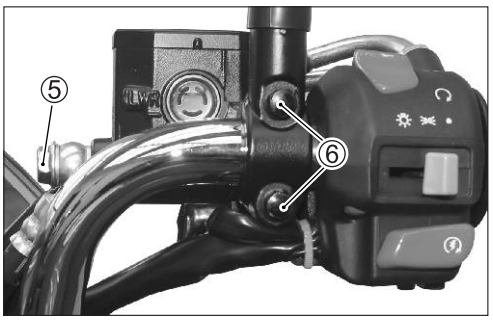
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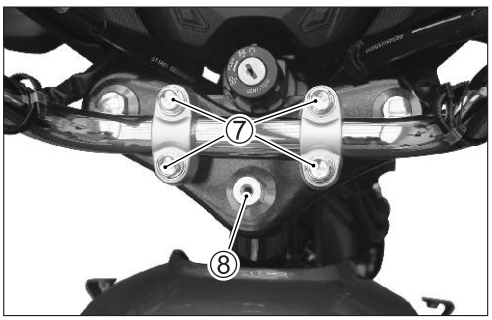
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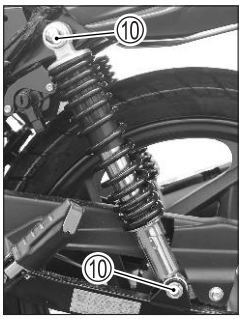
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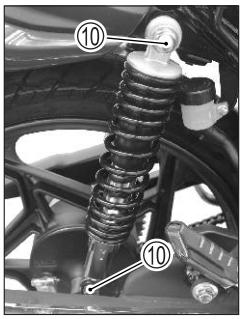
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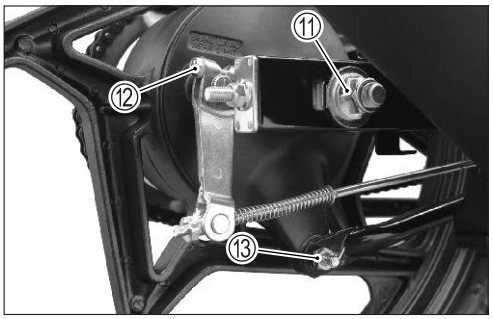
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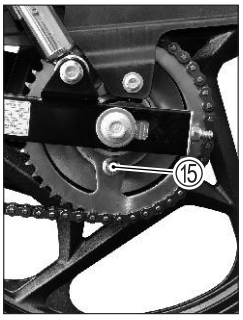
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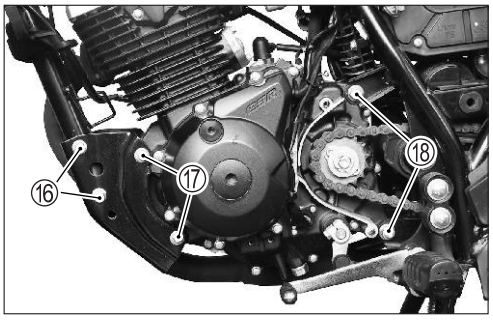
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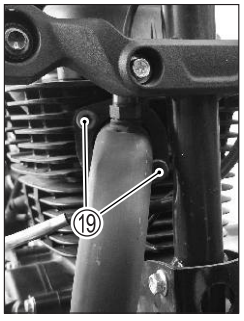
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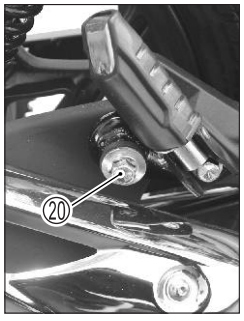
⑳

COMPRESSION PRESSURE

The cylinder compression is an indicator of its inner state. The necessity of repair depends on the result of the inspection. The record of your authorized dealer should include the compression pressure readings obtained in every maintenance.

NOTE:

* *Before testing the engine for compression pressure, make sure that the cylinder head nuts and bolts are tightened to specified torque values and valves clearance are properly adjusted.*
* *Have the engine warmed up by idling before testing it.*
* *Ensure that the battery is fully charged.*

Remove the related parts and inspect the compression pressure in the following process:

- Warm up the engine.
- Remove the spark plug. (Refer to page 2-7)
- Fit the pressure gauge and adapter to the spark plug hole, paying attention to the connector tightening.
- Turn the throttle to full opening.
- Electrically start the motorcycle, take the maximum reading in the recording as the cylinder compression pressure.

09915-64510: Compression gauge

Compression pressure

Standard	Limit
1 000-1 400 kPa 10-14 kgf/cm^2	800 kPa 8 kgf/cm^2

When the compression pressure noted is down to or below the limit indicated above, the engine must be disassembled, inspected and repaired as required.

A low compression pressure may indicate any of the following malfunctions:

* Excessively worn cylinder wall.
* Worn piston or piston rings.
* Piston rings stuck in the grooves.
* Poor seating contact of the valves.
* Defective the cylinder head gasket.

ENGINE

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REMOVING ENGINE PARTS WITH ENGINE IN PLACE

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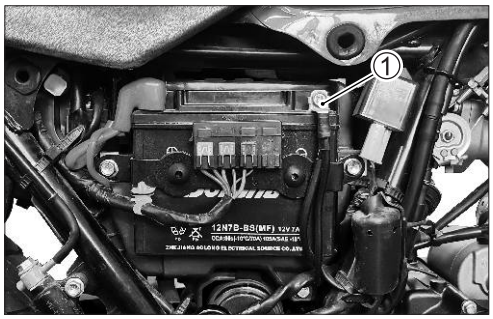
ENGINE REMOVAL AND REASSEMBLY

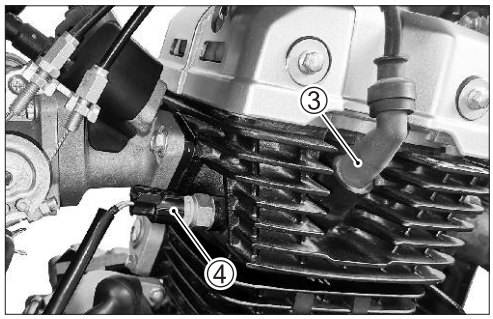
ENGINE REMOVAL

Before taking the engine out of the chassis, wash the engine with a steam cleaner. The procedure of engine removal is sequentially explained in the following steps:

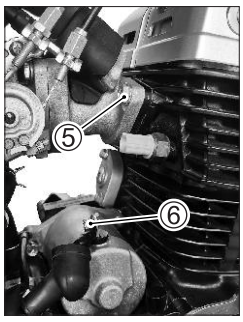


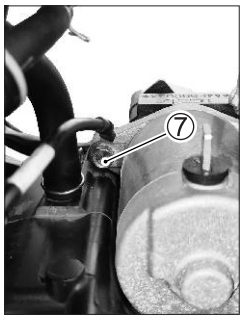
- Remove the saddle, the left side covers and the fuel tank left decoration cover, remove the right side covers and fuel tank right decoration cover. (Refer to page 6-1)
- Remove the battery ⊖ lead wire ①.
- Loosen the clutch cable lock nut and adjuster nut, take off the clutch cable② from the clutch lever.

- Disconnect the spark plug cap ③ and engine temp. sensor coupler ④.

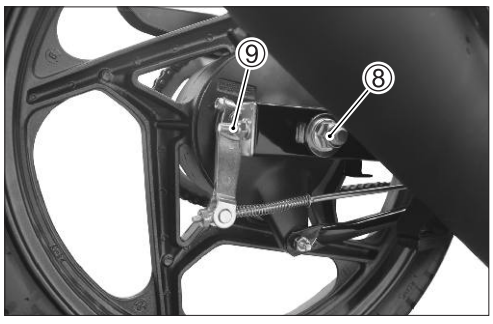


- Remove the throttle body right bolt ⑤.
- Remove the starter motor wire ⑥.
- Remove the engine ground wire ⑦.

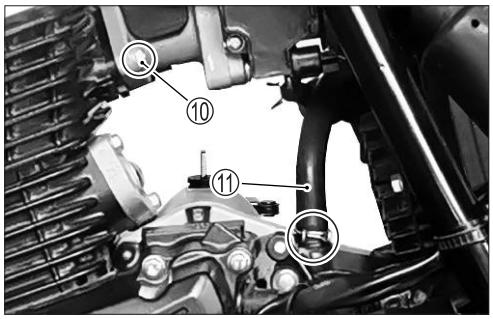




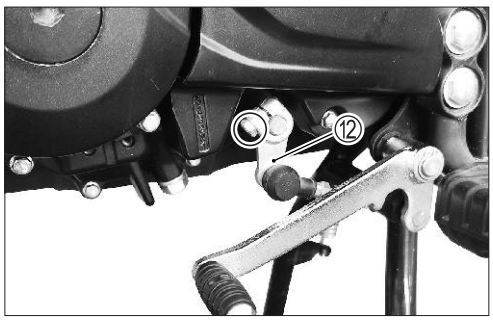
- Loosen the rear axle nut ⑧ and chain adjuster bolt ⑨.



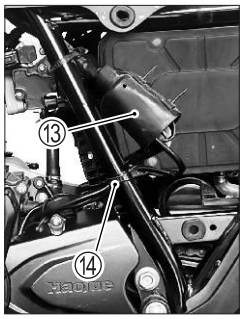
- Remove the throttle body left bolt ⑩.
- Remove the engine breath tube ⑪.

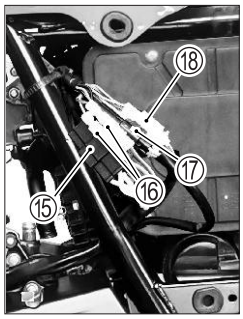


- Remove the gear shifting lever rocker arm ⑫.

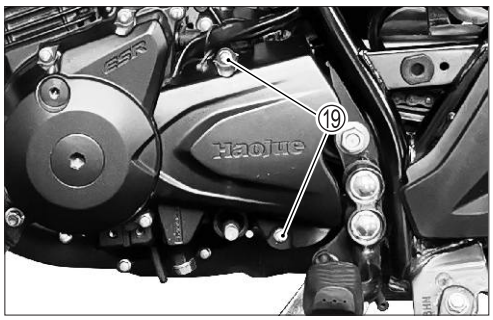


- Remove the PVC jacket ⑬ and the clamp ⑭.
- Disconnect the generator charge coupler ⑮, CKPS couplers ⑯, speed sensor coupler ⑰, gear position switch coupler ⑱.

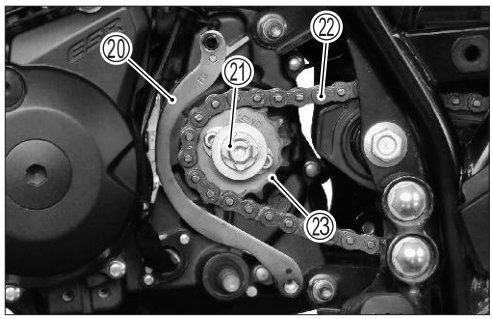




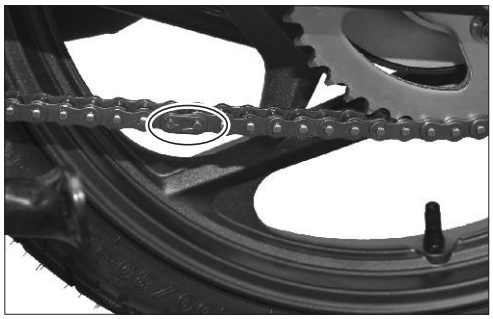
- Remove the rear sprocket cover ⑲ with the speed sensor wire.



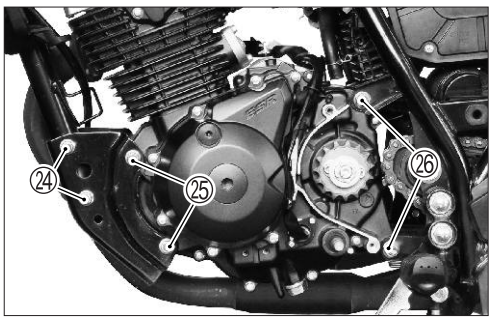
- Remove the retainer ⑳.
- Depress the rear brake pedal, remove the speed sensor ㉑, drive chain ㉒ and engine sprocket ㉓.



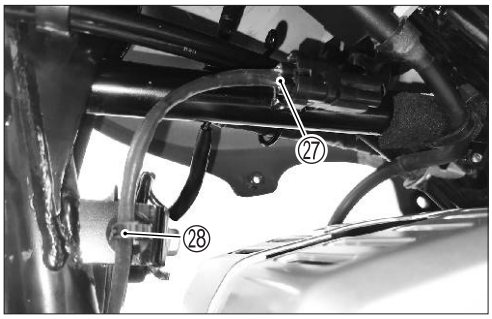
- Remove the drive chain clip.



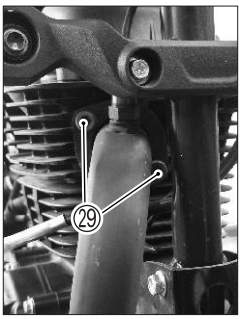
- Remove the engine front mounting nuts ㉔and engine front hanging plate nuts ㉕.
- Remove the engine rear mounting nuts ㉖.



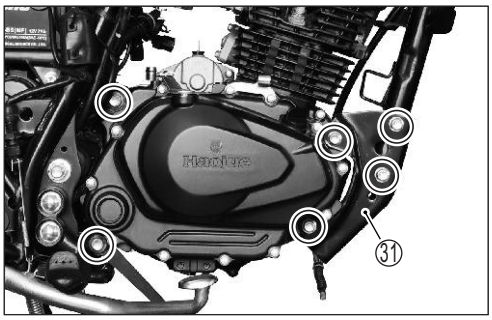
- Disconnect the oxygen sensor coupler ㉗.
- Loose the oxygen sensor wire clamp ㉘.



- Remove the exhaust pipe bolts ㉙.
- Remove the muffler mounting bolt and nuts, take off the muffler ㉚.



- Draw out the engine front mounting bolts and remove the engine front hanging plate ㉛.
- Draw out the engine rear mounting bolts, take off the engine.

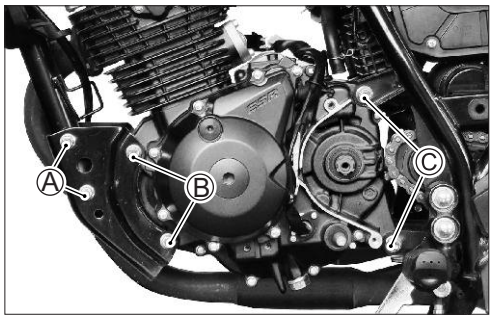


ENGINE REMOUNTING

The engine can be mounted in the reverse order of removal.

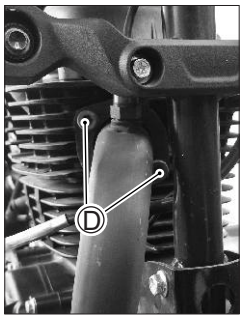
NOTE:

The engine mounting nuts are self-lock nuts. Once the nut has been removed, it is no longer of any use. Be sure to use new nuts and tighten them to the specified torque.



Tightening torque for engine mounting bolts:

Engine front hanging plate nut Ⓐ : 22-33 N•m
Engine front mounting nut Ⓑ : 33-39 N•m
Engine rear mounting nut Ⓒ: 50-60 N•m
Exhaust pipe bolt Ⓓ : 11-15 N•m
Muffler mounting bolt Ⓔ : 35-45 N•m



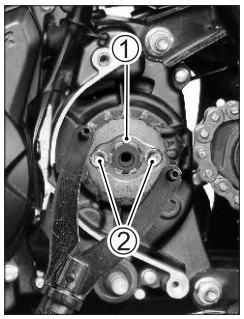
- Hold the sprocket with special tool, install the sprocket retainer ① and the speed sensor rotor ③, apply THREAD LOCK to the mounting bolts, tighten the bolts to the specified torque.

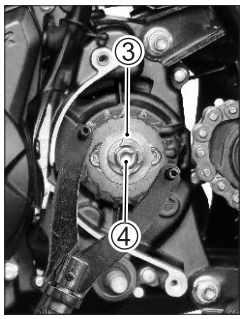
99000-32050 : THREAD LOCK "1342"

Engine sprocket bolts ②: 8-12 N•m

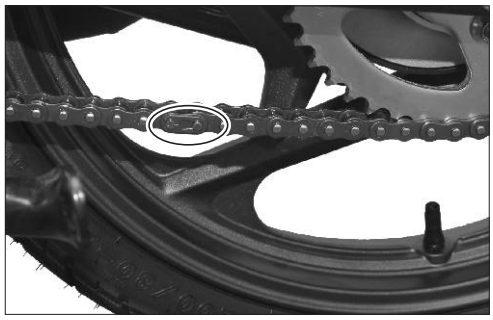
Speed sensor rotor bolt ④: 18-28 N•m

09930-40113 : Sprocket holder





- When install the drive chain clip, the clip opening faces to reverse direction of traveling as shown.

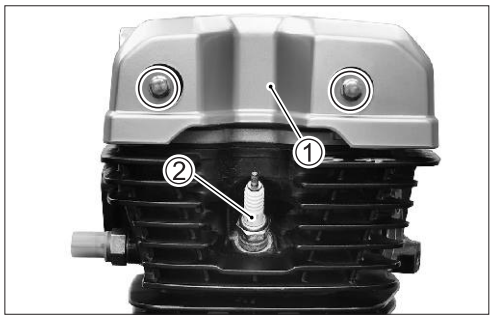


- After assembling the engine, add the engine oil and check the engine oil level. (Refer to page 2-7)
- After remounting the engine, following adjustments are necessary.
  - ∗ Clutch cable (Refer to page 2-8)
  - ∗ Idle (Refer to page 2-8)
  - ∗ Throttle cable (Refer to page 2-8)
  - ∗ Drive chain (Refer to page 2-10)

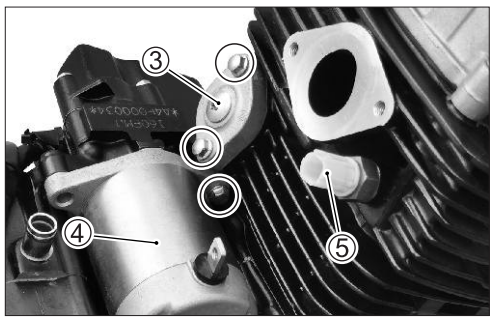
CYLINDER HEAD AND VALVE

REMOVAL

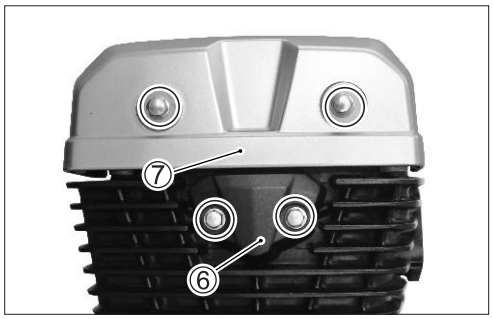
- Drain the engine oil, remove the engine. (Refer to page 3-2)
- Remove the cylinder head right shell ① and spark plug ②.



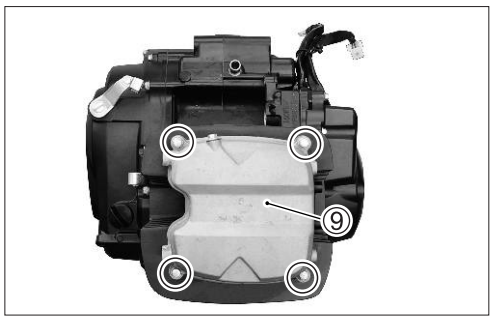
- Remove the camshaft chain tensioner ③ by removing the mounting bolts.
- Remove the starter motor ④ and engine temp. sensor ⑤.



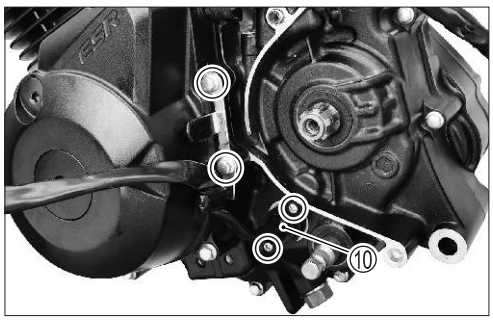
- Remove the cover ⑥ and cylinder head cover left shell ⑦.



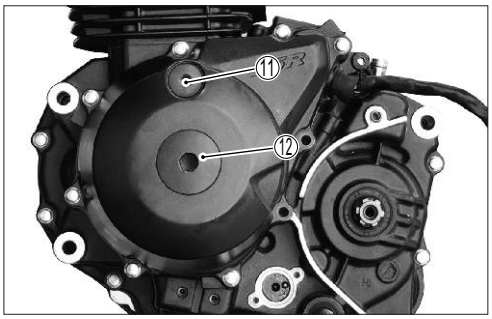
- Remove the cylinder head mounting bolts, take off the cylinder head cover ⑨ and gasket.



- Remove the gear position sensor ⑩ by removing mounting bolts and screws.

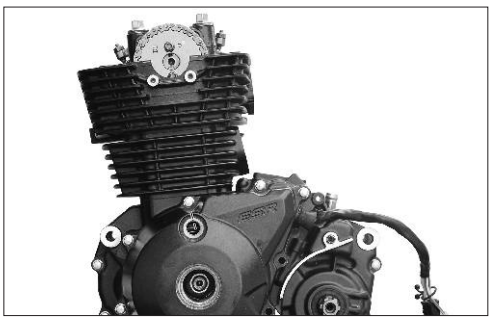


- Remove the valve timing inspection plug ⑪ and generator cover cap ⑫.



- Rotate the generator rotor counter clockwise until the white line on the rotor is aligned with center (mark line) of hole on the crankcase to set the piston at TDC of the compression stroke.

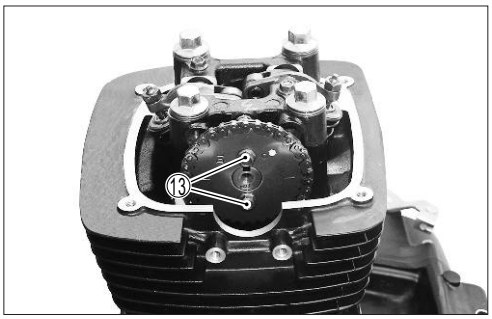
***NOTE*:**
The valve clearance is to be checked when the engine is cold.



- Remove the cam shaft sprocket mounting bolt ⑬, take off the cam shaft sprocket.

⚠ CAUTION

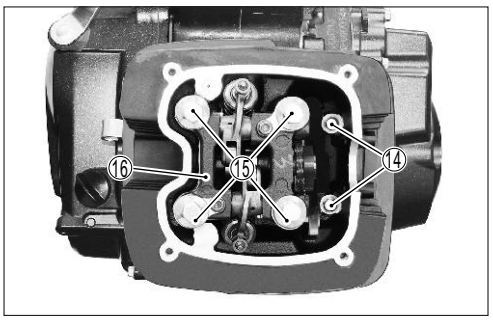
Do not drop camshaft sprocket, drive chain and bolt into the crankcase.



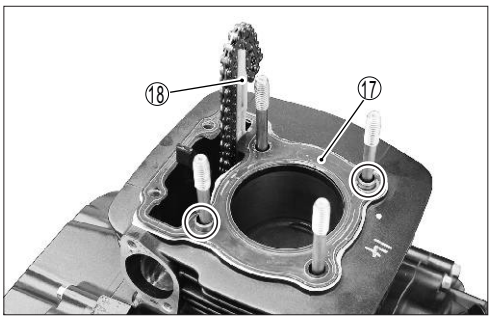
- Remove the cylinder head side bolts ⑭ and cylinder head nuts ⑮, take off the cylinder head assembly ⑯.

⚠ CAUTION

If it is difficult to remove the cylinder head, gently pry it off while tapping the finless portion of the cylinder head with a plastic hammer. Be careful not to break the fins.

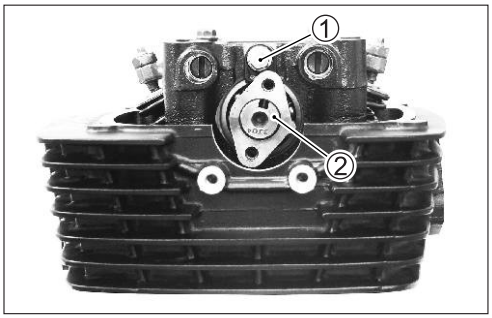


- Remove the cylinder head gasket ⑰, the dowel pins and the cam chain guide ⑱.



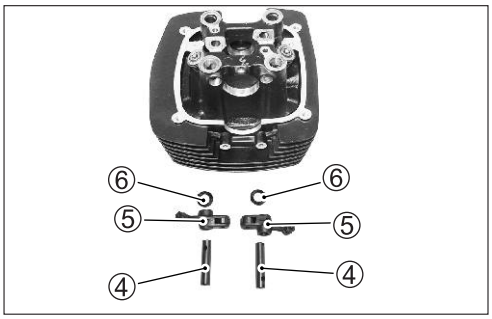
DISASSEMBLY

- Remove the camshaft retainer bolt and retainer① , take off the camshaft ②.



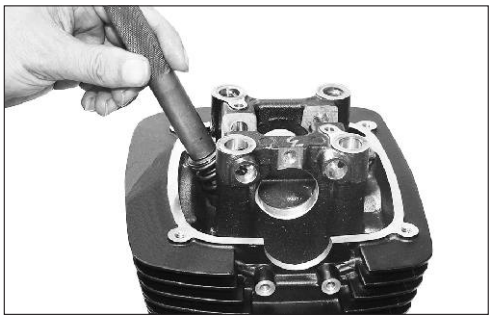
- Remove the rocker arm shaft locating bolt ③.

- Draw out the rocker arm shaft ④, remove the rocker arm ⑤ and flexuous washer ⑥.



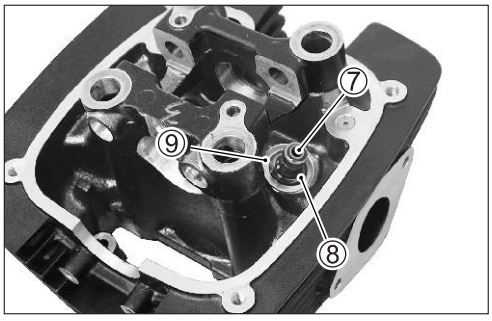
- Take off the valve cotters from the valve stem with the special tool.

TOOL **11F14-018 : Valve installer and remover**



- Take out the valve spring retainer, inner spring and outer spring.
- Pull out the valves from the other side.

- Remove the oil seal ⑦ and spring inner seat ⑧ and outer seat ⑨.



CAUTION

Always use new oil seal.

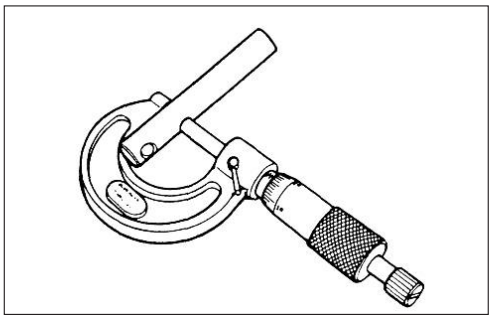
INSPECTION

ROCKER ARM SHAFT

Measure O.D. of the rocker arm shaft.

DATA Rocker arm shaft O.D. (IN. & EX.): 9.972-9.987 mm

TOOL 09900-20205: Micrometer (0~25 mm)

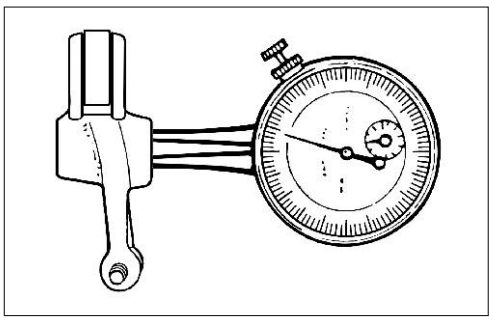


ROCKER ARM

Check I.D. of the rocker arm and wear of bearing's contacting surface.

DATA Rocker arm I.D. (IN. & EX.): 10.0-10.015 mm

TOOL 09900-20605 : Caliper gauge

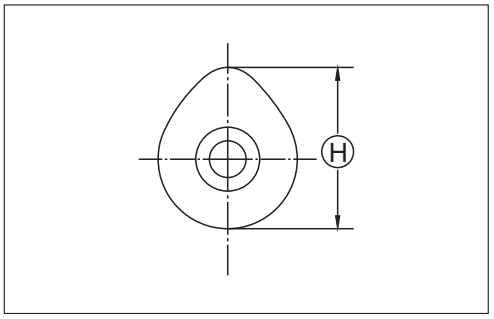


CAMSHAFT

The camshaft should be checked for run-out and also for wear of cams and journals. The cam height Ⓗ can be measured by a micrometer. Replace the camshafts if found it worn down to limit.

DATA CAM HEIGHT Ⓗ limit: IN.: 32.1 mm EX.: 32 mm

TOOL 09900-20202: Micrometer (25~50 mm)

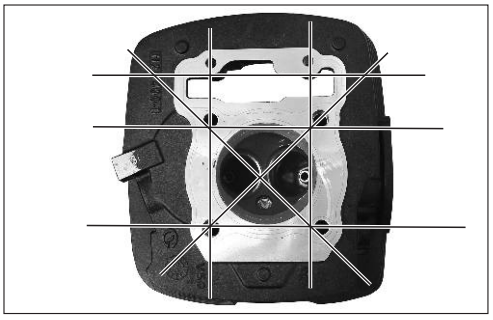


CYLINDER HEAD

- Visual inspect the spark plug hole and valve seat for scratch or wear.
- Check the surface of the cylinder head for distortion with a straightedge and thickness gauge.

DATA Cylinder head distortion limit : 0.05 mm

TOOL 09900-20803 : Thickness gauge

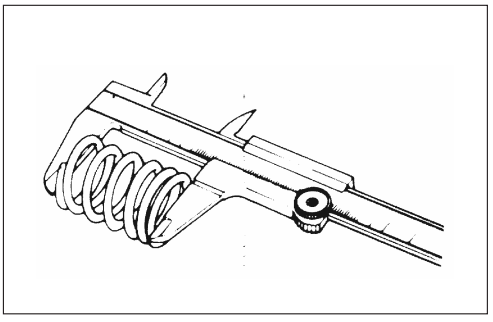


VALVE SPRING

- Measure the valve spring free length.

DATA Valve spring free length limit:
IN.: 34.8 mm EX.: 35.38 mm

TOOL 09900-20103: Vernier calipers



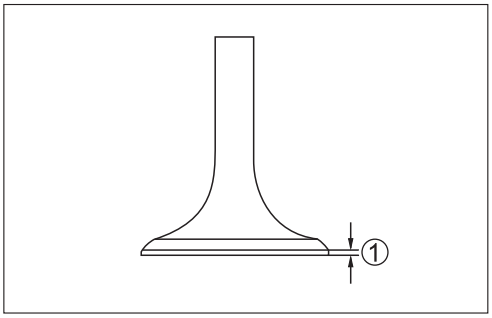
VALVE

- Measure the thickness ①, if the thickness is found to have been reduced to the limit, replace the valve.

DATA **Valve face thickness limit: 0.5 mm**

⚠ CAUTION

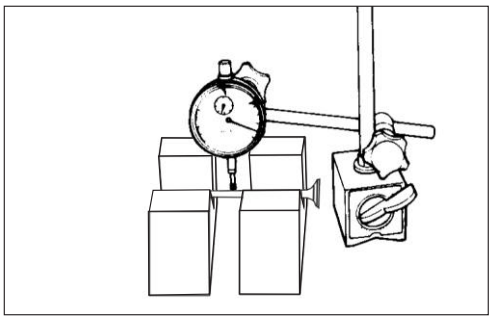
Visually inspect each valve for wear of its seating face. Replace any valve with an abnormally worn face.



- Support the valve with "V" blocks, as shown, and check its run-out with a dial gauge. The valve must be replaced if the run-out exceeds the limit.

DATA **Valve stem run-out limit: 0.05 mm**

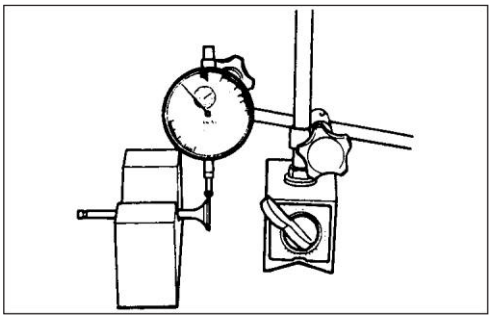
TOOL **09900-20606: Dial gauge (0.1 mm)**
09900-20701: Magnetic stand
09900-21304: V-block (100 mm)



- Place the dial gauge at right angles to the valve head, and measure the valve head radial runout. If measures more than limit, replace the valve.

DATA **Valve head run-out limit: 0.03 mm**

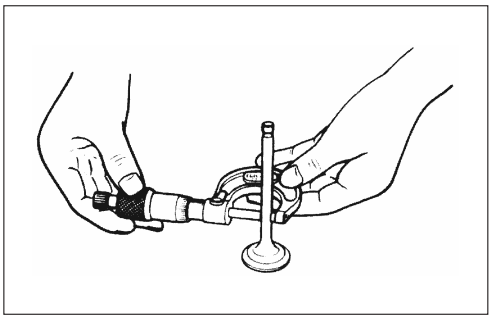
TOOL **09900-20606: Dial gauge (0.1 mm)**
09900-20701: Magnetic stand
09900-21304: V-block (100 mm)



- Check the valve for bend, scrape, ablation or abnormal wear.
- Measure the valve stem O.D..

DATA **Valve stem O.D.:**
IN.: 4.975-4.99 mm EX.: 4.955-4.97 mm

TOOL **09900-20205 : Micrometer (0~25 mm)**

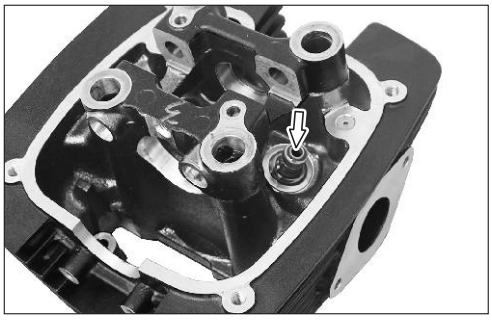


VALVE SEAL

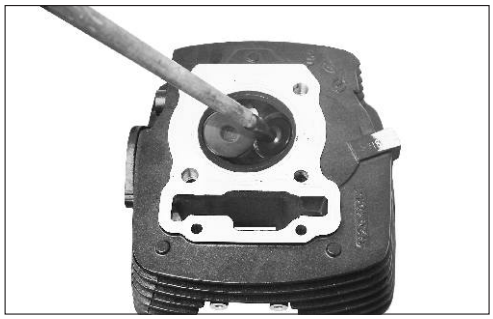
- Oil each seal, and drive them into the position.

⚠ CAUTION

Always use new oil seal.



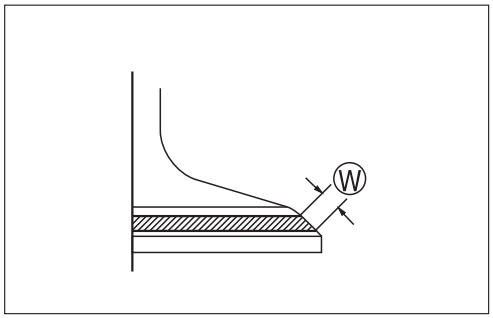
VALVE SEAT WIDTH



Coat the valve seat with prussian blue uniformly. Fit the valve and tap the coated seat with the valve face in a rotating manner, in order to obtain a clear impression of the seating contact. The ring-like dye impression left on the vale face must be continuous-without any break. In addition, the width of the dye ring, which is the visualized seat “width”, must be within the specification.

09916-10911: Valve lapper set

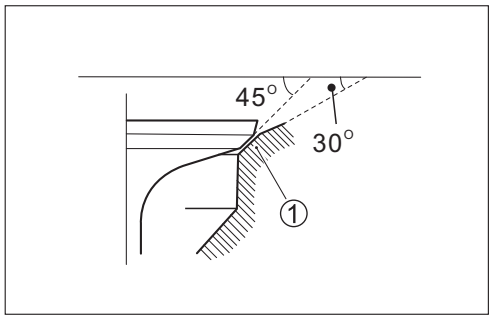
Valve seat width Ⓦ standard: 0.9-1.1 mm



If either requirement is not met, correct the seat by servicing it as follows.

- The valve seats① for both intake and exhaust valve are angles. The valve seat contact surface is 45°.

	INTAKE	EXHAUST
Valve seat angle	30°, 45°	30°, 45°
Valve seat width	0.9-1.1 mm	0.9-1.1 mm
Valve diameter	29.64 mm	25.64 mm
Valve guide I.d.	5.000-5.012 mm	



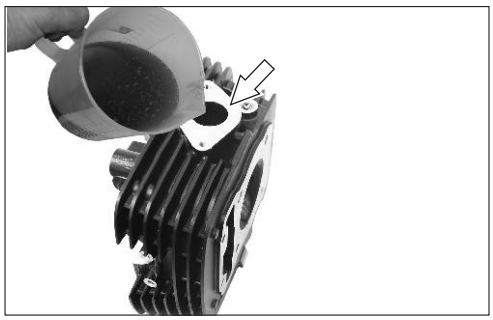
CAUTION

The valve seat contact area must be inspected after each cut. Do not use lapping compound after the final cut is made. The finished valve seat should have a velvety smooth finish and not a highly polished or shiny finish. The will provide a soft surface for the final seating of the valve which will occur during the first few seconds of engine operation.

CAUTION

Be sure to adjust the valve clearance after reassemble the engine.

- Clean and assemble the cylinder head and valve components Fill the intake and exhaust ports with gasoline to check for leaks.
- If any leaks occur, check the valve seat and face for burrs or other things that could prevent the valve from sealing.



WARNING

Always use extreme caution when handling gasoline.

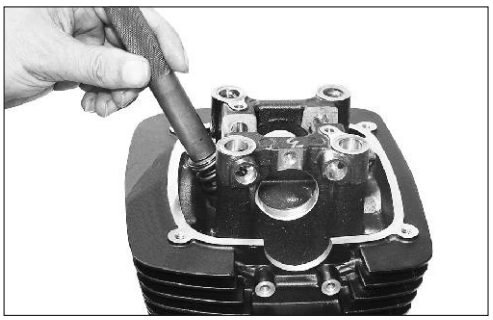
CYLINDER HEAD ASSEMBLY

- Clean the cylinder head with detergent, clean all oil passages with compressed air.
- Clean each valve with Moly paste.
- Insert the valves into valve guide.
- Install the washer, the valve spring and the spring retainer.

99000-25140: Moly paste

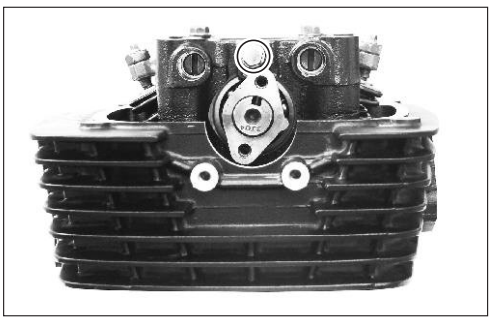
CAUTION

Install each valve spring with the small-pitch portion facing the cylinder head.
Turn the valve slowly while installing the valve oil seal to prevent the valve oil seal from damage.



- Install the valve cotters with special tool.

TOOL **11F14-018: Valve remover and installer**



- Apply a small quantity moly paste to the camshaft journal and place camshaft on the cylinder head.
- Install the rocker arm retainer and tighten the bolt.

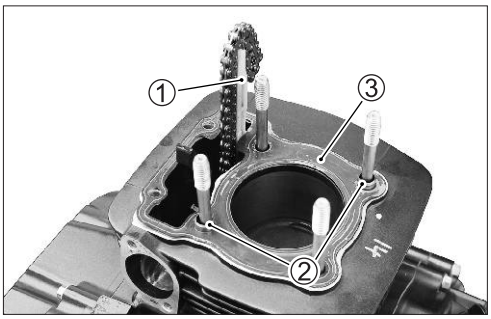
99000-25140: Moly paste

- Oil the rocker arm shaft.
- Mount the rocker arm, the wave washer and the rocker arm shaft to the cylinder head.

- Turn the rocker arm shaft until the screw hole of rocker arm shaft is aligned with the locating bolt hole, then install the rocker arm shaft locating bolts.

REASSEMBLY

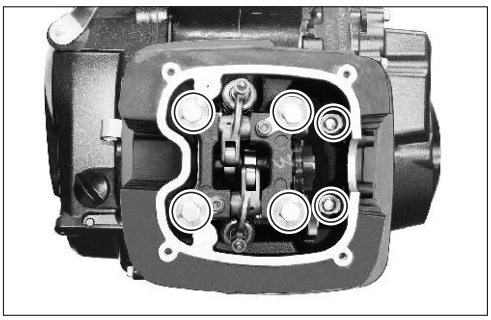
Reassemble the cylinder head in the reverse order of removal.
Pay attention to the following points.



- Install the guide rod ①, dowel pins ② and new gasket ③.

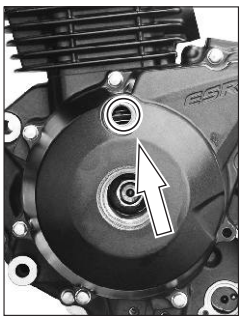
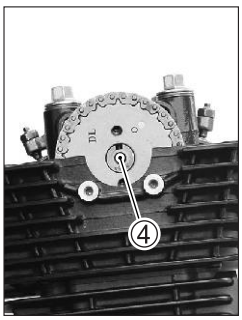
⚠ CAUTION

Always use new gasket.



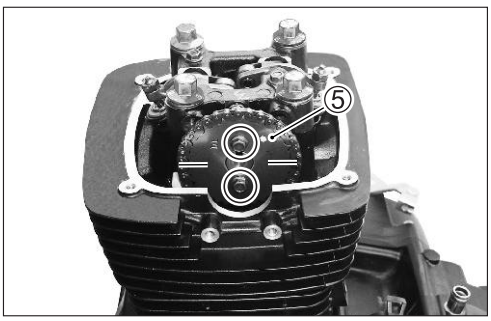
- Instal the four cylinder head nuts, side bolts and washer, tighten the bolts and nuts to the specified torque.

Cylinder head bolts: 30-34 N•m
Cylinder head side bolts: 8-12 N•m

- Turn the generator rotor anticlockwise until the mark line on the rotor is aligned with the mark of hole on the crankcase keeping the camshaft notch ④ faces upright.

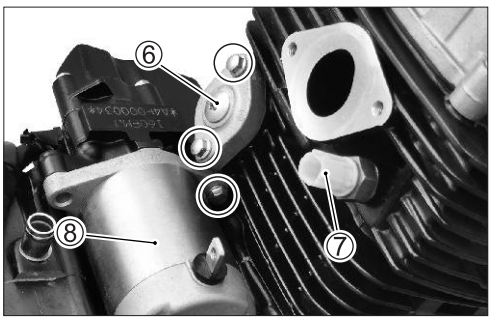
NOTE:
When installing the cylinder head cover, the piston must be at the top dead center on the compression stroke.



- Align the marks ⑤ on the sprocket so it is parallel with the surface of the cylinder head.
- Apply a few THREAD LOCK to the sprocket bolt and tighten the bolt to the specified torque.

99000-32030 : THREAD LOCK "1303"

Cam sprocket bolt:8-10 N•m

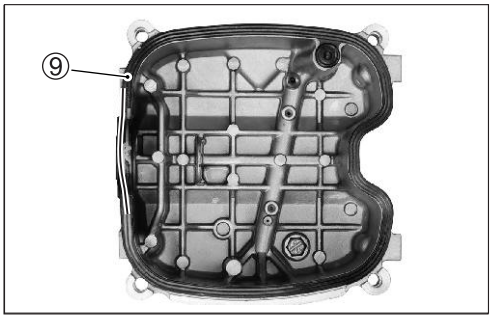


- Use a blade screw driver to turn the cam chain tensioner ⑥ to the end, tighten the bolts to the specified torque.
- Turn the screw driver anti-clockwire to release the tensioner.
- Tighten the engine temp. sensor ⑦ to the specified torque.
- Tighten the starter motor ⑧ bolt to the specified torque.

Chain tensioner mounting bolt: 8-12 N•m
Engine temp. sensor: 13-16 N•m
Starter motor mounting bolt: 8-12 N•m

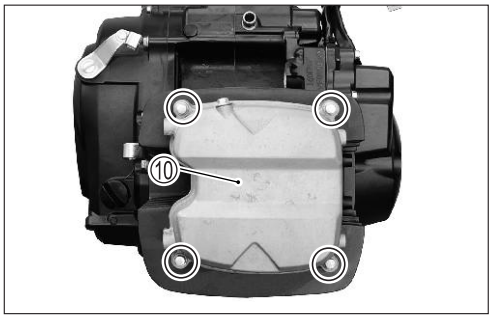
- Inspect the valve clearance. (Refer to page 2-5)
- Apply the bond to half circle of cylinder head cover mating surface ⑨ as shown.

1207B **99000-31140: Bond "1207B"**



- Install the cylinder head cover ⑩, tighten the cylinder head cover bolts to the specified torque.

Cylinder head cover bolt: 8-12 N•m



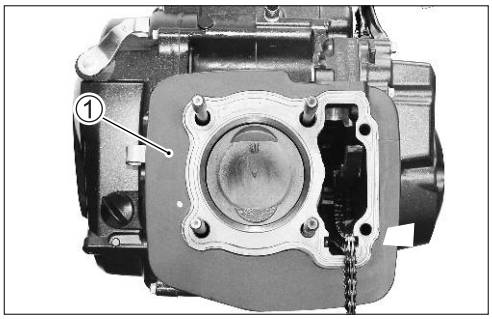
- Install the spark plug ⑪, tighten the spark plug to the specified torque.

Spark plug: **10-12 N•m**

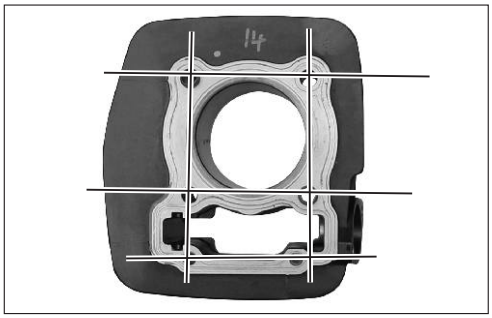
- Install the engine. (Refer to page 3-5)

CYLINDER AND PISTON

CYLINDER REMOVAL AND INSPECTION



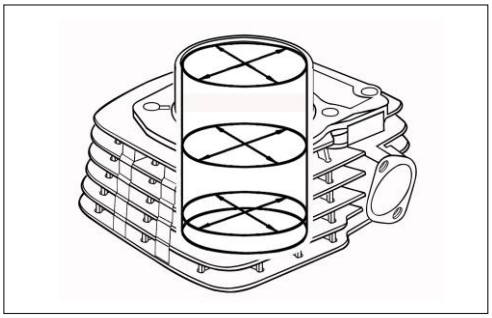
- Remove the cylinder head and cam chain guide. (Refer to page 3-7)
- Remove the cylinder ①.



- Remove the remaining gasket on the cylinder surface.
- Inspect the surface of cylinder for distortion with a straight edge and thickness gauge.

DATA Cylinder distortion limit : 0.05 mm

TOOL 09900-20803 : Thickness gauge



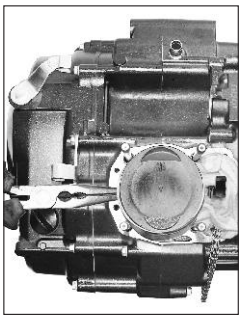
- Check the cylinder bore inner wall for scratches or wear.
- Measure the cylinder bore inner diameter at six places. If any one of measurements exceeds the limit, overhaul the cylinder and replace the piston with an oversize, or replace the cylinder.

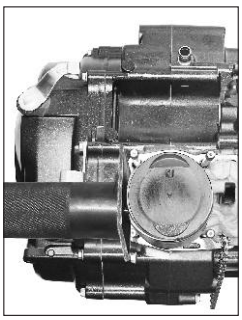
DATA Cylinder bore I.D. limit: 60.13 mm

TOOL 09900-20508: Cylinder gauge set

PISTON REMOVAL AND INSPECTION

REMOVAL



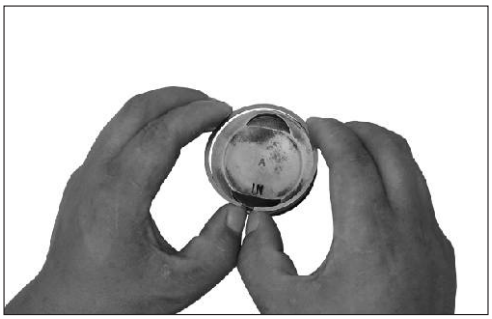


- Place a clean rag over the cylinder base to prevent piston pin circlip from dropping into crankcase and then remove the piston pin circlip with long-nose pliers.
- Remove the piston pin and piston with the special tool.

TOOL 09910-34510: Piston pin puller

⚠ CAUTION

Hold the piston stably when taking out the piston pin to prevent the damage to conrod big end bearing.



- Remove the piston rings one by one by expanding the gap symmetrically.

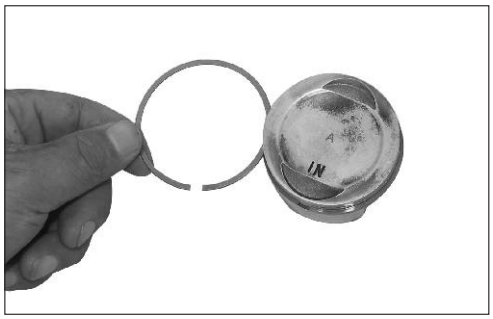
⚠ CAUTION

Do not expand the piston ring excessively since it is apt to be broken down.

- Decarbonize the piston.

⚠ CAUTION

Using a soft-metal scrapper or discarded piston ring to decarbon the ring grooves of the piston. Do not use the steel brush or the piston will be scratched.

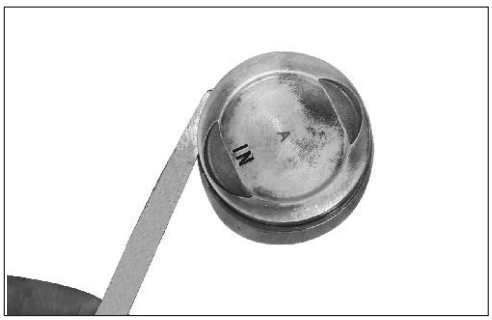


INSPECTION

- Temporarily fit the piston ring on the suitable position with the marked side to the top.
- Measure the piston ring-groove clearance with thickness gauge.

DATA **Piston ring-groove clearance limit:**
1st: 0.18 mm 2nd: 0.15 mm

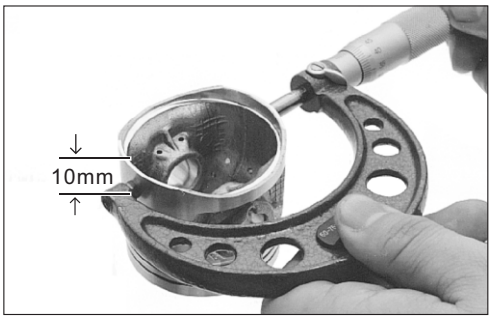
TOOL **09900-20803: Thickness gauge**



- Measure the piston outer diameter with a micrometer at the place 10mm from the skirt end.

DATA **Piston outer diameter limit:**
10 mm from the skirt end, 59.855 mm

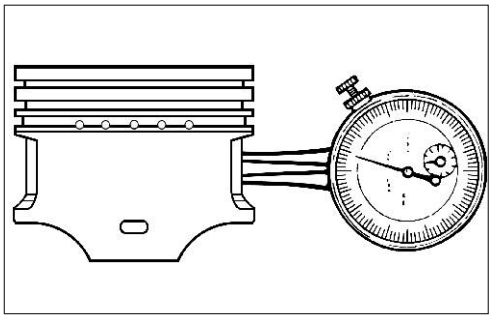
TOOL **09900-20203: Micrometer (50-75 mm)**



- Using a caliper gauge to measure the piston pin bore inner diameter.

DATA **Piston pin bore inside diameter limit: 14.03 mm**

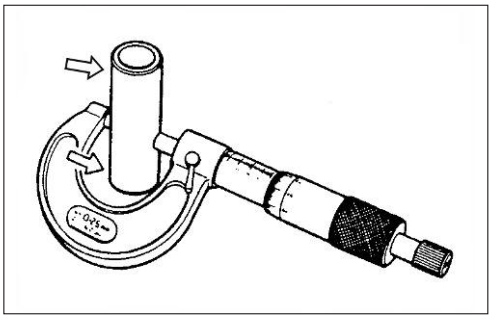
TOOL **09900-20605: Caliper gauge**



- Using a micrometer measure the piston pin O.D..

DATA **Piston pin O.D. limit: 13.98 mm**

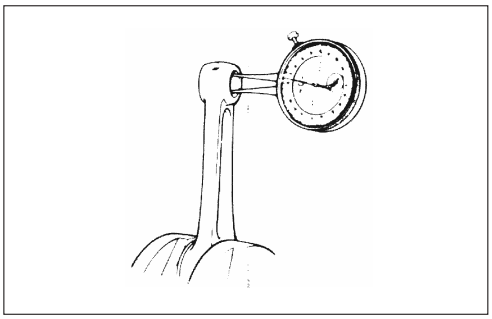
TOOL **09900-20205: Micrometer (0-25)**



- Measure the conrod small end I.D..

DATA **Conrod small end I.D. limit: 14.04 mm**

TOOL **09900-20605: Caliper gauge**

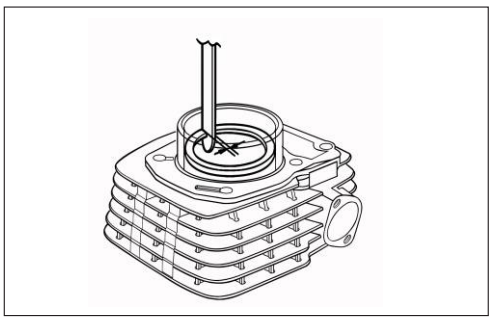


- Fit the ring in the cylinder, and measure each ring end gap using a thickness gauge. If any ring has an excess end gap, replace the ring.

DATA **Piston ring end gap limit (1st & 2nd): 0.45 mm**

TOOL **09900-20803: Thickness gauge**

NOTE:
Make sure fit the piston ring into the cylinder evenly with the piston head.



REASSEMBLY

Reassemble the piston and cylinder in the reverse order of removal, pay attention to the following points:

- Clean the piston top, skirt and groove.

⚠ CAUTION

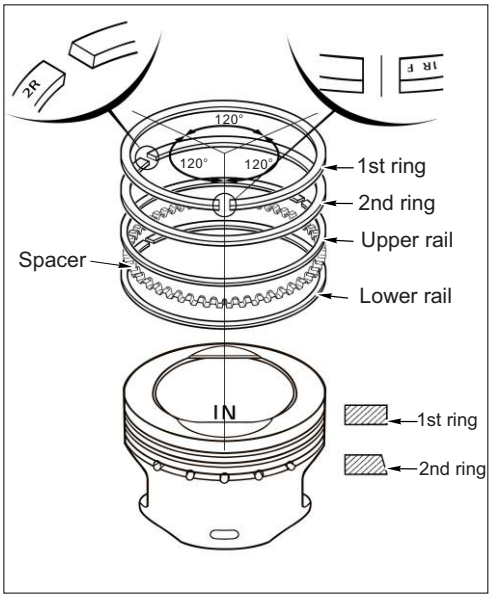
Fit the piston rings in the groove correctly, rotate the piston ring by hand to see if it turns smoothly.

- Install the piston ring carefully with marked side to the top.

⚠ CAUTION

Be careful not to damage the piston ring and piston. Position the gaps of the three rings as shown.

- The gaps of two piston rings should be staggered at 120°.



NOTE:

Install the piston rings in the order of oil ring, 2nd ring and 1st ring.

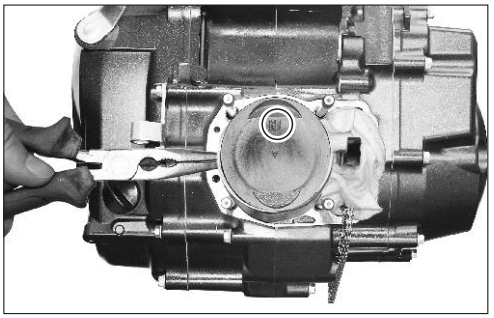
- Piston ring should turn smoothly in the piston groove after assembly.

- Apply engine oil to piston pin.
- Place a clean rag over the cylinder base to prevent piston pin circlip from dropping into crankcase, and then fit the piston pin circlip with long-nose pliers.

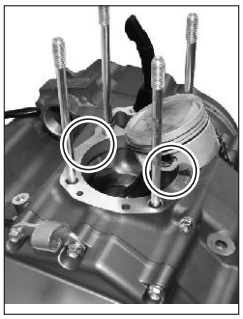
⚠ CAUTION

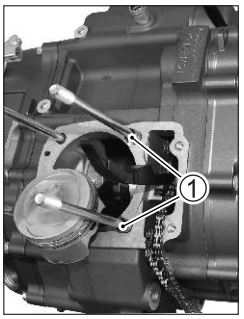
Use a new piston pin circlip.
Do not align the end gap of the circlip with the cut away of the piston pin bore.

- When fitting the piston, turn the "IN" on the piston head to intake side.



- Apply engine oil to the conrod big end, small end, and piston sliding part before reassembling cylinder.
- Apply sealant to the contacting surface between crankcase and cylinder, install the dowel pins ① and gasket.



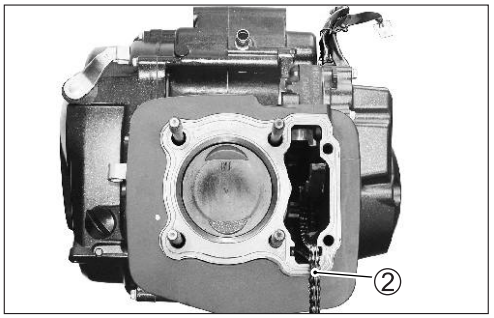


1215 **99000-31110: Sealant "1215"**

⚠ CAUTION

Always use new gasket to prevent leaking.

- Turn the opening of piston ring to the suitable position, apply the engine oil to the piston surface and then fit the piston in the cylinder, mark sure the piston is fitted in the cylinder completely.



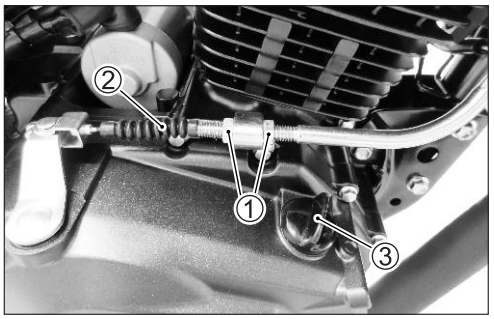
⚠ CAUTION

When mounting the cylinder, keep the cam chain ② taut, or the chain will be caught between the crankcase and cam drive sprocket.
There are the guide holder for the bottom end of cam chain guide cast in the crankcase. Be sure that the cam chain guide are inserted properly.

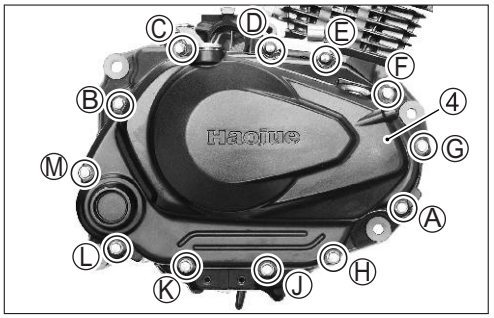
- Install the cylinder head. (Refer to page 3-13)

CLUTCH

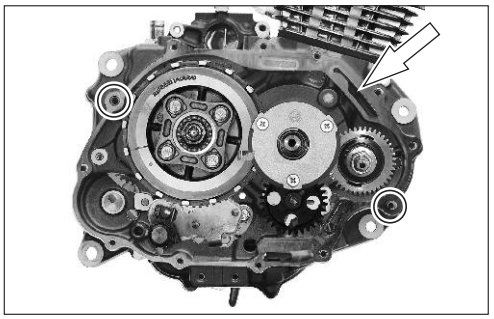
REMOVAL



- Drain the engine oil.
- Loosen the clutch lock and adjuster nut ①, remove the clutch cable ②from clutch lever.
- Remove the oil level gauge ③.



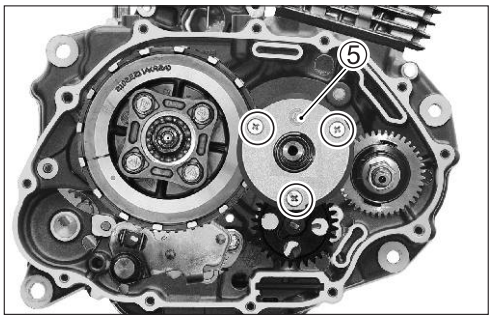
- Remove the clutch cover bolts in alphabetical order, take off the clutch cover ④.



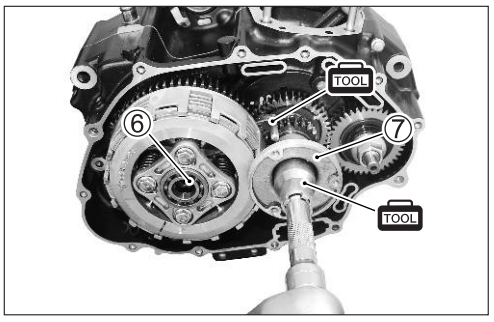
- Remove the gasket and dowel pins.

⚠ CAUTION

Always use new gasket.



- Remove the oil filter rotor cover ⑤.

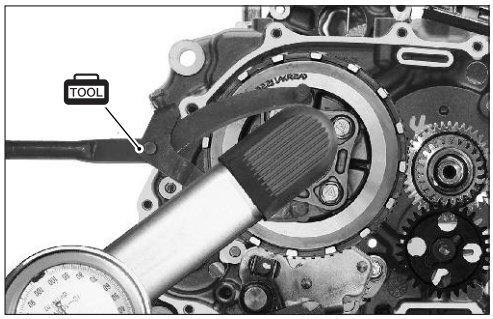


- Remove the bearing ⑥.
- Hold the crankshaft with the special tool, remove the oil filter rotor lock nut and the filter rotor ⑦.

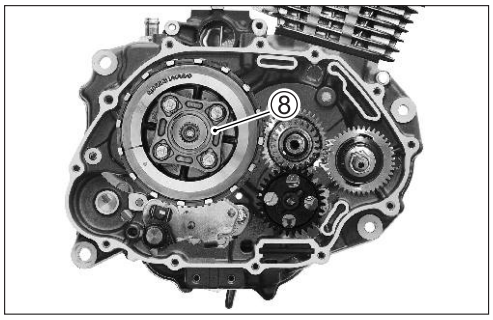
TOOL **11F14-036: Gear stopper**
11F14-011: Four claws sleeve

- Hold the clutch with the special tool, remove the clutch hub lock nut.

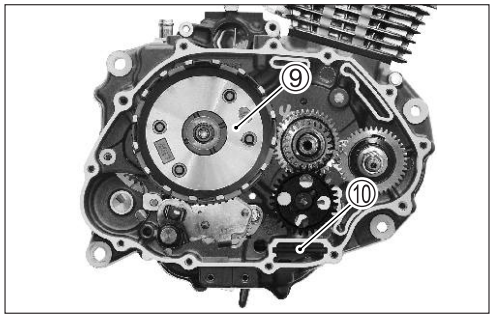
TOOL **09930-40113: Sprocket holder**



- Remove the washer and clutch driven hub assembly ⑧.



- Remove the clutch washer and primary driven gear ⑨ and oil strainer ⑩.



- Disassemble the clutch driven hub assembly, take off the clutch pressure plate, spring, drive and driven plates.

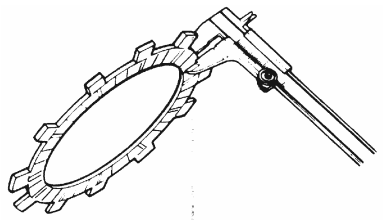
INSPECTION

- Measure the thickness and claw width of each drive plate with vernier calipers. Replace drive plates found to have worn down to the limit.

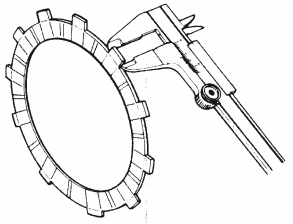
TOOL **09900-20103: Vernier calipers**

DATA **Drive plate thickness limit: 2.62 mm**

DATA **Drive plate claw width limit: 15.25 mm**



Checking thickness



Checking claw width

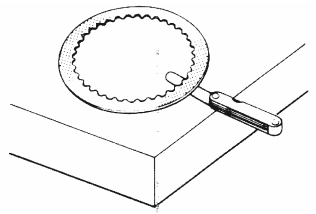
- Measure each driven plate for distortion with a thickness gauge. Replace driven plates which exceed the limit.

TOOL **09900-20803: Thickness gauge**

DATA **Driven plate distortion limit: 0.1 mm**

⚠ CAUTION

Replace the drive and drive plates as a whole set which exceed the limit.



Checking distortion

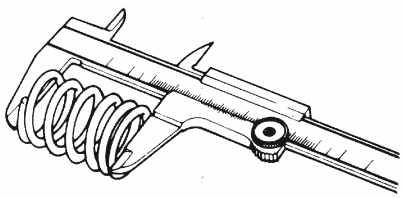
- Measure the clutch spring free length.

DATA **Clutch spring free length limit: 36.2 mm**

TOOL **09900-20103: Vernier calipers**

⚠ CAUTION

Replace all the springs if any spring is not within the limit.



- Inspect the clutch release bearing for any abnormality, particularly cracks, to decide whether it can be reused or should be replaced.
- Smooth engagement and disengagement of the clutch depends on the condition of this bearing.

- Inspect clutch outer disk notch for flaws or dents caused by drive plates, replace as necessary.

- Inspect the clutch driven hub for notches or abnormal wear caused by driven plates, replace as necessary.

REASSEMBLY

- Apply the engine oil to clutch drive and driven plates, reassemble the clutch driven hub assembly, clutch springs, pressure plate and bolts.

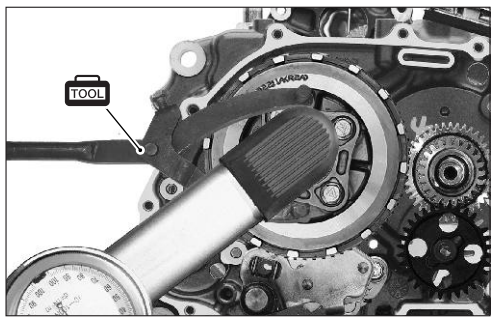
⚠ CAUTION

The color of top and bottom clutch drive plates is different from that of the 3 middle drive plates, install them correctly. When install the clutch drive plates, fit the top one to the shallow groove of clutch sleeve hub.

REASSEMBLY

Reassemble the clutch in the reverse order of removal. Pay attention to the following point:

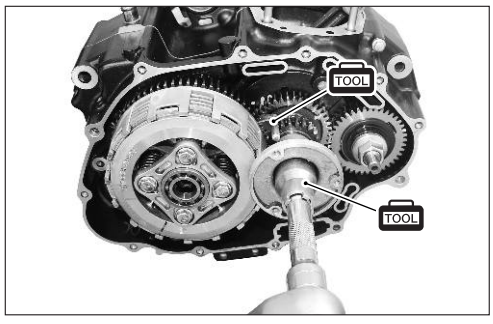
- Hold the clutch driven hub with the special tool, tighten the clutch lock nut to the specified torque.



TOOL **11F14-030: Gear stopper**

Clutch lock nut: 69-79 N•m

- Install the oil filter rotor and washer, tighten the filter rotor lock nut to the specified torque.

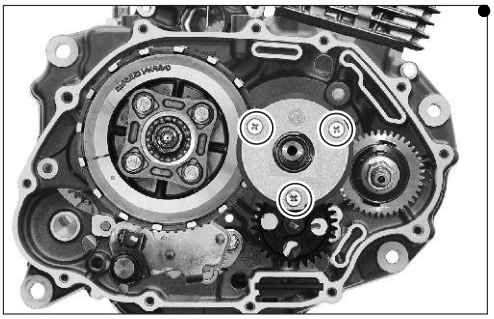


TOOL **11F14-036: Gear stopper**
11F14-011: 4 claws sleeve

Oil filter rotor lock nut: 45-55 N•m

- Install the engine oil filter rotor cover, apply the THREAD LOCK “1342” to the bolt.

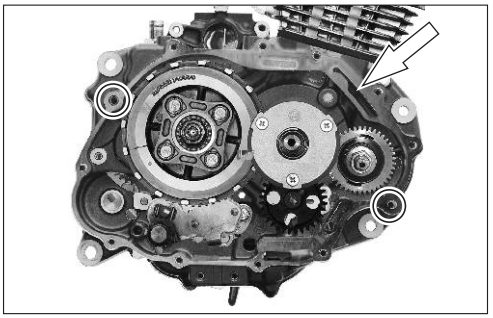
1342 **99000-32050: Bond “1342”**



⚠ CAUTION

Always use new gasket.

- Install the dowel pins and new gasket.

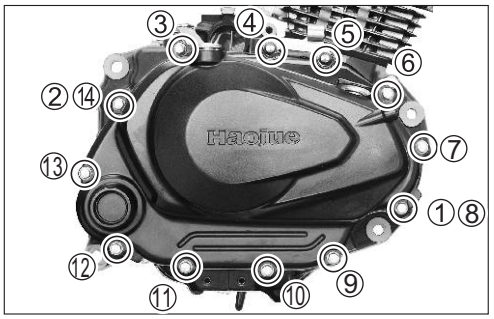


⚠ CAUTION

Always use new gasket.

- Install the clutch cover, tighten bolts to the specified torque according to the numerical sequence.

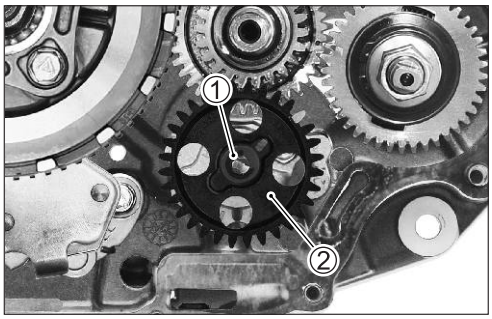
Clutch cover bolt: 9-13 N•m



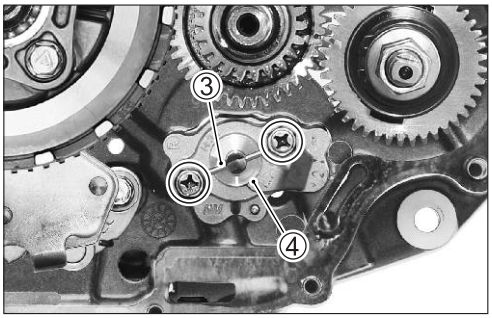
- Adjust the clutch cable clearance. (Refer to page 2-8).
- Add engine oil and check oil level. (Refer to page 2-7).

OIL PUMP AND SUMP FILTER

REMOVAL AND INSPECTION



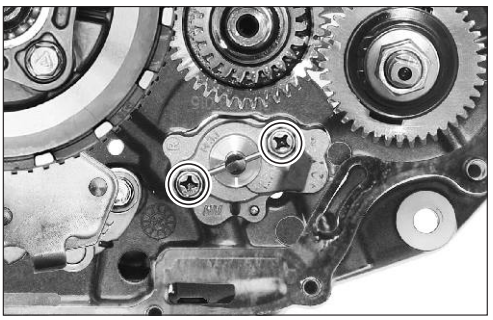
- Remove the clutch cover and oil filter rotor. (Refer to page 3-19)
- Remove the oil pump drive gear circlip ① and washer.
- Remove the oil pump driven gear ②.



- Draw out the pin ③, remove the oil pump ④.

- Check the oil pump if turns smoothly. Replace the oil pump if there is something unusual.

INSTALLATION



Install the oil pump in the reverser order of removal. Pay attention to the following points:

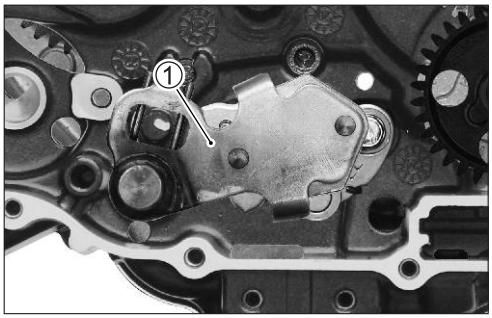
- Apply a small quantity of engin oil to the inside of oil pump before installing the oil pump.
- Apply thread lock “1342” to the oil pump mounting screws and tighten.

1342 **99000-32050: Thread lock “1342”**

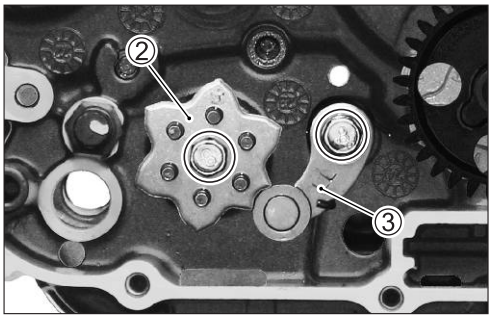
GEAR SHIFTING MECHANISM

REMOVAL

- Remove gear shifting lever, the clutch cover and clutch. (Refer to page 3-3 and page 3-19).
- Remove the gear shifting shaft assembly ① and washer.



- Remove the fixing cam ② and pins.
- Remove the cam stopper ③.



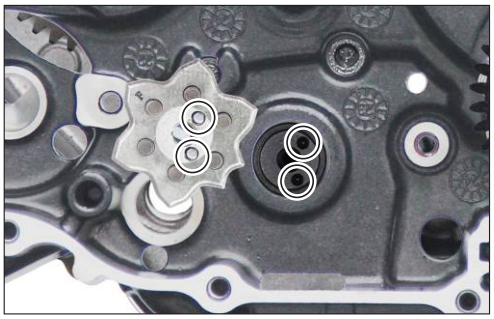
INSPECTION

- Inspect the gear shifting shaft ⑤ for wear or bend.

INSTALLATION

Install the gear shifting mechanism, pay attention to the following points.

- Fit the end of pins to the fixing cam and gear shifting cam.

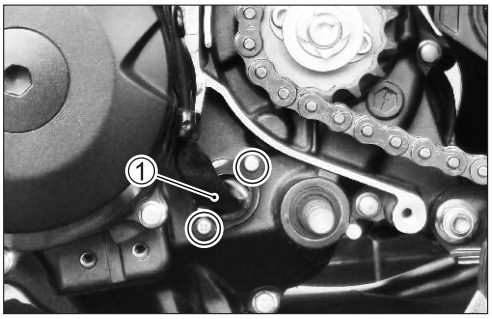


- Apply THREAD LOCK “1342” to the threaded part of bolts and tighten the bolts.

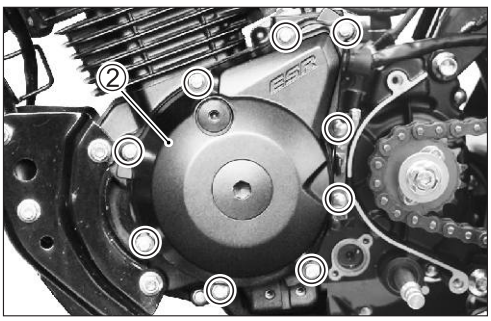
1342 **99000-32050: THREAD LOCK “1342”**

GEAR POSITION SWITCH, GENERATOR AND STARTER CLUTCH

REMOVAL



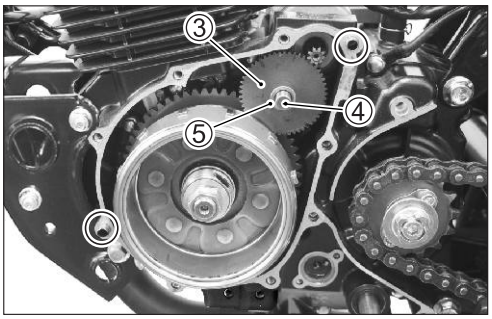
- Drain the engine oil, remove the saddle and left side cover. (Refer to page 6-1)
- Disconnect the generator coupler, remove the wire clamp and the sprocket cover. (Refer to page 3-2)
- Remove the gear position switch wire from the clamp.
- Remove the gear position switch ① , pin and spring.
- Remove the bolts, take off the generator cover ②.



⚠ CAUTION

Loosen the generator cover bolts diagonally. Be careful to the magnetic force between the generator cover (stator) and rotor during the removal.

- Remove the gasket and dowel pins.
- Remove the starter motor idle gear ③, shaft ④ and bush ⑤.

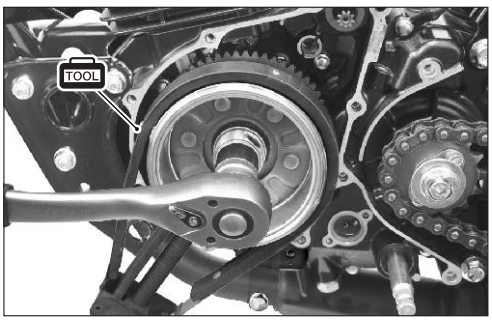


⚠ CAUTION

Be careful do not drop or lose the bush ⑤.

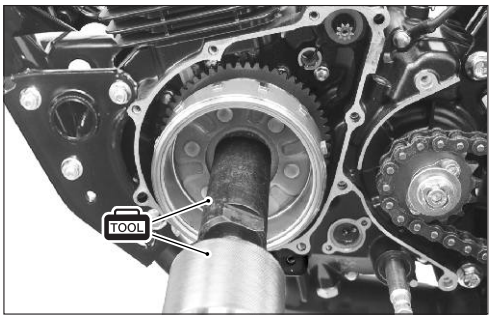
- Hold the rotor with the special tool, remove the generator rotor bolts.

TOOL 11F14-001 : Rotor holder



- Remove the generator rotor and key with the special tool.
- Remove the starter clutch gear assembly and washer.

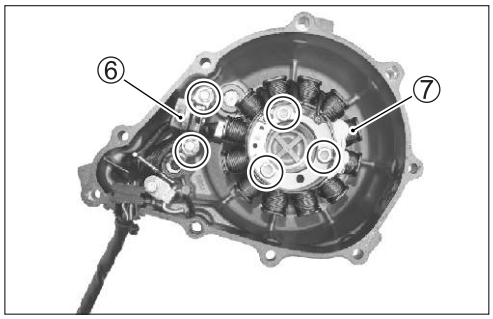
TOOL 11F14-006 : Generator rotor remover M33
09930-30102 : Slide shaft



⚠ CAUTION

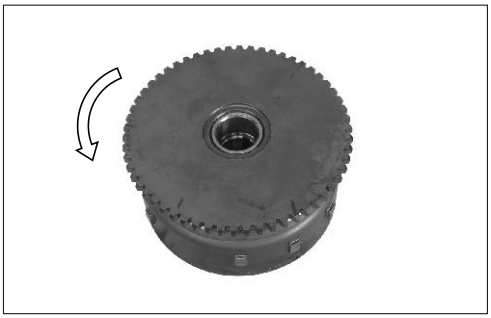
Do not damage the key slot and the crankshaft when removing the key.

- Remove the bolts, take off the crankshaft position sensor⑥.
- Remove the generator stator bolts, take off the generator stator ⑦.

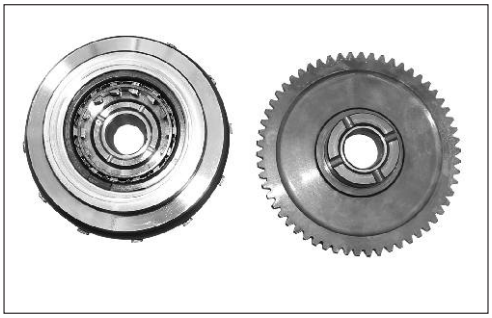


INSPECTION

- Rotate the starter clutch gear by hand in the direction as shown. The rotation should be smooth and continuous.
- Check gear engagement in the opposite direction. If the rotation is difficult or noise occurs, check the starter clutch gear wear. Replace as necessary.



- Inspect the starter driven gear spacer for damage, replace the spacer if there is something unusual.

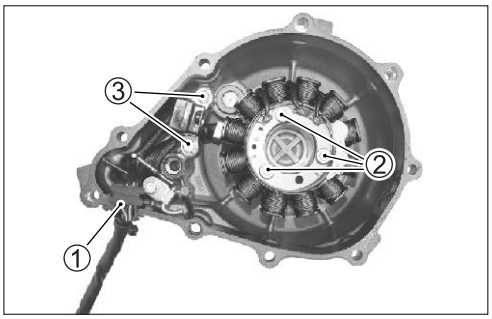


INSTALLATION

Install the generator cover in the reverse order of removal.
Pay attention to the following points:

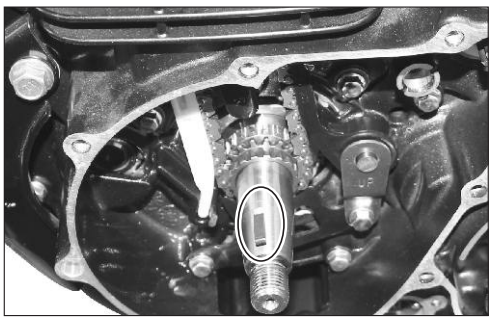
- Apply the Sealant "1207B" to the generator stator grommet ① , then install it to the generator cover.

99000-31140 : SEALANT "1207B"



- Install the stator, apply the THREAD LOCK "1342" to the bolts ② and tighten the bolts.
- Install the crankshaft position sensor, apply the THREAD LOCK to the bolts ③, then tighten the bolts.

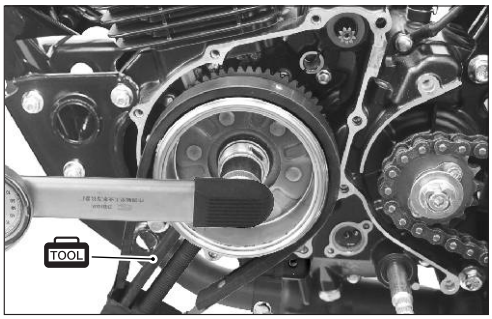
99000-32050 : THREAD LOCK "1342"



- Install the key to the crankshaft key slot.
- Degrease to the rotor cone surface and crankshaft cone surface, make sure the surface is clean and dry without oil or grease.

⚠ CAUTION

Do not damage the key slot and crankshaft when installing the key.



- Align the key slot on the rotor with the key on the crankshaft, install the rotor.
- Apply the THREAD LOCK to the threaded part of bolt, tighten the bolt to the specified torque.

99000-32050: THREAD LOCK "1342"

Generator bolt: 70-90 N•m

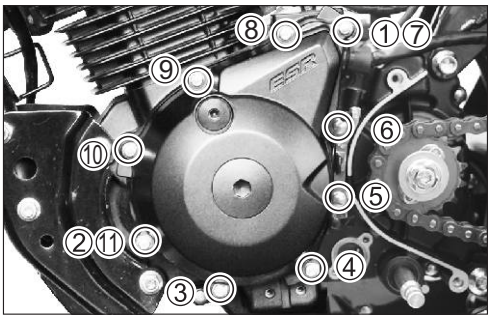
11F14-001: Rotor holder

- Install the gasket and dowel pins.

⚠ CAUTION

Replace the gasket with a new one to prevent leakage.

- Install the generator cover, tighten bolts to the specified torque according to the numerical sequence.

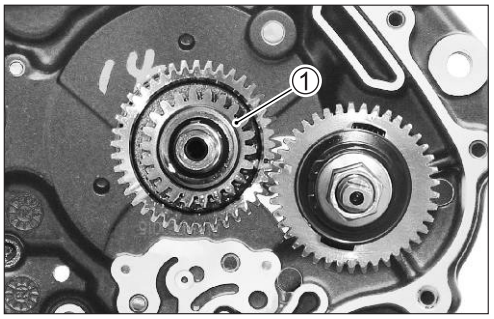


Clutch cover bolt: 9-13 N•m

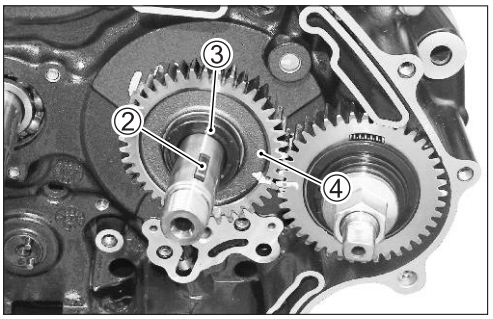
CRANKSHAFT AND TRANSMISSION

REMOVAL

- Drain the engine oil and remove the engine from chassis. (Refer to page 3-2)
- Remove the cylinder head, cylinder and piston. (Refer to page 3-7 and page 3-15)
- Remove the clutch. (Refer to page 3-19)
- Remove the oil pump. (Refer to page 3-24)
- Remove the gear shifting mechanism. (Refer to page 3-25)
- Remove the gear position switch, the generator rotor and starter clutch. (Refer to page 3-26)
- Remove the primary drive gear assembly ①.

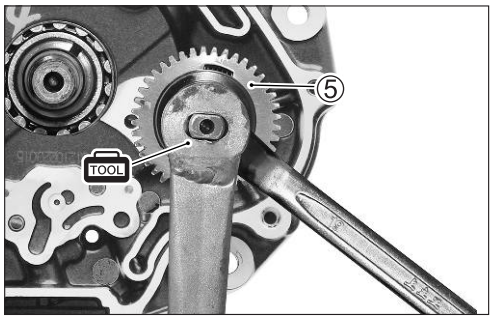


- Remove the key ②, cushion ③, balancer primary drive gear ④ and square key.

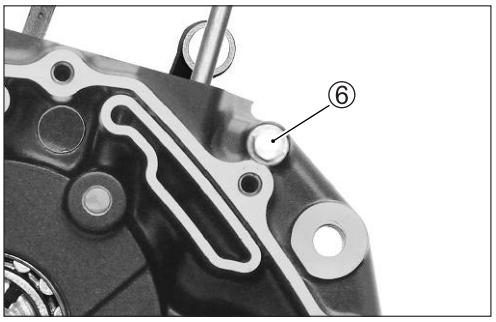


- Hold the balancer shaft with the special tool, remove the balancer nut.
- Remove the balancer shaft driven gear assembly ⑤, key and washer.

TOOL **11F14-030: Balancer shaft holder**

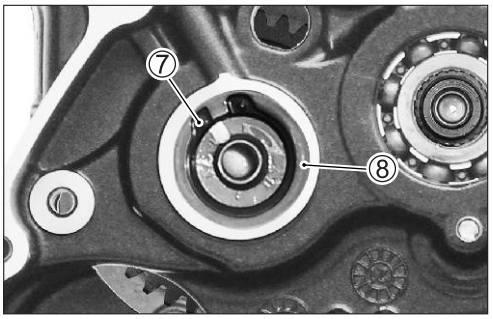


- Remove the crankcase mounting bolt ⑥.

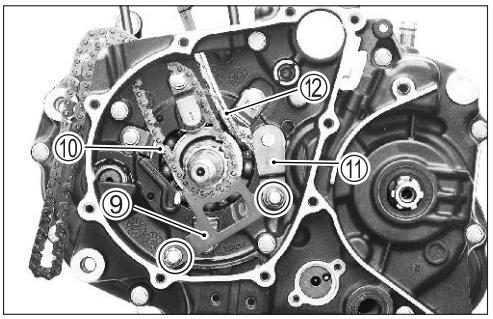


- Remove the circlip ⑦and washers ⑧.

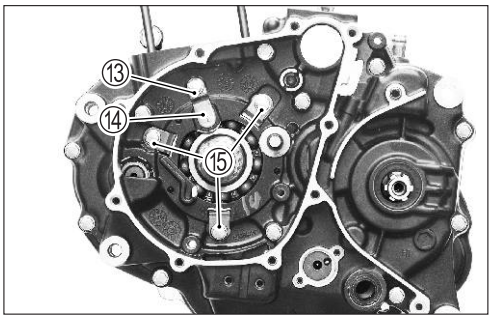
TOOL **09900-06107: Snap ring pliers (opening type)**



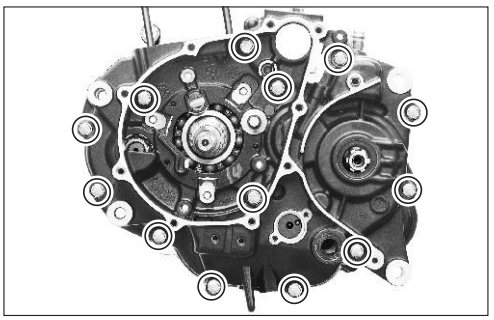
- Loosen the cam chain holder bolt, remove the cam chain holder ⑨, cam chain ⑩, tensioner ⑪and chain guide ⑫.



- Remove the crankshaft bearing pin bolt ⑬, pressure plate ⑭, spring and pin.
- Remove the crankshaft bearing retainer bolt ⑮ and retainer.

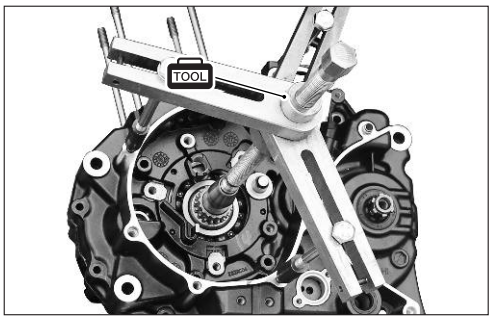


- Remove the crankcase mounting bolts.

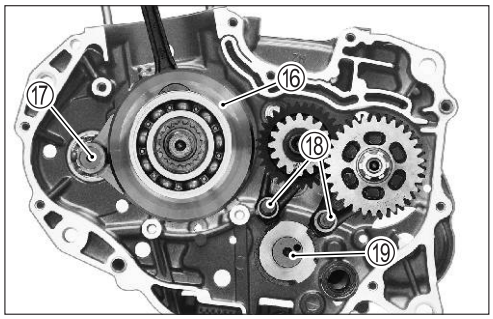


- Remove the left crankcase with the special tool.
- Fit the crankcase separating tool, so that the tool plate is parallel with the end face of the crankcase.

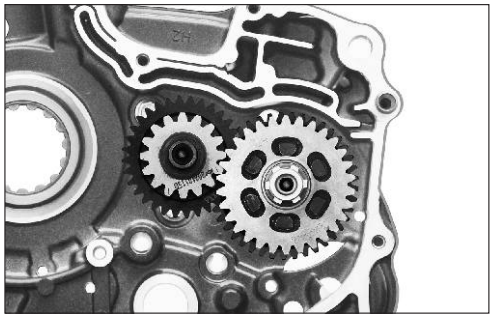
TOOL **09920-13120: Crankcase separating tool**



- Remove the crankshaft ⑯ and balancer shaft ⑰.
- Remove the gear shifting fork shaft ⑱, take off the forks.
- Remove the gear shifting cam ⑲.



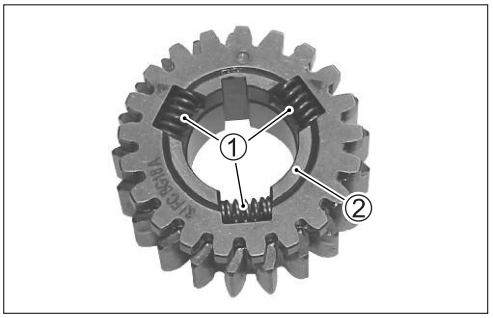
- Remove the transmission assembly.



DISASSEMBLY

PRIMARY DRIVE GEAR

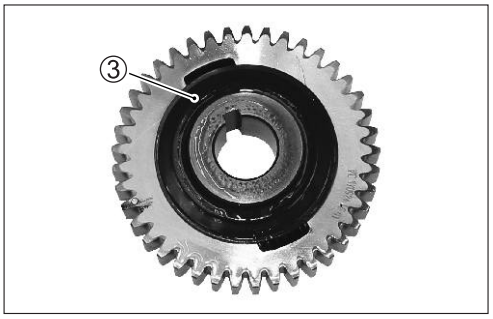
- Remove the spring ① and primary drive gear I ②.



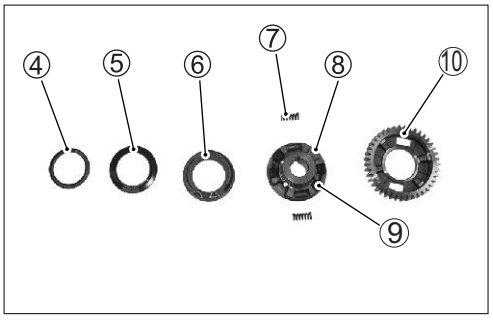
BALANCER DRIVEN GEAR ASSEMBLY

- Remove the balancer shaft driven gear circlip ③ with the special tool.

TOOL **09900-06107: Snap ring pliers (Opening type)**



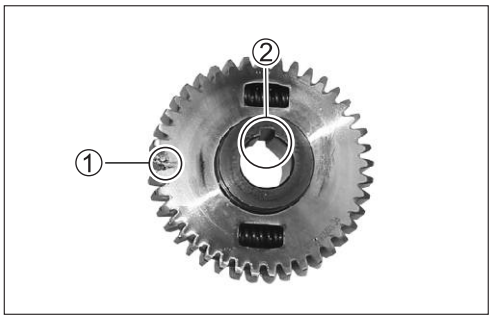
- Remove the thrust washer ④, washer ⑤, wave washer ⑥, compression spring ⑦, balancer shaft driven gear inner race ⑧, damper ⑨ and balancer shaft driven gear ⑩.



REASSEMBLY

Reassemble the balancer shaft driven gear assembly. Pay attention to the follow points:

BALANCER SHAFT DRIVEN GEAR ASSEMBLY



- When install the balancer shaft driven gear and balancer shaft driven gear inner race, make sure the key slot ① face upward, the mark ② faces left.

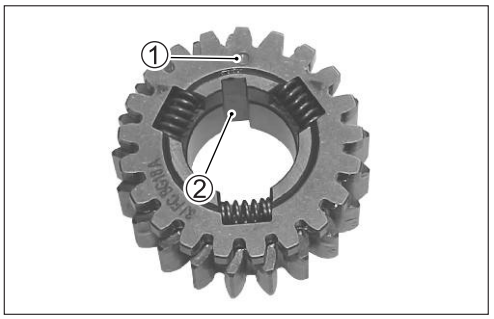
- Install the wave spring, washer, thrush washer and spring circlip.

⚠ CAUTION

When installing the wave spring, the wide opening faces balancer shaft driven gear.

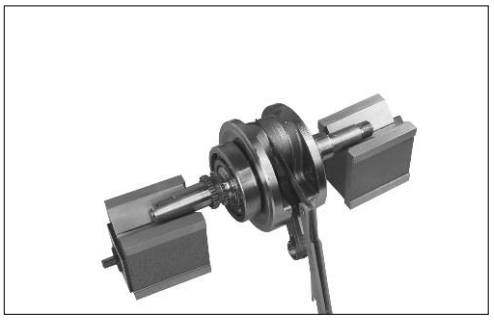
TOOL 09900-06107 : Snap ring pliers (Opening type)

PRIMARY DRIVE GEAR ASSEMBLY



- When reassemble the primary drive gear assembly, make sure align the mark point ① on the primary drive gear I with notch ② on the primary drive gear II as shown, than install the compression spring.

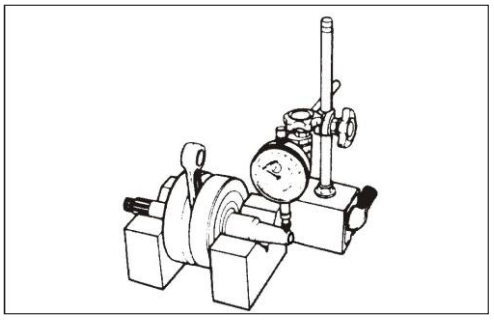
INSPECTION



- Measure the big end of conrod side clearance with a thickness gauge.

DATA Conrod big end side clearance limit: 0.6 mm

TOOL 09900-20803 : Thickness gauge



- Support the crankshaft with “V” blocks as shown, with the two end journals resting on the blocks. Measure the crank-shaft runout with the dial gauge.

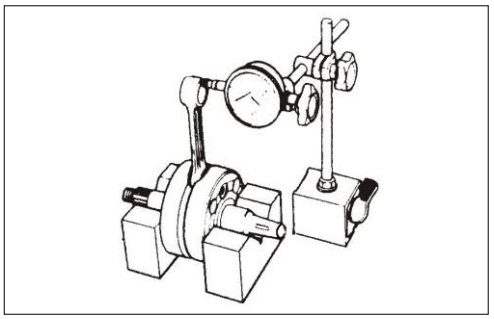
DATA Crankshaft runout limit: 0.05 mm

TOOL 09900-20606: Dial gauge (0.1 mm)
09900-20701: Magnetic stand
09900-21304: V-block (100 mm)

- Wear on the big end of the conrod can be estimated by checking the movement of the small end of the rod. This method can also check the extend of wear on the parts of the conrod's big end.

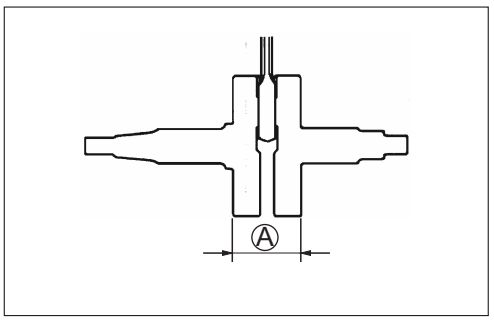
DATA Conrod small end deflection limit: 3.0 mm

TOOL 09900-20606:Dial gauge (0.1 mm)
09900-20701:Magnetic stand
09900-21304:V-block (100 mm)



- Measure the width between the crankshaft webs Ⓐ referring to the figure when rebuilding the crankshaft.

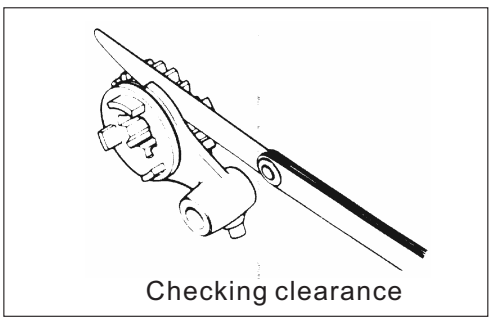
DATA Crank web to web width Ⓐ: 48.0-48.1 mm



- Using a thickness gauge, check the shift fork clearance in the groove of its gear. If the clearance limit is exceeded by any of the three gears, determine whether the gear of the gear shift fork should be replaced by measuring the thickness and groove width.

DATA Shifting fork to groove clearance: 0.44 mm

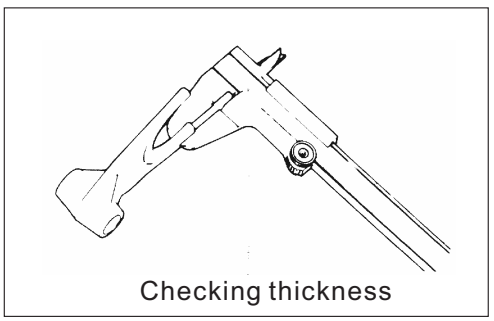
TOOL 09900-20803 : Thickness gauge


Checking clearance

- Measure the thickness of gear shift fork.

DATA Gear shift fork thickness: 4.93-5.0 mm

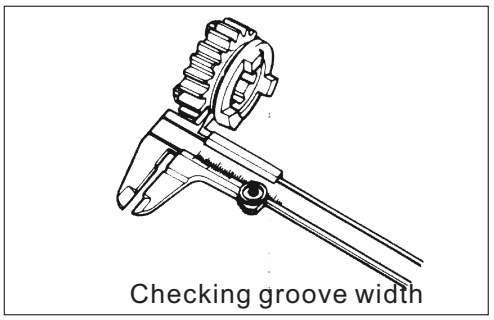
TOOL 09900-20103: Vernier calipers


Checking thickness

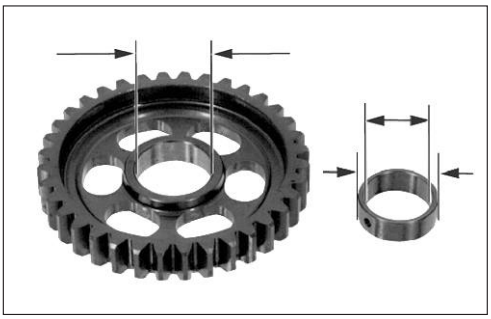
- Measure the gear shift fork groove width.

DATA The gear shift fork groove width: 5.1-5.17 mm

TOOL 09900-20103: Vernier calipers


Checking groove width

- Check if abnormal wear or poor lubrication exist on each bush, inner race of gear and gear teeth surface.



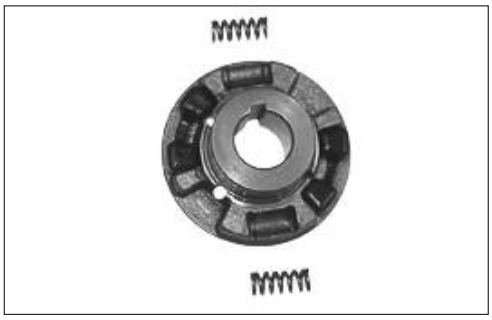
- Inspect the gear shifting cam guide slot for damage.
- Replace the gear shifting cam if the damage is found.

- Visual inspect the primary drive gear I , primary drive gear II and compression spring for wear or damage. If any damage are found, replace the defective with a new one.

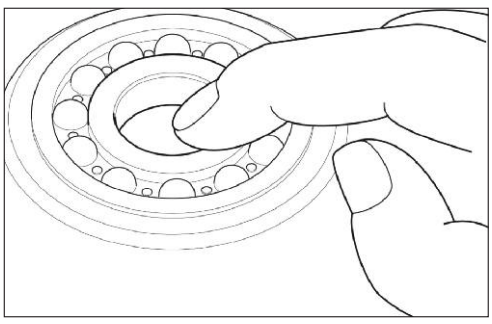
⚠ CAUTION

If no defects found in the visual inspect for primary drive gear, do not disassemble it.

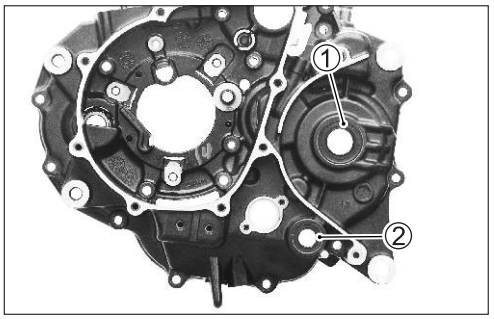
- Inspect the balancer shaft driven gear damper and compression spring for damage. If any defects are found, replace the defective part.



- Rotate the inner races of bearings with finger.
- Bearing rotation should be smooth and silent.
- Simultaneously inspect if the bearing outer race rotates in crankcase.
- In case of difficult rotation of the bearing inner or outer race, replace the bearing with a new one.



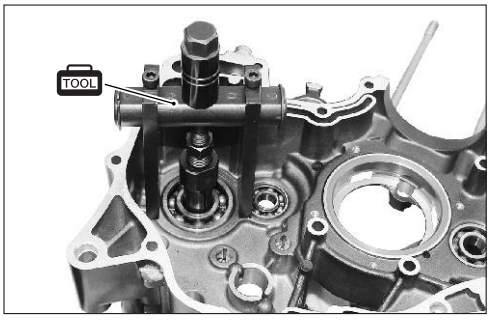
OIL SEAL AND BEARING REMOVAL



- Remove the left crankcase drive shaft oil seal ① and gear shifting shaft oil seal ②.

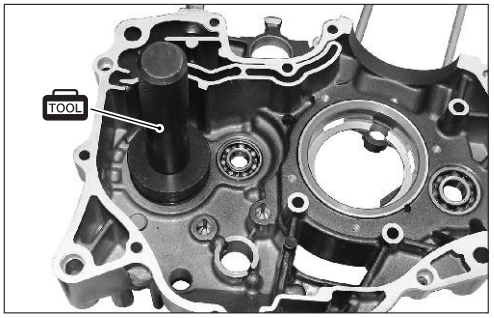
TOOL **09913-50121: Bearing removal tool**

- Remove all the bearings from the left and right crankcases with the special tool.



TOOL **09921-20240: Bearing removal tool**

BEARING INSTALLATION



Install the bearings in the reverse order of removal. Pay attention to the following points:

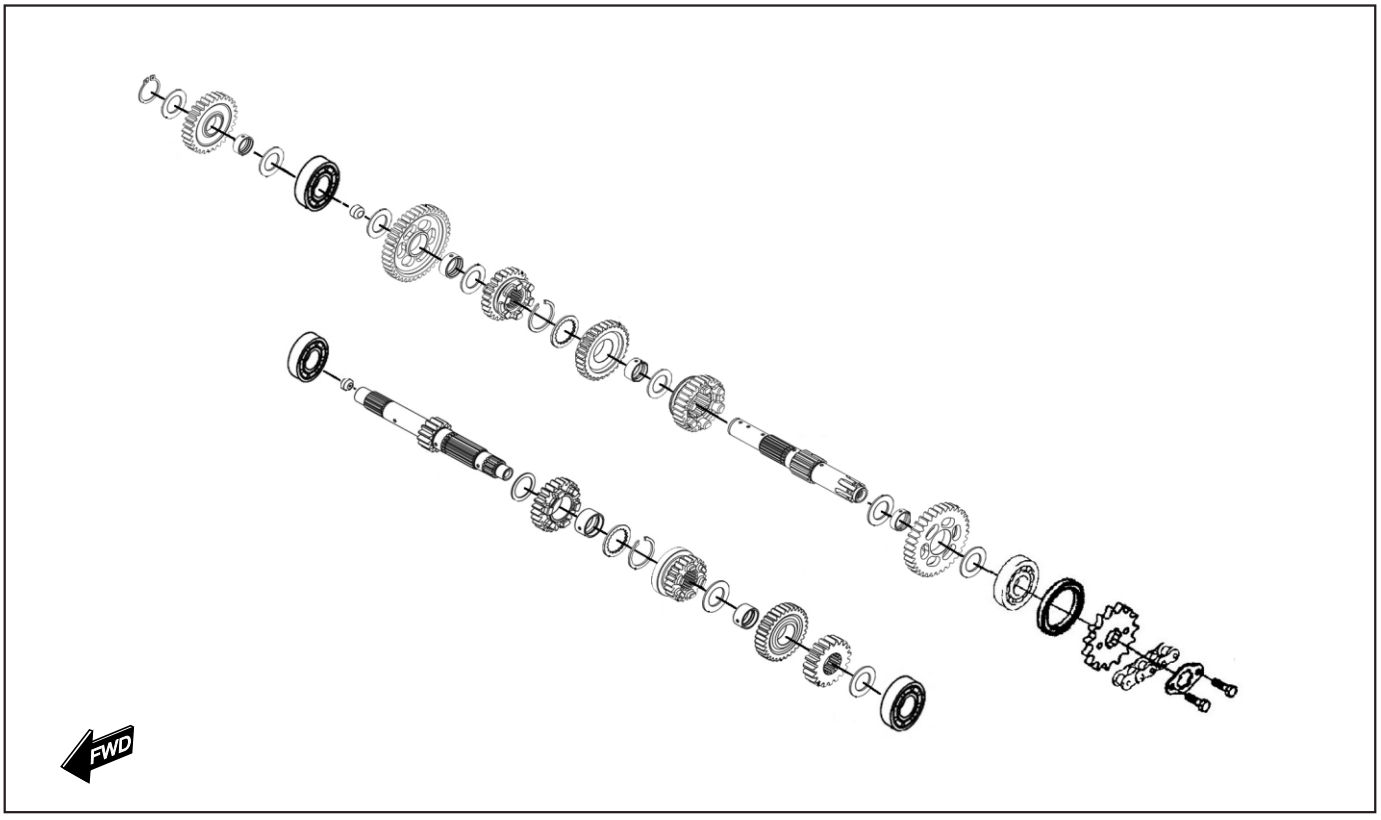
- Install the bearings on the left and right crankcases with the special tool.

TOOL **09913-70210: Bearing installer set**

ASSEMBLY

Assemble the removed parts in the reverse order of removal. Pay attention to the following points.

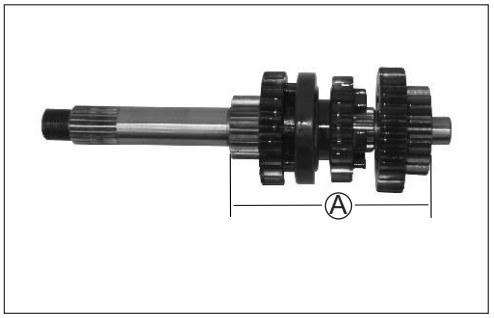
TRANSMISSION



⚠ CAUTION

Never reuse a circlip. After a circlip has been removed from a shaft, it should be discarded and a new circlip must be installed. When installing a new circlip, care must be taken not to expand the end gap larger than required to slip the circlip over the shaft. After installing a circlip, always insure that is is completely seated in its groove and securely fitted.

- Press-fit 2nd drive gear onto the counter shaft. Before reassembling, coat the inner face of the 2nd drive gear with Thread Lock “1303" and install it so that the length Ⓐ as shown in Fig.



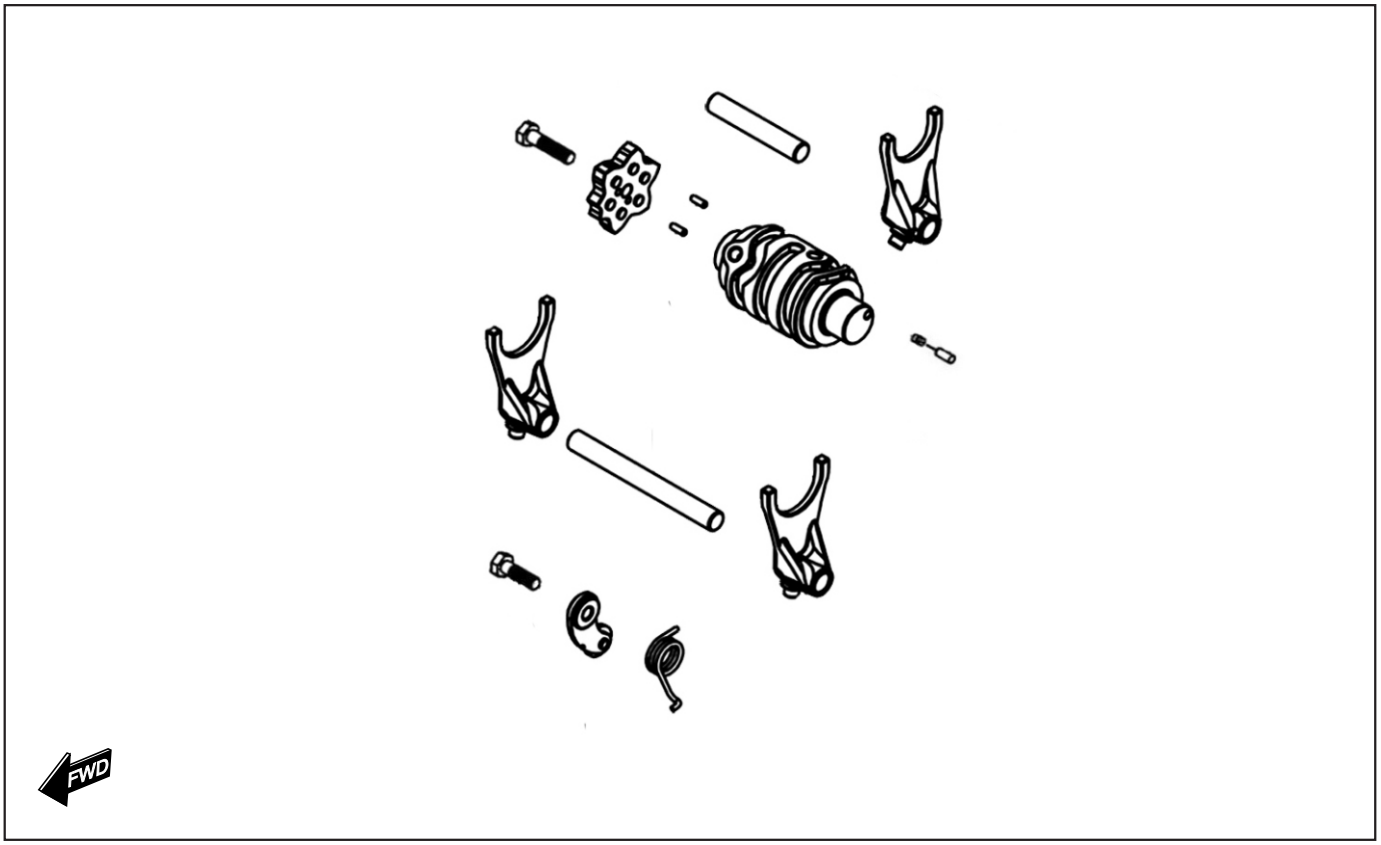
DATA Counter shaft length(Low to 2nd) Ⓐ: 83.15-83.35 mm

1303 99000-32030: Thread lock “1303”

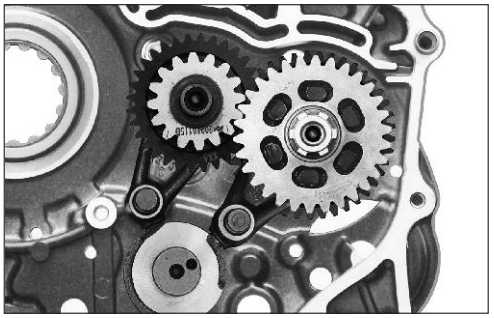
⚠ CAUTION

This procedure may be performed only twice before shaft replacement is required.

GEARING SHIFTING CAM AND FORK



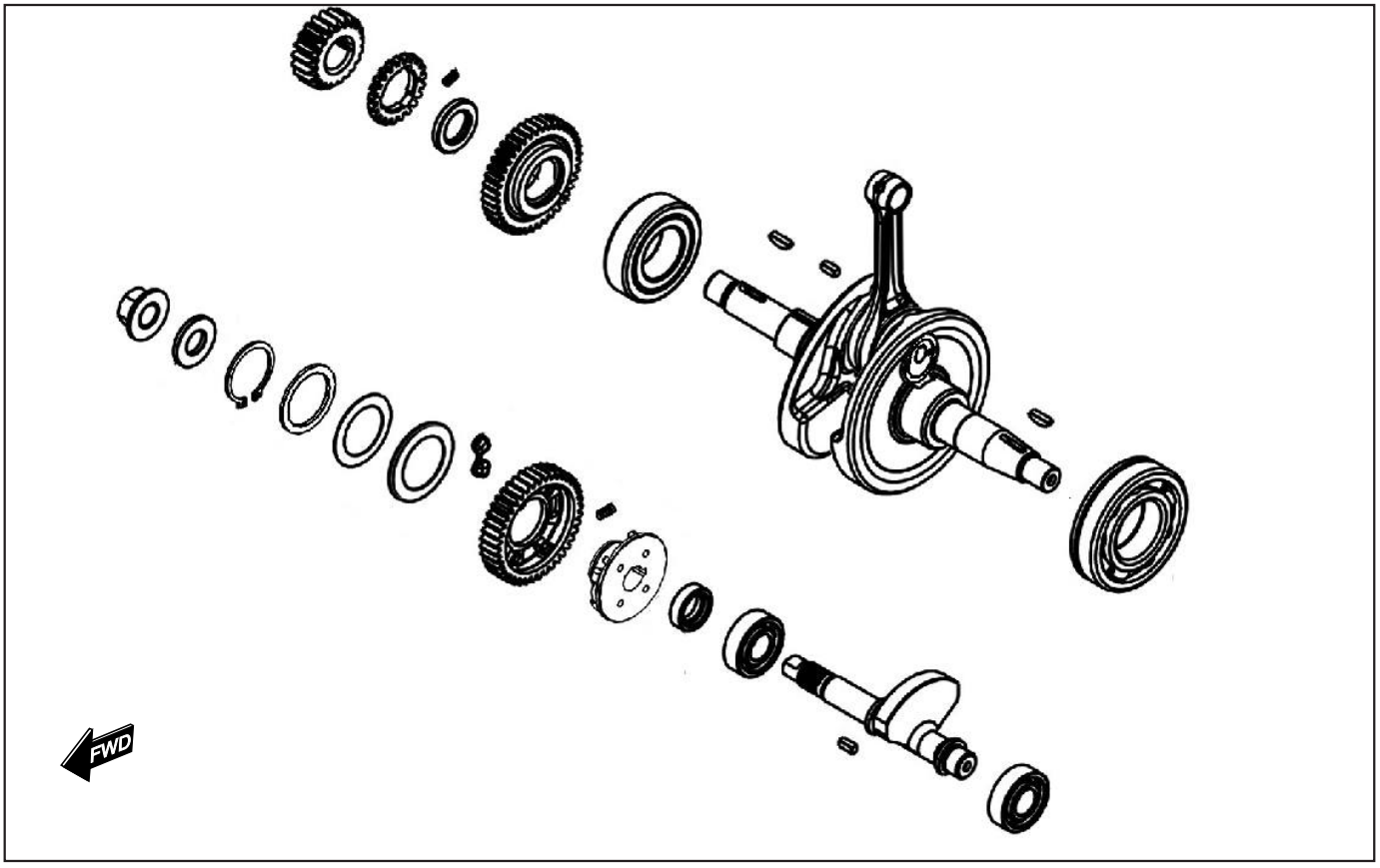
- Install the gear shifting cam to crankcase and install the gear shifting forks to the cam.



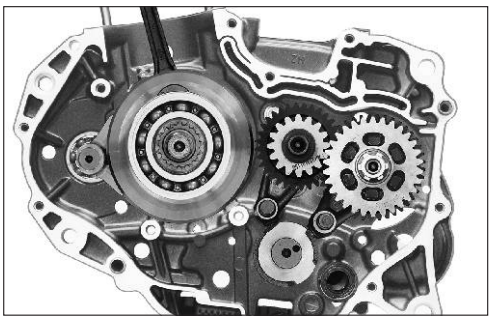
⚠ CAUTION

The forks resemble each other very closely in external appearance and configuration, install the fork with mark "3JC" to the short shaft beside the crankshaft. Install the fork with mark "3J R/L" to the long shaft. Check if driven shaft and counter shaft rotate smoothly after forks are installed.

CRANKSHAFT AND BALANCER SHAFT



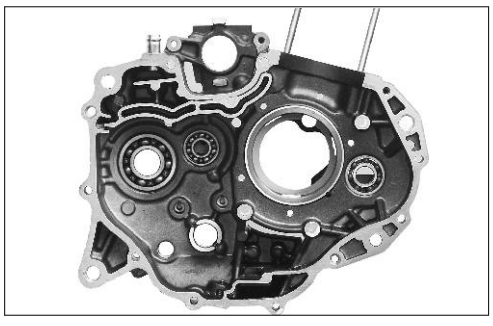
- Install the balancer shaft and crankshaft in the right crankcase.



⚠ CAUTION

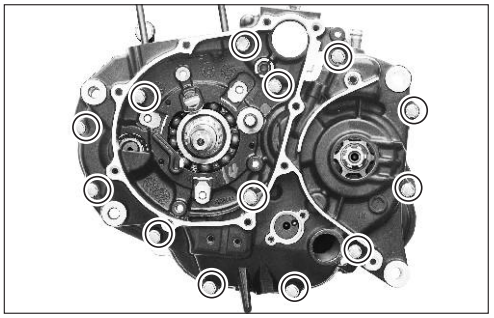
Mare sure the roller bearing inner race is fitted properly before installing the crankshaft.

- Apply the bond to the left crankcase contacting surface as shown.



- Install the left crankcase to the right crankcase, tighten the mounting bolts to the specified torque.

Crankcase mounting bolt: 8-12 N•m



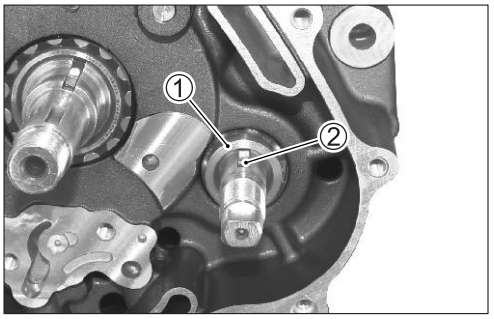
CAUTION

Tighten the bolts in 2-3 steps diagonally. If it is difficult to coupling the crankcases, use a plastic hammer to tap the crankcase to make the crankcase coupling properly.

- Install the bush ① and key ②.

CAUTION

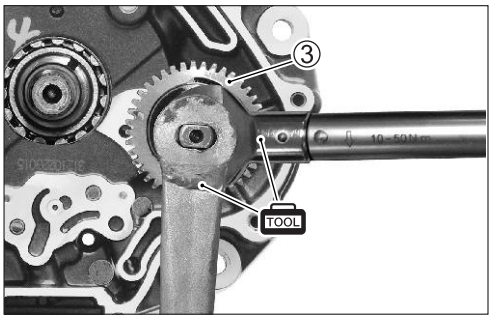
When installing the bush ①, the groove side faces outward.



- Install the balancer shaft driven gear assembly ③ and washer.
- Hold the balancer shaft with the special tools, tighten the nut to the specified torque.

TOOL **11F14-030: Balancer shaft holder 8**
11J14-001: Balancer nut wrench 19

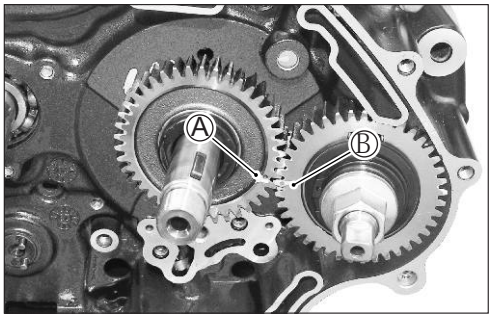
Balancer shaft nut: 40-60 N•m



- Install the balancer shaft drive gear, key and washer.

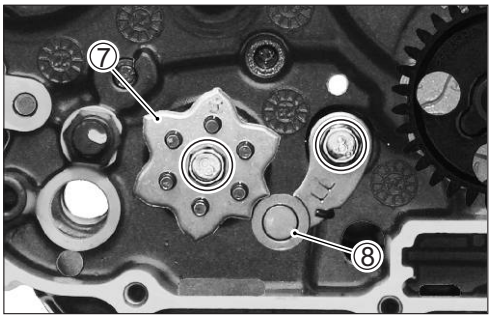
CAUTION

Align the mark Ⓐ on the gear with the mark Ⓑ on the gear.



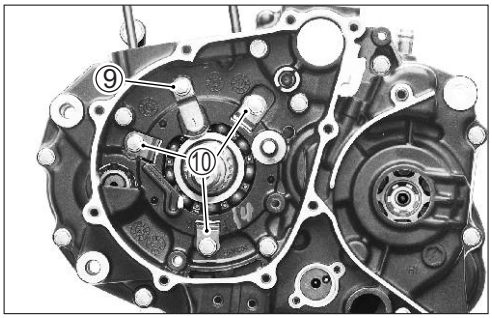
- Fit the end of pin to the fixing cam ⑦and gear shifting cam. Apply the THREAD LOCK "1342" to the threaded part of bolt and tighten the bolt.
- Install the cam stopper ⑧, apply the THREAD LOCK "1342" to the threaded part of bolt and tighten the bolt.

99000-32050: THREAD LOCK "1342"



- When install the crankshaft left bearing retainer, tighten the bolt ⑨ to the specified torque.

Crankshaft left bearing retainer bolt: 8-12 N•m

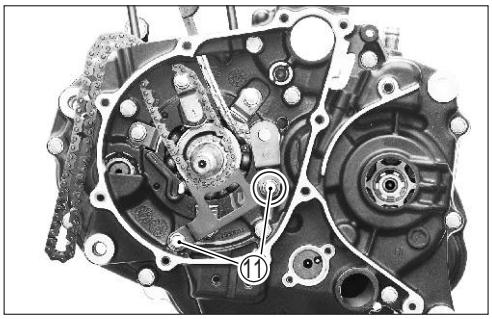


- When install the bearing fixing plate, tighten the bolts ⑩ to the specified torque.

Retainer plate bolt: 8-12 N•m

- When install the cam chain tensioner fixing plate, tighten the bolts ⑪ to the specified torque.

Cam chain retainer bolt: 8-12 N•m

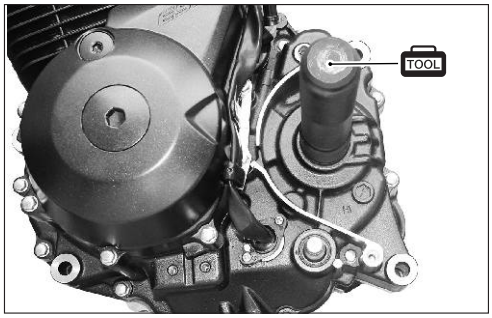


⚠ CAUTION

When installing the tensioner fixing plate, set the "UP" mark as shown in the right illustration.

- Install the oil seal on the left crankcase with the special tool.

09913-70210: Bearing installer set



FI

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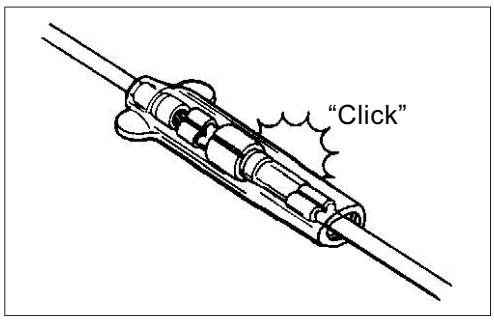
PRECAUTIONS IN SERVICING

When handling the component parts or servicing the FI system, observe the following points for the safety of the system.

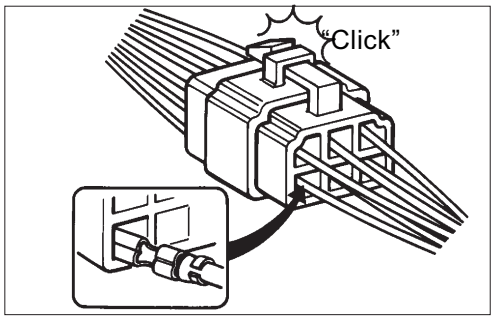
ELECTRICAL PARTS

CONNECTOR/COUPLER

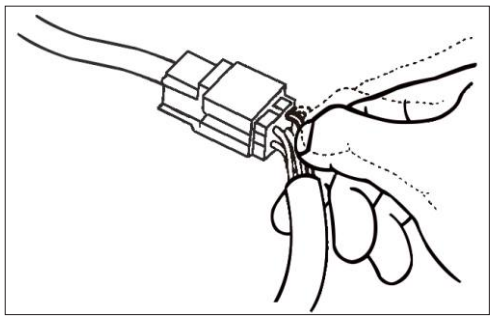
- Faulty FI system is often related to poor electrical contact of connector/coupler. Before servicing individual electronic part, check electrical contact of connector/coupler.
  When connecting a connector, be sure to push it in until a click is felt.



- With a lock type coupler, be sure to release the lock when disconnecting, and push in fully to engage the lock when connecting.
- When disconnecting the coupler, be sure to hold the coupler body and do not pull the lead wires.
- Inspect each terminal on the connector/coupler for looseness or bending. Push in the coupler straightly. An angled or skewed insertion may cause the terminal to be deformed, possibly resulting in poor electrical contact.
- Inspect each terminal for corrosion and contamination.
- The termnals must be clean and free of any foreign material which could impede proper terminal contact.
- Before refitting the sealed coupler, make sure its seal rubber is positioned properly. The seal rubber may possibly come off the position during disconnecting work and if the coupler is refitted with the seal rubber improperly positioned, it may result in poor water sealing.

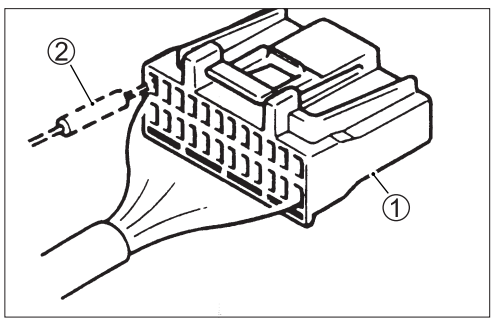


- Inspect each lead wire circuit for poor connection by shaking it by hand lightly. If any abnormal condition is found, repair or replace.

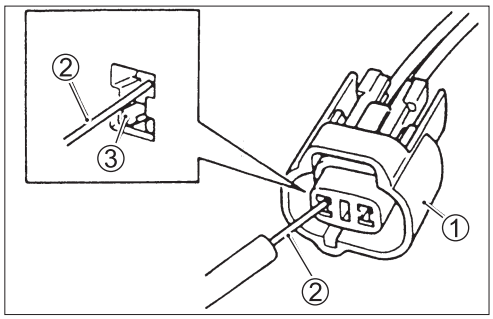


- When taking measurements at electrical connectors using a tester probe, be sure to insert the probe from the wire harness side (rear) of the connector/coupler.

① Coupler
② Probe



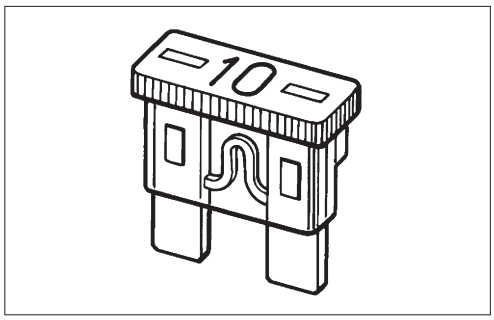
- When connecting meter probe from the terminal side of the coupler (where connection from harness side not being possible), use extra care not to force and cause the male terminal to bend or the female terminal to open.
  Connect the probe as shown to avoid opening of female terminal.
- Never push in the probe where male terminal is supposed to fit.
- Check the male connector for bend and female connector for excessive opening. Also check the coupler for locking (looseness), corrosion, dust, etc.
- Avoid applying grease or other similar material to connector/coupler terminals to prevent electric trouble.



① Coupler
② Probe
③ Where male terminal fits

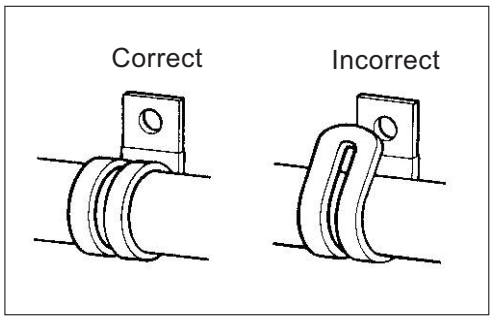
FUSE

- When a fuse blows, always investigate the cause to correct it and then replace the fuse.
- Do not use a fuse of a different capacity.
- Do not use wire or any other substitute for the fuse.



CLAMP

- Clamp the wire harness at such positions as indicated in the“WIRING HARNESS ROUTING”.
- Bend the clamp properly so that the wire harness is clamped securely.
- In clamping the wire harness, use care not to allow it to hang down.
- Do not use wire or any other substitute for the band type clamp.



SWITCH

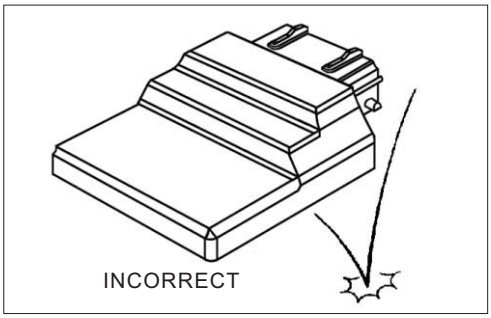
- Never apply grease material to switch contact points to prevent damage.

SEMI-CONDUCTOR EQUIPPED PART

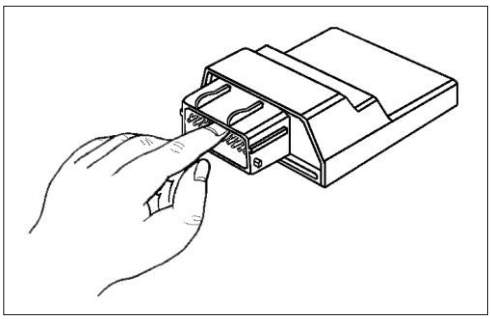
- Be careful not to drop the part with a semi-conductor built in such as a regulator/rectifier.
- When inspecting this part, follow inspection instruction strictly. Neglecting proper procedure may cause damage to this part.

ECM / VARIOUS SENSORS

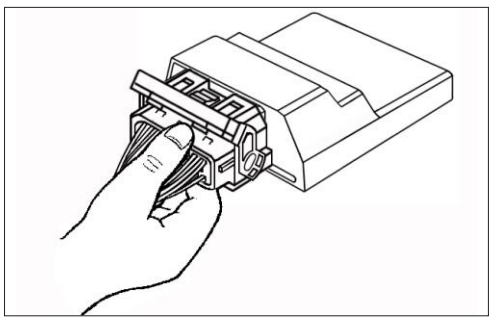
- Since each component is a high-precision part, great care should be taken not to apply any severe impacts during removal and installation.



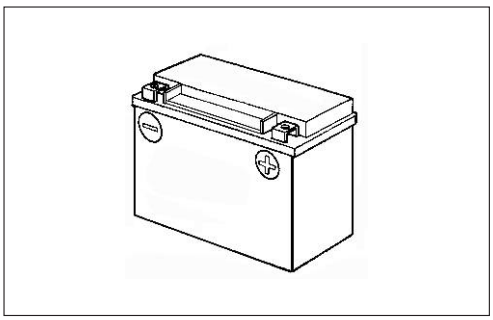
- Be careful not to touch the electrical terminals of the electronic parts (ECM, etc.). The static electricity from your body may damage them.



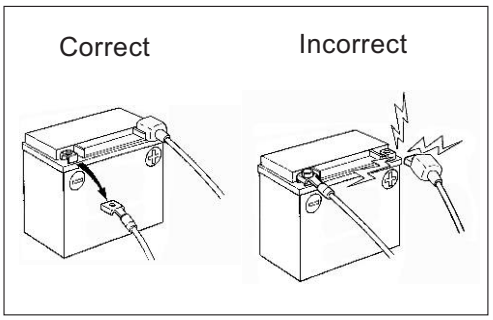
- When disconnecting and connecting the coupler, make sure to turn off the ignition switch, or electronic parts may get damaged.



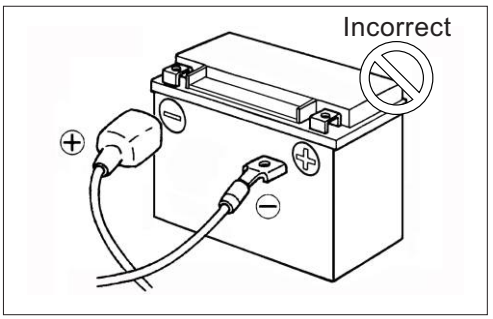
BATTERY



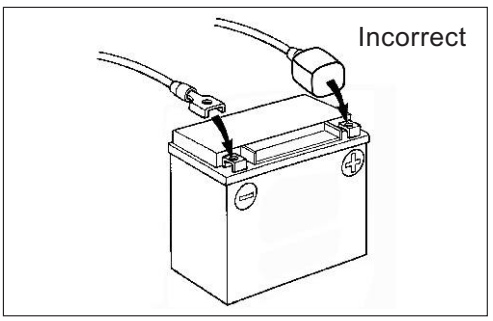
- The MF battery used in this motorcycle does not require maintenance (e.g., electrolyte level inspection, distilled water replenishment).
- During normal charging, no hydrogen gas is produced.
- However, if the battery is overcharged, hydrogen gas may be produced. Therefore, be sure there are no fire or spark sources (e.g., short circuit) nearby when charging the battery. Be sure to recharge the battery in a well-ventilated and open area.
- Removing any battery terminal of a running engine is strictly prohibited.
- The moment such removal is made, damaging counter electromotive force will be applied to the electronic unit which may result in serious damage.
- When disconnecting terminals from the battery for disassembly or servicing, be sure to disconnect the battery ⊖ lead wire first.



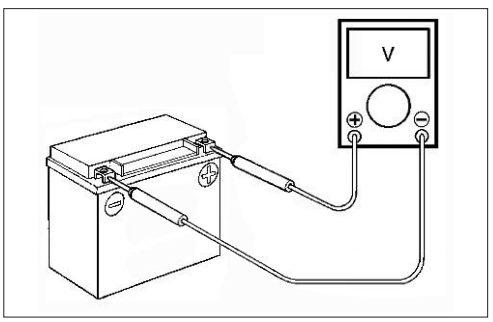
- Battery connection in reverse polarity is strictly prohibited. Such a wrong connection will damage the components of the FI system instantly when reverse power is applied.



- When connecting the battery lead wires, be sure to connect the battery ⊕ lead wire first.
- If the terminal is corroded, remove the battery, pour warm water over it and clean it with a wire brush.
- After connecting the battery, apply a light coat of grease to the battery terminals.
- Install the cover over the ⊕ battery terminal.



- Before measuring the voltage at each terminal, make sure that battery voltage is 12.4 V or higher. Terminal voltage check with a low voltage battery will lead to erroneous diagnosis.



ELECTRICAL CIRCUIT INSPECTION

While there are various methods for electrical circuit inspection, described here is a general method to check for open and short circuit using an ohmmeter and a voltmeter.

- Never connect any tester (voltmeter, ohmmeter, or what-ever) to the electronic unit when its coupler is disconnected. Otherwise, damage to electronic unit may result.
- Never connect an ohmmeter to the electronic unit with its coupler connected. If attempted, damage to ECM or sensors may result.
- Be sure to use a specified voltmeter/ohmmeter. Otherwise, accurate measurements may not be obtained and personal injury may result.

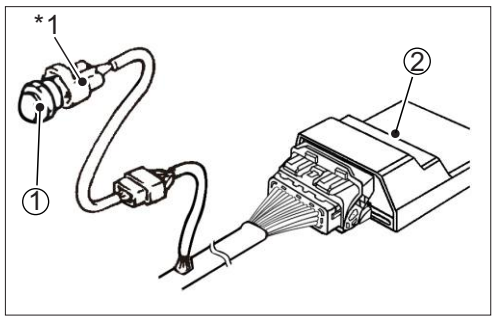
OPEN CIRCUIT CHECK

Possible causes for the open circuits are as follows. As the cause can exist in the connector/coupler or terminal, they need to be checked carefully.

- Loose connection of connector/coupler.
- Poor contact of terminal (due to dirt, corrosion or rust, poor contact tension, etc.).
- Wiring harness being open.
- Poor terminal-to-wire connection

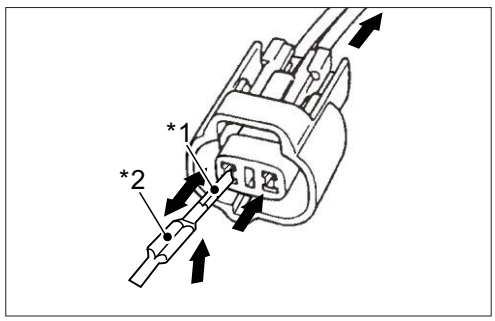
When checking system circuits including an electronic control unit such as ECM, etc., it is important to perform careful check, starting with items which are easier to check.

- Disconnect the negative cable from the battery.
- Check each connector/coupler at both ends of the circuit being checked for loose connection. Also check for condition of the coupler lock if equipped.

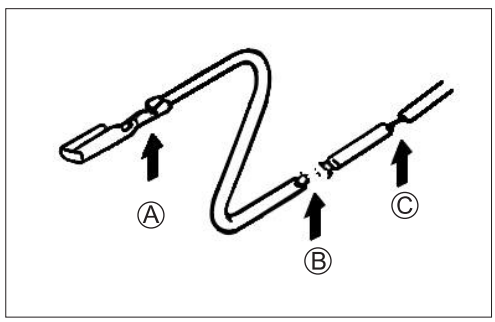


① Sensor
② ECM
*1 Check for loose connection.

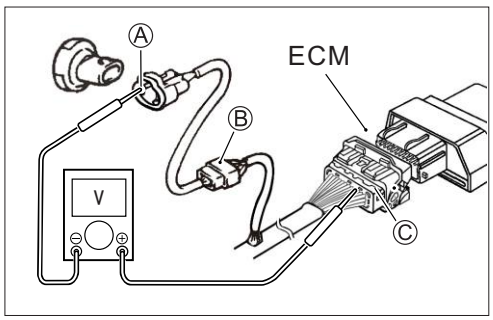
- Using a test male terminal, check the female terminals of the circuit being checked for contact tension.
  Check each terminal visually for poor contact (possibly caused by dirt, corrosion, rust, entry of foreign object, etc.).
  At the same time, check to make sure that each terminal is fully inserted in the coupler and locked.
  If contact tension is not enough, rectify the contact to increase tension or replace.
  The terminals must be clean and free of any foreign material which could impede proper terminal contact.



*1 Check contact tension by inserting and removing.
*2 Check each terminal for bend and proper alignment.

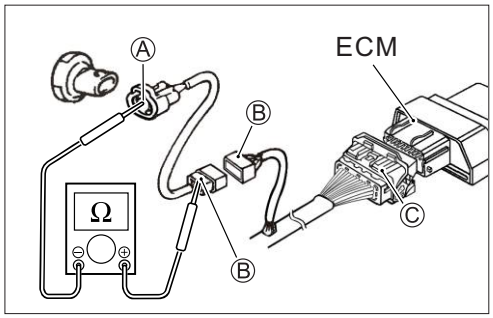


- Using continuity inspect or voltage check procedure as described below, inspect the wire harness terminals for open circuit and poor connection. Locate abnormality, if any.



Ⓐ Looseness of crimping
Ⓑ Open
Ⓒ Thin wire (a few strands left)

- Measure resistance across coupler Ⓑ (between Ⓐ and Ⓒ in the figure).
  If no continuity is indicated (Infinity or over limit), the circuit is open between terminals Ⓐ and Ⓒ.
- Disconnect the coupler Ⓑ and measure resistance between couplers Ⓐ and Ⓑ.
- If no continuity is indicated, the circuit is open between couplers Ⓐ and Ⓑ. If continuity is indicated, there is an open circuit between couplers Ⓑ and Ⓒ or an abnormality in coupler Ⓑ or coupler Ⓒ.



VOLTAGE CHECK

If voltage is supplied to the circuit being checked, voltage check can be used as circuit check.

- With all connectors/couplers connected and voltage applied to the circuit being checked, measure voltage between each terminal and body ground.

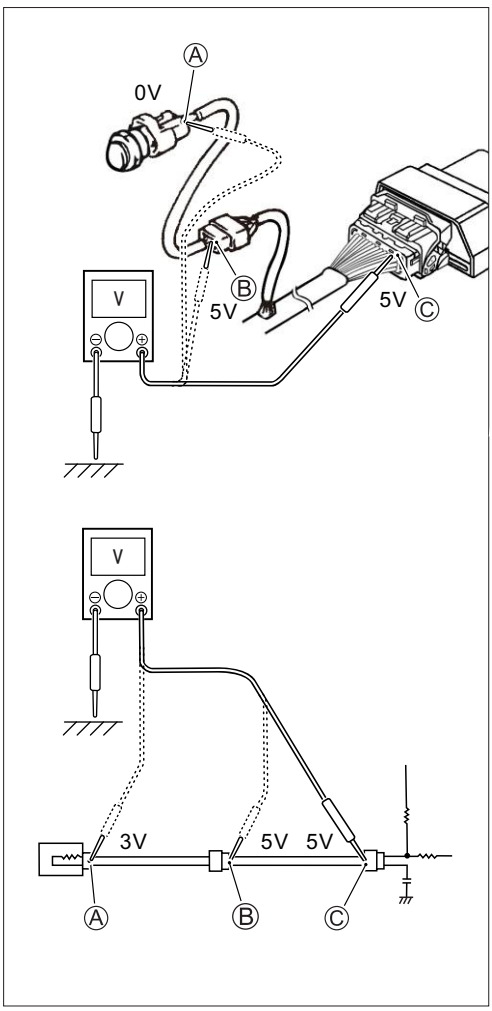
If measurements were taken as shown in the figure at the right and results are as listed below, it means that the circuit is open between terminals Ⓐ and Ⓑ.

Voltage Between:

Ⓒ and body ground: Approx. 5 V

Ⓑ and body ground: Approx. 5 V

Ⓐ and body ground: 0 V



Also, if measured values are as listed below, a resistance (abnormality) exists which causes the voltage drop in the circuit between terminals Ⓐ and Ⓑ.

Voltage Between:

Ⓒ and body ground: Approx. 5 V

Ⓑ and body ground: Approx. 5 V ——— 2 V voltage drop

Ⓐ and body ground: 3 V

SHORT CIRCUIT CHECK (WIRE HARNESS TO GROUND)

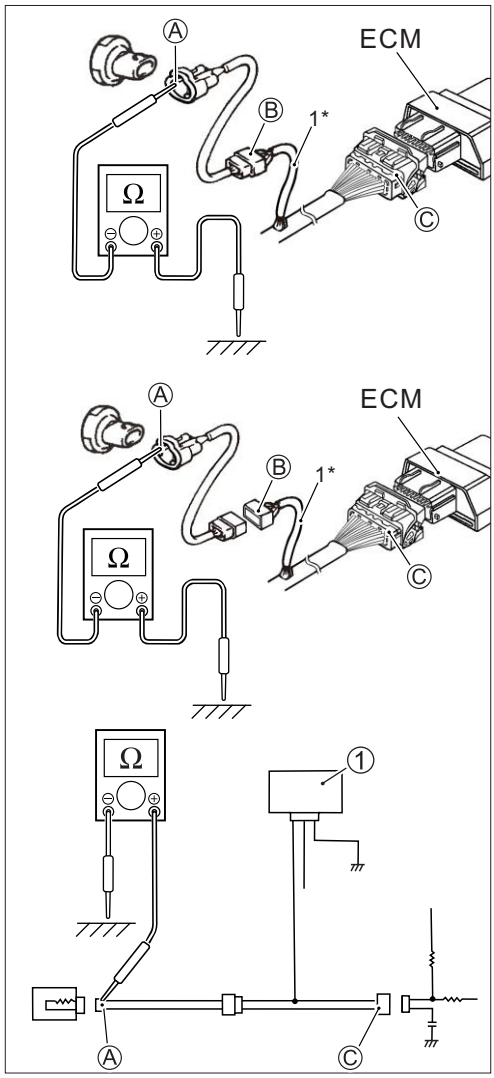
- Disconnect the negative ⊖ cable from the battery.
- Disconnect the connectors/couplers at both ends of the circuit to be checked.

NOTE:
If the circuit to be checked branches to other parts ① as shown, disconnect all connectors/couplers of those parts. Otherwise, diagnosis will be misled.

- Measure resistance between terminal at one end of circuit (Ⓐterminal in figure) and body ground. If continuity is indicated, there is a short circuit to ground between terminals Ⓐ and Ⓒ.

- Disconnect the connectors/couplers included in circuit (coupler Ⓑ) and measure resistance between terminal Ⓐ and body ground.

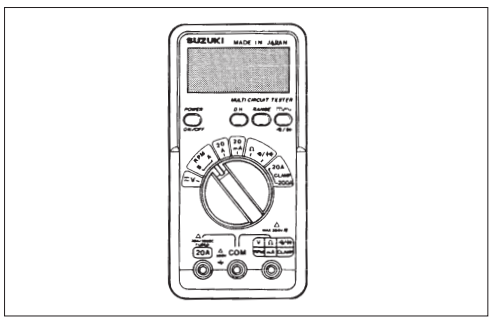
If the continuity is indicated, the circuit is shorted to the ground between terminal Ⓐand Ⓑ.



① Other parts
*1 To other parts

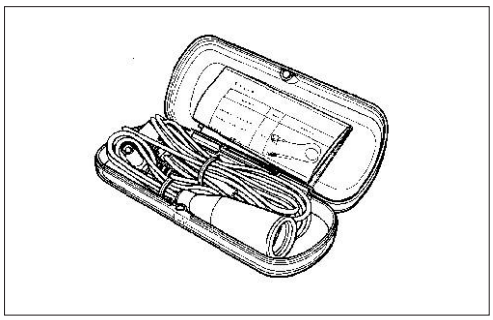
USING THE TESTER

- Incorrectly connecting the + and - probes may cause the inside of the tester to burnout.
- If the voltage and current are not known, make measurements using the highest range.
- When measuring the resistance with the tester ①, it will be shown as 10.00 M and “1” flashes in the display.
- Check that no voltage is applied before making the measurement. If voltage is applied the tester may be damaged.
- After using the tester, turn the power off.



TOOL **09900-25008: Multi circuit tester set**

NOTE:
When connecting the tester, use the needle-point probe to the back side of the lead wire coupler and connect the probes of tester to them.
Use the needle-point probe to prevent the rubber of the water proof coupler from damage.
When using the tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent the terminal damage.

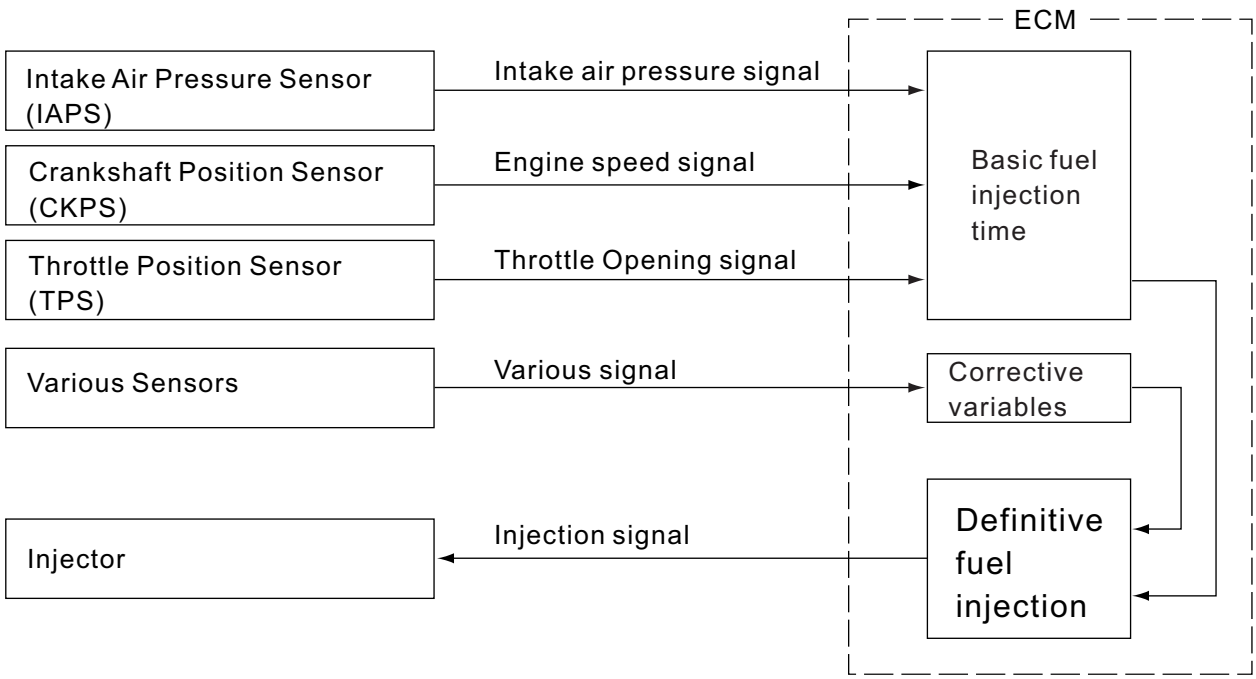


TOOL **09900-25009: Needle-point probe set**

FI SYSTEM TECHNICAL FEATURES

INJECTION TIME (INJECTION VOLUME)

The factors to determine the injection time include the basic fuel injection time, which is calculated on the basis of intake air pressure, engine speed and throttle opening angel, and various compensations. These compensations are determined according to the signals from various sensors that detect the engine and driving conditions.



COMPENSATION OF INJECTION TIME (VOLUME)

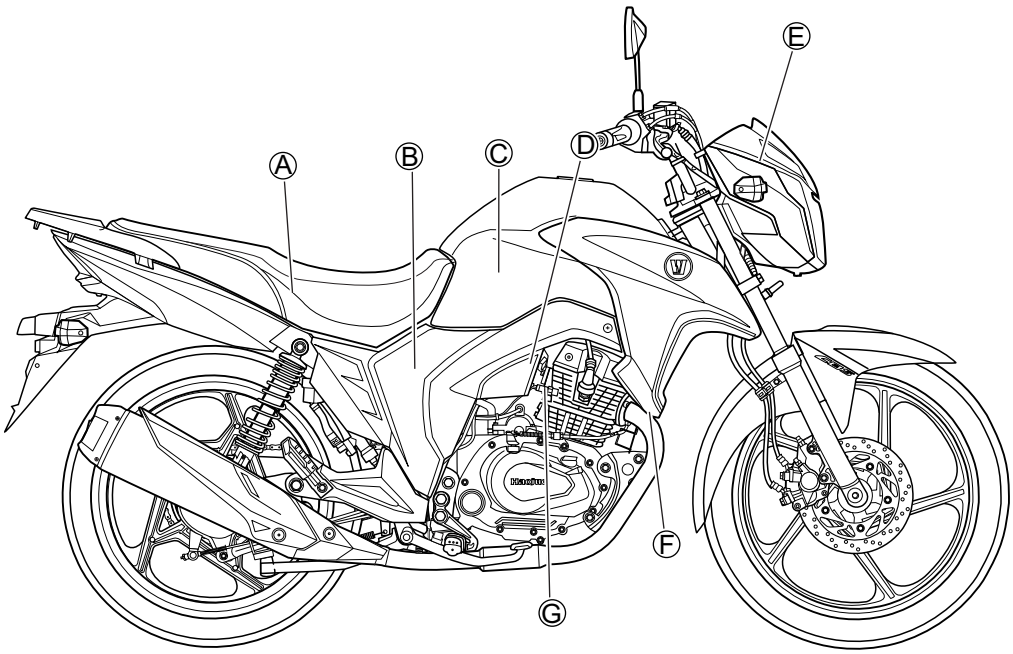
The following different signals are output from the respective sensors for compensation of the fuel injection time (volume).

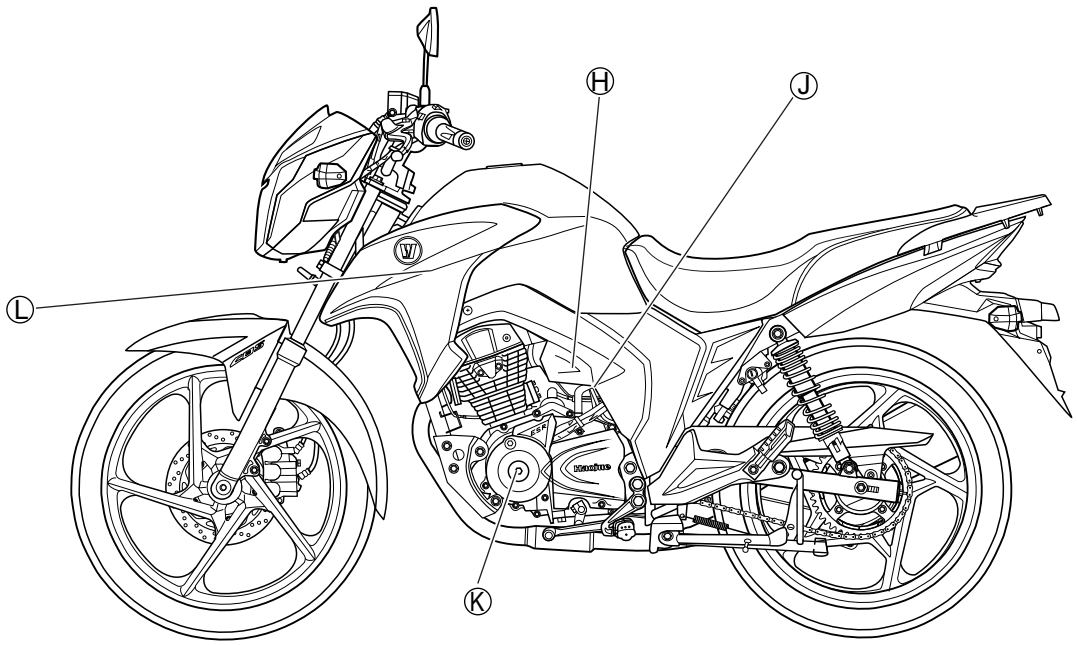
SIGNAL	DESCRIPTION
Engine temperature sensor signal	When engine temperature is low, injection time (volume) is increased.
Intake air temperature sensor signal	When intake air temperature is low, injection time (volume) is increased.
Oxygen sensor signal	Air/fuel ratio is compensated to the theoretical ratio from density of oxygen in exhaust gas. The compensation occurs in such a way that more fuel is supplied if detected air/fuel ratio is lean and less fuel is supplied if it is rich.
Battery voltage signal	ECM operates on the battery voltage and at the same time, it monitors the voltage signal for compensation of the fuel injection time (volume). A longer injection time is needed to adjust injection volume in the case of low voltage.
Engine speed signal	At high speed, the injection time (volume) is increased.
Starting signal	When starting engine, additional fuel is injected during cranking engine.
Acceleration signal/deceleration signal	During acceleration, the fuel injection time (volume) is increased in accordance with the throttle opening speed and engine rpm. During deceleration, the fuel injection time (volume) is decreased.

INJECTION STOP CONTROL

SIGNAL	DESCRIPTION
Over-rev. limit signal	The fuel injectors stop operation when engine speed reaches 10 000 r/min, this circuit cuts off fuel at the fuel injectors.

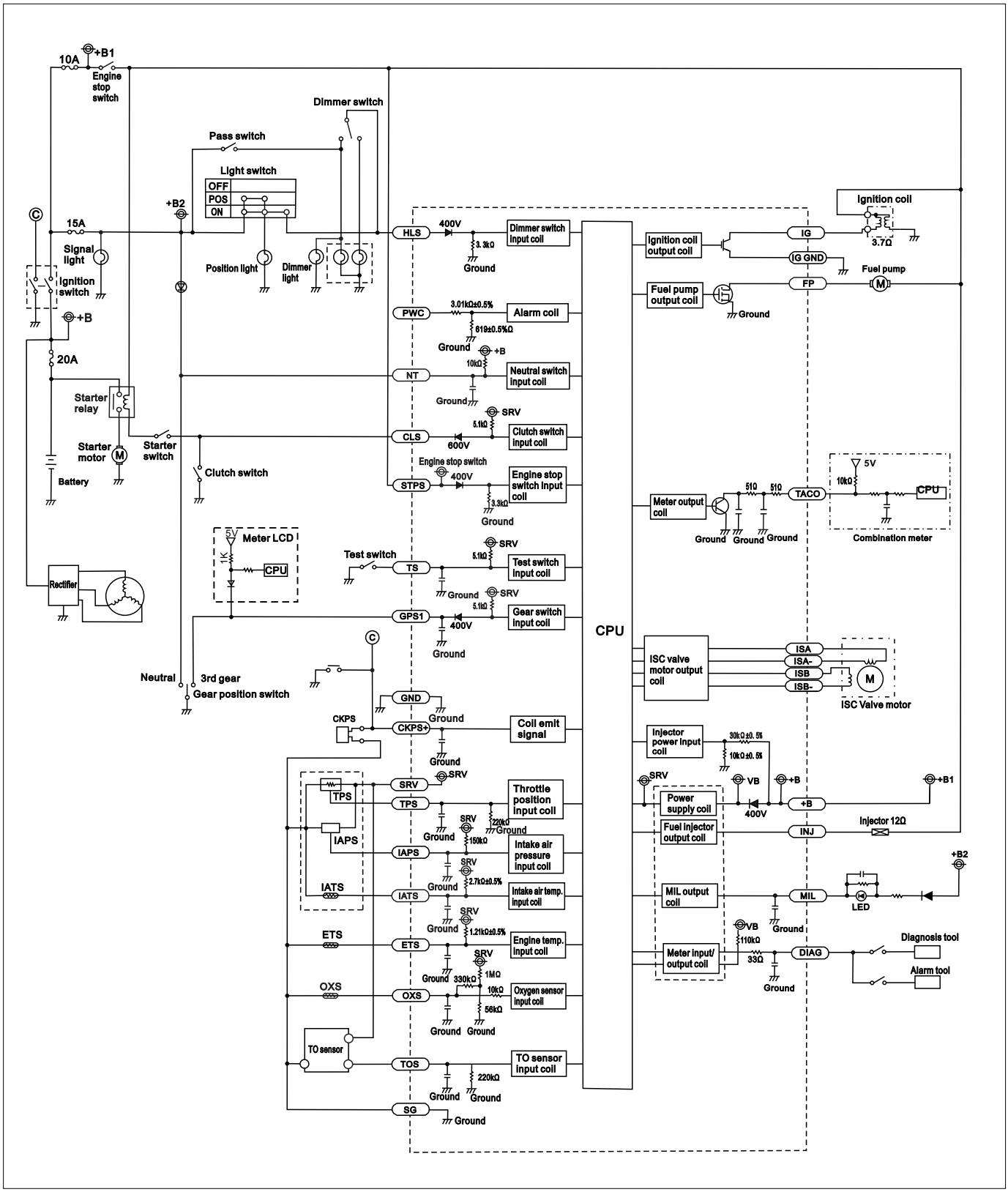
FI SYSTEM PARTS LOCATION



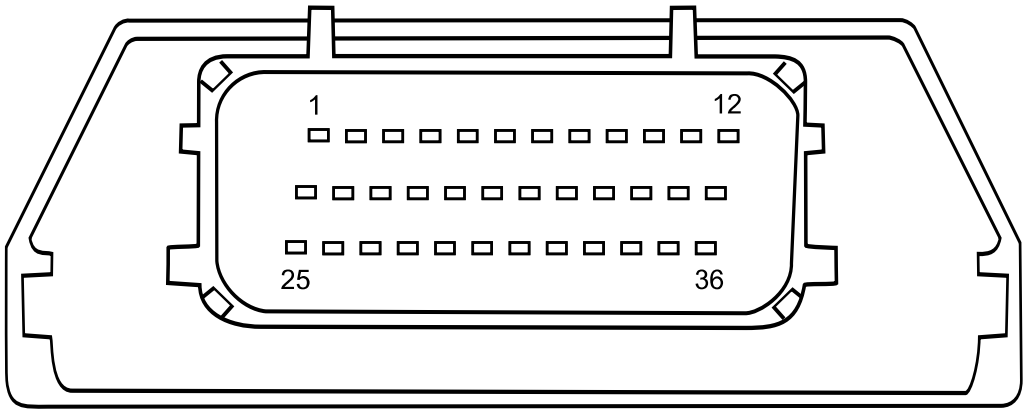


Ⓐ ECM
Ⓑ Fuse case
Ⓒ Fuel pump
Ⓓ Fuel injector
Ⓔ Meter
Ⓕ Oxygen sensor
Ⓖ Engine temp.sensor
Ⓗ ISC valve
Ⓙ 3 in 1 sensor
Ⓚ CKP sensor
Ⓛ Ignition coil

FI SYSTEM WIRING DIAGRAM



ECM TERMINALS



NO.	Circuit	NO.	Circuit
①	IG Ignition coil	⑲	IATS Intake air temperature sensor
②	MIL Malfunction indication light	⑳	OXS Oxygen sensor
③	FP Fuel pump	㉑	TPS Throttle position sensor
④	NT Neutral	㉒	ETS Engine temperature sensor
⑤	GND Ground	㉓	ISA Idle speed stepping motor A+
⑥	SG Sensor ground	㉔	ISA- Idle speed stepping motor A-
⑦	+B Battery	㉕	IG GND Ignition coil ground
⑧	CKPS+ Crankshaft position sensor +	㉖	TACO Tachometer
⑨	SRV Sensor voltage	㉗	————
⑩	TS Test switch	㉘	————
⑪	ISB- Idle speed stepping motor B-	㉙	STPS Engine stop switch
⑫	ISB Idle speed stepping motor B+	㉚	————
⑬	INJ Injector	㉛	HLS Headlamp switch
⑭	————	㉜	GPS1 Gear position switch1
⑮	————	㉝	IAPS Intake air pressure sensor
⑯	PWC Power check	㉞	DIAG Diagnosis tool
⑰	————	㉟	————
⑱	TOS Tip-over sensor	㊱	CLS Clutch switch

SELF-DIAGNOSIS FUNCTION

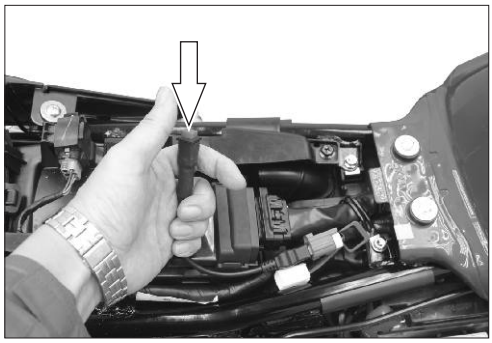
The self-diagnosis function is incorporated in the ECM. When the FI system has fault, ECM will turn on the "" (malfunction indication light) and save the DTC to erasable memory. When the voltage is too low, ECM no longer perform self-diagnosis.

- Turn on the ignition switch, after self-diagnosis, the "" light turns on, then start the engine, the "" turns off, it means no ECM defect is detected.
- If after engine is started, the "" light remains lit, it means the DTC or past DTC exists.
- Push the start button several times but the staring is no successful, the "" light is staying on, it means DTC exists.
- A two-digit DTC is shown through the flashing pattern of the "" light and also can be read by a defect diagnose device.

"" (MALFUNCTION INDICATION LIGHT) INSPECTION

When turn on the ignition switch, if the "" light doesn't perform self-diagnose, check if the ignition switch circuit 15A fuse burnt or "" open circuit.

CURRENT DTC / PAST DTC



There are two kinds of DTC:

CURRENT DTC

- Connect the testing switch and switch it on.
- Turn on the ignition switch, after self-diagnosis completed, the "" light starts output DTC.
- After the defect is removed, check and delete the past DTC.

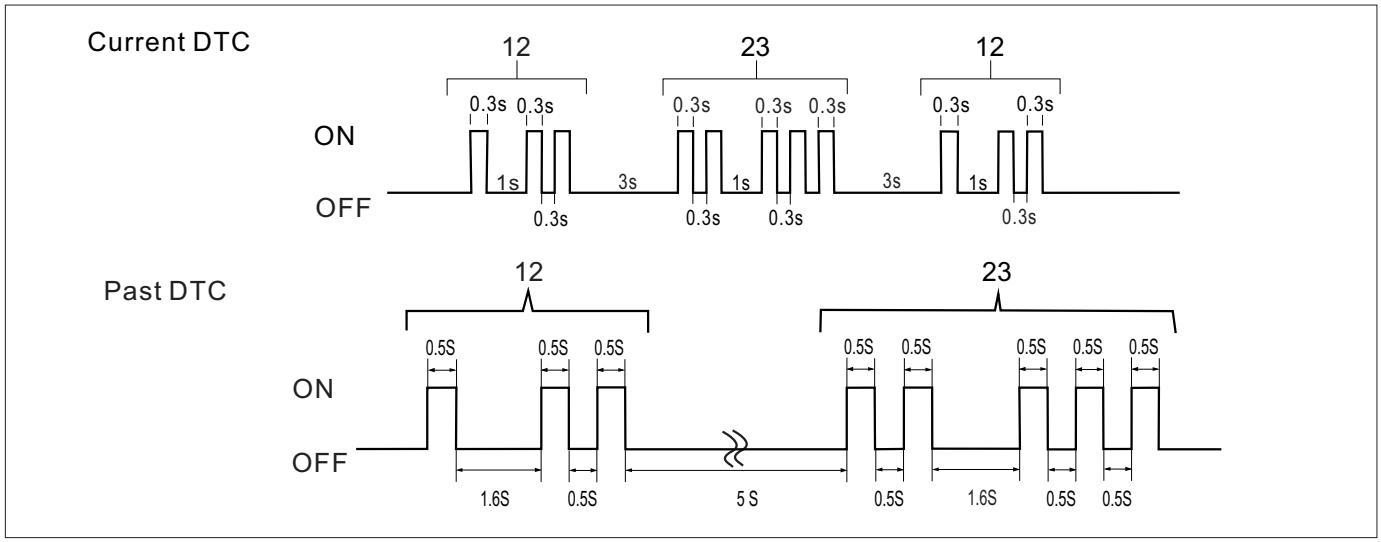
PAST DTC

- Connect the testing switch and switch it on.
- Open the throttle fully.
- Turn on the ignition switch, after the "" being off for 3's, the "" light starts flashing to output past DTC.
- Delete the past DTC with the testing switch.
  (Refer to page 4-14)

TOOL 09930-82720: Testing switch

UNDERSTANDING THE DTC (DIAGNOSTIC TROUBLE CODE)

A two-digit DTC is shown through the flashing pattern of the MIL. For current DTC, a number between 1 and 9 is represented by the number of times that the "" lights up in interval of 0.3 seconds and the separation between the tens and ones are indicated by the light staying off for 1 seconds. The separation between each DTC is indicated by the light being off for 3 seconds.
For past DTC, a number between 1 and 9 is represented by the number of times that the "" lights up in interval of 0.5 seconds and the separation between the tens and ones are indicated by the light staying off for 1.6 seconds. The separation between each DTC is indicated by the light being off for 5 seconds.

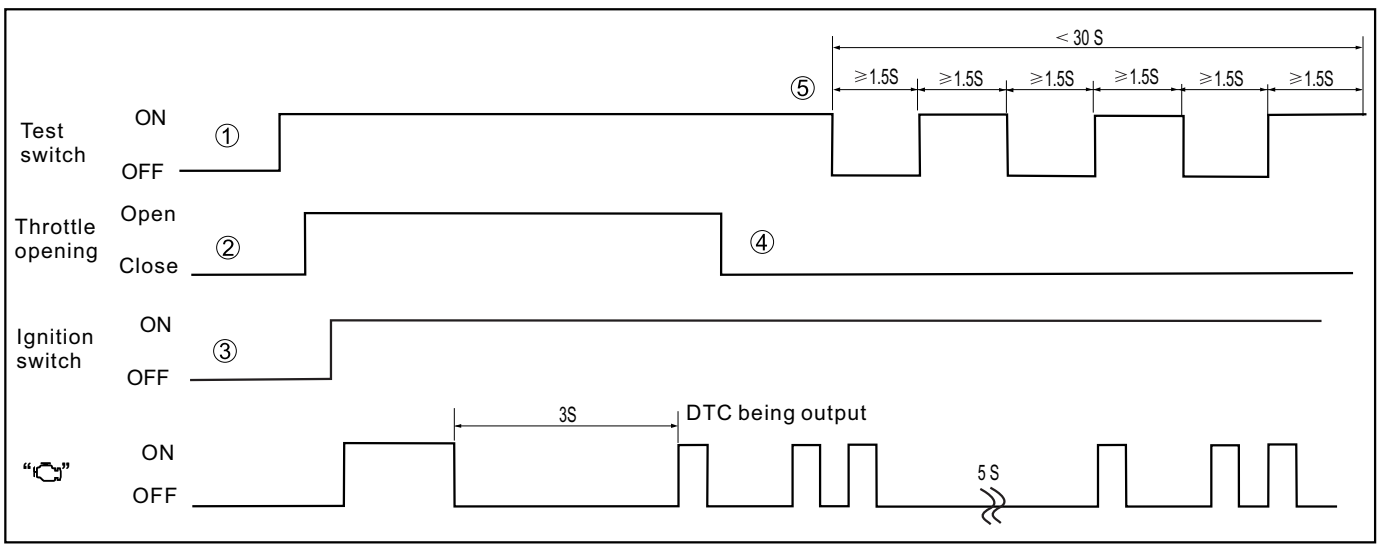


PAST DTC DELETING

1. Connect the test switch to test coupler, and turn on the test switch.
2. Open the throttle fully.
3. Turn on the ignition switch.
4. After the "" being off for 3s, "" starts flashing to output DTCs, close the throttle.
5. Turns off the test switch for more than1.5s, and switch on it for more than 1.5s, switch it from on to off three times, past DTCs are deleted.

⚠ CAUTION

Step 5 must be completed in 30 seconds.



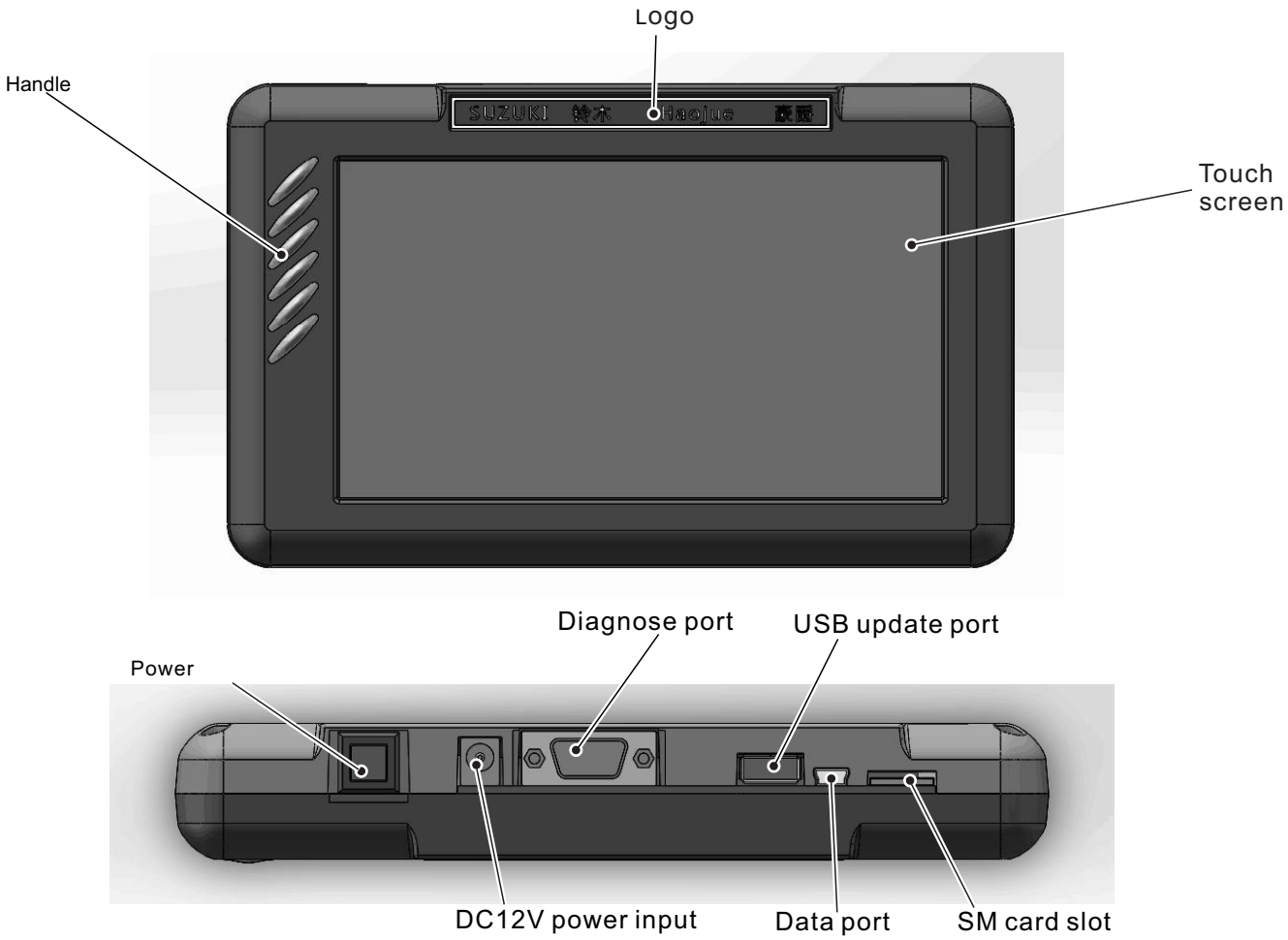
DTC TABLE

DTC	DTC	Malfunction part	Malfunction descriptions	MIL	Starting and running ability
12	P0335	Crankshaft position sensor(CKPS)	Open / short circuit to SRV	Blinks	Starting No Running No
13	P0105	Intake air pressure sensor (IAPS)	Open / short circuit to SRV	Turns on	Starting Yes Running Yes
	P0107		Short circuit to GND		
14	P0123	Throttle position sensor (TPS)	Open / short circuit to SRV	Turns on	Starting Yes Running Yes
	P0120		Short circuit to GND		
15	P0115	Engine temperature sensor(ETS)	Open / short circuit to SRV	Turns on	Starting Yes Running Yes
	P0117		Short circuit to GND		
21	P0110	Intake air temperature sensor(IATS)	Open / short circuit to GND	Turns on	Starting Yes Running Yes
	P0112		Short circuit to GND		
23	P1502	Tip-over sensor(TOS)	Open / short circuit to SRV	Turns on	Starting No Running No
	P1503		Short circuit to GND		
24	P0350	Ignition coil	Open / short circuit to GND	Turns on	Starting No Running No
32	P0200	Injector(INJ)	Open / short circuit to GND	Turns on	Starting No Running No
40	P0505	Idle speed control valve(ISC valve)	Short circuit to GND	Turns on	Starting Yes Running Yes
	P0507		Open / short circuit to +B/ ISC valve is fixed.		
41	P0230	Fuel pump(FP)	Open / short circuit to GND	Turns on	Starting No Running No
42	P1501	Power check	Open circuit/lower voltage	Turns on	Starting No Running No
44	P0130	Oxygen sensor(OXS)	Short circuit to GND	Turns on	Starting Yes Running Yes
	P0132		Short circuit to SRV		
	P0131		Short circuit to GND		
99	P0560	Battey(high voltage)	Defective rectifier	Turns on	Starting Yes Running Yes

Note: PXXXX DTC would only be read by the diagnosis device.

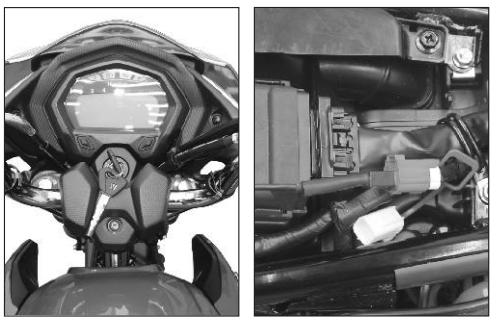
DIAGNOSIS DEVICE

INTRODUCTION

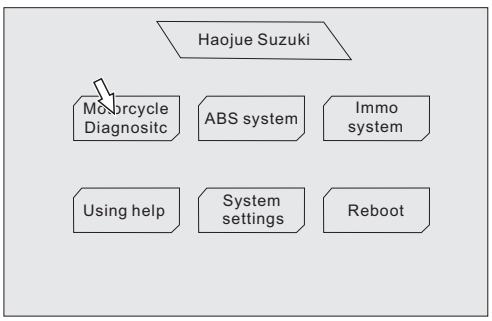


USE THE DIAGNOSE DEVICE

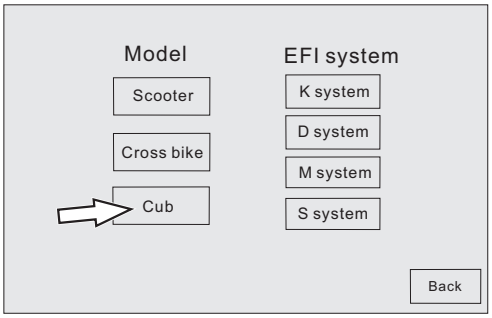
- Turn off the ignition switch.
- Connect the diagnosis device to battery, then connect the diagnose device coupler.



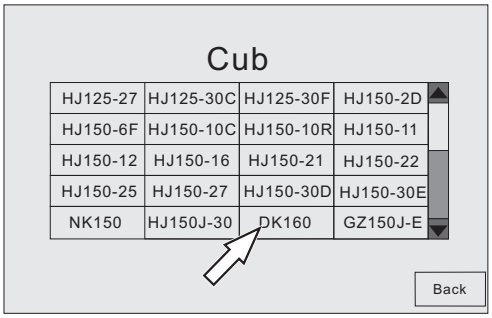
- Turn on the ignition switch.
- Turn on the diagnosis device.



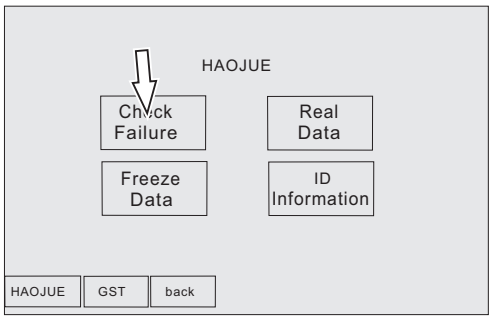
- Click “Cross bike”, enter the model selection page



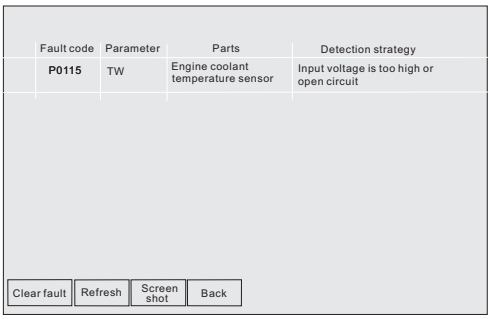
- Click “DR160” on the model selection page.



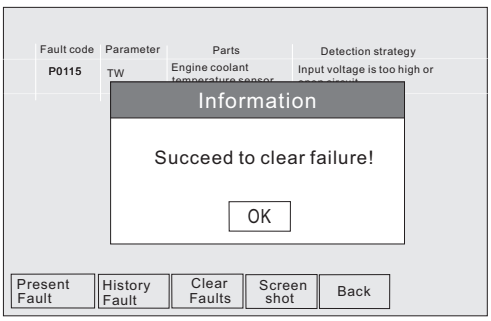
- Click “Check Failure”.



- Enter the defect diagnosis page, display the defect content.



- After the defect is removed, click” Delete DTC”.
- After the DTCs are deleted, click”Refresh”, check if the defect is removed again. After confirming defect is removed, exit the diagnosis system, turn off the ignition switch.



FI SYSTEM TROUBLESHOOTING

INSPECTION PRIOR TO DIAGNOSIS

Prior to diagnosis, perform the following visual inspections. The reason for visual inspection is that mechanical failures(such as oil leakage) cannot be displayed on the screen.

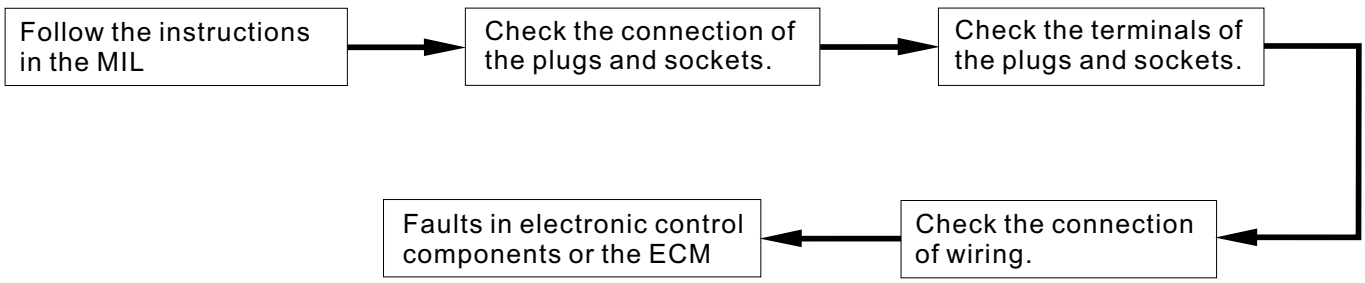
* Working condition of the MIL
* Engine oil volume and leakage
* Battery condition
* Connection of plugs and sockets
* Hose crack, damage and loose
* Melted fuse
* Throttle cable clearance
* Fuel volume and leakage
* Air cleaner blockage

FI SYSTEM FAULT ANALYSIS

◇ Fault with specific DTC: System faults that have specific fault codes corresponding to it.
Features: Malfunction indication light displays a single DTC.
Causes: Short or broken circuits in the signal circuit or control circuit of electronic control components; short or broken circuits in the electronic control components; Short or broken circuits in the ECM.

For example: defective ET sensor and its wire.

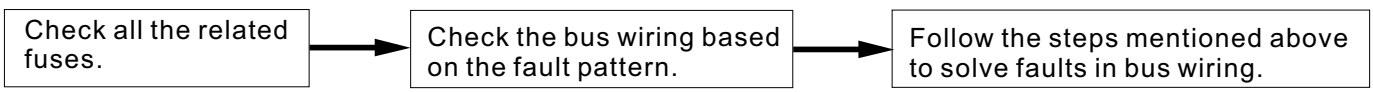
PROCESS PROCEDURE:



◇ **Faults without specific DTC**: The actual fault is not the same one indicated by the MIL or the diagnosis device.
Features: The MIL shows several DTC at the same time.
Causes: Short or broken circuits in the bus wiring.

For example: Faults in the central relay, relay fuse, sensor power supply or sensor wiring.

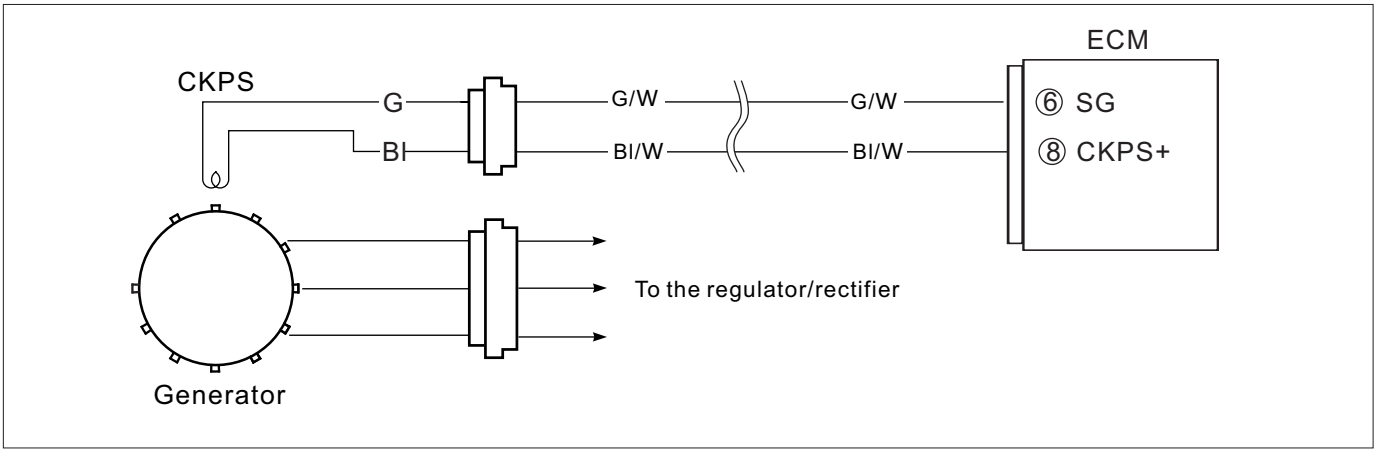
PROCESS PROCEDURE:



◇ The motorcycle tests “normal” using a diagnosis device but will not start or will not work normally after starting.
Features: The MIL doesn't display any DTC.
Causes: Non-electrical faults or inaccurate signals in the electrical controlled components.

DTC "12" : CKPS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

⚠ CAUTION

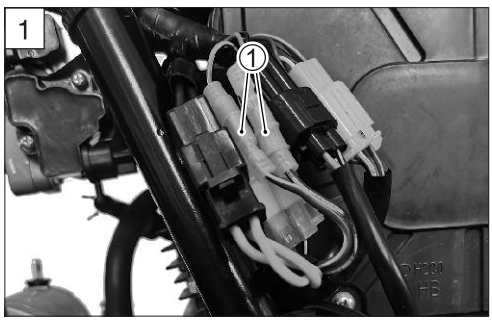
When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

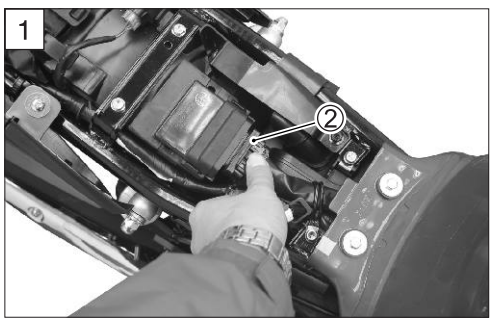
Step 1

CKP sensor signal circuit check

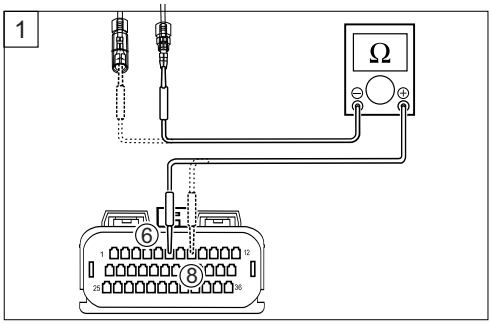
1) Turn off the ignition switch.
2) Remove the saddle and left side cover. (Refer to page 6-1)
3) Disconnect the CKP sensor coupler ①.



4) Disconnect the ECM coupler ②.



5) Check for proper terminal connection to the CKPS coupler and ECM couplers.
6) If connections are OK, check the following points.

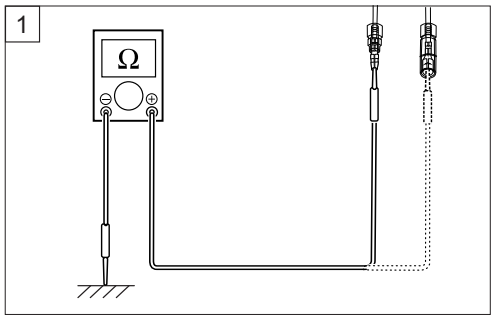


＊Resistance

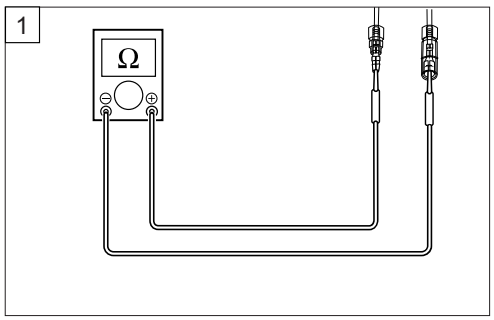
TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

DATA **Between the sensor's coupler and ECM coupler: less than 1Ω**

DATA **Between G/W wire or Bl/W wire with ground: infinity.**



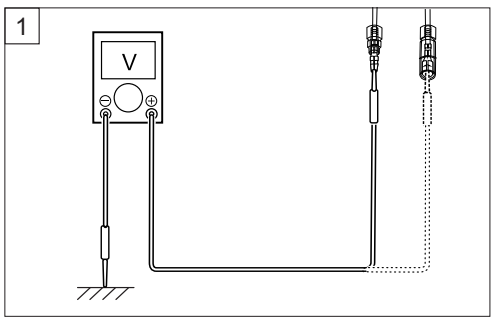
DATA **Between the G/W wire and Bl/W wire: infinity**



＊Voltage

TOOL **09900-25008: Multi-circuit tester set**

DATA **Between the G/W wire or Bl/W wire with ground: approx 0 V. (When the ignition switch is turned on).**



Is check result ok ?

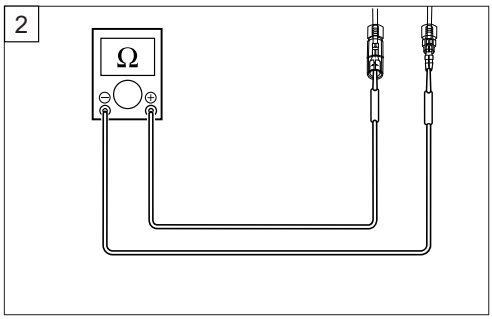
Yes Go to step 2.
No Repair or replace the defective wire.

Step 2
CKPS check

1) Measure the CKPS resistance.

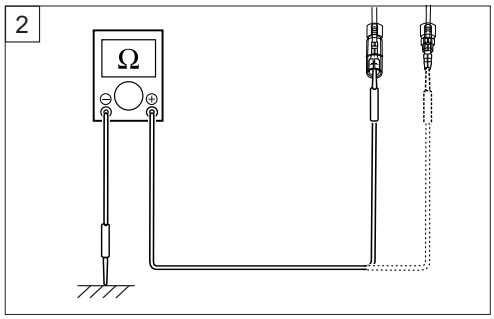
09900-25008: Multi circuit tester set

CKPS resistance: 115-155 Ω



2) Check the continuity between each terminal and ground.

CKPS continuity: infinity

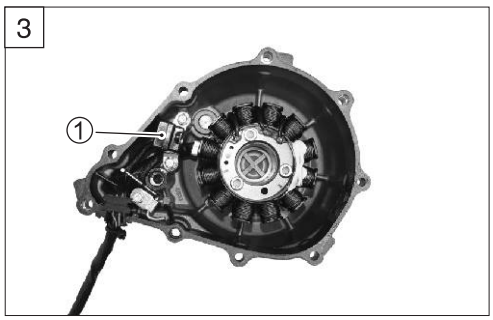


Is check result OK?

Yes Go to step 3.
No Replace CKPS.

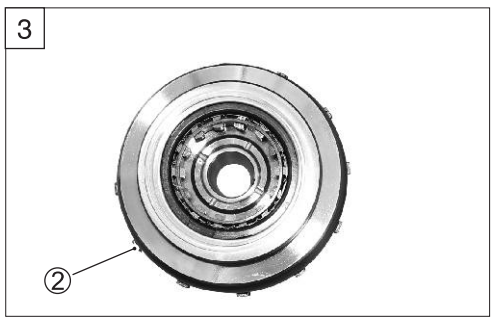
Step 3
CKPS and signal rotor check

1) Remove the generator cover. (Refer to page 3-27)
2) Check the end face of the CKPS ① and generator teeth ② are free from any metal particles and damage.



Is check result OK ?

Yes Go to step 4.
No Clear or replace defective parts.

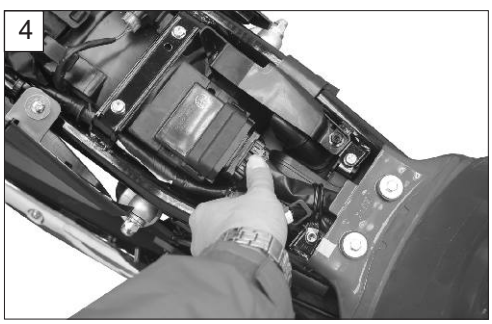


Step 4
ECM check

1) Reconnect a new ECM.

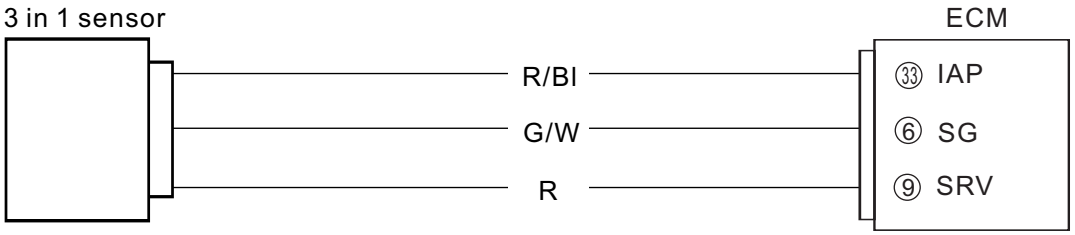
Is check result OK ?

Yes End.
No Replace CKPS. (Refer to page 3-27)



DTC "13" : IAPS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

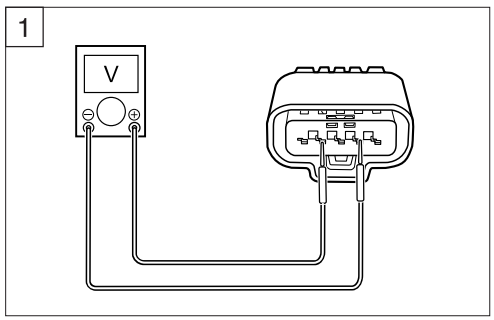
Step 1
IAPS power supply circuit check

1) Turn off the ignition switch.
2) Remove the rear and front saddles. (Refer to page 6-1)
3) Disconnect the 3 in 1 sensor coupler ①.
4) Check for proper terminal connection to the IAPS coupler.

5) If connections are OK, turn on the ignition switch.
6) Measure the voltage between the red wire and G/W wire.

TOOL **09900-25008: Multi-circuit tester set**

DATA **TPS power supply voltage: 4.5-5.5V**
(⊕terminal red wire – ⊖terminal G/W wire)



Is check result ok ?

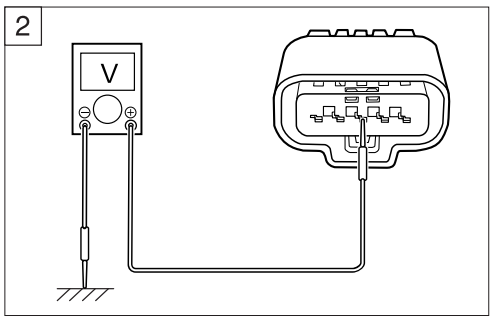
Yes Go to step 3.
No Go to step 2.

Step 2
IAP sensor ground circuit check

1) Measure the voltage between the red wire and ground.

09900-25008: Multi-circuit tester set

IAP sensor voltage: 4.5-5.5V
(⊕terminal: red wire – ⊖terminal: ground)

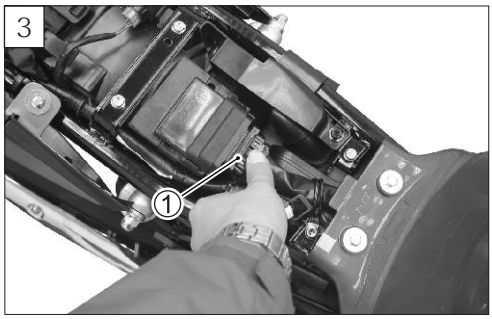


Is check result ok ?

Yes Repair or replace G/W wire.
No Repair or replace red wire.

Step 3
IAPS signal wire check

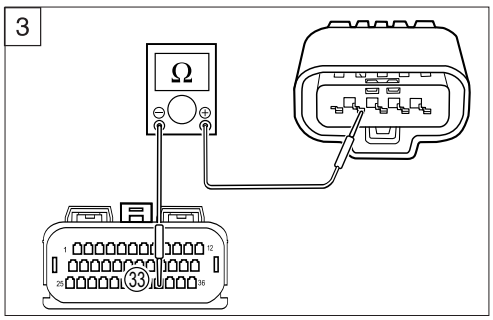
1) Turn off the ignition switch.
2) Disconnect the ECM coupler ①.
3) Check for proper terminal connection to the ECM coupler.
4) If connections are OK, check the following points.



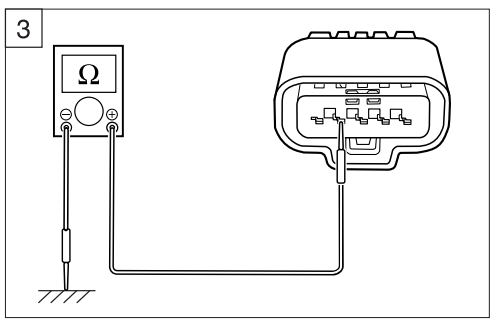
∗ Resistance

09900-25008: Multi-circuit tester set
09900-25009: Needle-point probe set

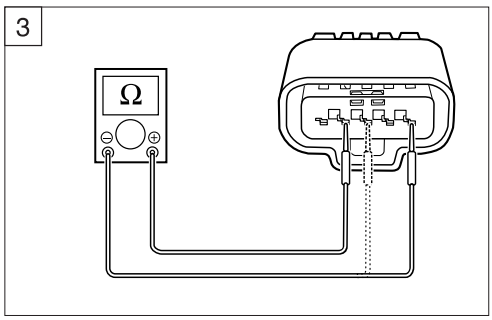
Between the R/Bl wire of IAP sensor coupler and ECM coupler: less than:1Ω



Between the R/Bl wire and ground: infinity



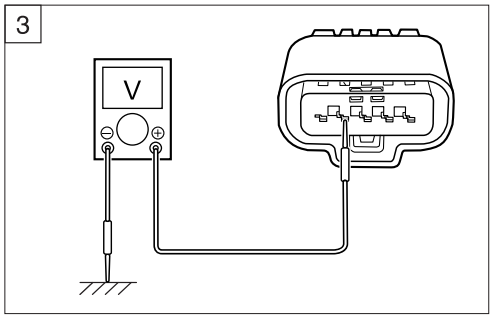
DATA **Between the R/Bl wire and G/W wire or red wire: infinity.**



*** Voltage**

TOOL **09900-25008: Multi circuit tester set**

DATA **Between the R/Bl wire and ground: approx 0 V. (When the ignition switch is turned on)**



Is check result ok ?

Yes Go to step 4.
No Repair or replace the R/Bl wire.

Step 4
ECM check

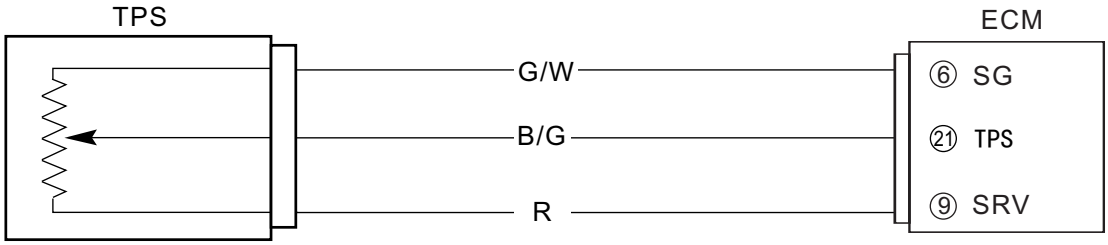
1) Reconnect a new ECM.

Is check result ok ?

Yes End.
No Replace 3 in 1 sensor and check again.

DTC "14": TPS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

⚠ CAUTION

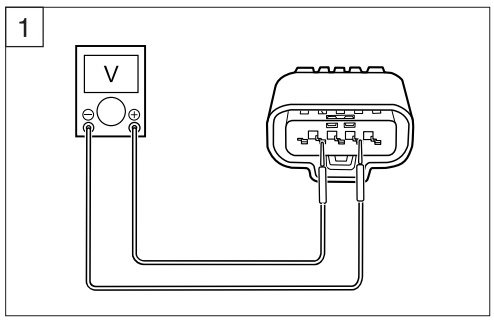
When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Step 1
TP sensor power supply circuit check

1) Turn off the ignition switch.
2) Remove the saddle and right side cover. (Refer to page 6-1)
3) Disconnect the 3 in 1 sensor coupler ①.
4) Check for proper terminal connection to the TPS coupler.

5) If connections are OK, turn on the ignition switch.
6) Measure the voltage between the red wire and G/W wire.



TOOL **09900-25008: Multi-circuit tester set**

DATA **TPS power supply voltage: 4.5-5.5V**
(⊕terminal red wire – ⊖terminal G/W wire)

Is check result ok ?

Yes Go to step 3.
No Go to step 2.

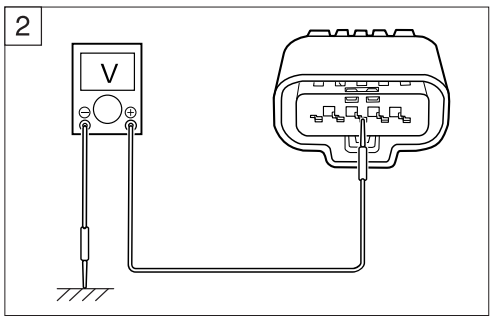
Step 2

TPS ground circuit check

1) Measure the voltage between the red wire and ground.

TOOL **09900-25008: Multi-circuit tester set**

DATA **TP sensor voltage: 4.5-5.5V**
(⊕ terminal: red wire – ⊖ terminal: Ground)



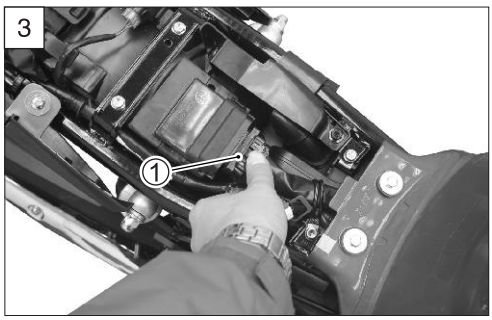
Is check result ok ?

Yes Repair or replace the G/W wire.
No Repair or replace the red wire.

Step 3

TPS signal wire check

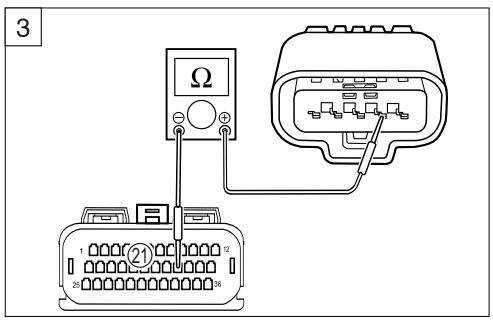
1) Turn off the ignition switch.
2) Disconnect the ECM coupler ①.
3) Check for proper terminal connection to the ECM coupler.
4) If connections are OK, check the following points.



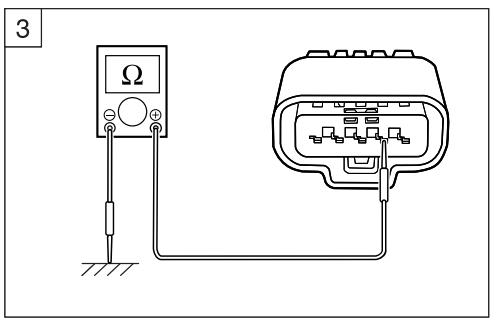
∗ **Resistance**

TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

DATA **Between the B/G wire of TPS coupler and ECM coupler: less than 1 Ω.**

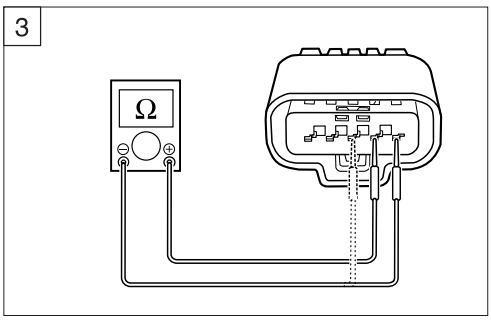


DATA **Between the B/G wire and ground: infinity**



DATA **Between the B/G wire and G/W wire with red wire: infinity**

3

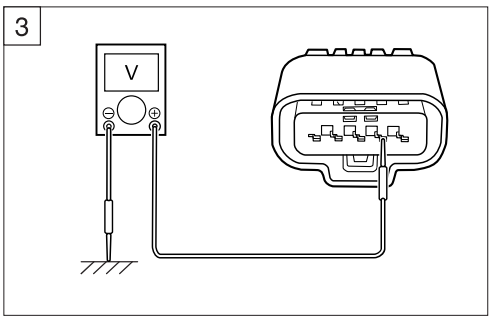


∗ **Voltage**

TOOL **09900-25008: Multi-circuit tester set**

DATA **Between the B/G wire and ground: approx 0 V. (When the ignition switch is turned on.)**

3

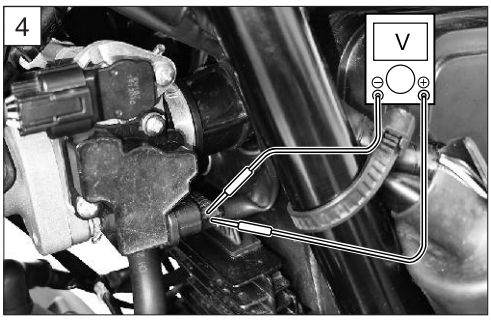


Is check result ok ?

Yes Go to step 4.
No Repair or replace B/G wire.

Step 4
TP sensor check

4



1) Turn off the ignition switch.
2) Connect the ECM couplers and TP sensor coupler.
3) Insert the needle-point probes to the lead wire coupler.
4) Turn on the ignition switch.
5) Measure the TP sensor output voltage between the B/G wire terminal ⊕ and G/W wire terminal ⊖ with turning the throttle grip open and close.

TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

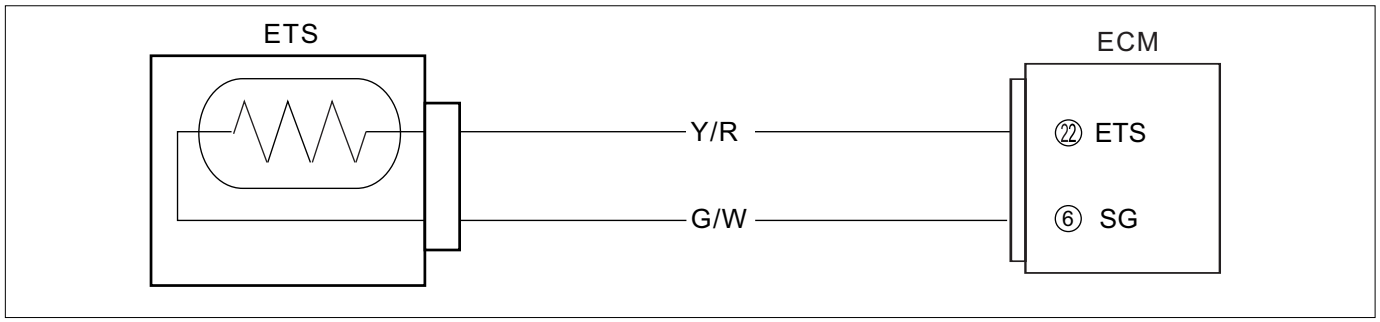
DATA **TP sensor output voltage: Full close: 0.4-0.9V**
Full open: 3.5-4.2V
(⊕ terminal: B/G wire – ⊖ terminal: G/W wire)

Is check result ok ?

Yes Replace the ECM and check again.
No Replace 3 in 1 sensor and check again.

DTC "15": ETS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

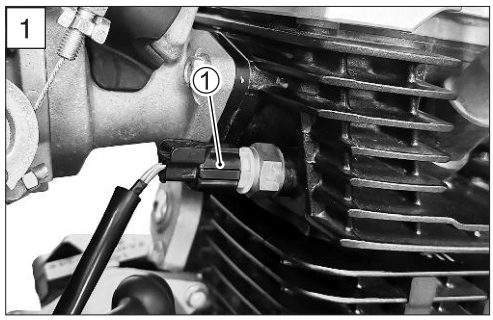
⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

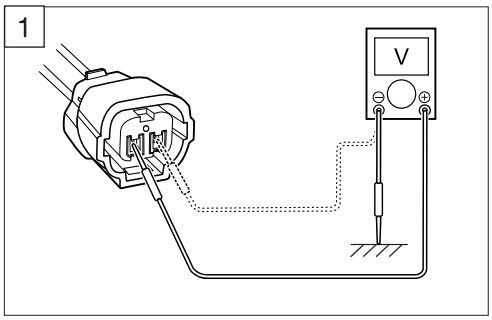
NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Step 1
ETS input voltage check

1) Turn off the ignition switch.
2) Disconnect the ETS coupler ①.
3) Check for proper terminal connection to the ET sensor coupler.



4) If ok, turn on the ignition switch.
5) Measure the voltage between the Y/R wire with ground and G/W wire.



TOOL **09900-25008: Multi-circuit tester set**

DATA **ETS input voltage: 4.5-5.5V**
(⊕ terminal: Y/R wire – ⊖ terminal: Ground)
(⊕ terminal: Y/R wire – ⊖ terminal: G/W wire)

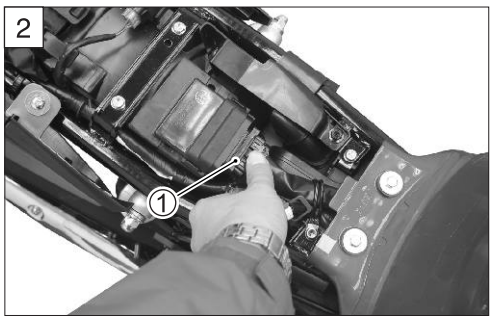
Is check result ok ?

Yes Go to step 3.
No Go to step 2.

Step 2

ETS circuit check

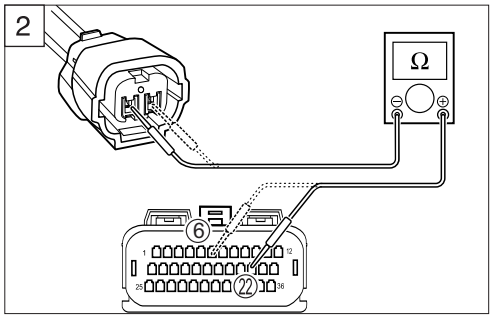
1) Turn off the ignition switch.
2) Remove the saddles.
3) Disconnect the ECM coupler ①.
4) Check for proper terminal connection to the ECM coupler.
5) If connections are OK, check the following points.



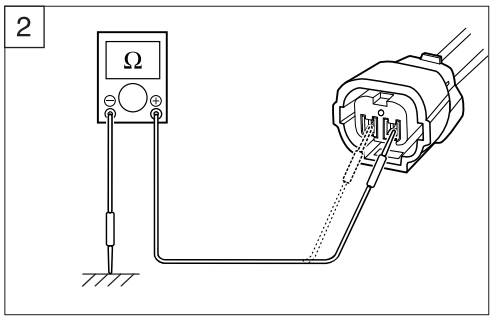
＊Resistance

TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

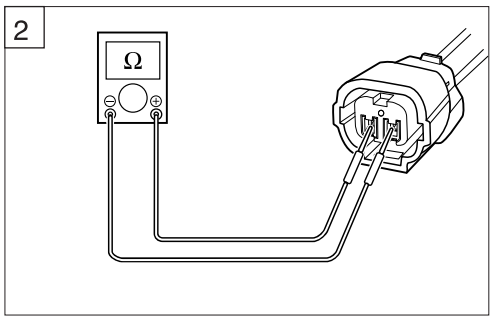
DATA **Between the sensor's coupler and ECM coupler: less than 1Ω**



DATA **Between the Y/R wire or G/W wire and ground: infinity.**



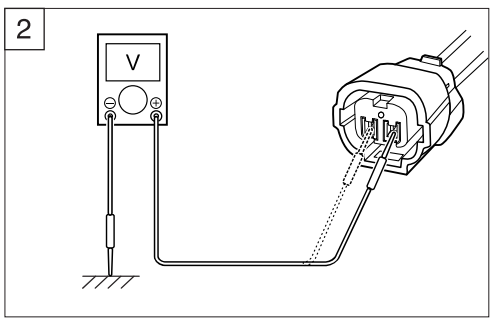
DATA **Between the Y/R wire or G/W wire: infinity.**



＊Voltage

TOOL **09900-25008: Multi-circuit tester set**

DATA **Between the Y/R wire or G/W wire with ground: approx. 0 V. (When the ignition switch is turned on.)**



Is check result ok ?

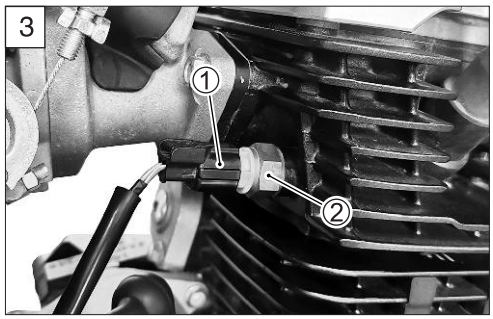
Yes Replace the ECM and check again.
No Repair or replace the defective wire harness.

Step 3

ETS check

1) Turn off the ignition switch.

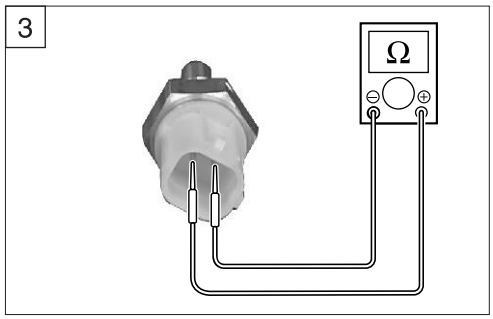
2) Disconnect the ETS coupler ①, remove the ETS ②.



3) Measure the ETS resistence.

09900-25008: Multi-circuit tester set

ET sensor resistence: approx 2.645 kΩ, at 20°C (Terminal-Terminal)



Is check result ok ?

Yes Replace the ECM and check again.

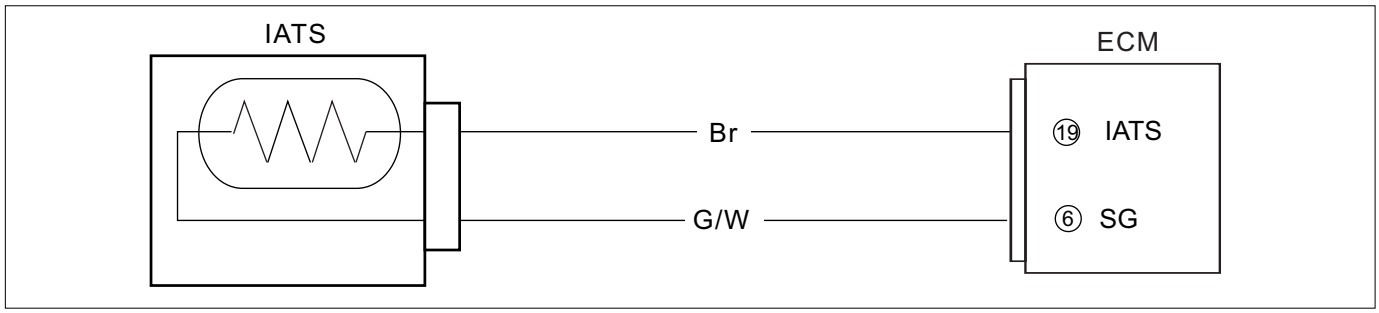
No Replace the ETS and check again.

CAUTION

Before installing the ETS, check the seal ring for damage or scratch, if any defect is found, replace the seal ring with a new one.

DTC "21": IATS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

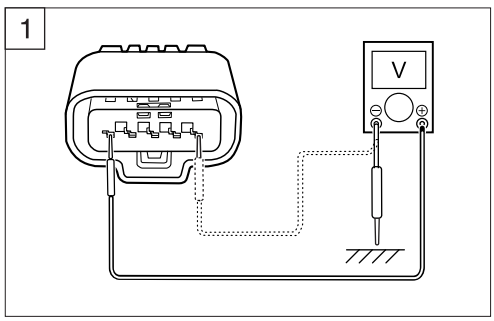
Step 1
IATS power circuit check

1) Turn off the ignition switch.
2) Remove the saddle and left side cover. (Refer to page 6-1)
3) Disconnect the 3 in 1 sensor coupler ①.
4) Check for proper terminal connection to the IATS coupler.
5) If connections are OK, turn on the ignition switch.

6) Measure the voltage between the brown wire with ground and G/W wire.

TOOL **09900-25008: Multi-circuit tester set**

DATA **IATS voltage: 4.5-5.5V**
(⊕ terminal: brown wire – ⊖ terminal: Ground)
(⊕ terminal: brown wire – ⊖ terminal: G/W wire)



Is check result ok ?

Yes Go to step 3.
No Go to step 2.

Step 2

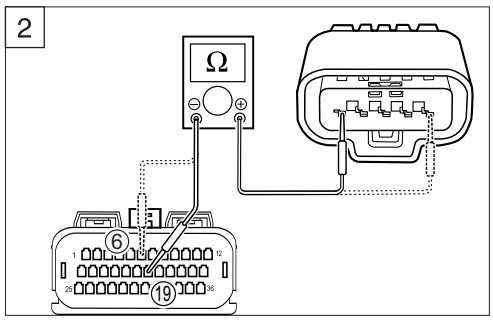
IATS circuit check

1) Turn the ignition switch off.
2) Disconnect the ECM coupler ①.
3) Check for proper terminal connection to the ECM coupler.
4) If connection is OK, check the following points.

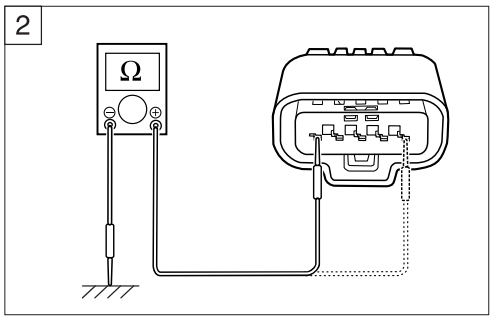
＊Resistance

TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

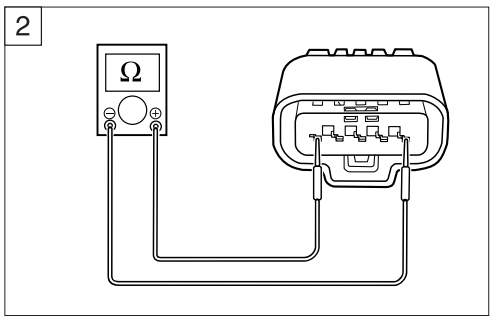
DATA **Between the sensor's coupler and ECM coupler: less than 1Ω**



DATA **Between the brown wire or G/W wire with ground: infinity.**



DATA **Between the brown wire and G/W wire: infinity.**



Step 3

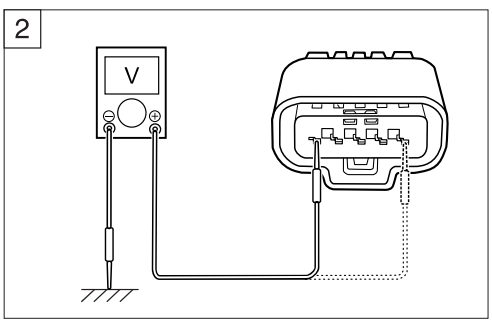
＊Voltage

TOOL **09900-25008: Multi-circuit tester set**

DATA **Between the brown wire or G/W wire with ground: approx 0 V. (When the ignition switch is turned on)**

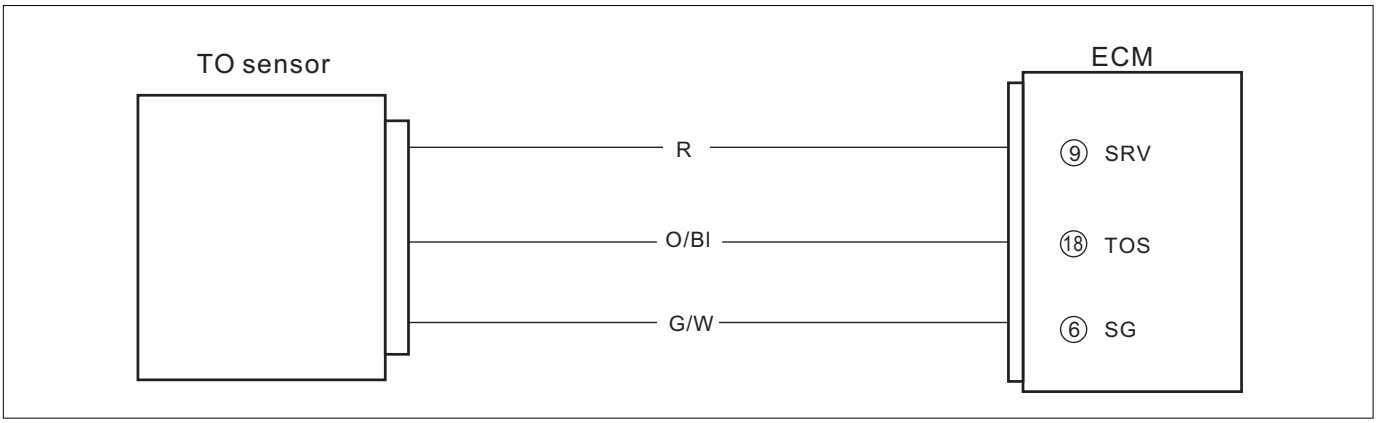
Is check result ok ?

Yes Replace the ECM and Check again.
No Replace 3 in 1 sensor and check again.



DTC "23": TOS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

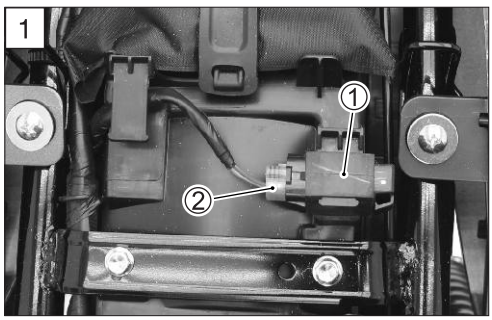
⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Step 1
TOS power supply circuit check

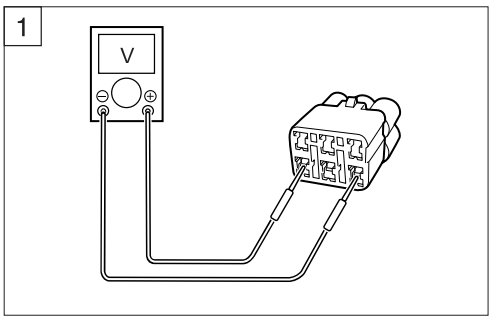
1) Turn off the ignition switch.
2) Remove the saddle.
3) Remove the TOS ① and disconnect the TOS coupler ②.
4) Check for proper terminal connection to the TOS coupler.



5) If connections are OK, turn on the ignition switch.
6) Measure the input voltage between the red wire and G/W wire.

TOOL **09900-25008: Multi-circuit tester set**

DATA **TOS input voltage: 4.5-5.5V**
(⊕terminal red wire – ⊖terminal G/W wire)



Is check result OK ?

Yes Go to step 3.
No Go to step 2.

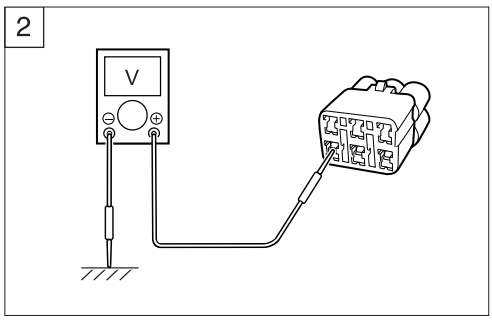
Step 2

TOS ground circuit check

1) Measure the voltage between the red wire and ground.

TOOL **09900-25008: Multi-circuit tester set**

DATA **TOS input voltage: 4.5-5.5 V.**
(⊕ terminal: red wire – ⊖ terminal: Ground)



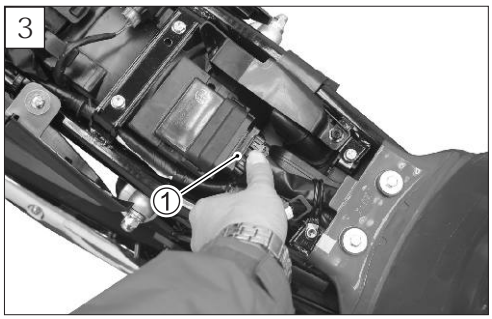
Is check result OK ?

Yes Repair or replace G/W wire
No Repair or replace red wire.

Step 3

TOS circuit check

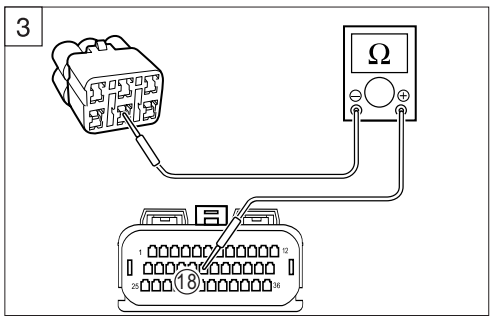
1) Turn off the ignition switch.
2) Disconnect the ECM coupler ①.
3) Check for proper terminal connection to the ECM coupler.
4) If connections are OK, check the following points.



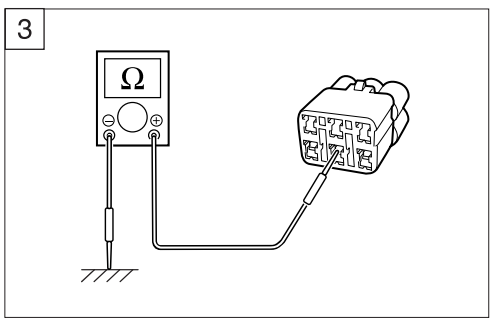
∗ **Resistance**

TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

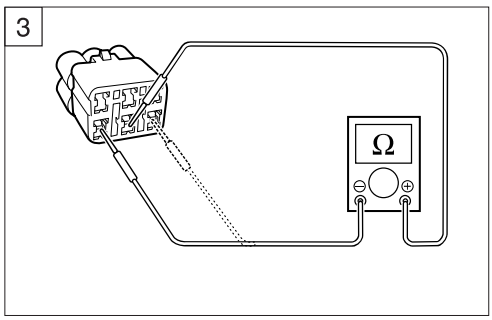
DATA **Between the O/Bl wire of TOS coupler and ECM coupler: less than 1 Ω.**



DATA **Between the O/Bl wire and ground: infinity.**



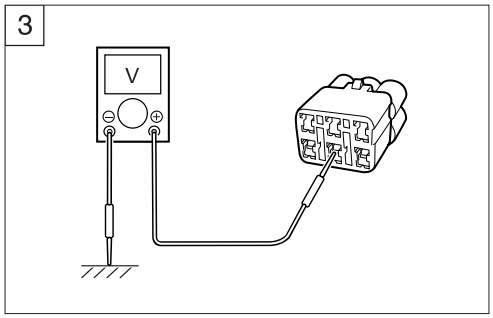
DATA **Between the O/BI wire and other terminals: infinity**



∗ **Voltage**

TOOL **09900-25008: Multi circuit tester set**

DATA **Between the O/BI wire and ground: approx 0 V.**
(When the ignition switch is turned on.)

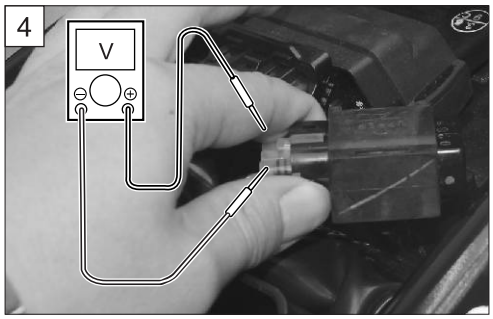


Is check result OK ?

Yes Go to step 4.
No Repair or replace the O/BI wire.

Step 4
TOS check

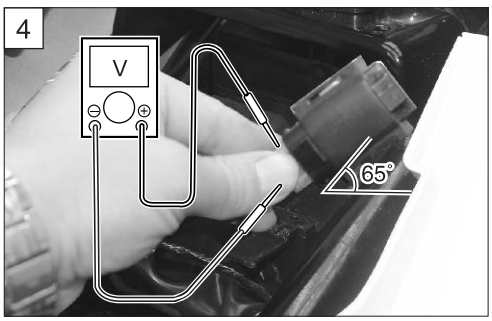
1) Turn off the ignition switch.
2) Connect the ECM couplers and TOS coupler.
3) Insert the needle-point probes to the lead wire coupler.
4) Turn on the ignition switch.
5) Measure the voltage between the O/BI wire and G/W wire.



TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

DATA **TOS voltage: 0.4-1.4V**
(⊕ terminal: O/BI wire – ⊖ terminal: G/W wire)

6) Measure the voltage when the TOS is leaned 45° and more, left and right, from the horizontal level.



TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

DATA **TOS voltage: 3.7-4.4V**
(⊕ terminal: O/BI wire – ⊖ terminal: G/W wire)

Is check result ok ?

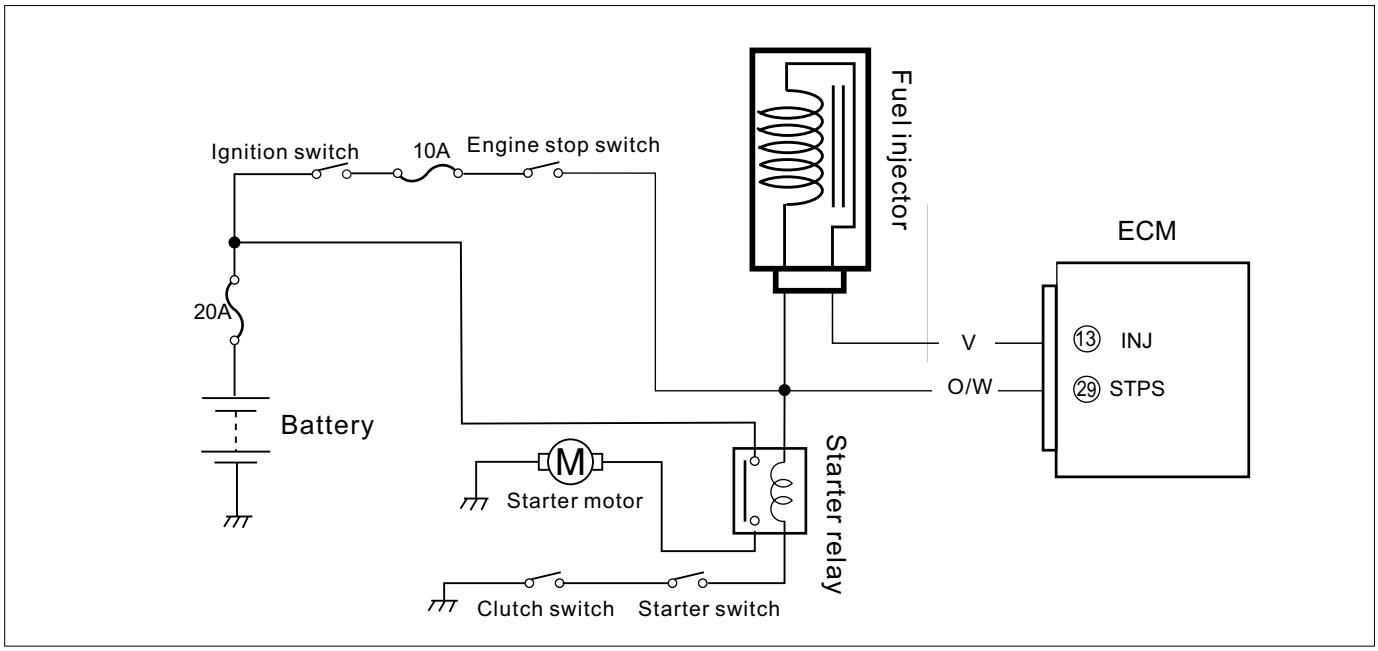
Yes Replace the ECM and check again.
No Replace the TOS and check again.

DTC “24”: IGNITION SYSTEM CIRCUIT MALFUNCTION

(Refer to page 7-8)

DTC “32”: INJECTOR CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

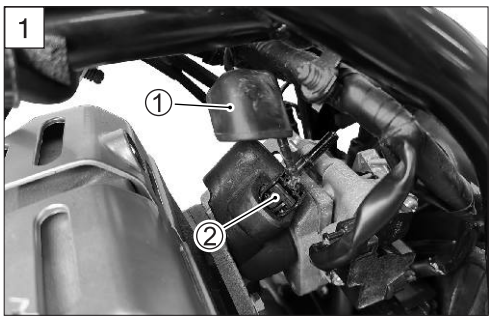
⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

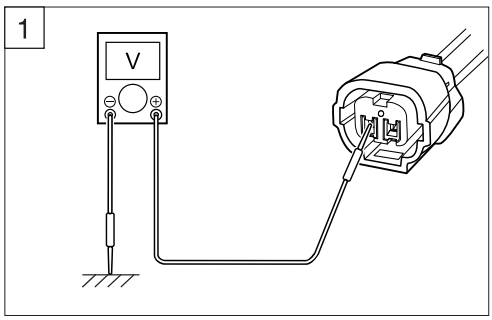
NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Step 1
Injector power supply circuit check

1) Turn off the ignition switch.
2) Remove the fuel tank. (Refer to page 5-1)
3) Remove the jacket, disconnect the injector coupler ①.
4) Check for proper terminal connection to the injector coupler.
5) If connections are OK, turn on the ignition switch.



6) Measure the voltage between the violet wire and ground.



NOTE:
Voltage can be detected only for 3 seconds after ignition switch is turned on.

TOOL **09900-25008: Multi-circuit tester set**

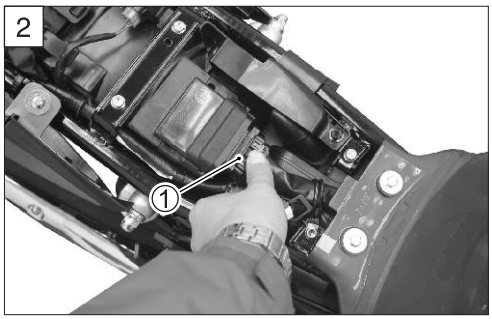
DATA **Injector voltage: battery voltage.**
(⊕terminal: violet wire – ⊖terminal; Ground)

Is check result ok ?

Yes Go to step 2.
No Repair or replace the violet wire.

Step 2
Injector drive circuit check

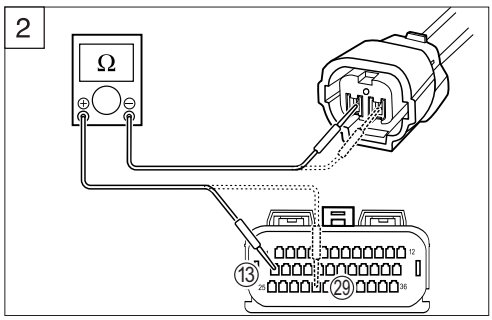
1) Turn off the ignition switch.
2) Disconnect the ECM coupler ①.
3) Check for proper terminal connection to the ECM coupler.
4) If connections are OK, check the following points.



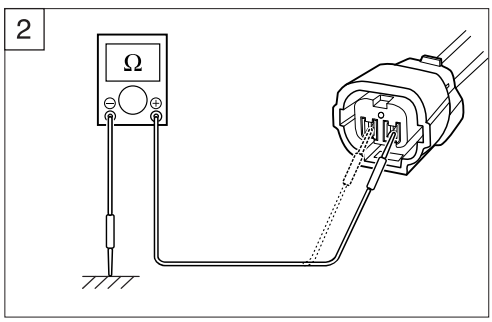
∗ **Resistance**

TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

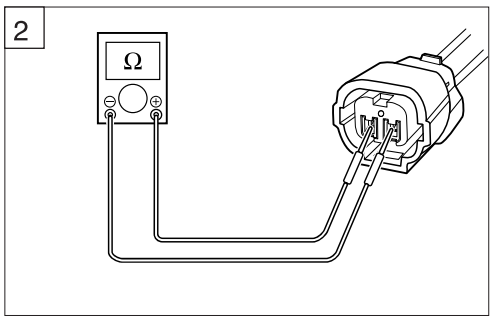
DATA **Between the sensor's coupler and ECM coupler:**
less than 1Ω



DATA **Between the violet wire or O/W wire with ground:**
infinity



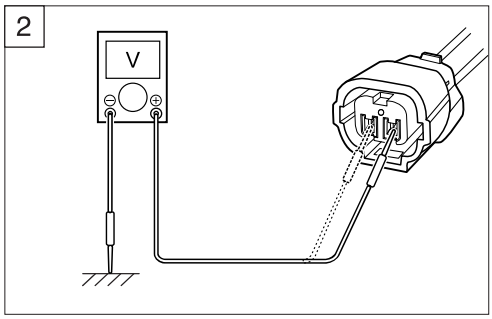
DATA **Between the violet wire and O/W wire of injector coupler: infinity.**



∗ **Voltage**

TOOL **09900-25008: Multi-circuit tester set**

DATA **Between the violet wire or O/W wire with ground: approx. 0 V. (When the ignition switch is turned on.)**



Is check result ok ?

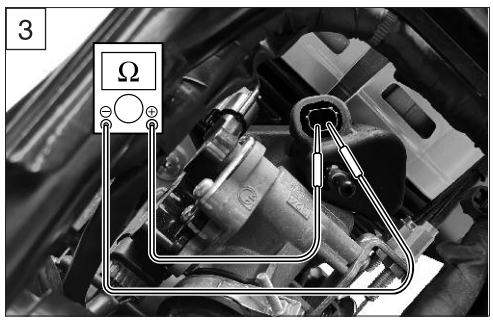
Yes Go to step 3.
No Repair or replace the defective wire harness.

Step 3
Injector resistence check

1) Turn off the ignition switch.
2) Measure the injector resistance between terminals.

TOOL **09900-25008: Multi-circuit tester set**

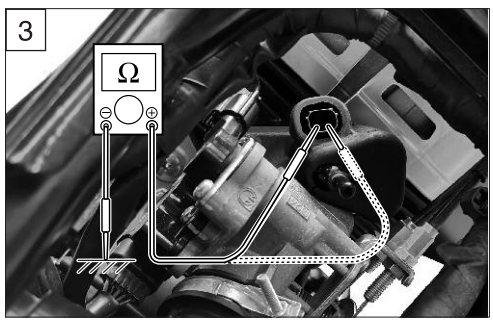
DATA **Injector resistance: 12 Ω at 20°C (Terminal – Terminal)**



3) If OK, check the continuity between each terminal and ground.

TOOL **09900-25008: Multi-circuit tester set**

DATA **Injector continuity: Infinity**

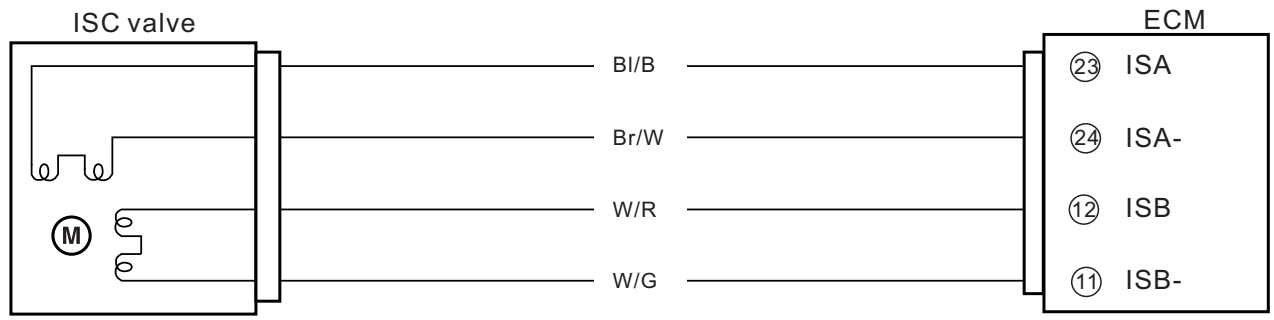


Is check result ok ?

Yes Replace a new ECM and check again.
No Replace the injector and check again. (Refer to page 5-10)

DTC “40”: ISC VALVE CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

⚠ CAUTION

*** Be careful not to disconnect the ISC valve coupler within 5 seconds after ignition switch is turned off. If the ECM coupler or ISC valve coupler is disconnected within 5 seconds after ignition switch is turned off, there is a possibility of an unusual value being written in the ECM and causing an error of ISC valve operation.**

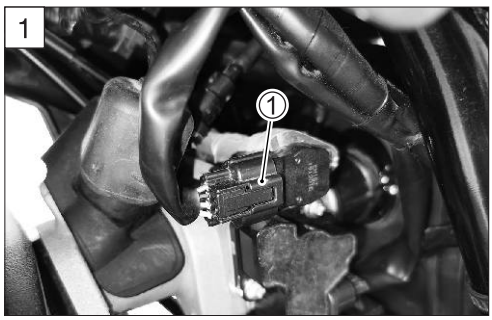
*** When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.**

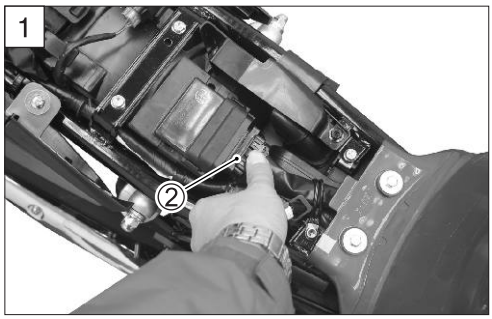
NOTE:
After repairing the trouble, clear the DTC. (Refer to page 4-14)

Step 1

ISC valve drive circuit check

1) Turn off the ignition switch.
2) Remove the saddle and left side cover.(Refer to page 6-1)
3) Disconnect the ISC valve coupler ① and ECM coupler ②.
4) Check for proper terminal connection to the ISC valve coupler and ECM coupler.



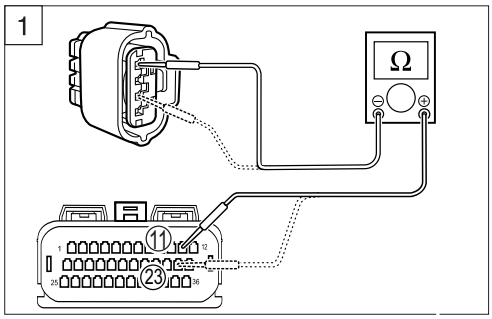


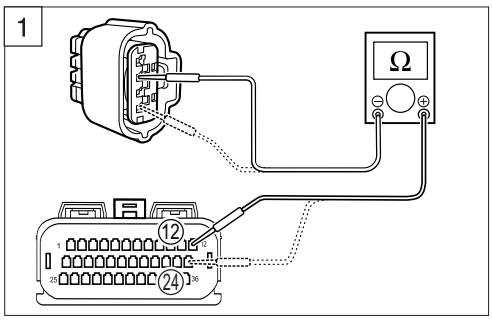
5) If connections are OK, check the following points.

***Resistance**

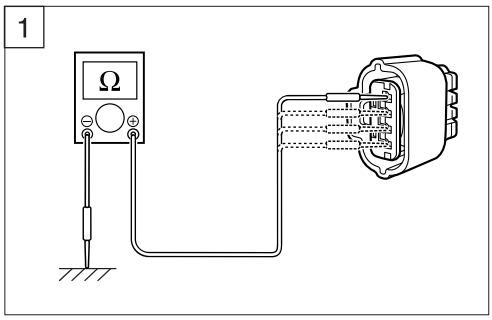
TOOL **09900-25008: Multi-circuit tester set**
09900-25009: Needle-point probe set

DATA **Between the Bl/B, Br/W, W/R and W/G wires of ISC valve coupler with ECM coupler: less than 1 Ω.**

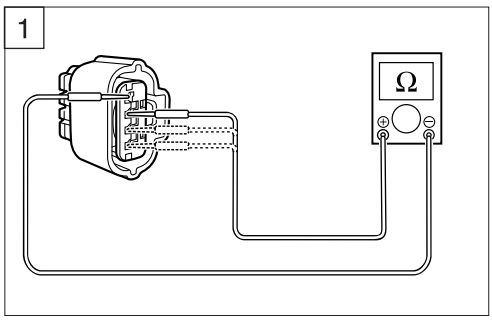




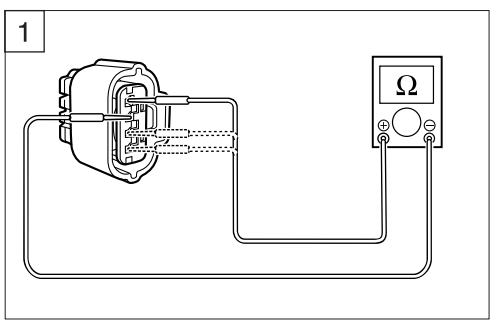
DATA **Between the Br/W, Bl/B, W/R and W/G wires with ground: infinity.**



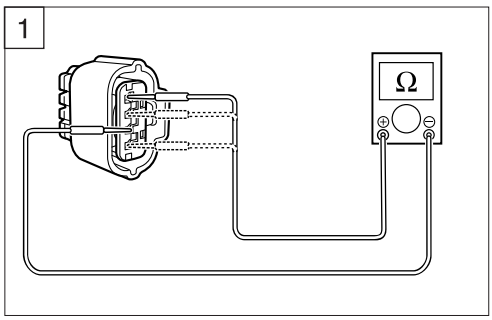
DATA **Between the W/G wire terminal and other terminal at ISC valve coupler: infinity.**



DATA **Between the W/R wire terminal and other terminal at ISC valve coupler: infinity.**



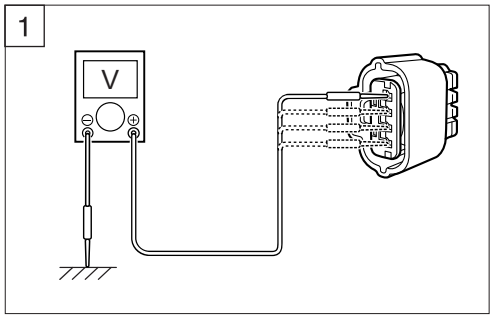
DATA **Between the BI/B wire terminal and other terminal at ISC valve coupler: infinity.**



***Voltage**

TOOL **09900-25008: Multi-circuit tester**

DATA **Between the Br/W, BI/B, W/R and W/G wires of ISC valve coupler with ground: approx 0 V. (When the ignition switch is turned on.)**

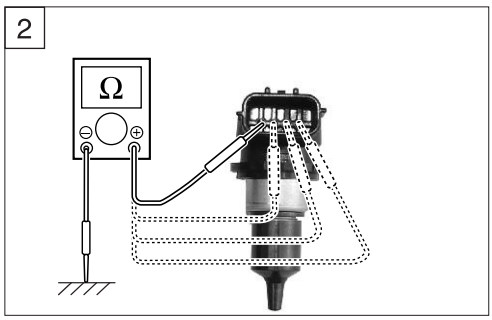


Is check result ok ?

Yes Go to step 2.
No Repair or replace defective wire.

Step 2
ISC valve check

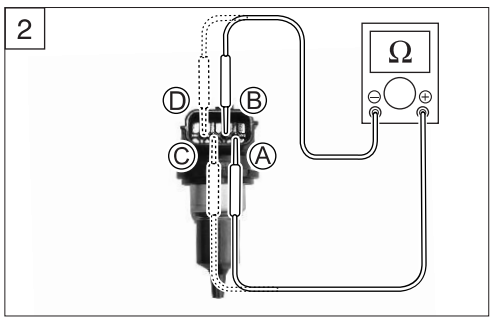
1) Remove the ISC valve. (Refer to page 5-11)
2) Check the continuity between each ISC valve terminal and ground.



TOOL **09900-25008: Multi-circuit tester**

DATA **ISC valve continuity: infinity. (Terminal-Ground)**

DATA **ISC valve resistence: approx 20Ω at 20°C**
Terminal Ⓐ– Terminal Ⓑ
Terminal Ⓒ– Terminal Ⓓ

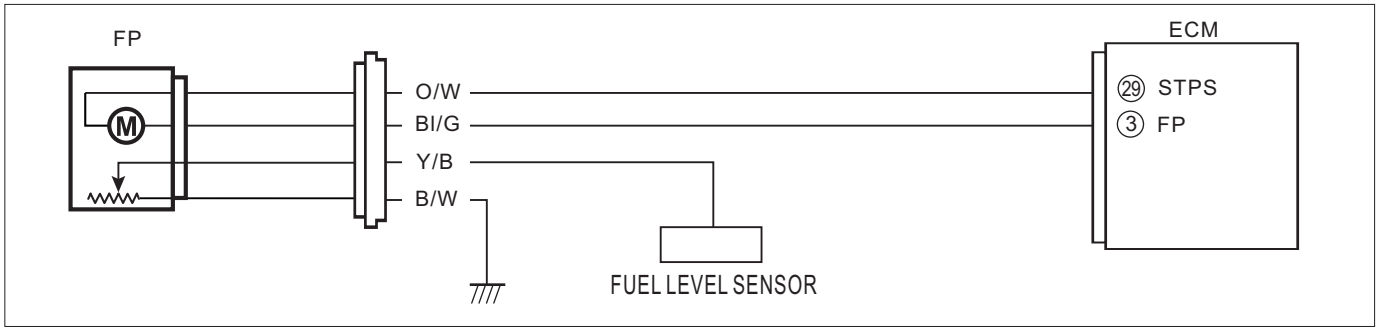


Is check result ok ?

Yes Replace the ECM and check again.
No Replace the ISC valve and check again.

DTC “41”: FUEL PUMP CIRCUIT MALFUNCTION

WIRING DIAGRAM



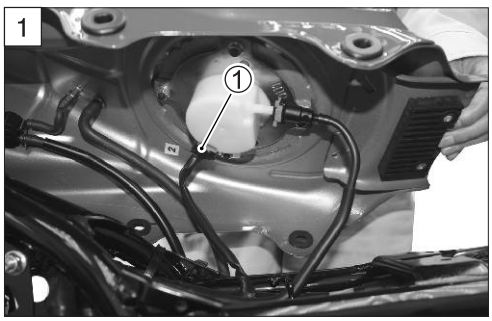
⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

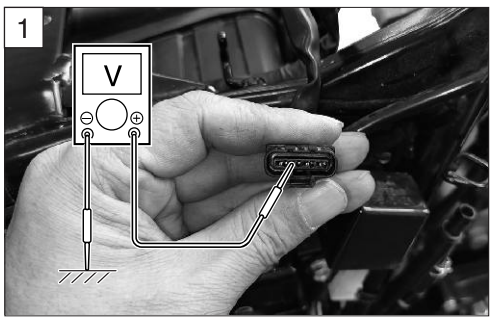
NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Step 1
Fuel pump power supply circuit check

1) Turn off the ignition switch.
2) Remove the fuel tank mounting bolts. (Refer to page 6-3)
3) Lift the fuel tank and disconnect the fuel pump coupler ①.
4) Check for proper terminal connection to the FP coupler.



5) If connections are OK, turn on the ignition switch.
6) Check the voltage between the O/W wire and ground.



TOOL **09900-25008: Multi-circuit tester**

DATA **Fuel pump voltage: battery voltage**
(⊕terminal: O/W wire – ⊖terminal: Ground)

Is check result ok ?

Yes Go to step 2.
No Repair or replace the defective wire harness.

Step 2

Fuel pump check

Inspect the fuel pump. (Refer to page 5-4)

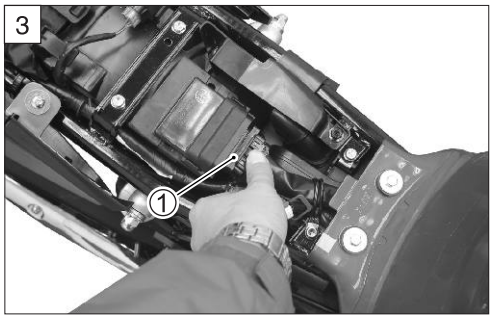
Is check result ok ?

Yes Go to step 3.
No Replace the fuel pump.

Step 3

Fuel pump circuit check

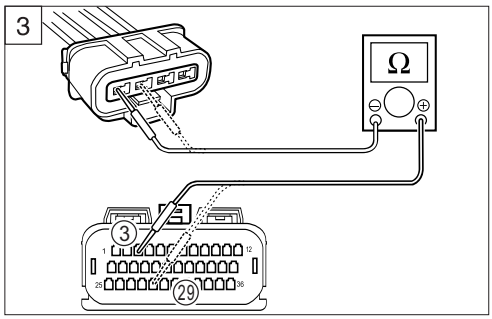
1) Turn off the ignition switch.
2) Disconnect the ECM①.
3) Check for proper terminal connection to the ECM coupler.
4) If connections are OK, check the following points.



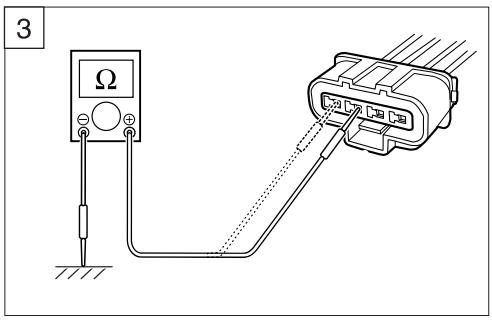
***Resistance**

TOOL **09900-25008: Multi-circuit tester**
09900-25009: Needle-point probe set

DATA **Between the O/W wire and Bl/G wire of FP coupler and ECM coupler: less than 1Ω.**



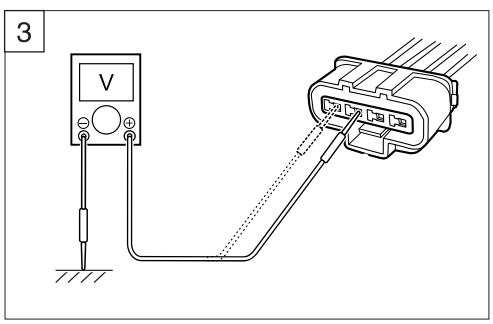
DATA **Between the Bl/G wire or O/W wire and ground: infinity.**



***Voltage**

TOOL **09900-25008: Multi-circuit tester**

DATA **Between the O/W wire or Bl/G wire and ground: approx 0 V. (When the ignition switch is turned on.)**

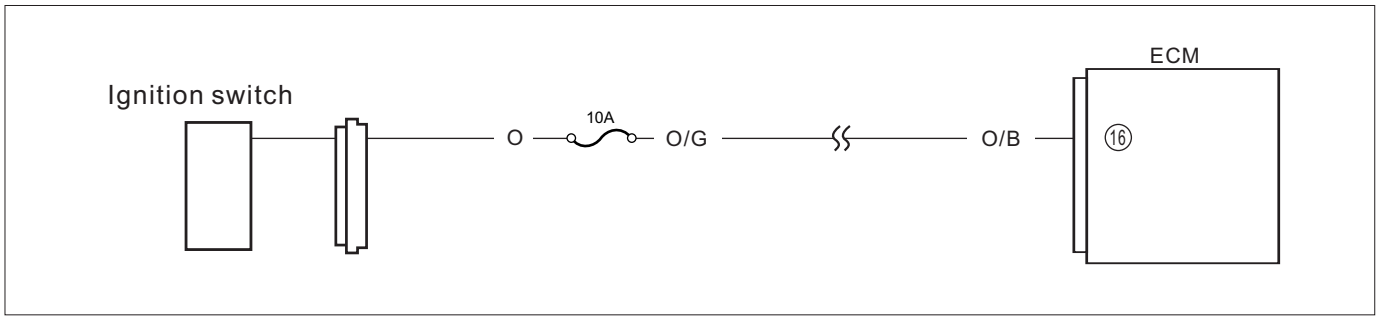


Is check result ok ?

Yes Replace the ECM and check again.
No Repair or replace the defective wire harness.

DTC “42”: POWER CHECK CIRCUIT MALFUNCTION

WIRING DIAGRAM



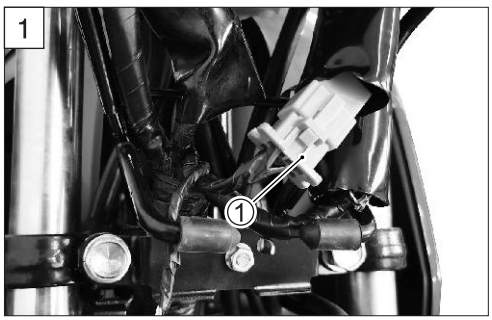
⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

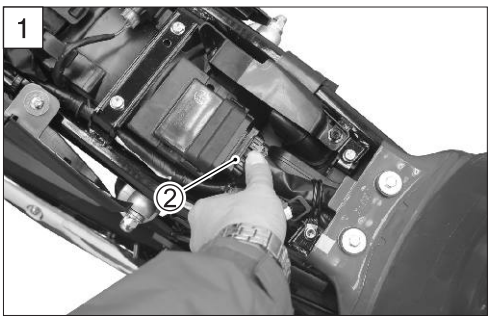
NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Ignition switch power supply circuit check

1) Turn off the ignition switch.
2) Remove the headlight assembly. (Refer to page 6-2)
3) Disconnect the ignition switch coupler ①.



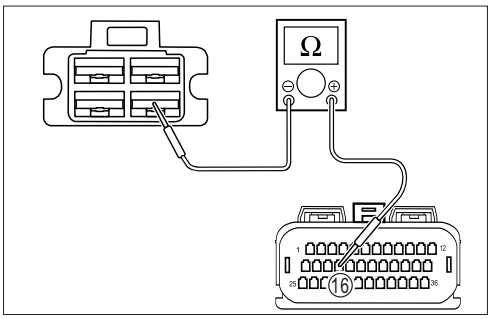
4) Remove the saddle.
5) Disconnect the ECM coupler ②.
6) Check for proper terminal connection between the couplers.
7) If connections are OK, check the following points.



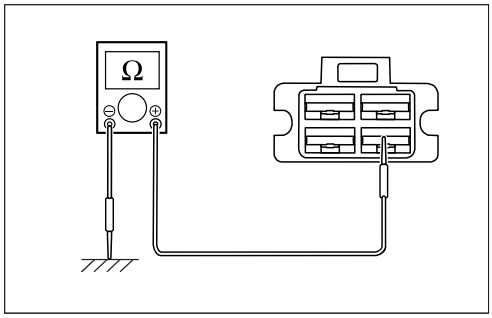
*** Resistance**

TOOL **09900-25008: Multi-circuit tester**
09900-25009: Needle-point probe set

DATA **Between the orange wire of ignition switch and ECM coupler: less than 1 Ω.**



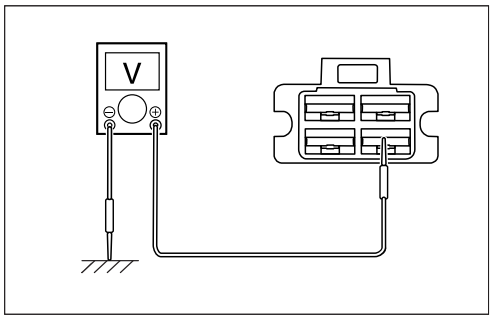
DATA **Between the orange wire and ground: infinity.**



* **Voltage**

TOOL **09900-25008: Multi-circuit tester**

DATA **Between the orange wire and ground: approx 0 V.**
(When the ignition switch is turned on.)

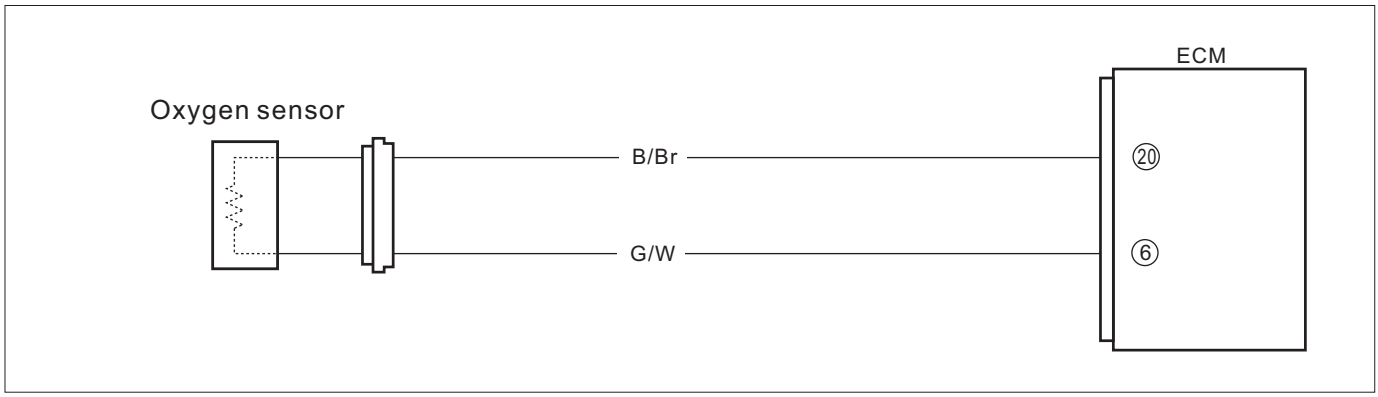


Is check result ok ?

Yes Replace the ECM and check again.
No Repair or replace the defective wire harness.

DTC “44”: OXS CIRCUIT MALFUNCTION

WIRING DIAGRAM



TROUBLESHOOTING

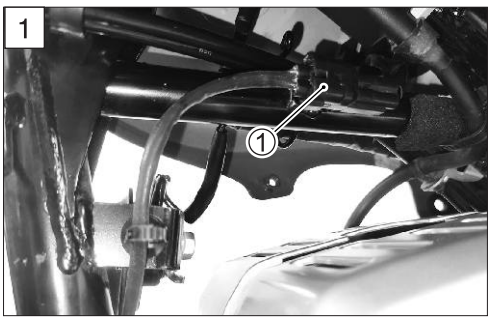
⚠ CAUTION

When using the multi circuit tester, do not strongly touch the terminal of the ECM coupler with a needle-point tester probe to prevent terminal damage.

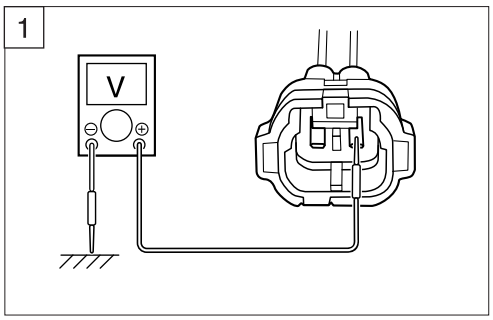
NOTE:
After repairing the trouble, clear the DTC (Refer to page 4-14).

Step 1
OXS power supply circuit check

1) Turn off the ignition switch.
2) Remove the fuel tank cover.(Refer to page 6-3)
3) Disconnect the OXS coupler ①.
4) Check for proper terminal connection to the OXS coupler.
5) If connections are OK, turn on the ignition switch.



6) Measure the voltage between B/Br wire and ground.



TOOL **09900-25008: Multi-circuit tester**

DATA **OXS power supply voltage: 4.5-5.5V**
(⊕terminal: B/Br wire – ⊖terminal: Ground)

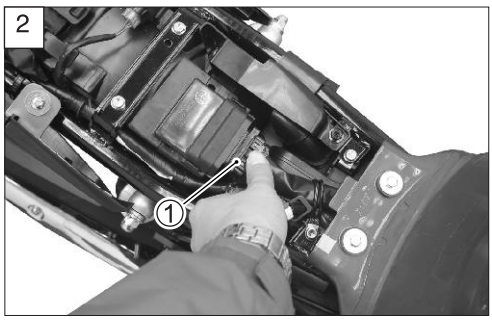
Is check result ok ?

Yes Go to step 2.
No Repair or replace B/Br wire.

Step 2

OXS circuit check

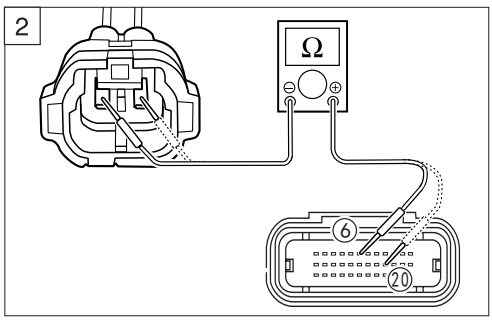
1) Turn off the ignition switch.
2) Remove the rear and front saddles. (Refer to page 6-1)
3) Disconnect the ECM coupler ①.
4) Check for proper terminal connection to the ECM couplers.
5) If connections are OK, check the following points.



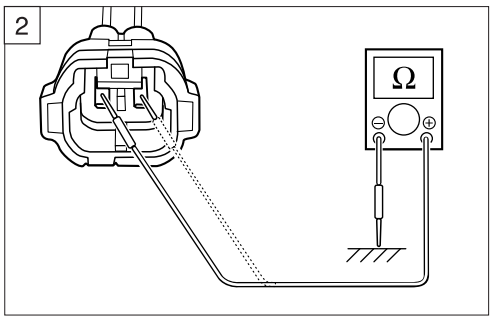
***Resistance**

TOOL **09900-25008: Multi-circuit tester**
09900-25009: Needle-point probe set

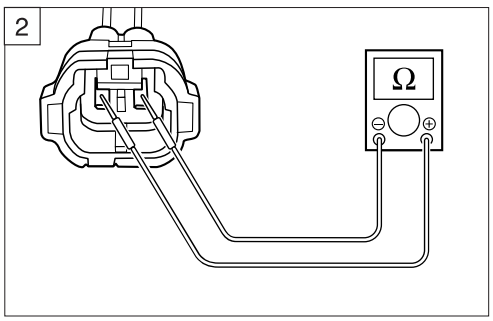
DATA **Between the G/W or B/Br wire of injector coupler and ECM coupler: less than 1 Ω .**



DATA **Between the G/W wire and B/Br wire with ground: infinity**



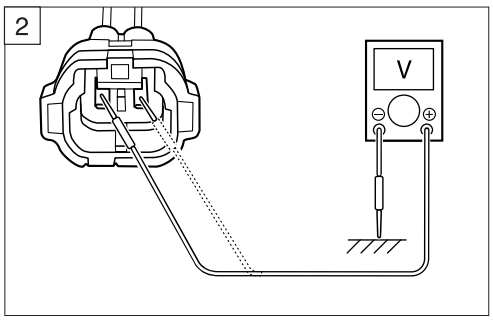
DATA **Between the G/W and B/Br wire of fuel pump coupler: infinity.**



***Voltage**

TOOL **09900-25008: Multi-circuit tester**

DATA **Between the G/W wire or B/Br wire with ground: approx. 0 V (When the ignition switch is turned on.)**



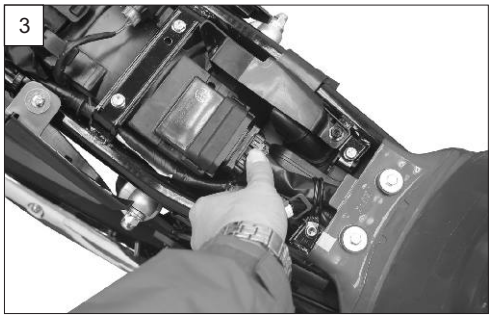
Is check result ok ?

Yes Go to step 3.
No Repair or replace defective wire.

Step 3
ECM check
1) Reconnect a new ECM.

Is check result ok ?

Yes End.
No Replace the OX sensor.



DTC “99”: BATTERY HIGH VOLTAGE MALFUNCTION

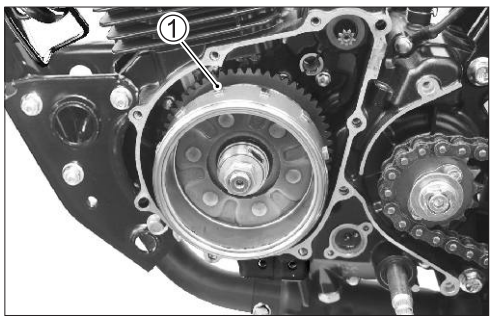
If DTC "99" exists, replace the rectifier with a new one.

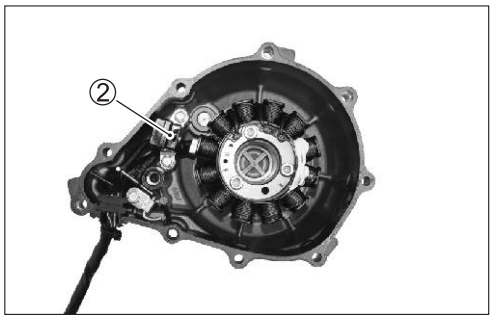
NOTE:
After repairing the trouble, clear the DTC. (Refer to page 4-14)

SENSORS

CKPS

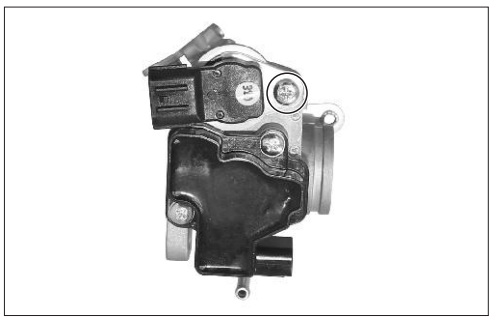
- Remove the CKPS (trigger coil). (Refer to page 3-27)
- The signal rotor is mounted on the generator rotor ① and CKPS ② is installed at the inside of the generator cover.
- Install the CKPS in the reverse order of removal. (Refer to page 3-28)





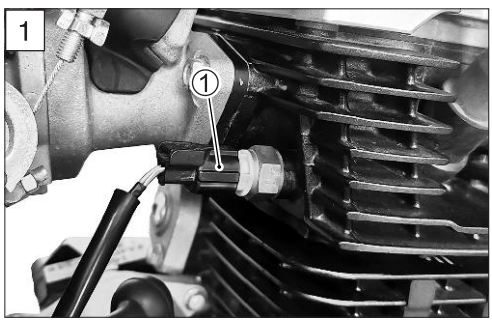
ISC VALVE

(Refer to page 5-12)

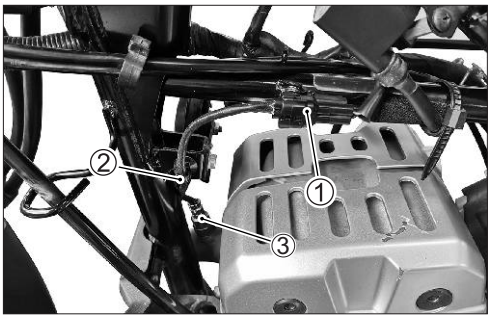


ETS

- Remove the ETS①. (Refer to page 3-7)
- Install the ETS in the reverse order of removal.



OXS



REMOVAL

- Remove the fuel tank cover. (Refer to page 6-2)
- Disconnect the OXS coupler ①, remove the clamp.
- Remove the OXS ② from the exhaust pipe.

⚠ WARNING

Do not remove the OXS while it is hot.

⚠ CAUTION

Be careful not to expose the OXS to excessive shock.
Do not use an impact wrench when removing or installing the OXS.
Be careful not to twist or damage the sensor lead wires.

INSTALLATION

Install the OXS is in the reverse order of removal.
Pay attention to the following points:

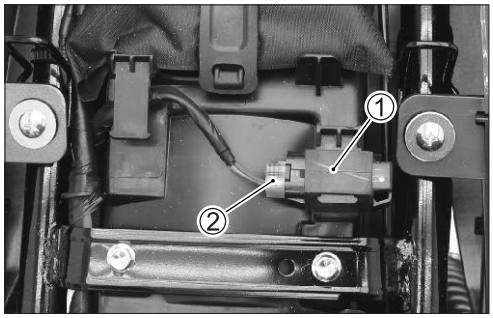
⚠ CAUTION

Do not apply oil or other materials to the sensor hole.

- Tighten the OXS to the specified torque.

OXS tighten torque: 20-30 N•m

TOS



- Remove the rear and front saddles. (Refer to page 6-1)
- Remove the TOS ①, disconnect the TOS coupler ②.
- Install the TOS in the reverse order of removal.

FUEL SYSTEM

CONTENT

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FUEL TANK 5-2
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5

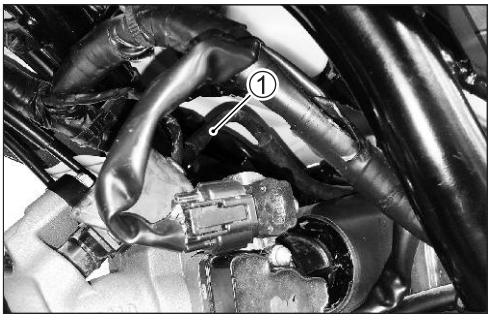
⚠ WARNING

* Far away from fire or spark.
* Take care to the overflow gasoline during the disassembly.
* Wipe up the overflow gasoline.
* Keep the area you work in is well-ventilated.

⚠ CAUTION

* Do not remove part blindly to prevent the impurities getting into fuel system.
* After the throttle body is removed, use the adhesive tape to seal the cylinder intake hole to prevent the dust from getting into the cylinder.

FUEL PRESSURE INSPECTION



- Remove the saddle and left side covers. (Refer to page 6-1)
- Place a cloth below the fuel hose ①, loosen the fastener, remove the fuel hose.

- Install the special tool to the fuel pump nozzle and fuel hose.

TOOL **11F14-023**: **Pressure inspection tool**

- Turn on the ignition switch, check the fuel pressure.

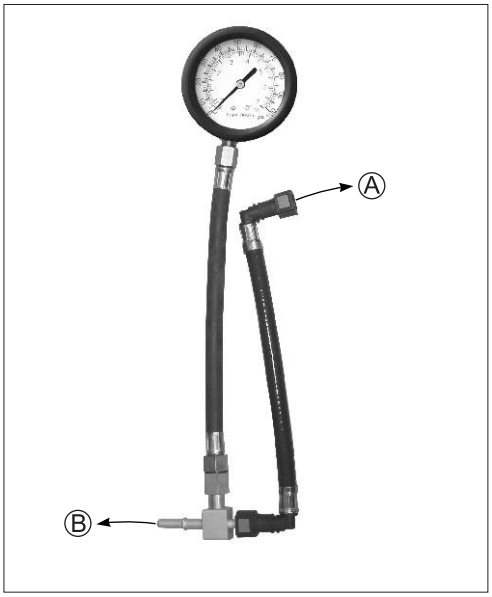
DATA **Fuel pressure: Approx 300 kPa (3.0kgf/cm²)**

A low compression pressure may indicate any of following malfunctions:

* Fuel hose leakage
* Fuel strainer clogged
* Pressure adjuster
* Fuel pump

If the fuel pressure is higher than the limit may indicate any of following malfunctions:

* Fuel pump
* Pressure adjuster



Ⓐ To fuel injector nozzle
Ⓑ To fuel hose

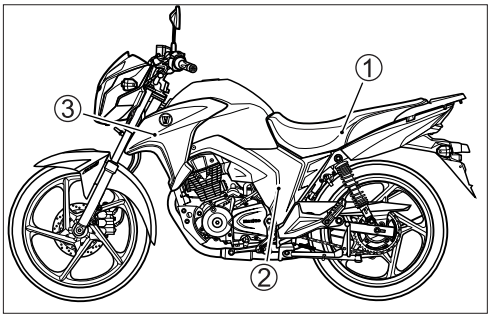
⚠ WARNING

Before removing the special tool, turn off the ignition switch and release the pressure slowly.

FUEL PUMP INSPECTION

Turn on the ignition switch, check if the fuel pump has running sound,
If the running sound is not heard, check the connection for fuel pump coupler.
If connection is ok, the fuel pump may have defect, replace it with a new one.

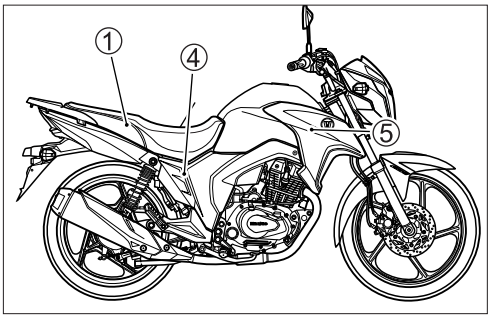
FUEL TANK



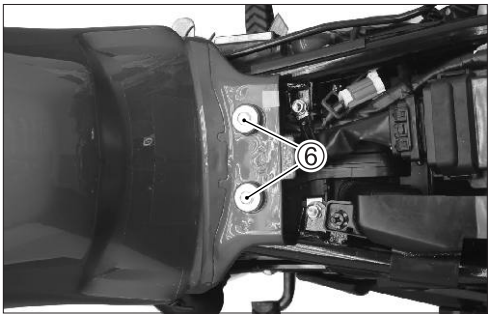
- Drain the gasoline in the fuel tank.
- Remove the saddle ①, left side cover ② and fuel tank left side shield ③. (Refer to page 6-1)

⚠ WARNING

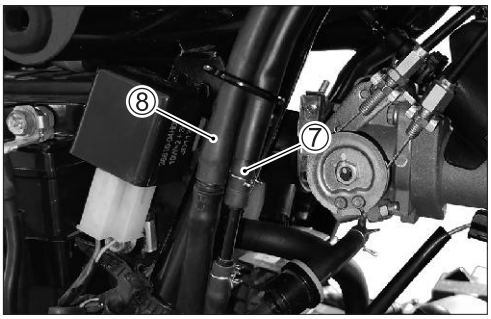
*** Far away from fire or spark.**
*** Wipe up the overflow gasoline.**
*** Keep the area you work in is well-ventilated.**



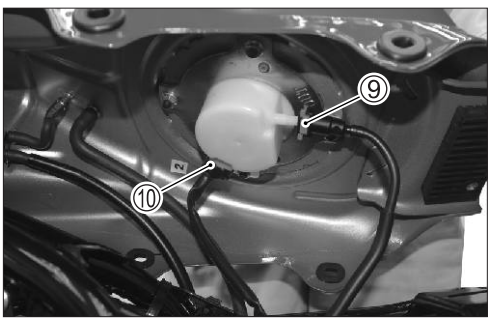
- Remove the right side cover ④ and fuel tank right side shield ⑤. (Refer to page 6-1)



- Remove the fuel tank mounting bolts ⑥.



- Disconnect the canister absorber hose ⑦ and overflow hose ⑧.



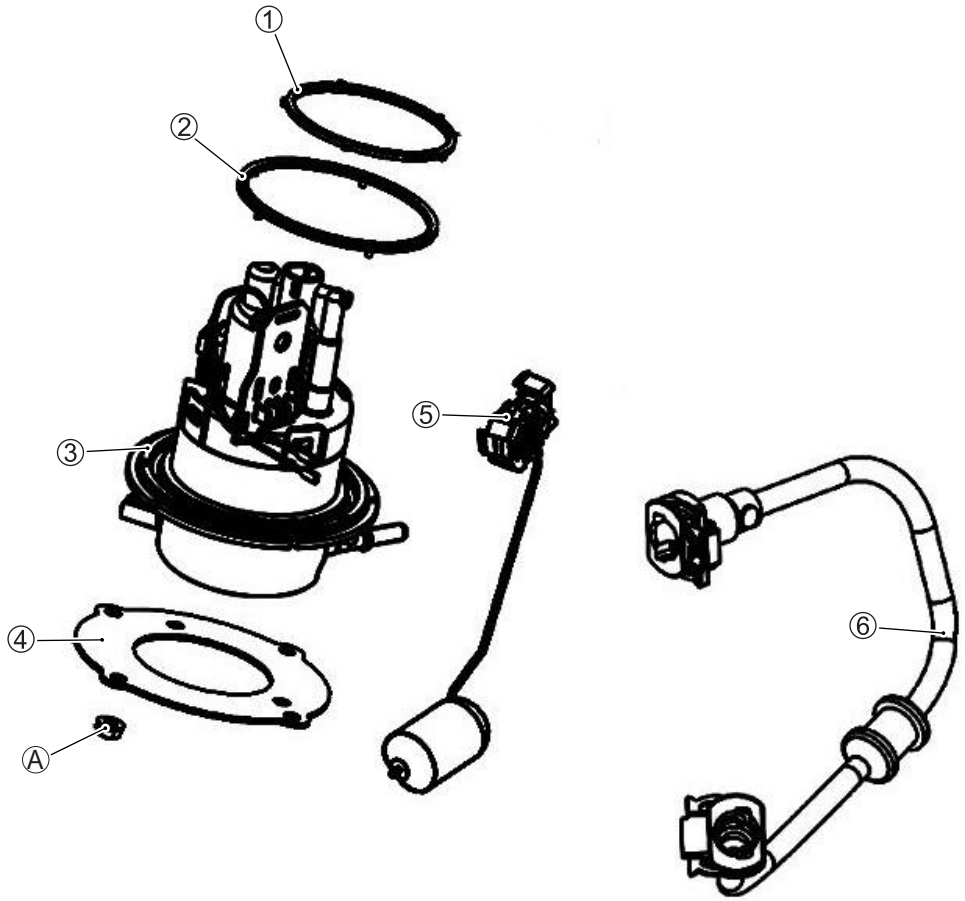
- Take off the fuel hose ⑨ and disconnect the fuel pump coupler ⑩, take off the fuel tank.
- Install the fuel tank in the reverser order of removal.

⚠ CAUTION

Connect the fuel hose to the fuel pump, lock the safe lock.

FUEL PUMP AND FUEL LEVEL SENSOR

CONSTRUCTION



①	Seal ring	⑤	Fuel level sensor
②	Dust ring	⑥	High pressure fuel hose
③	Fuel pump	Ⓐ	Nut
④	Fuel pump retainer plate		

ITEM	N•m	Kgf•m
Ⓐ	8-12	0.8-1.2

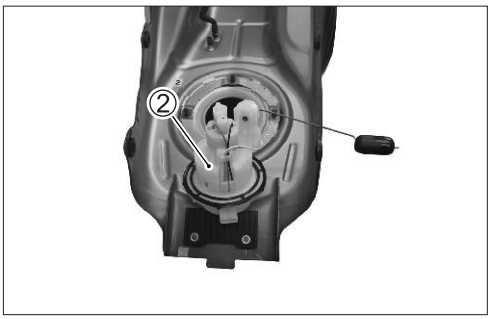
FUEL PUMP REMOVAL

- Remove the fuel tank. (Refer to page 5-1)
- Turn over the fuel tank, remove the fuel pump mounting nuts, take off the fuel pump retainer plate ①.

⚠ DANGER

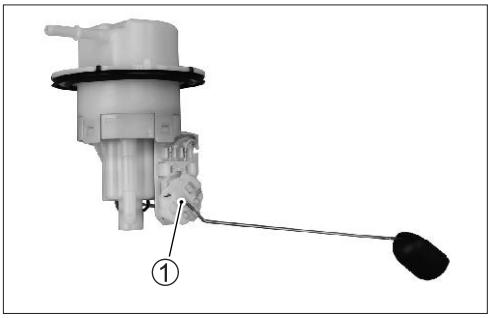
*** Far away from fire or spark.**
*** Wipe up the overflow gasoline.**
*** Keep the area you work in is well-ventilated**

- Remove the fuel pump assembly ②.

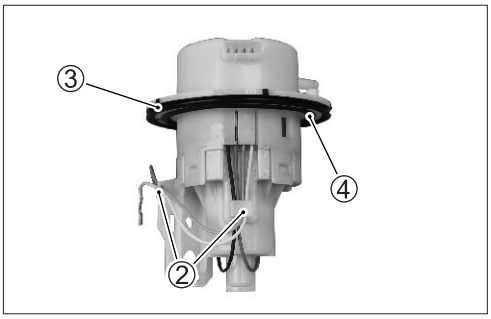


FUEL PUMP DISASSEMBLY

- Disconnect fuel level sensor coupler, remove the fuel level sensor ①.



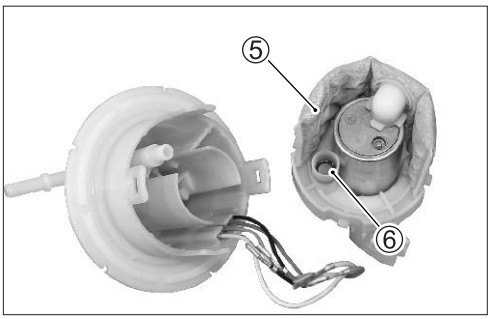
- Disconnect the fuel pump coupler ②, remove the wire from the clamp.
- Remove the fuel pump dust ring ③and seal ring ④.



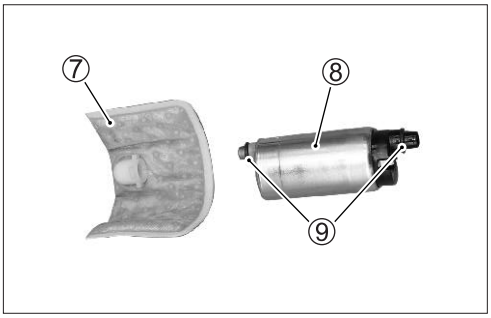
- Open the fastener ④, remove the fuel pump and fuel strainer assembly ⑤.

⚠ CAUTION

Replace the O-ring ⑥ with a new one.



- Remove the fuel stainer ⑦ and fuel pump ⑧.



⚠ CAUTION

Replace the O-ring ⑦ with a new one.

FUEL LEVEL SENSOR INSPECTION

(Refer to page 7-16)

FUEL STAINER INSPECTION AND CLEANING

If the fuel strainer is dirty with sediment, fuel will not flow smoothly and loss in engine power may result.

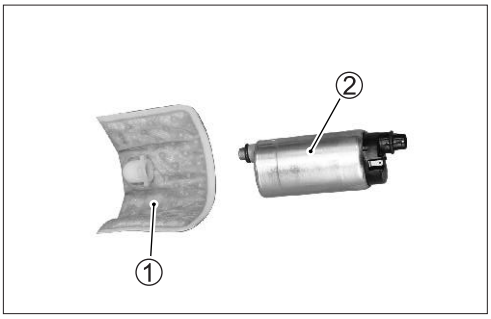
⚠ CAUTION

Replace the fuel strainer as necessary.

FUEL PUMP AND FUEL LEVEL SENSOR REASSEMBLY

Reassemble the fuel pump in the reverse order of removal, pay attention to the following point:

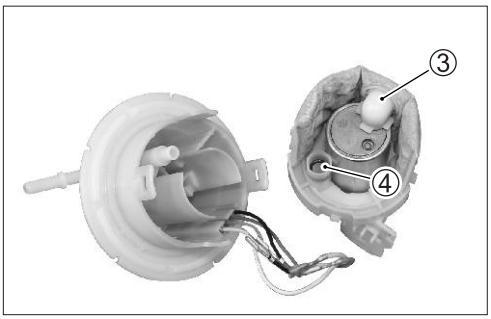
- Reinstall the fuel pump ① and fuel strainer ②.



⚠ CAUTION

Replace the O-ring when reassemble the fuel pump.

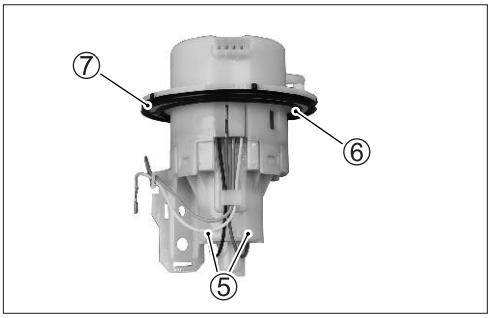
- Install the fuel pump ③ to the fuel pump seat.



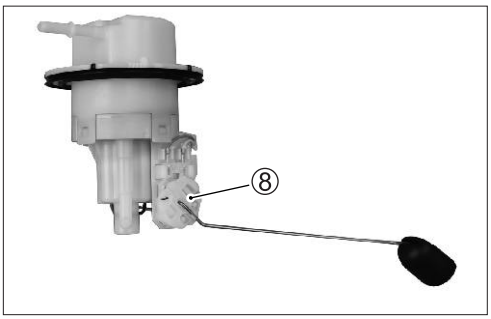
⚠ CAUTION

Replace the seal ring ④ if deformation or damage is found.

- Fit the wires in the clamp orderly, connect the fuel pump coupler ⑤.
- Install the new fuel pump seal ring ⑥ and dust ring ⑦.



- Install the fuel level sensor ⑧connect the fuel level sensor coupler.



- Install the fuel pump assembly to the fuel tank as shown.

- Install the fuel pump fixing plate, tighten the fuel pump mounting nut to the specified torque.

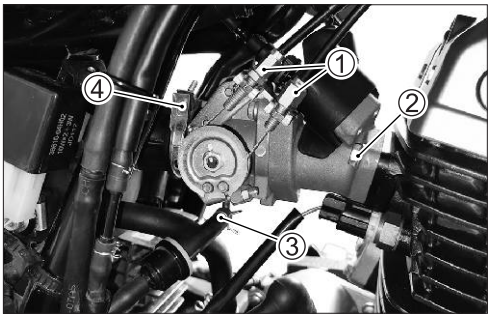
Fuel pump mounting nut: 8-12 N•m

CAUTION

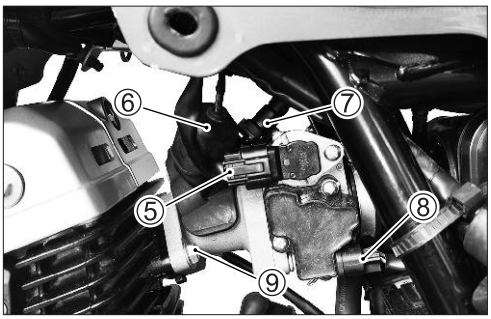
Align the fuel pump boss ⑨ with the hole on the fuel pump fixing plate hole.

THROTTLE BODY

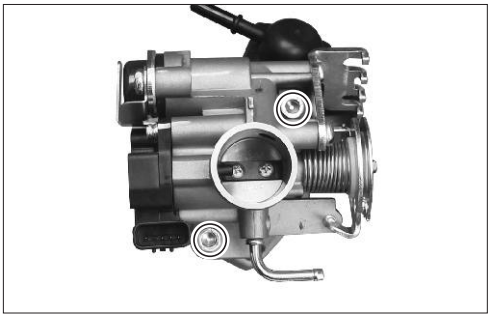
THROTTLE BODY REMOVAL



- Remove throttle cable①.
- Remove the intake hose mounting bolt ②.
- Disconnect the absorber hose ③.
- Loosen the air filter-throttle clamp bolt ④.



- Disconnect the ISC valve sensor coupler ⑤.
- Remove the water-proof cap ⑥, disconnect the fuel injector coupler.
- Disconnect fuel hose coupler ⑦.
- Disconnect 3 in 1 sensor coupler ⑧.
- Remove the intake hose mounting bolt ⑨.
- Pull the intake hose-thorttle assembly rearward, take off it from left side.



- Remove the mounting bolts, take off the throttle body from the intake hose-throttle body assembly.

THROTTLE BODY CLEANING

⚠ WARNING

Some carburetor cleaning chemicals, especially dip-type soaking solutions, are very corrosive and must be handled carefully. Always follow the chemical manufacturer's instructions on proper use, handling and storage.

- Use a swab moistened with a throttle-body cleaner to clean the passageways, throttle valve, and main bore, then dry the cleaned surfaces with compressed air.

⚠ CAUTION

Do not apply cleaning chemicals to the throttle body inner and rubber and plastic products.
Do not use wire or compressed air to clean passage ways.
Do not use a dip-type cleaning solution and do not allow them to soak.
Use cleaning solvent contains chlorine or ketone element is prohibited.

THROTTLE BODY INSPECTION

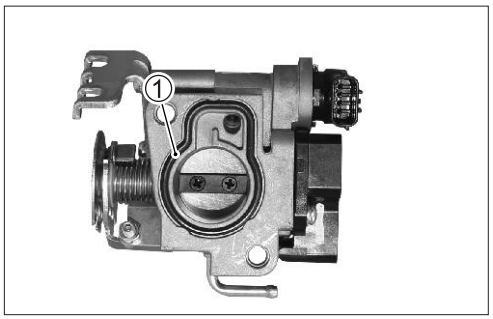
Check following items for any damage or clogging.
Replace the damaged part if necessary.

* O-ring
* Throttle valves
* Absorber hose
* Fuel hose

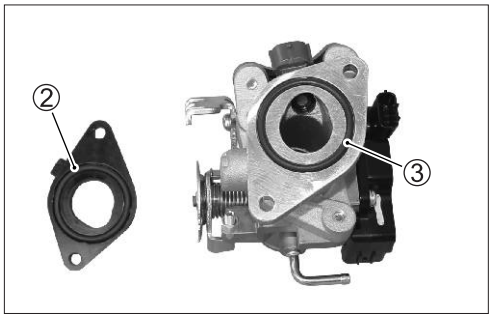
THROTTLE BODY REASSEMBLY

Reassemble the throttle body in the reverse order of removal.
pay attention to the following points:

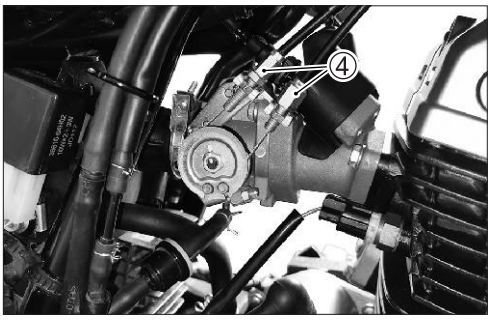
- Replace the O-ring ① before the installing the throttle body to engine intake hose.



- Replace the O-ring ② and ③ before installing the engine intake hose.



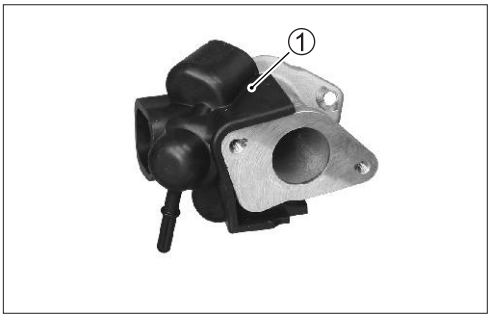
- Install the throttle cable ④.
- Adjust the throttle cable clearance. (Refer to page 2-9)



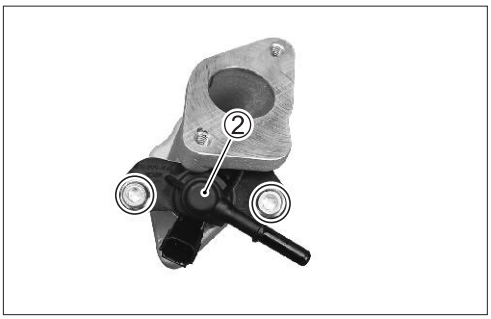
FUEL INJECTOR

FUEL INJECTOR REMOVAL

- Remove the fuel injector jacket ①.

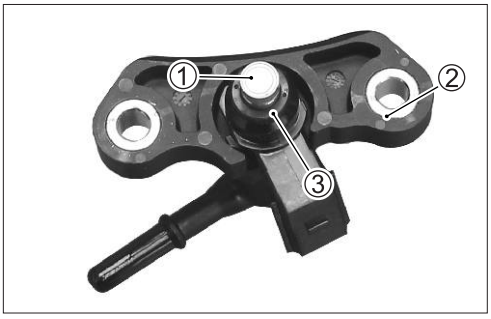


- Remove the fuel injector ② assembly.



FUEL INJECTOR INSPECTION

- Inspect the fuel injector ① and injector seat ② if dust or dirt exists, if any defects are found, clean the filter, check the fuel hose and fuel tank.
- Inspect the fuel injector rubber seal ③ for damage or aging, if any unusual is found, replace the rubber seal.



FUEL INJECTOR REASSEMBLY

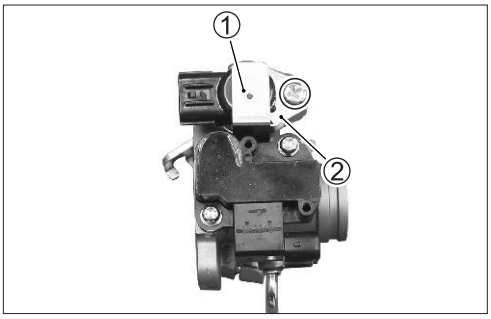
- Reassemble the fuel injector in the reverse order of removal.

ISC VALVE

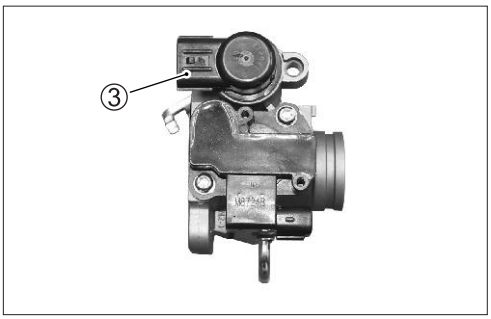
ISC VALVE REMOVAL

NOTE:
When only remove the ISC valve, remove throttle body from the engine is not necessary.

- Remove the ISC valve bracket screw, retainer ①and bracket ②.



- Remove the ISC valve ③.



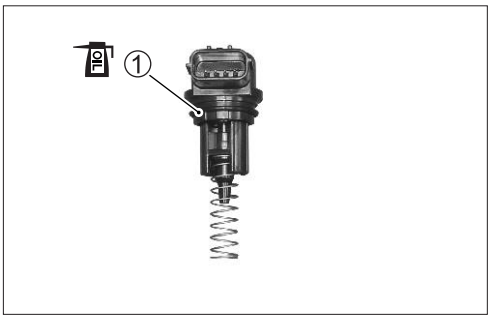
ISC VALVE INSPECTION

- Check the ISC valve if carbon deposits exists, clean or replace the ISC valve as necessary.
- Check the ISC valve.(Refer to page 4-39)

ISC VALVE REASSEMBLY

Reassemble the ISC valve in the reverse order of removal, pay attention to the following points:

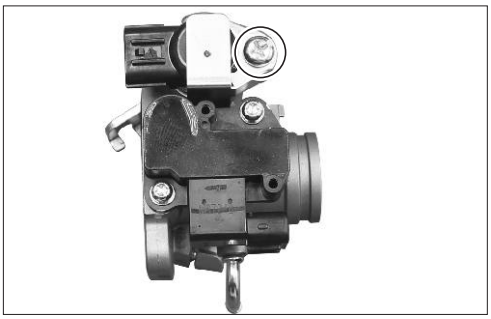
- Apply a small quantity engine oil to the new O-ring ①.



- Install the ISC valve retainer, bracket and mounting screws.

⚠ CAUTION

The holder position should be correct.



CHASSIS

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EXTERIOR PARTS REMOVAL

SADDLE

- Support the motorcycle with main stand.
- Open the saddle lock using the key, and take off the saddle.

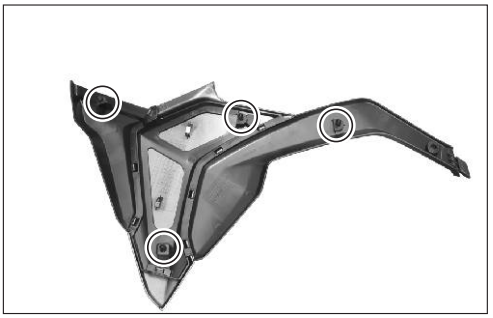
- Install the saddle in the reverse order of removal.

SIDE COVER

NOTE:

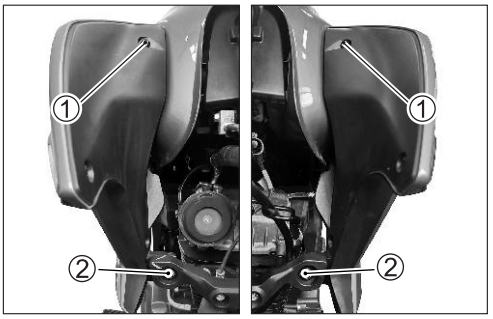
The left and right covers are installed symmetrically and therefore the removal procedure for one side is the same as that for the other side.

- Remove the side cover mounting screw, take off the side cover.



- Install the side cover in the reverse order of removal.

FUEL TANK LEFT AND RIGHT SHIELDS



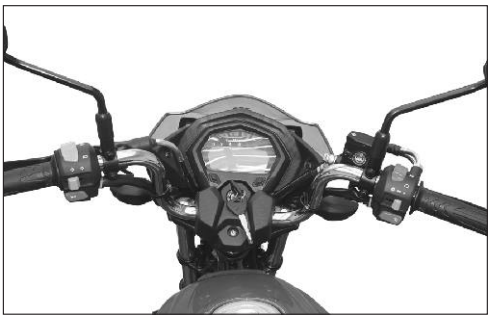
NOTE:

The left and right shields are installed symmetrically and therefore the removal procedure for one side is the same as that for the other side.

- Remove the fuel tank shield mounting screw ① and bolt ②.

- Turn the shield ③ as the direction of the arrow as shown, remove the shield from the fuel tank.

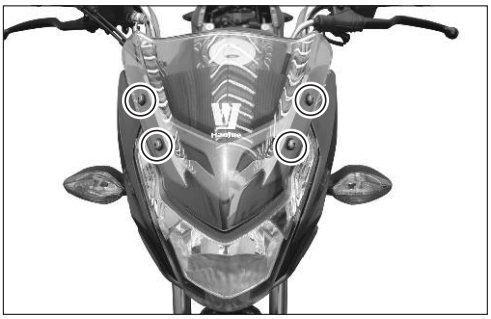
REARVIEW MIRROR



- Remove the left and right rear view mirrors.

- Install the rearview mirror in the reverse order of removal.

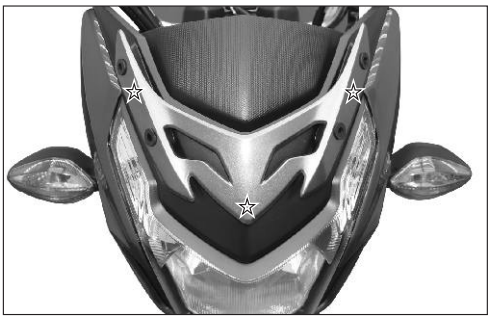
WIND SHIELD



- Remove the wind shield.

- Install the wind shield in the reverse order of removal.

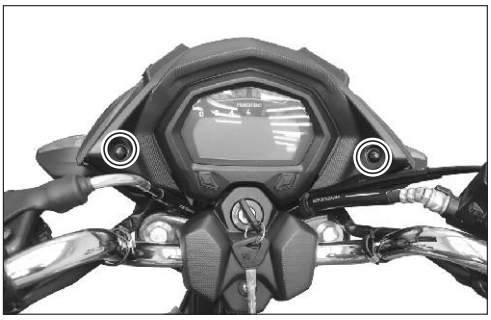
WIND SHIELD BRACKET



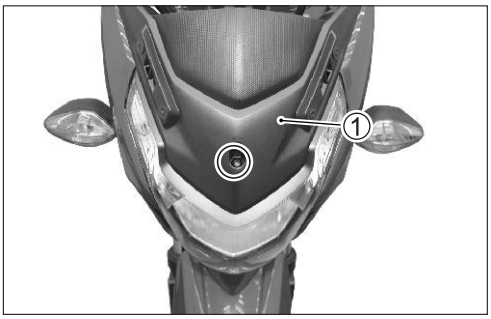
- Remove the wind shield bracket.

- Install the wind shield bracket in the reverse order of removal.

HEADLAMP UPPER COVER

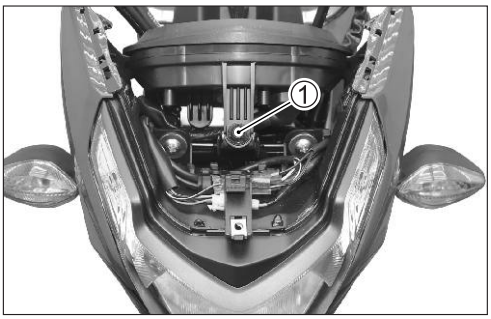


- Remove the headlamp upper cover mounting screws, take off the headlamp upper cover ①.

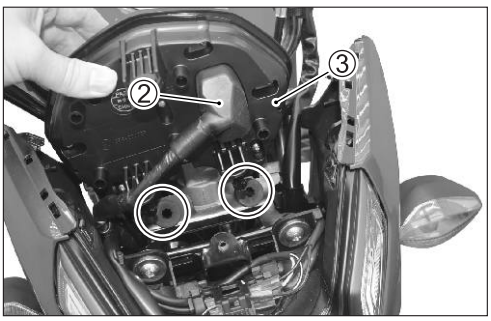


- Reassemble the headlamp upper cover in the reverse order of disasembly.

METER



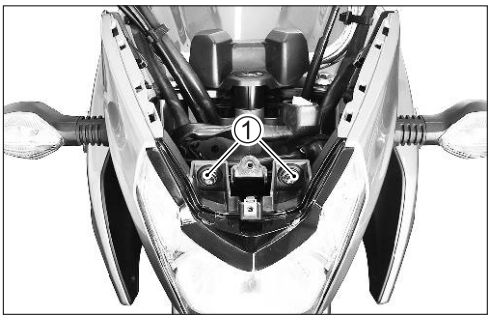
- Remove the meter mounting bolt ①.



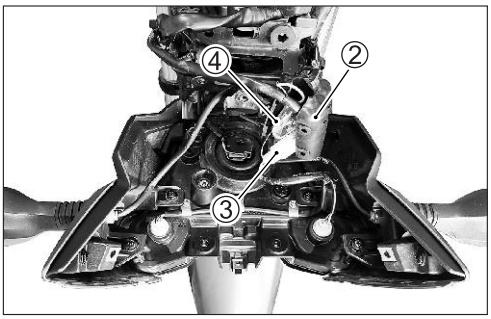
- Disconnect the meter coupler ②, take off the meter ③ from the bracket.

- Install the meter in the reverse order of removal.

HEADLAMP ASSEMBLY



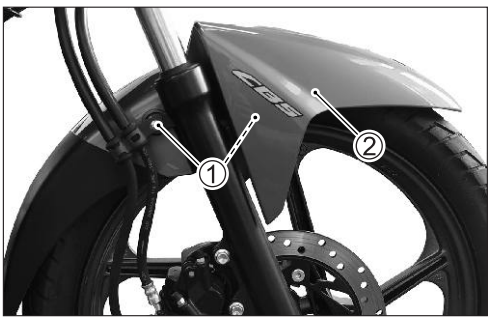
- Remove the headlamp assembly mounting bolts ①.



- Open the PVC jacket②.
- Disconnect the headlamp coupler ③ and front turn signal lights coupler ④.
- Remove the head-lamp assembly from the bracket.
- Install the headlamp in the reverse order of removal.

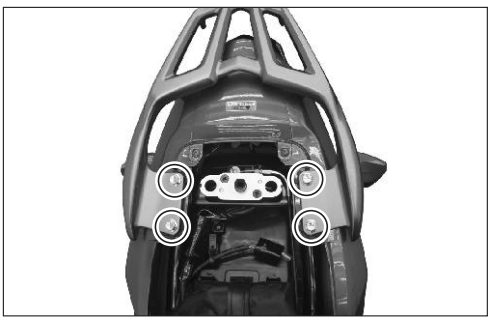
FRONT FENDER

- Remove the front fender mounting bolts ① , remove the front fender ③.



- Install the front fender in the reverse order of removal.

REAR HANDRAIL

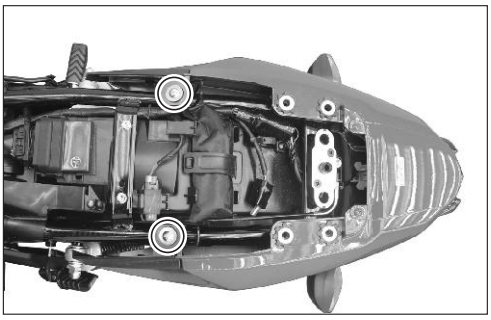


- Remove the rear handrail by removing the mounting bolts.
- Install the rear handrail in the reverse order of removal.

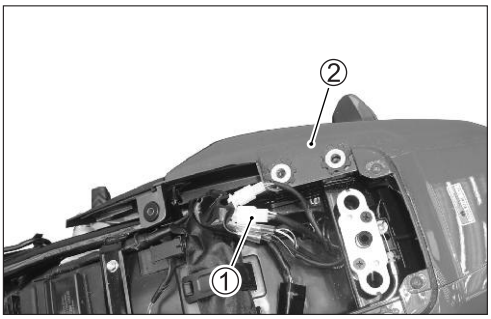
TAIL COVER ASSEMBLY

REMOVAL

- Remove the tail cover assembly and mounting bolts.

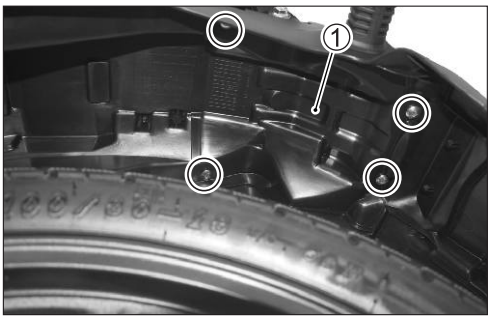


- Remove the wire clamp, open the wire jacket, disconnect the tail light/brake light coupler ①, take off the tail cover assembly ②.



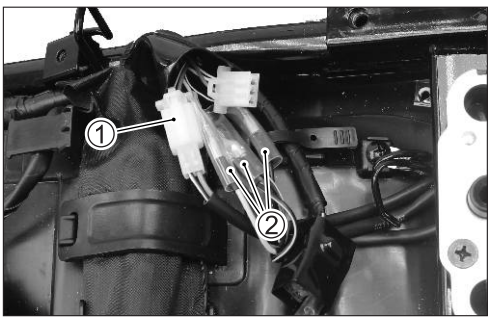
REAR FENDER INNER PLATE

- Remove the rend fender inner plate screws, take off the rear fender inner plate ① .

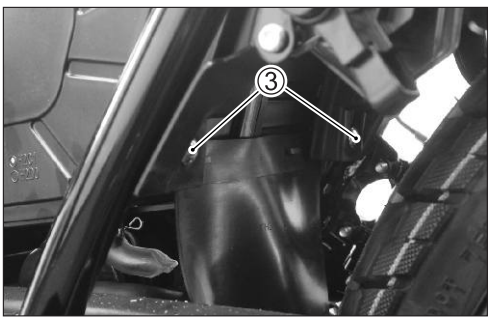


REAR FENDER

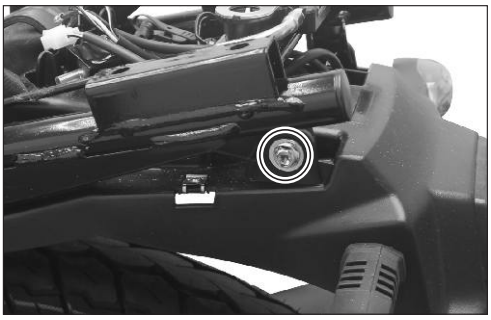
- Disconnect the rear license light coupler ① and rear left and right turn signal light coupler ②.



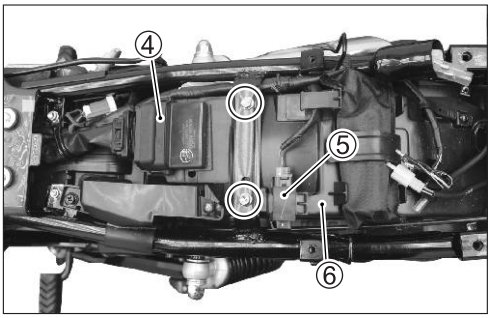
- Remove the rear fender mounting screws ③.



- Remove the rear fender mounting bolts.



- Remove the ECM ④ and TO sensor ⑤, take off the rear fender ⑥.

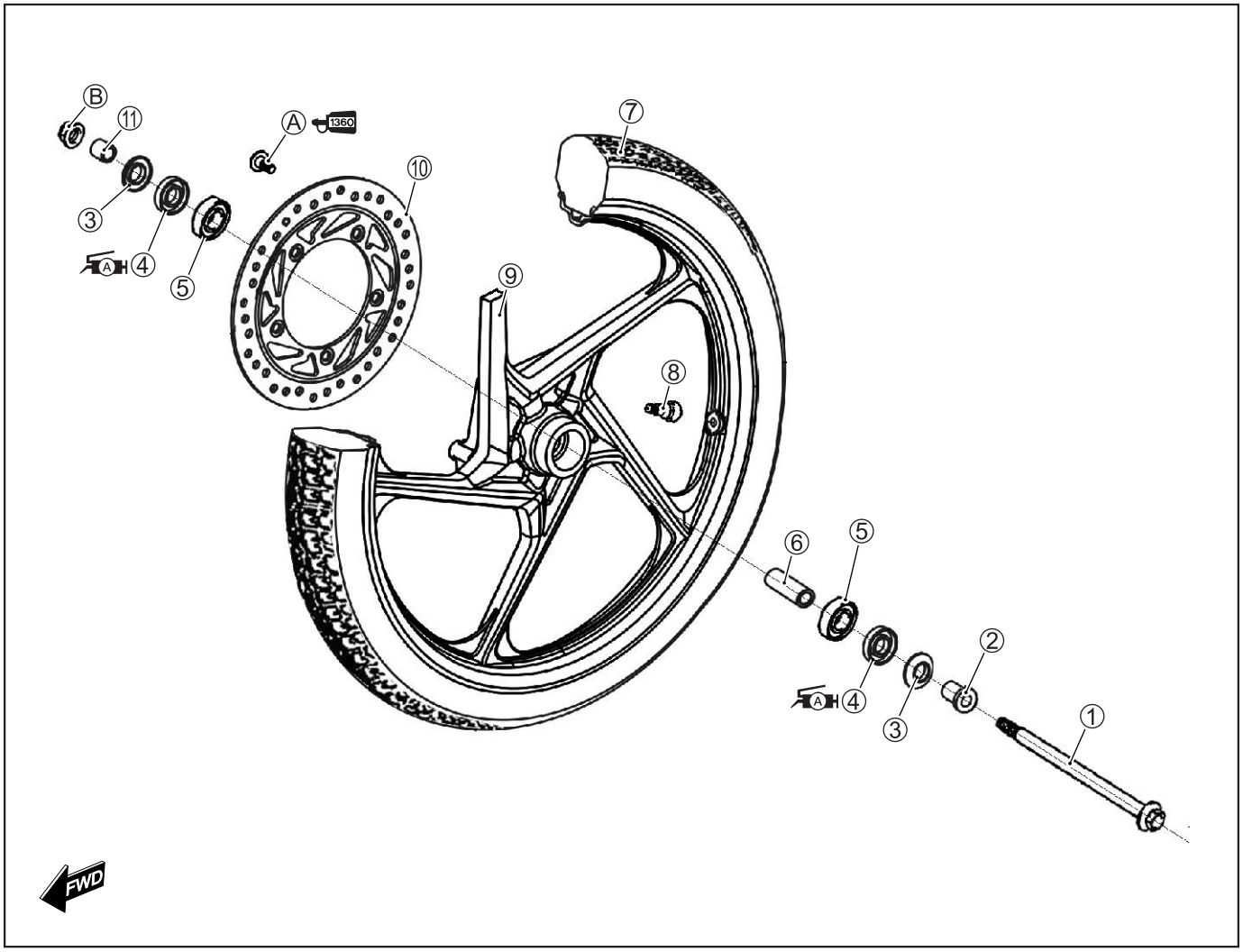


INSTALLATION

Install the rear fender in the reverse order of removal.

FRONT WHEEL

CONSTRUCTION



①	Front axle	⑧	Air valve
②	Sleeve	⑨	Front wheel rim
③	Dust-proof cover	⑩	Front brake disc
④	Oil seal	⑪	Spacer
⑤	Roller bearing	Ⓐ	Brake disc bolt
⑥	Spacer	Ⓑ	Front axle
⑦	Front tire		

ITEM	N•m
Ⓐ	18-28
Ⓑ	36-52

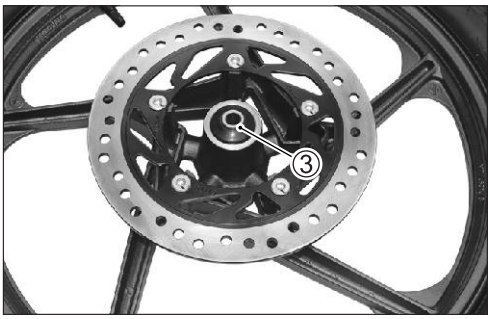
REMOVAL AND DISASSEMBLY

- Support the motorcycle with main stand.
- Remove the front axle nut ①, draw out the front axle shaft, remove the front wheel ②.

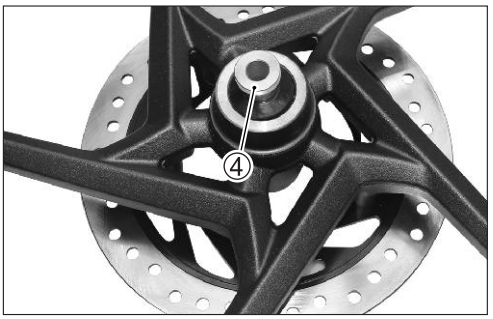
⚠ CAUTION

Do not operate the front brake lever when removing the front wheel.

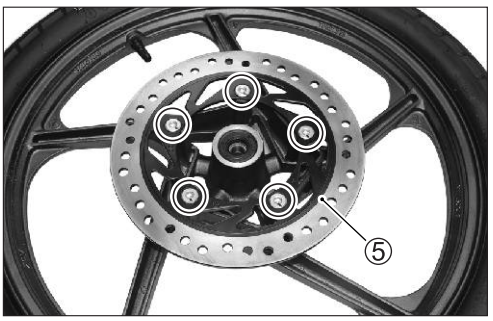
- Remove the front axle right spacer and dust-proof cover ③.



- Remove the front axle left spacer and dust-proof cover ④.



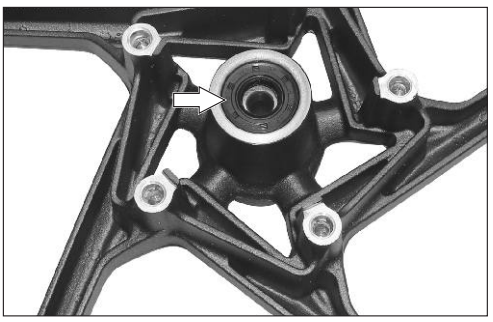
- Remove the brake disc bolts, take off the brake disc ⑤.



INSPECTION AND DISASSEMBLY

OIL SEAL

Inspect the oil seal for wear or damage. Replace the oil seal if there is something unusual.



- Remove the oil seal with the special tool.

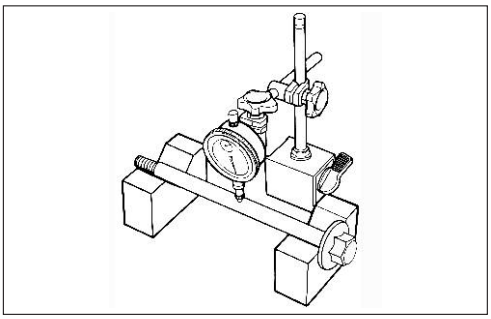
TOOL **09913-50121: Oil seal remover**

FRONT WHEEL AXLE

Using the special tools, check the axle shaft for runout and replace it if the runout exceeds the limit.

TOOL **09900-20606 : Dial gauge (1/100)**
09900-20701 : Magnetic stand
09900-21304 : V-block (100mm)

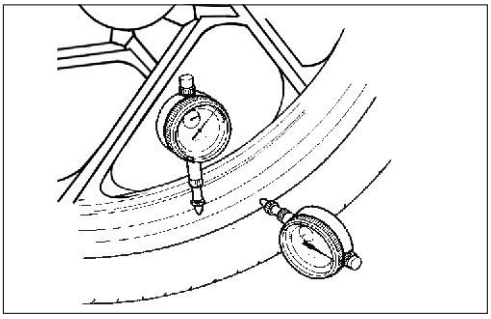
DATA **Wheel axle runout limit: 0.25 mm**



WHEEL RIM

Make sure that the wheel rim runout does not exceed the service limit when checked as shown. An excessive amount of runout is usually due to loose spokes or a bent wheel rim. If properly tightening the spokes will not correct the runout, replace the wheel rim.

DATA **Wheel rim runout limit: 2.0 mm**



WHEEL BEARING

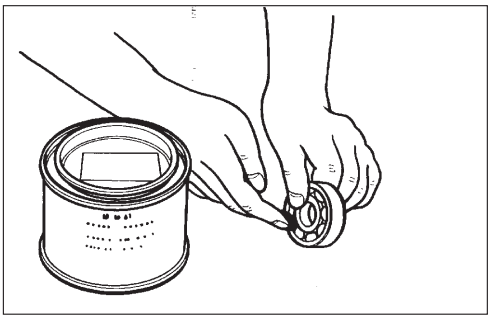
Inspect the wheel bearings for play by hand. Rotate the inner race by hand to inspect whether abnormal noise occurs and it rotate smoothly. Replace the bearing if there are any defects.

- Remove the bearings on both side with the special tool, take off the spacer.

TOOL **09921-20240:Bearing remover tool**

REASSEMBLY

Reassemble and remount the front wheel in reverse order of removal. Pay special attention to the following points.

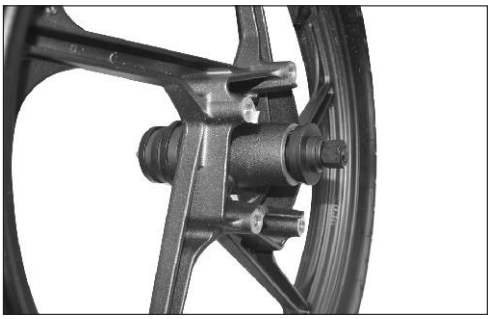


WHEEL BEARING

- Apply super grease “A” to bearings before reassembly.

99000-25010: Super grease “A”

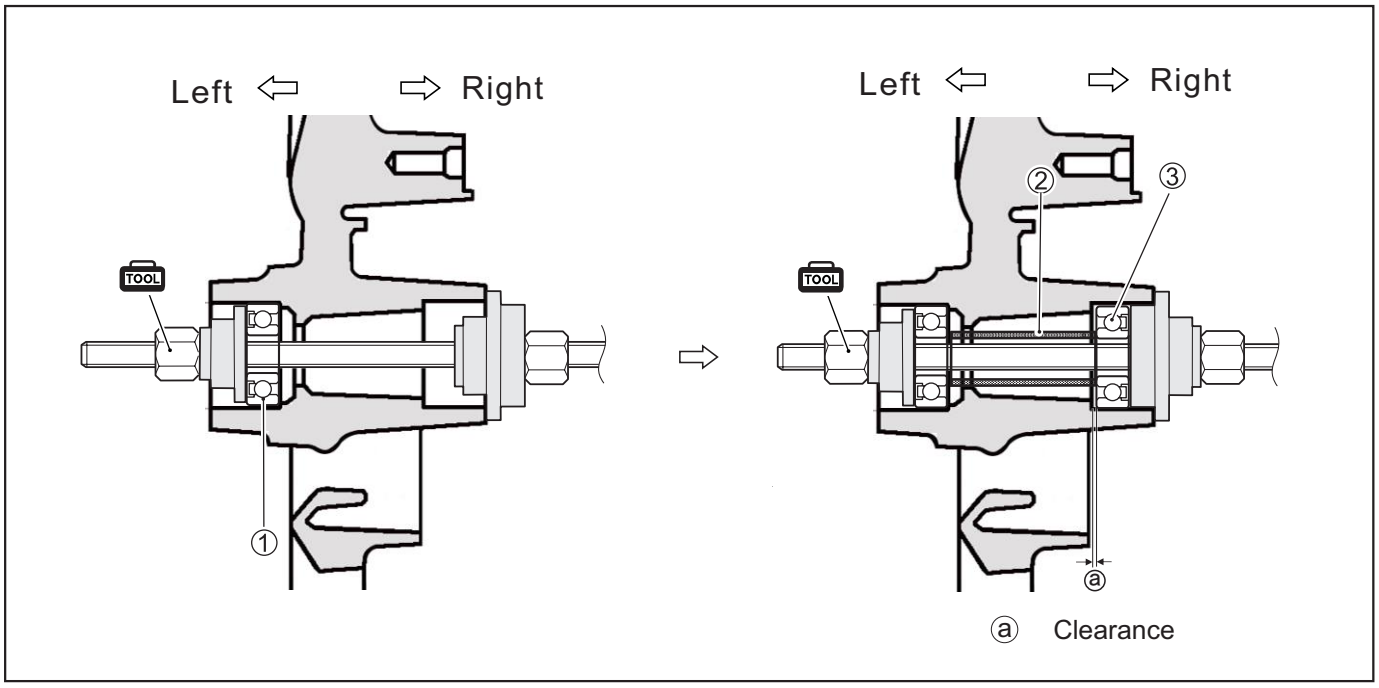
- Install left wheel bearing ① first, then install the spacer ② and right wheel bearing ③ using the special tool.



09924-84510: Bearing installer

⚠ CAUTION

The sealed cover of the bearing must face outside.



OIL SEAL

- Apply the super grease “A” to the oil seal lips.
- Install the new oil seal using the special tool.

99000-25010 : SUPER GREASE “A”

09913-70210 : Bearing installing set (Φ10-75)

BRAKE DISC

- Make sure that the brake disc is clean and free of any greasy matter.
- Install the brake disc, apply the bond to the thread, tighten the brake disc bolts to the specified torque.

99000-32130 : Thread Lock "1360"

Front brake disc bolt: 18~28 N•m

FRONT AXLE SHAFT

- Install the front axle shaft spacer, dust-proof cover and front wheel.
- Tighten the front axle nut to the specified torque.

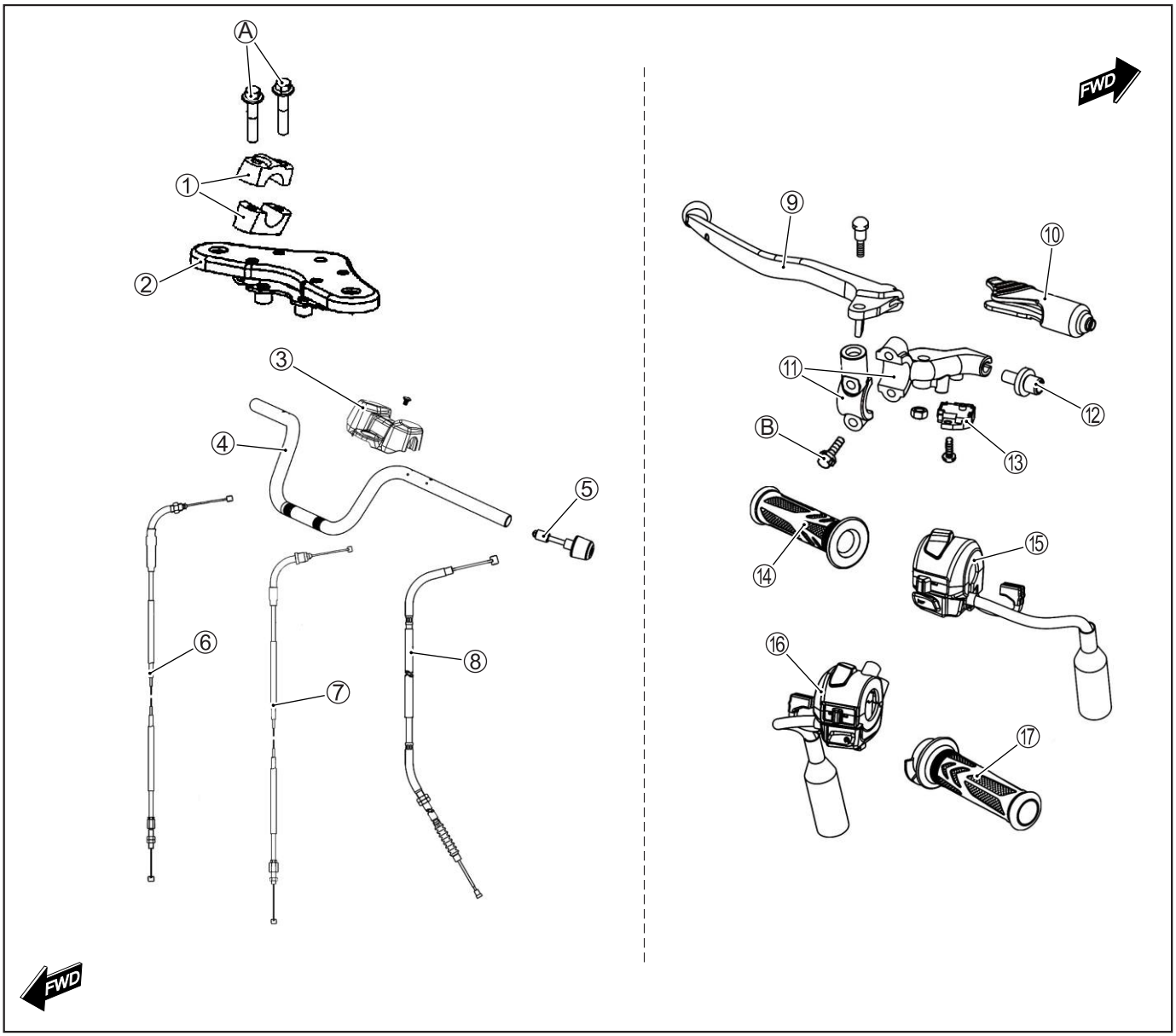
Front axle nut: 36-52 N•m

CAUTION

After installing the front wheel, pump the brake lever until the piston pushes the pad correctly.

HANDLEBAR

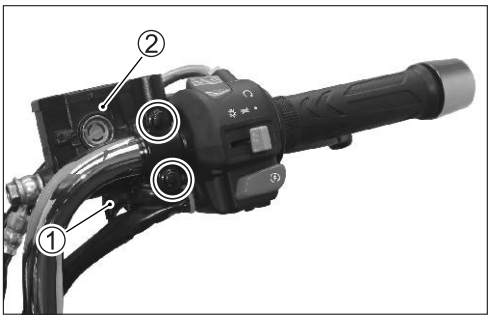
CONSTRUCTION



①	Handlebar holder	⑪	Clutch lever holder
②	Upper bracket	⑫	Adjuster nut
③	Handlebar cover	⑬	Clutch switch
④	Handlebar	⑭	Left grip
⑤	Balancer	⑮	Left handlebar switch assy
⑥	Throttle cable II	⑯	Right handlebar switch assy
⑦	Throttle cable I	⑰	Right handlebar switch assy
⑧	Clutch cable	Ⓐ	Bolt
⑨	Clutch lever	Ⓑ	Bolt
⑩	Clutch lever cover		

ITEM	N•m	kgf•m
Ⓐ	12-20	1.2-2.0
Ⓑ	8-12	0.8-1.2

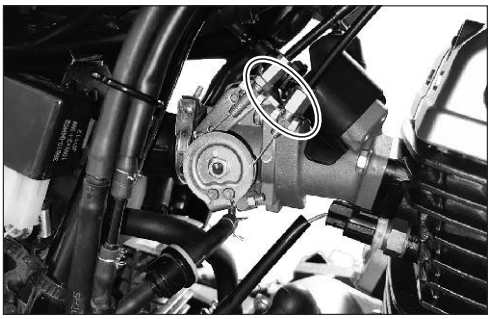
REMOVAL AND DISASSEMBLY



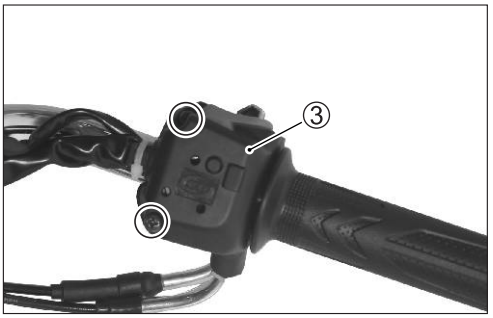
- Remove the rearview mirror. (Refer to page 6-2)
- Disconnect the front brake switch coupler ①.
- Loosen the master cylinder mounting bolts, take off the master cylinder ②.

⚠ CAUTION

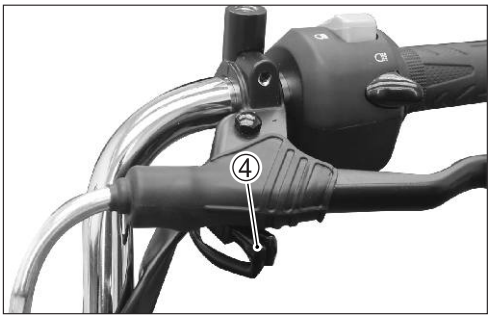
Do not operate the front brake lever during removal.



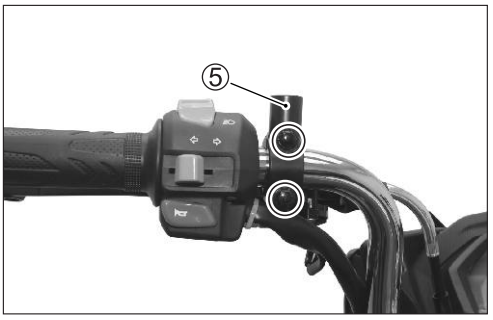
- Remove the right side cover.(Refer to page 6-1)
- Remove the throttle cable I and II from the throttle body.



- Remove the right handlebar switch assembly mounting screws, remove the right handlebar switch assembly ③.

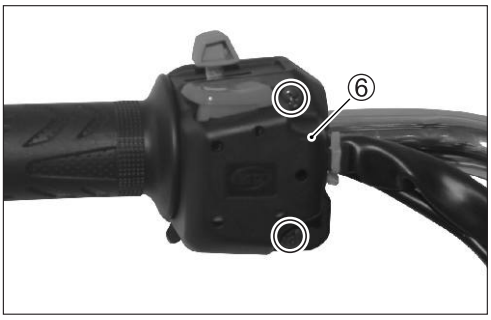


- Disconnect the clutch switch coupler ④.

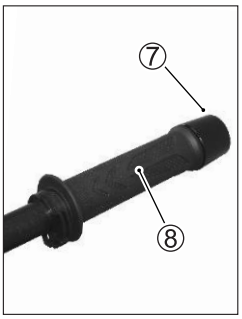


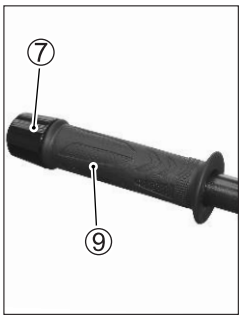
- Remove the clutch lever mounting bolt, take off the clutch lever ⑤.

- Loosen the left handlebar switch mounting screws, remove the left handlebar switch assembly ⑥.



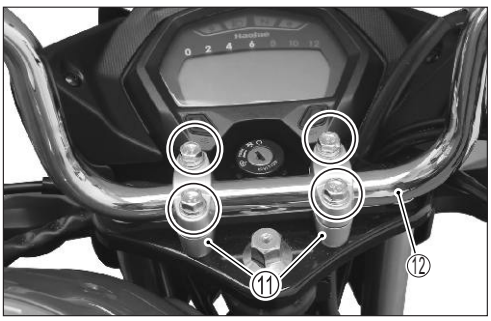
- Remove the left and right handlebar balancers ⑦, throttle grip ⑧, left handlebar grip ⑨.





- Remove the handlebar cover ⑩.

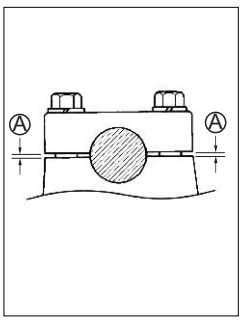
- Remove the handlebar clamp bolts, handlebar upper holder ⑪ and handlebar ⑫.



INSTALLATION

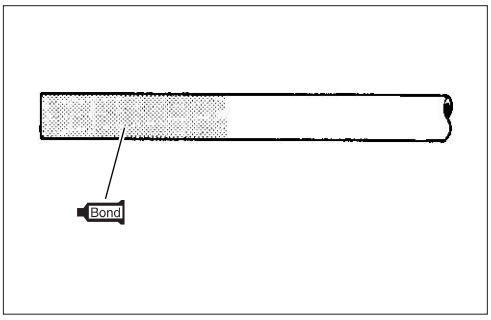
- Set the handlebars so that its punch mark aligns with the mating surface of the handlebar holder. Secure the each handlebars clamp in such a way that the clearance ahead of and behind the handlebar Ⓐ should be equalized.
- Tighten the handlebar holder clamp bolts to the specified torque.

Handlebar clamp bolt: 12-20 N•m

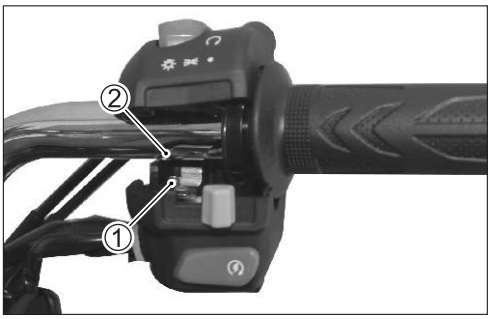


- Apply the handle grip bond onto the left handlebar before installing the handlebar grip.

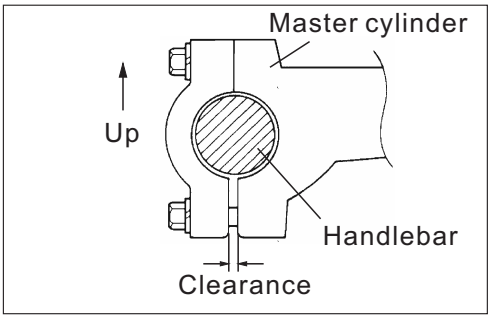
Bond **Handlebar grip bond**



- When installing right handlebar switch box, insert the projection ① of the right handlebar switch box into hole ② of the right handlebar.



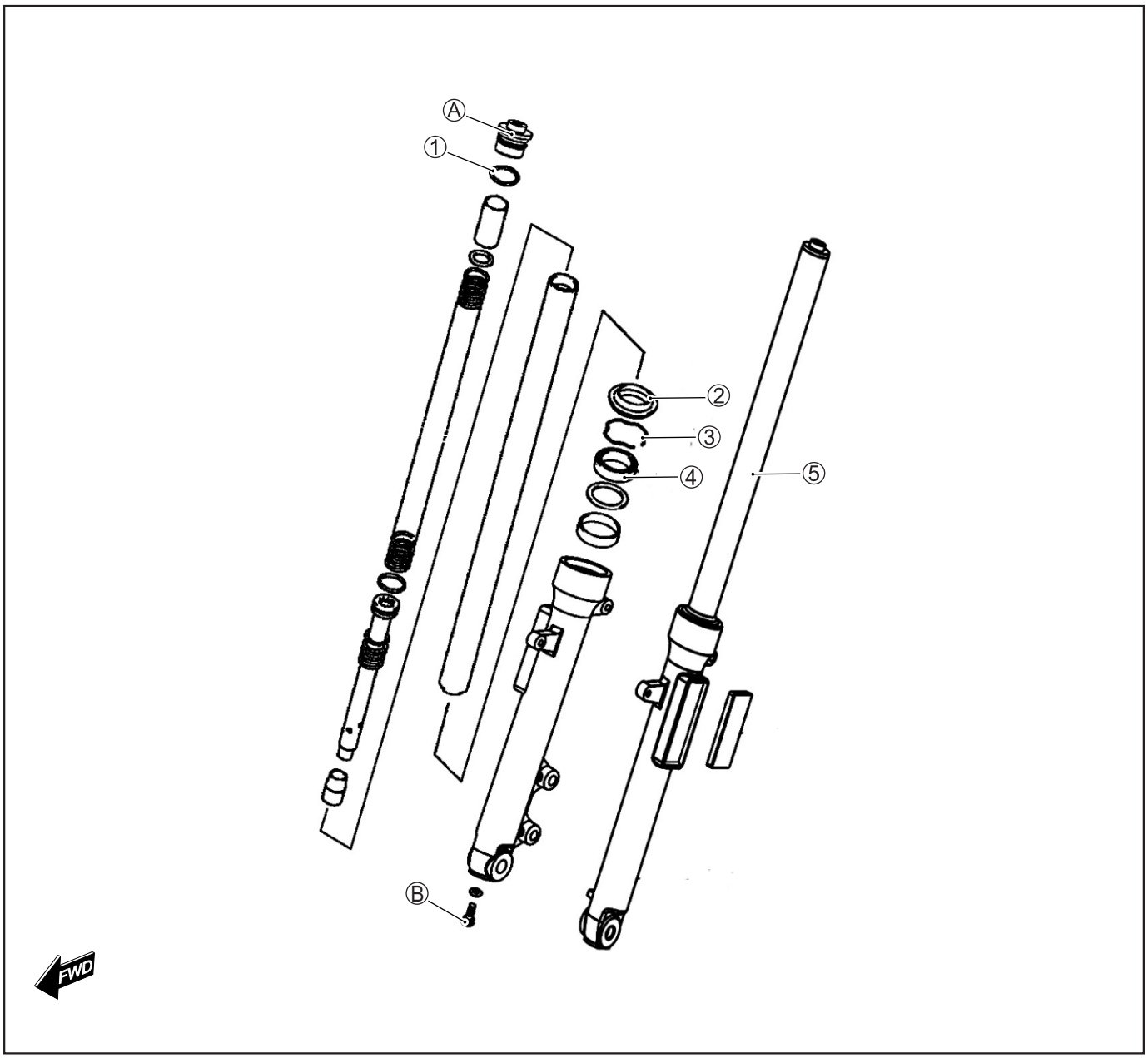
- Install the master cylinder with the punch mark aligned with the mating surface of master cylinder holder as shown, tighten the upper mounting bolt first, then the lower one.



- When install the clutch lever holder on the handlebar, align the clutch lever holder mating surface with the punched mark on the handlebar, tighten the upper mounting bolt first, then the lower one.

FRONT ABSORBER

CONSTRUCTION

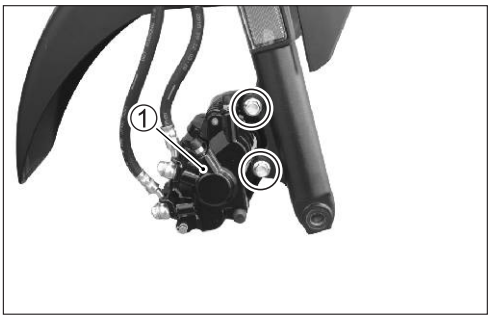


①	O-ring	⑤	Left front absorber
②	Dust-proof cover	Ⓐ	Cap bolt
③	Snap ring	Ⓑ	Bolt
④	Oil seal		

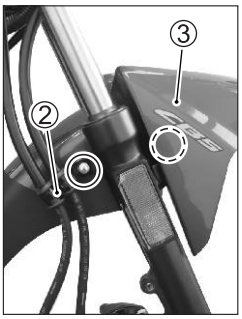
ITEM	N•m	kgf•m
Ⓐ	22-35	2.2-3.5
Ⓑ	20-26	2.0-2.6

REMOVAL

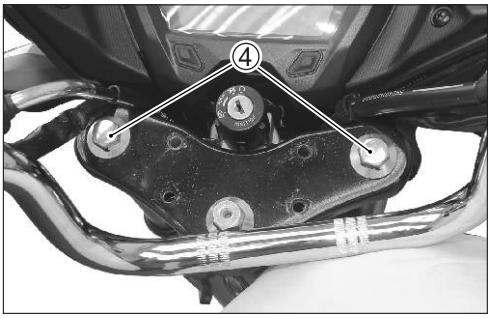
- Remove the front wheel. (Refer to page 5-8)
- Remove the 2 bolts, take off the front brake caliper ①.



- Remove the front fender ② by removing the mounting bolts.

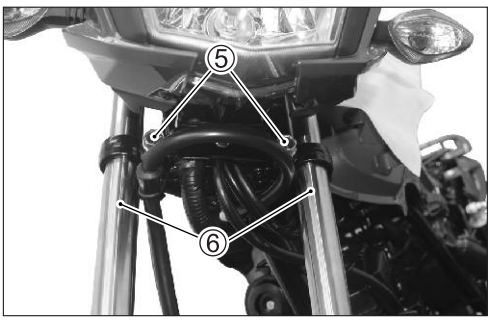


- Take off the handlebar. (Refer to page 6-14)
- Remove the front absorber upper cap bolt ④.

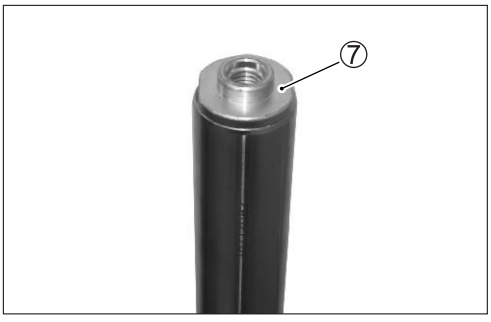


- Hold the front absorber with hand, loosen the lower bracket mounting bolts ⑤, remove the left and right front absorbers ⑥.

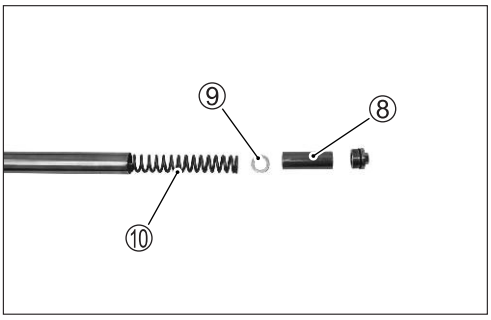
⑦



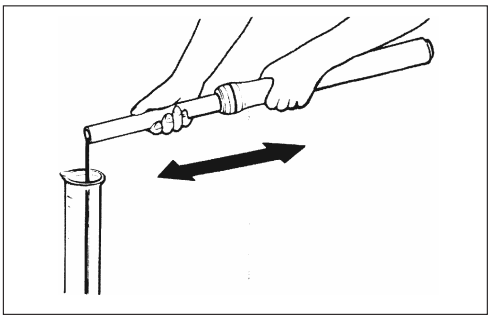
- Remove the upper cap bolt .



- Remove the spacer ⑧, washer ⑨ and spring ⑩.

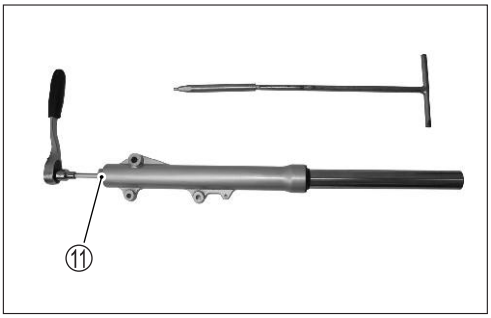


- Invert the absorber and stroke it several times to remove the oil.

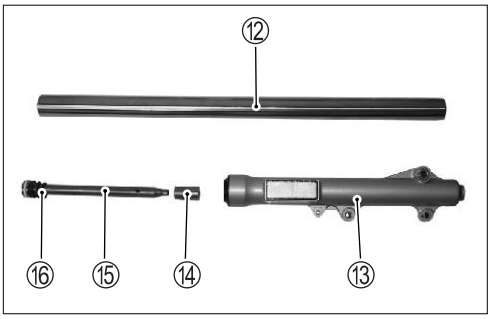


- Remove the piston rod bolt ⑪ with special tool.

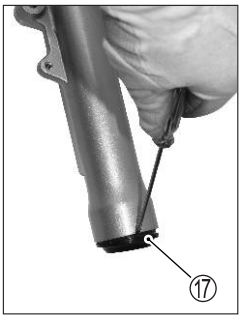
11F14-010 : Front absorber tools

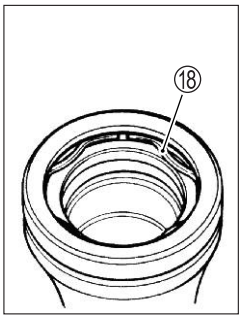


- Separate the inner tube ⑫ from the outer tube ⑬.
- Remove the oil lock piece ⑭ and damper rod ⑮ with the rebound spring ⑯.

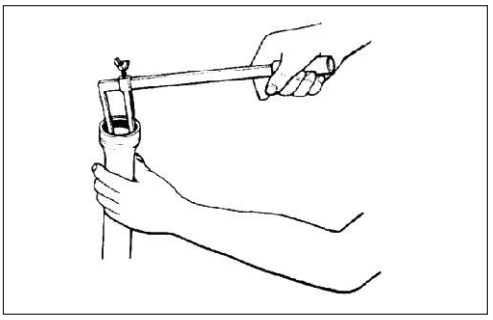


- Remove the dust-proof seal ⑰ and stopper ring ⑱.





- Remove the oil seal and washer by using the special tool.



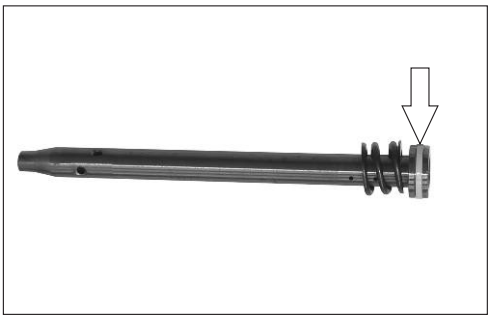
TOOL **09913-50121: Oil seal remover**

⚠ WARNING

Always use new oil seal.

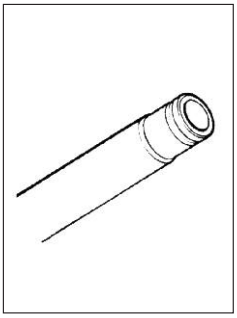
INSPECTION

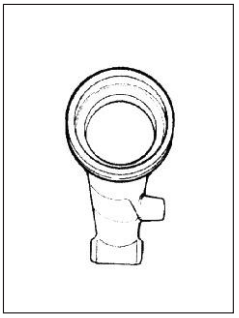
DAMPER ROD RING



- Inspect the damper rod ring for wear and damage.

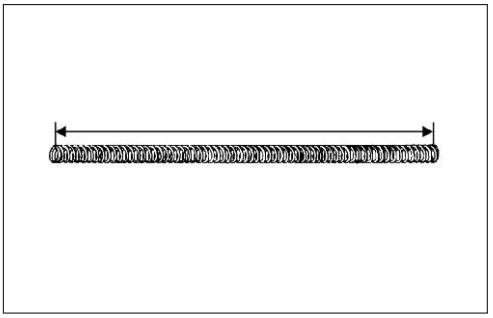
INNER TUBE AND OUTER TUBE





- Inspect the inner tube and outer tube sliding surface for any scuffing or flaws.

ABSORBER SPRING



- Measure the spring free length. If it is shorter than the service limit, replace it.

DATA **Spring free length limit: 413.6 mm**

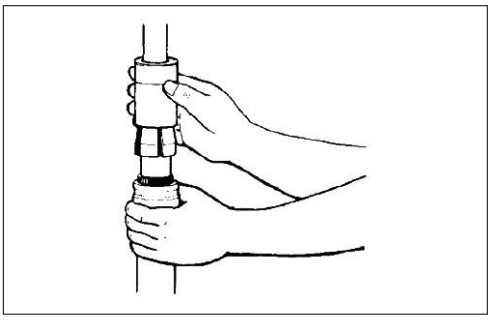
*** Cleaning: clean the parts with solvent and blow dry.**

REASSEMBLY

Reassemble the front absorber in the reverse order of disassembly. Pay attention to the following points.

- Insert the oil seal with the special tool.

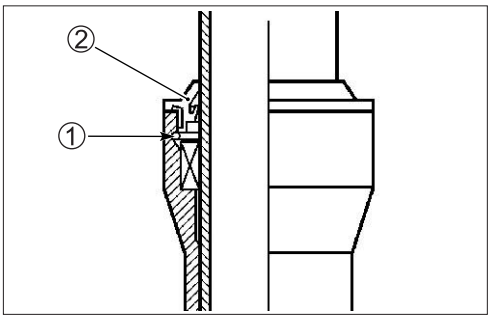
09940-50112: Absorber oil seal installer



CAUTION

Apply a few grease to the oil seal lip before installing.

- Install the stopper ring ①and new dust-proof seal ②.



CAUTION

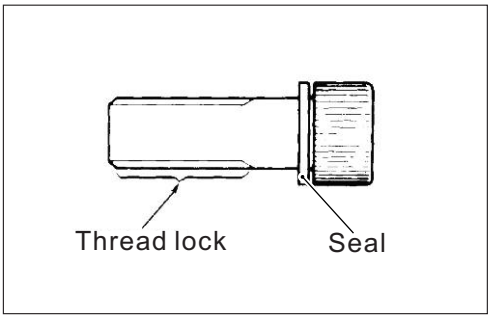
Make sure the stopper ring is fitted properly.

- Install the new seal ring.
- Apply a small quantity of thread lock "1303" to the damper rod bolt, tighten the bolt to the specified torque.

99000-32040: THREAD LOCK "1303"

11F14-010 : Front absorber tools

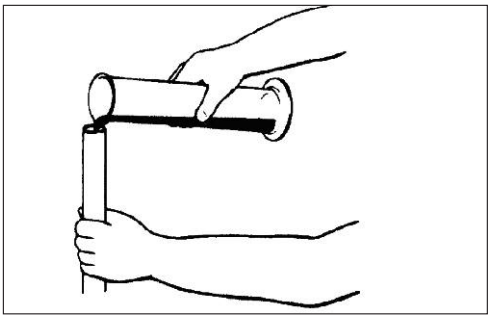
Damper rod bolt: 20~26 N•m



- Pour the specified fork oil into the inner tube.

Front absorber oil capacity (each leg): 152±1 ml

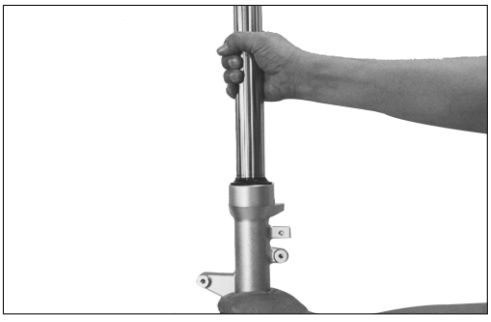
Absorber oil type: MX34 # or equivalent



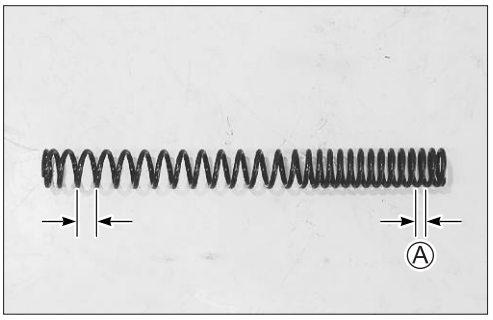
- Move the inner tube up and down until there are no air bubbles in the absorber oil.
- Keep the front absorber upright for 5-6 minutes.

CAUTION

Exhaust the air completely.



- Install the front absorber spring.



⚠ CAUTION

The close-pitch end Ⓐ of spring must face topside.

- Apply the absorber oil to the new O-ring.

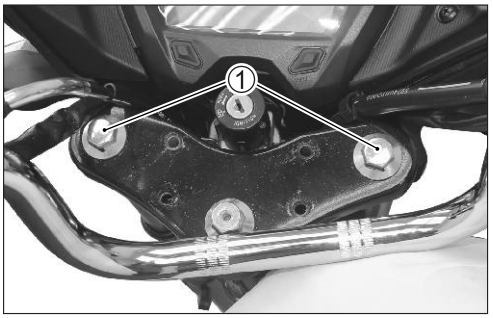
FORK **Absorber oil: 34#**

INSTALLATION

Install the front absorber in the reverse order of removal.
Pay attention to the following points.

- Install the front absorber to the upper bracket, tighten the front absorber cap bolt ① to the specified torque.

Upper cap bolt: 25-35 N•m

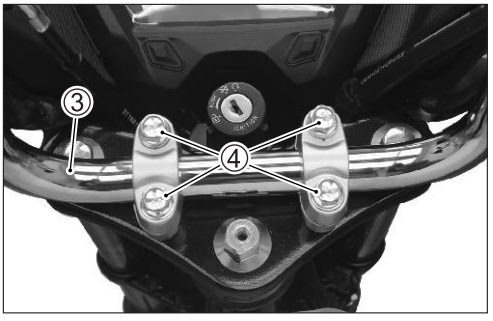


- Tighten the lower bracket clamp bolts ② to the specified torque.

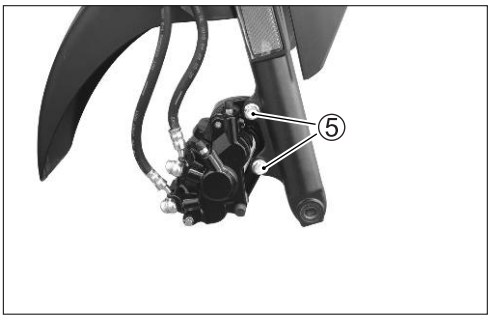
Lower bracket clamp bolt: 25~35 N•m

- Install the handlebar ③, tighten the handlebar holder clamp bolt ④ to the specified torque.

Handlebar holder bolt: 12-20 N•m



- Install the front brake caliper, tighten the mounting bolts ⑤ to the specified torque.

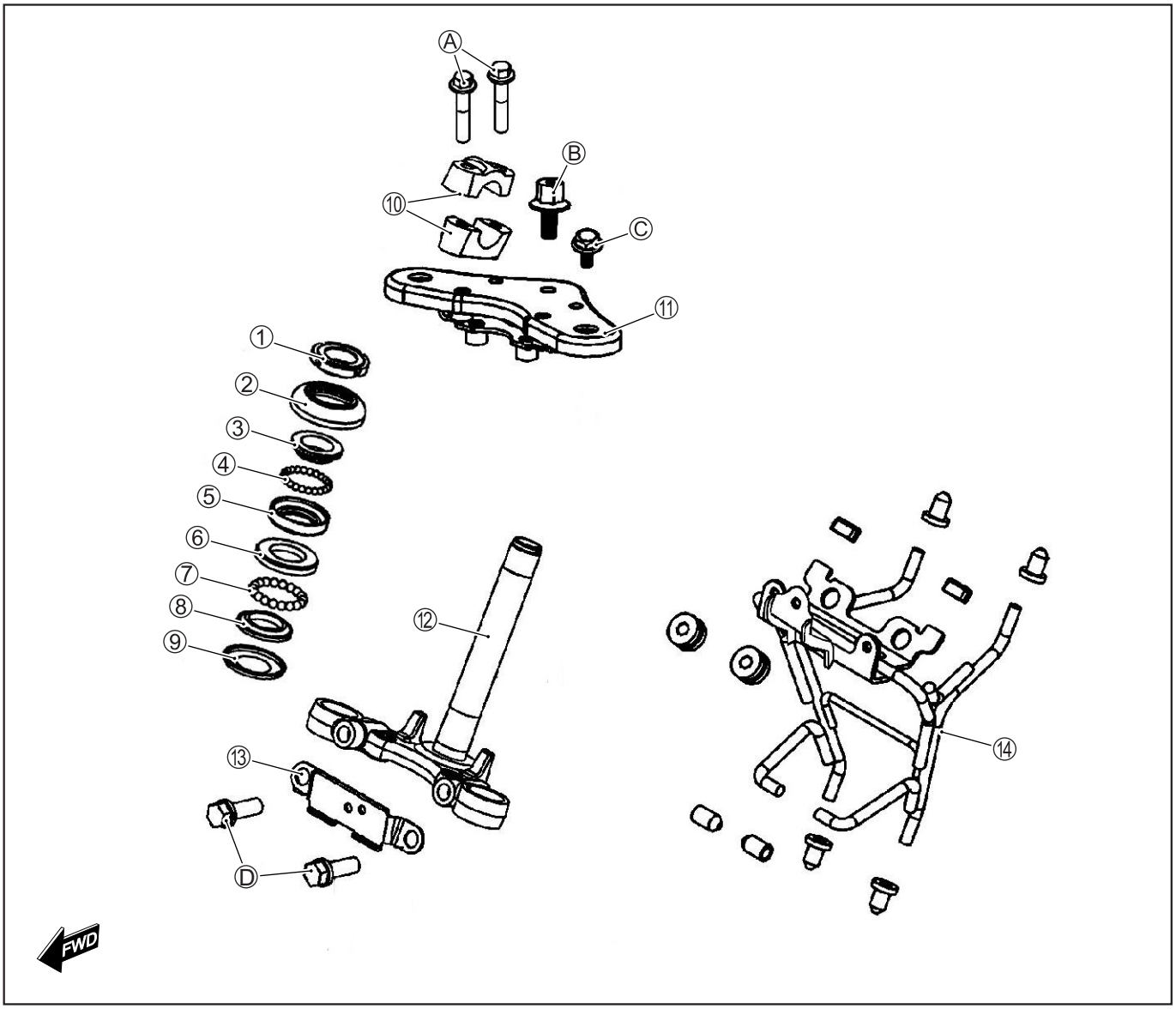


Front brake caliper mounting bolt: 18-28 N•m

- Install the front wheel.(Refer to page 5-11)

STEERING STEM

CONSTRUCTION



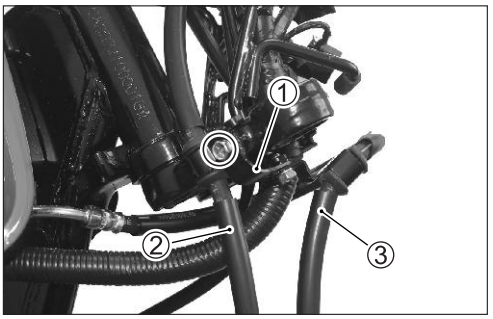
①	Steering stem nut	⑩	Handlebar holder
②	Dust seal	⑪	Upper bracket
③	Upper steering outer race	⑫	Lower bracket
④	Upper ball ring	⑬	Cover bracket
⑤	Upper steering inner race	⑭	Headlamp bracket
⑥	Lower steering inner race	Ⓐ	Handlebar clamp bolt
⑦	Lower ball ring	Ⓑ	Steering stem lock bolt
⑧	Lower steering outer race	Ⓒ	Cap bolt
⑨	Seal ring	Ⓓ	Bolt

ITEM	N•m	kgf•m
Ⓐ	12-20	1.2-2.0
Ⓑ	35-55	3.5-5.5
Ⓒ	22-35	2.2-3.5
Ⓓ	25-35	2.5-3.5

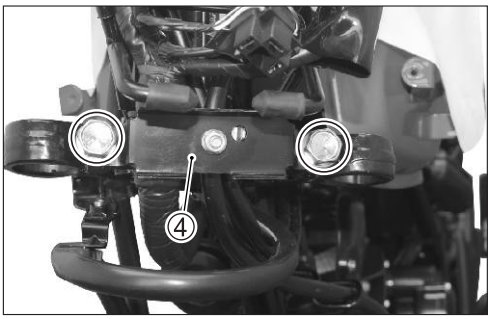
REMOVAL

- Remove the front absorber.(Refer to page 6-16)

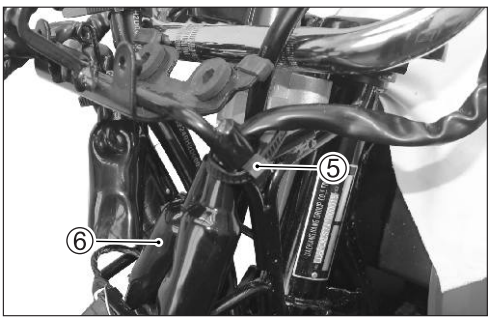
- Remove the front brake hose guide bolt, take off the front brake hose guide ①, front brake hose ② and rear brake hose ③.



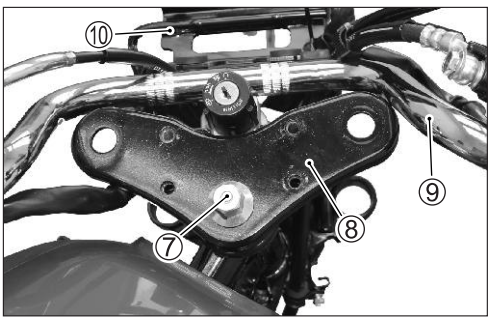
- Remove the front cover bracket ④.



- Loosen the wire clamp ⑤, open the PVC jacket ⑥, disconnect the ignition switch coupler.



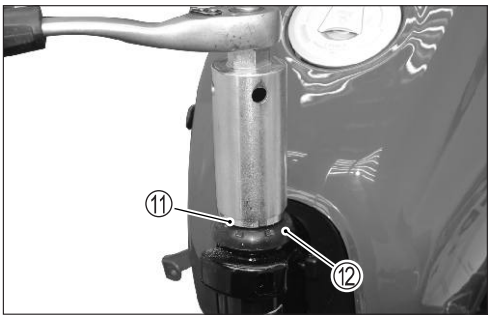
- Remove the steering stem lock bolt ⑦, remove the upper bracket ⑧, handlebar ⑨ and headlamp bracket ⑩.



⚠ CAUTION

Place the handlebar vertically to prevent the brake fluid from flowing out of master cylinder.

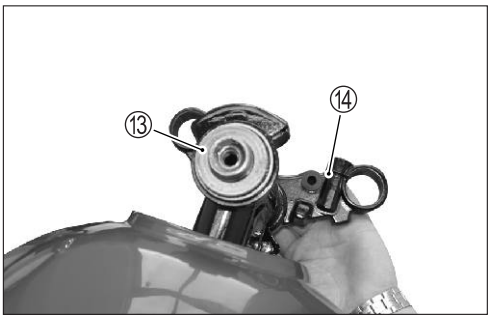
- Remove the steering stem nut ⑪ and dust-proof cover ⑫.



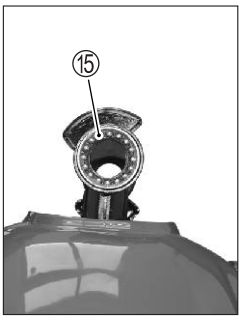
TOOL **09940-14911: Steering nut socket**

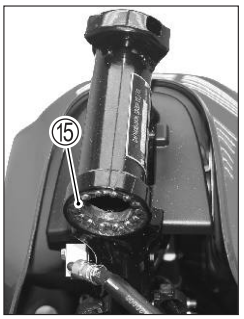
NOTE:
Hold the steering stem by hand to prevent from falling.

- Remove the upper steering outer race ⑬ and lower bracket ⑭.

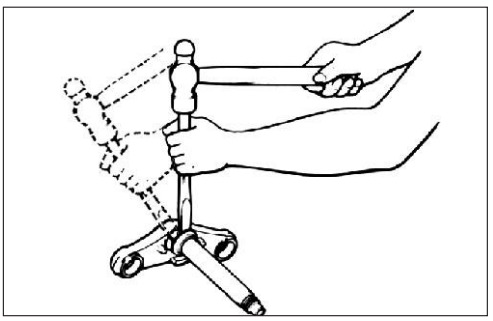


- Remove the upper and lower steel ball rings.



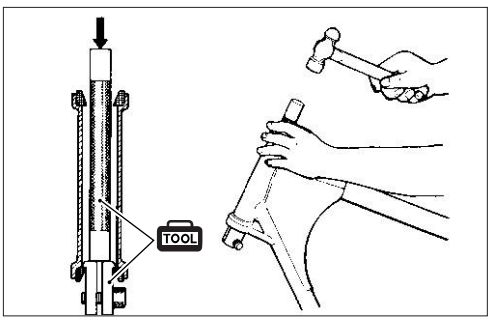


- Remove the outer race with a chisel.



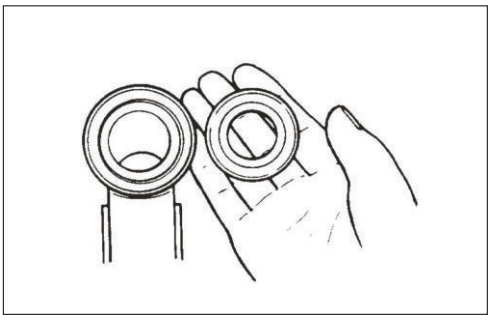
- Draw out the two inner races fitted to the top and bottom ends of the head pipe with the special tool.

TOOL **09941-54911: Bearing remover**
09941-74910: Steering bearing installer



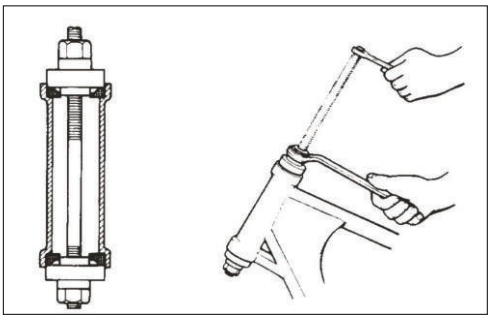
INSPECTION

- Inspect and check the removed parts for the following abnormalities.
  - — Race wear and brineling.
  - — Worn or damaged steel balls.
  - — Distortion of steering stem.



REASSEMBLY

Reassemble and remount the steering stem in the reverse order of disassembly and removal, and also carry out the following steps:



INNER RACES

- Press in the upper and lower inner races using the special tool.

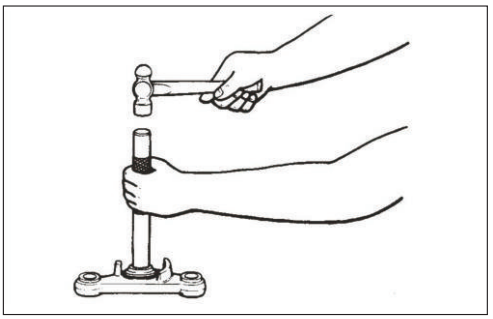
09941-34513: **Steering race installer**

OUTER RACE

- Apply grease on the new lower oil seal.
- Press in the lower outer race using the special tool.

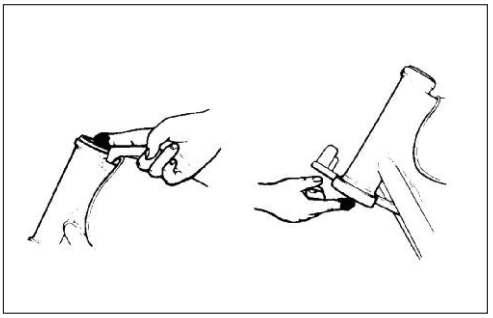
99000-25010 : SUPER GREASE "A"

09941-74910: Steering race installer



REASSEMBLY

Reassemble and remount the steering stem in the reverse order of disassembly and removal. and also carry out the following steps.



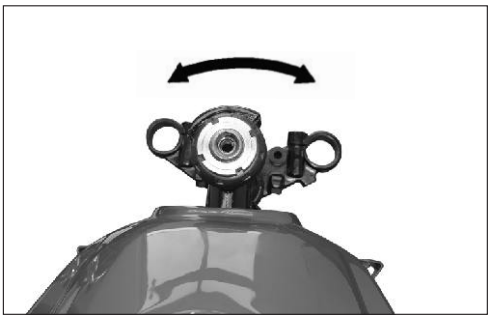
STEEL BALL

- Apply the super grease to the upper and lower inner races when installing the steel balls.

99000-25010 : SUPER GREASE "A"

STEERING STEM NUT

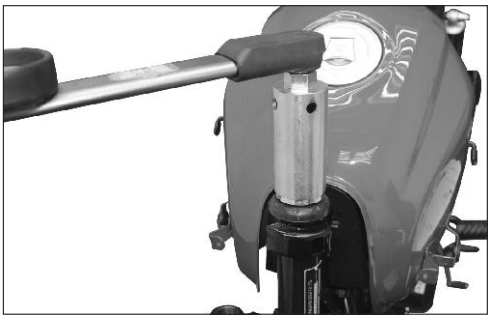
- Tighten the steering stem nut with the special tool.
- Turn the steering stem right and left, lock-to-lock, five or six times to "seat" the steel balls.



- Tighten the steering stem nut to the specified torque with the special tool.

09940-14911: Steering stem nut socket wrench

Steering stem nut: 25-40 N•m

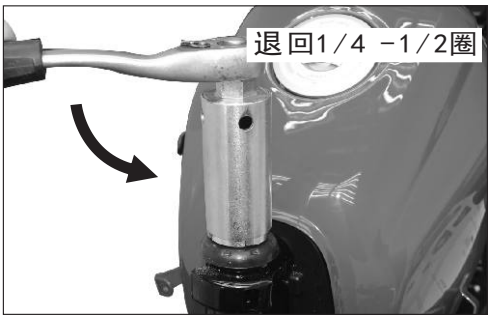


- Turn back the stem nut by 1/4-1/2 turn.

09940-14911: Steering stem nut socket wrench

⚠ CAUTION

The adjustment will vary from motorcycle to another.



- Install the upper bracket, tighten the steering stem lock bolt to the specified torque.

Steering stem lock nut: 35-55 N•m

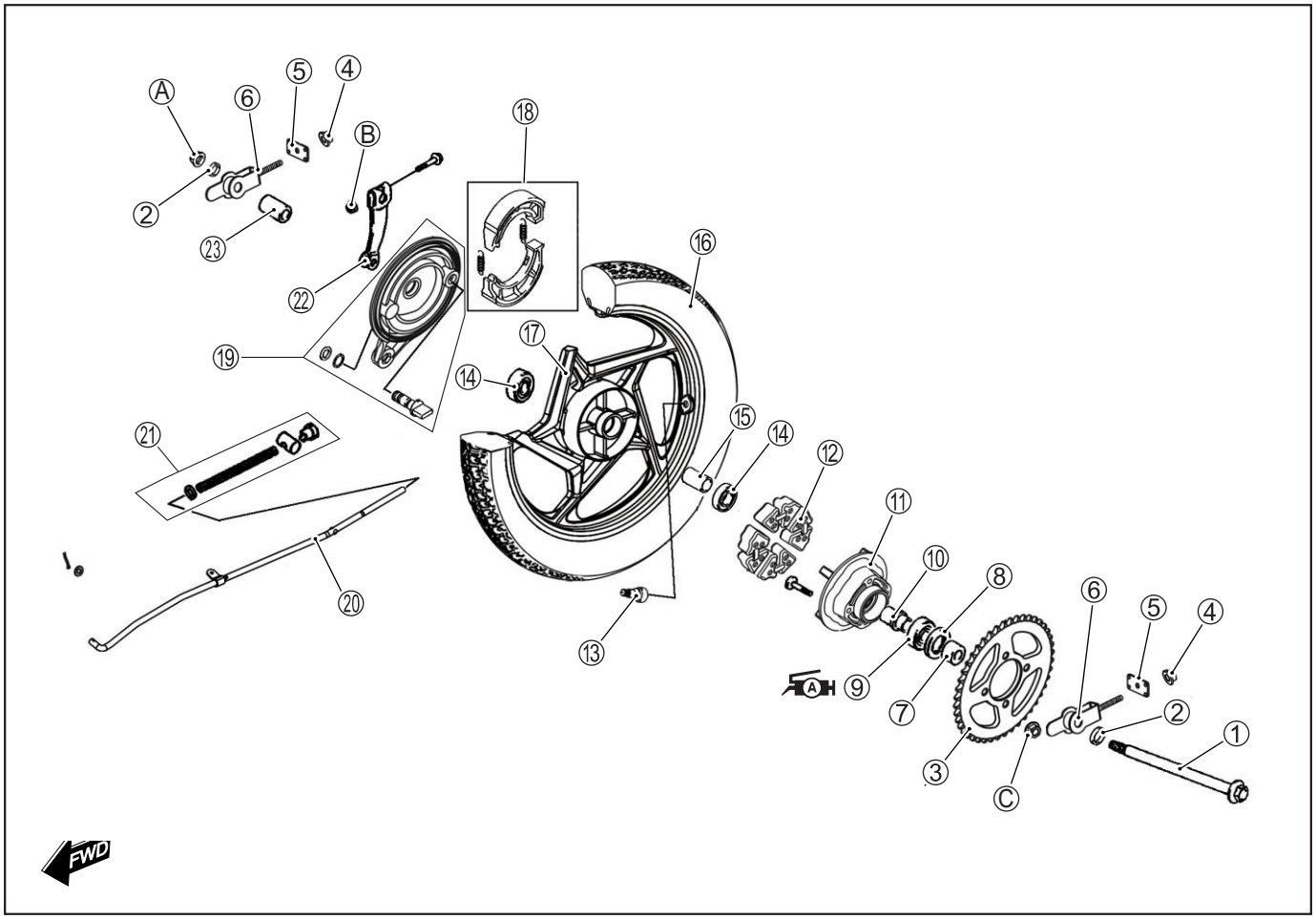
- Route the wiring harness, brake hose and cable. (Refer to page 8-21)
- Install the front absorbers. (Refer to page 6-21)
- Install the front wheel. (Refer to page 6-11)
- Install the meter and headlamp assembly. (Refer to page 6-3 to page 6-4)

⚠ WARNING

After performing the adjustment and installing the steering stem upper bracket, "rock" the front wheel assembly forward and backward to ensure that there is no play and that the procedure was accomplished correctly. Finally check to be sure that the steering stem moves freely from left to right with its own weight. If play or stiffness is noticeable, re-adjust the steering stem nut.

REAR WHEEL

CONSTRUCTION

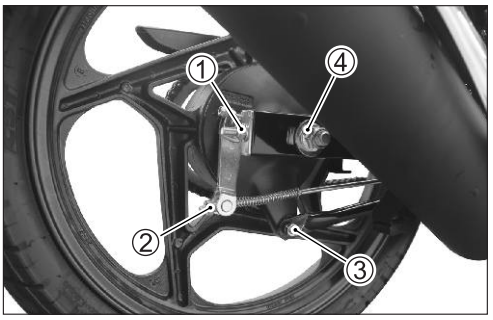


①	Rear axle shaft	⑭	Roller bearing
②	Chain adjuster washer	⑮	Spacer
③	Rear sprocket	⑯	Rear tire
④	Adjuster nut	⑰	Rear rim
⑤	Chain adjuster guide plate	⑱	Rear brake shoe
⑥	Chain adjuster	⑲	Rear brake drum
⑦	Spacer	⑳	Rear brake rod
⑧	Oil seal	㉑	Rear brake adjuster
⑨	Ball bearing	㉒	Rear brake cam lever
⑩	Rear sprocket drum retainer	㉓	Spacer
⑪	Rear sprocket drum	Ⓐ	Nut
⑫	Damper	Ⓑ	Nut
⑬	Air bleed valve	Ⓒ	Nut

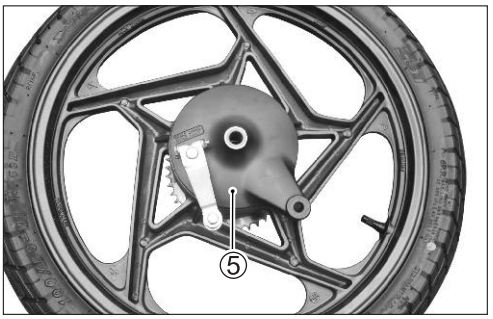
ITEM	N•m	kgf•m
Ⓐ	50-80	5.0-8.0
Ⓑ	8-12	0.8-1.2
Ⓒ	30-37	3.0-3.7

REMOVAL

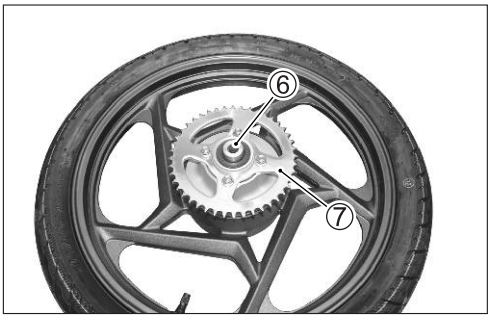
- Support the motorcycle with the main stand.
- Loosen the chain adjuster lock nut ①and rear brake adjuster nut ②.
- Remove the rear torque link③ and rear axle nut ④ , take off the rear wheel and rear axle right spacer.



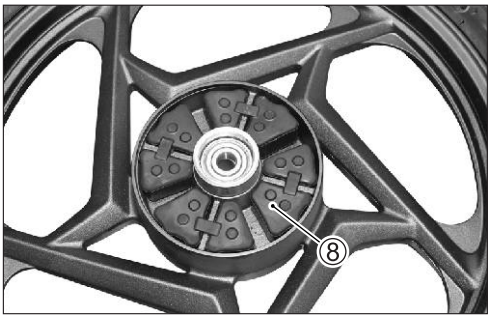
- Remove the rear brake panel ⑤.



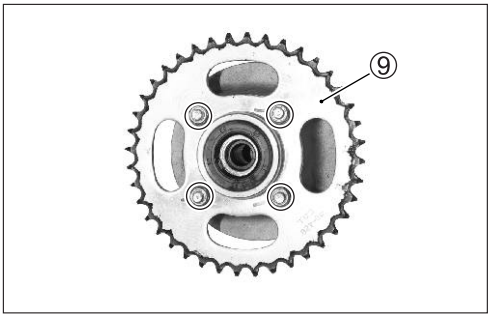
- Remove the rear axle left spacer ⑥ and rear sprocket drum ⑦.



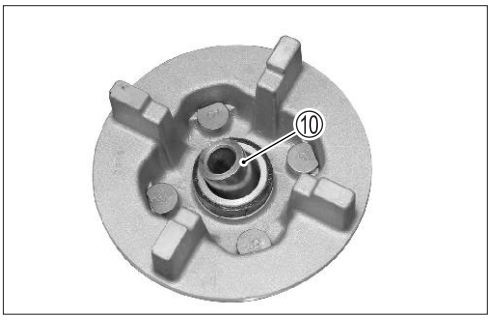
- Remove the damper ⑧.



- Remove the sprocket four mounting bolts.
- Remove the rear sprocket ⑨ from the rear sprocket drum.



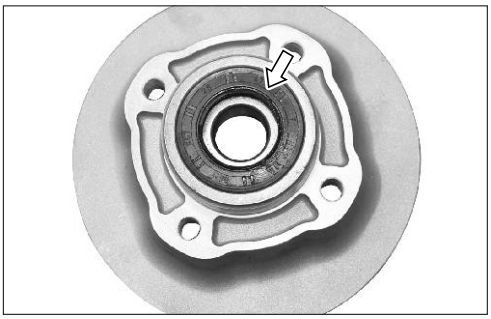
- Remove the rear brake drum retainer ⑩.



INSPECTION

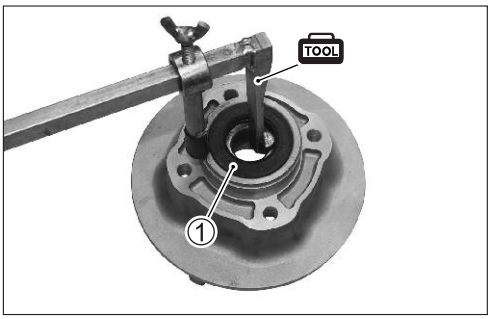
OIL SEAL

- Inspect the wheel and sprocket mounting drum oil seal for wear or damage. If any damage is found, replace the oil seal with a new one.



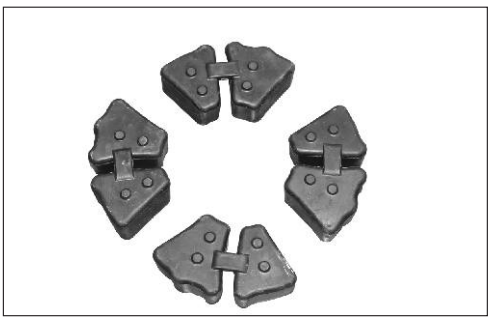
- Remove each oil seal ① with the special tool

09913-50121: Oil seal remover



DAMPER

- Inspect the dampers for wear and damage.
- Replace the damper if there is anything unusual.

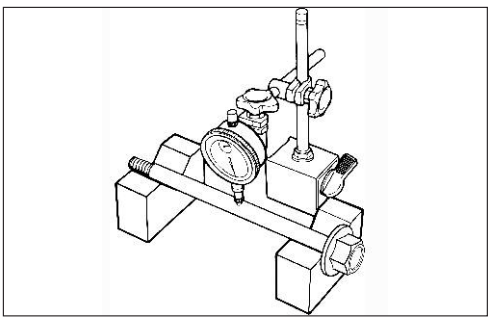


AXLE SHAFT

- Using the special tools, check the axle shaft for run-out and replace it if the run-out exceeds the limit.

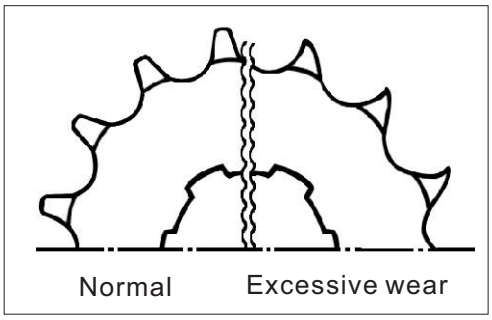
Axle shaft run-out limit: 0.25 mm

09900-20606: Dial gauge (1/100)
09900-20701: Magnetic stand
09900-21304: V-block



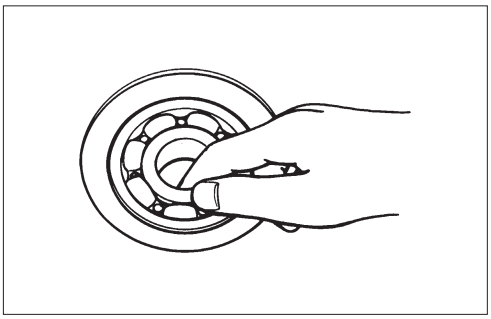
SPROCKET

- Inspect the sprocket teeth for wear. If they are worn as illustrated, replace the sprockets and drive chain.



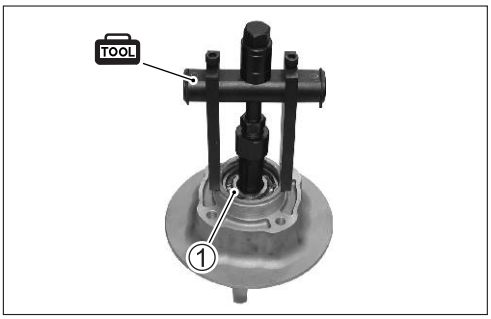
WHEEL BEARING

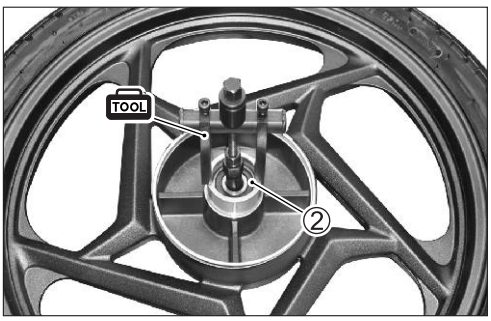
Inspect the wheel bearings for play by hand, Rotate the inner race by hand to inspect whether abnormal noise occurs and it rotates smoothly. Replace the bearing if there are any defects.



- Remove the sprocket mounting drum bearing ① and wheel bearing ② using the special tool.

TOOL **09921-20240: Bearing remover tool**

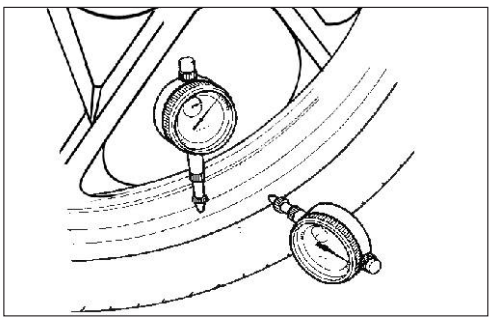




WHEEL

Make sure that the wheel runout checked as shown does not exceed the service limit. An excessive runout is usually due to worn or loosened wheel bearings and can be reduced by replacing the bearings. If bearing replacement fails to reduce the runout, replace the wheel.

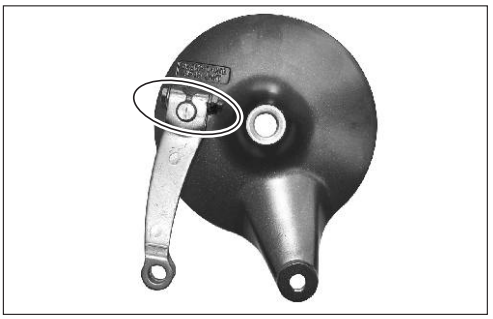
DATA **Wheel rim runoutl limit (Axial and Radial): 2.0 mm.**



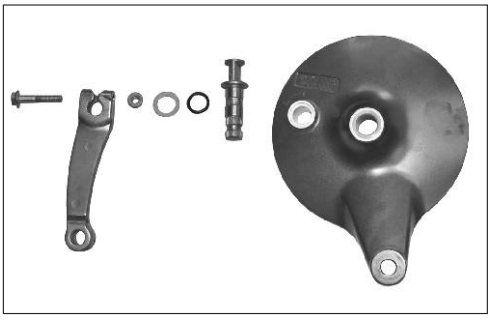
REAR BRAKE PANEL

- Remove the brake shoe.

- Remove the brake cam lever nut and bolt.



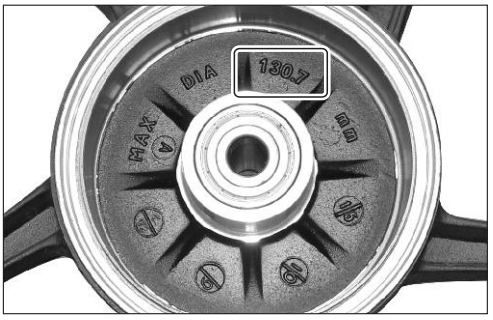
- Remove the brake cam, washer, O-ring and brake cam lever.



- Measure the brake drum I.D to determine the extent of wear, if the limit is exceeded by the wear noted, replace the drum. The value of the limit is indicated inside the drum.

DATA **Break drum I.D. limit: 130.7 mm**

TOOL **09900-20103**: **Vernier calipers**



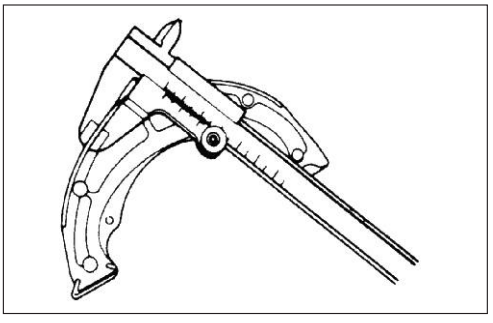
- Check the brake shoes and decide whether is should be replaced or not from the thickness of the brake shoe linings.

DATA **Brake shoe lining thickness limit: 1.5 mm**

TOOL **09900-20103**: **Vernier calipers**

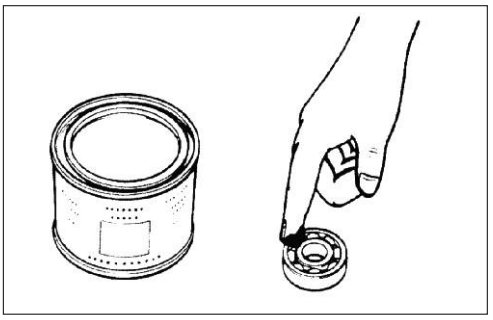
⚠ WARNING

Replace the brake shoe as a set, otherwise braking performance will adversely affected.



REASSEMBLY

Reassemble and remount the rear wheel and rear brake in the reverse order of disassembly and removal, and also carry out the following steps.



- Install the wheel bearings using the special tool.

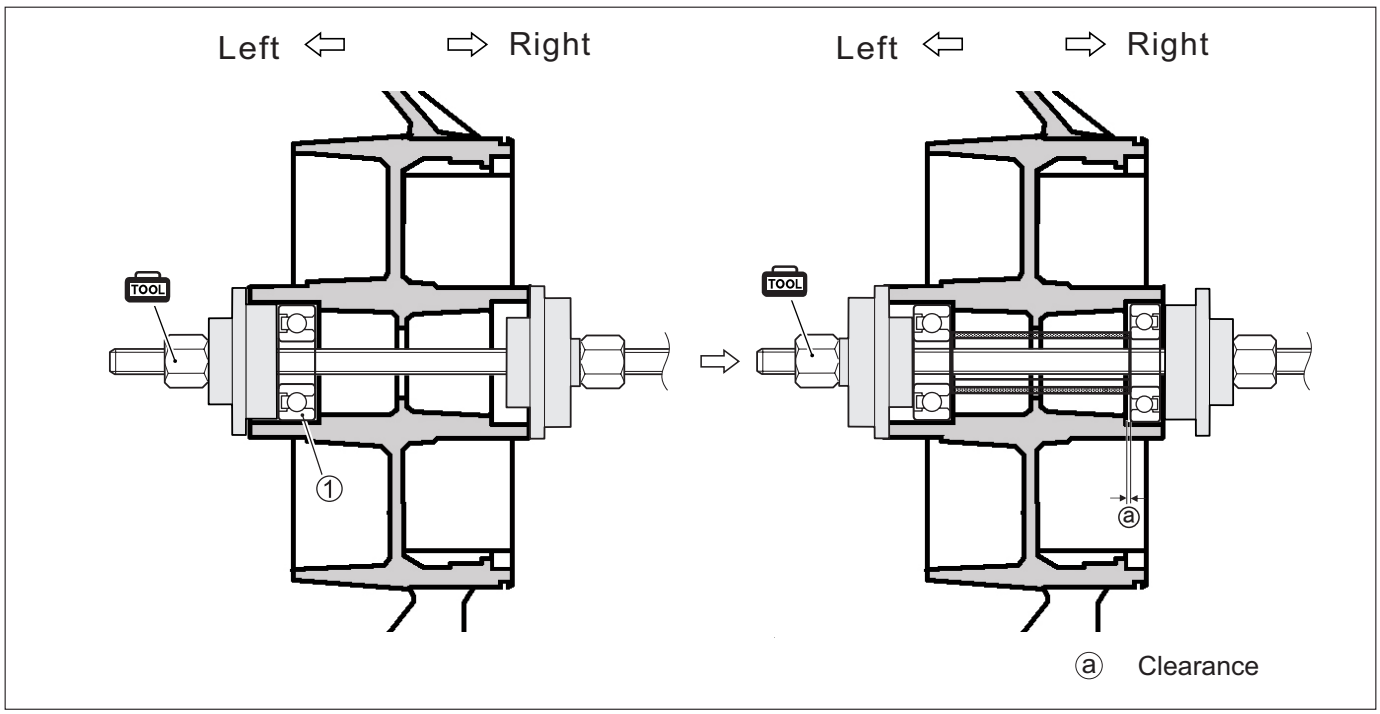
TOOL **09913-70210: Bearing installer**

- Install the left wheel bearing ① first, then the spacer ②and right wheel bearing③using the special tool.

TOOL **09924-84510: Bearing installer**

⚠ CAUTION

The sealed cover of the bearing must face outside.



OIL SEAL

- Install the new oil seals with the special tool.

09913-70210: Bearing installer

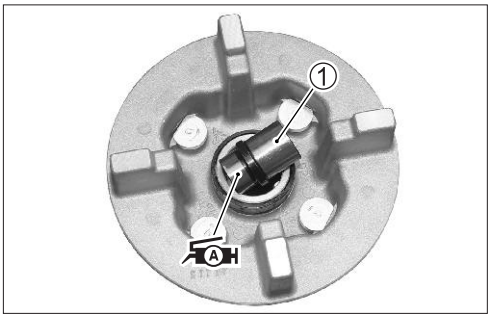
- Apply the super grease “A” to the oil seal lips.

99000-25010: SUPER GREASE “A”

REAR SPROCKET AND MOUNTING DRUM

- Apply the super grease “A” to the retainer ① before installing the rear sprocket mounting drum.

99000-25010: SUPER GREASE “A”

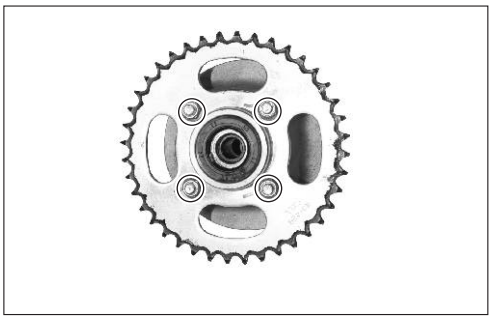


- Tighten the rear sprocket nuts to the specified torque.

Rear sprocket nut: 30-37 N•m

CAUTION

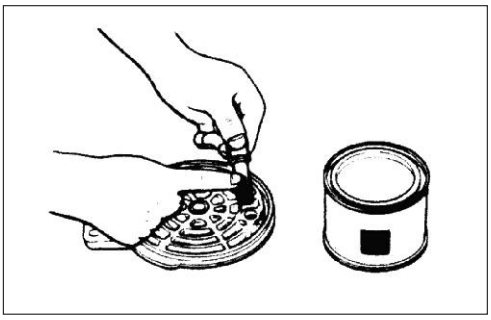
The mark Ⓐ side of sprocket must face outside.



BRAKE CAM

- Apply the super grease to the brake cam.

99000-25010: SUPER GREASE "A"



⚠ WARNING

Be careful not to apply too much grease to the brake cam. If grease gets on the lining, brake slippage will result.

BRAKE CAM LEVER

- Install the brake cam lever and tighten the nut to specified torque.

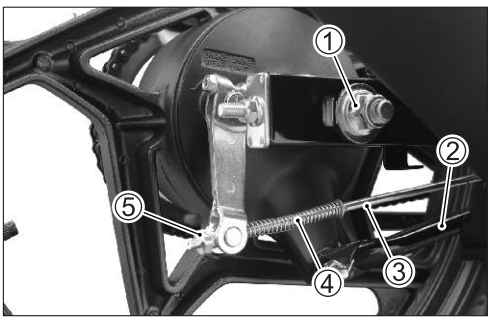
Cam lever nut: 8~12 N•m

- Apple the super grease "A" to the rear sprocket mounting drum before install the rear sprocket mounting drum to the rear wheel.

99000-25010: SUPER GREASE "A"

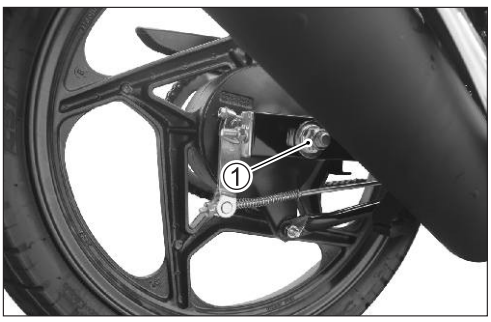
REAR WHEEL

- Install the rear wheel and rear axle, pre-tighten the rear axle nut ①.
- Install the rear torque link ②and rear brake rod③, spring④ and adjuster nut ⑤.



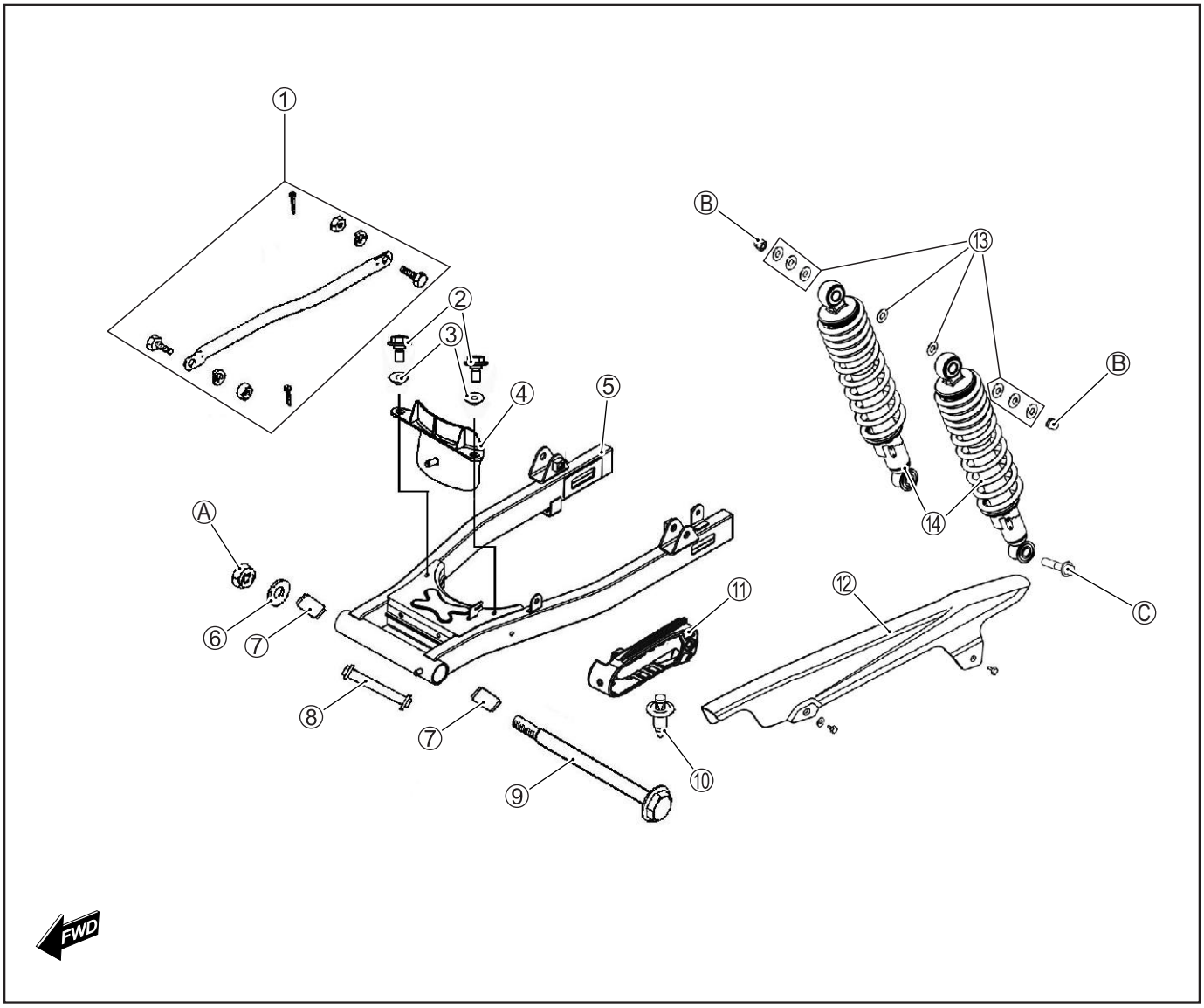
- Adjust the chain sag. (Refer to page 2-18)
- Tighten the rear axle nut ① to the special torque.

Rear axle nut: 50-80 N•m



REAR ABSORBER AND REAR SWING ARM, DRIVE CHAIN

CONSTRUCTION

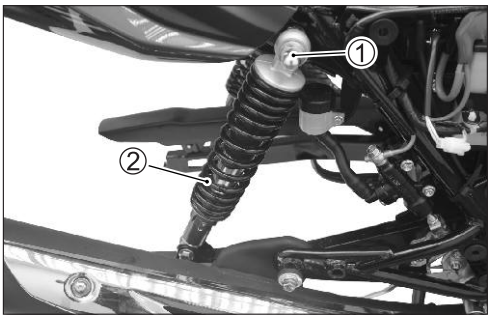


①	Rear torque link	⑩	Fastener
②	Flanged bolt	⑪	Chain buffer
③	Special washer	⑫	Chain case
④	Rear fender cover	⑬	Flat washer
⑤	Rear swing arm	⑭	Rear absorber
⑥	Flat washer	Ⓐ	Nut
⑦	Spacer	Ⓑ	Nut
⑧	Spacer	Ⓒ	Bolt
⑨	Rear swing arm shaft		

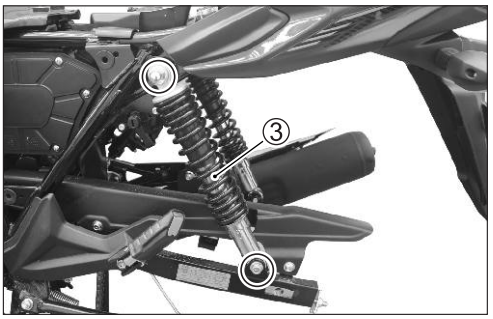
ITEM	N•m	kgf•m
Ⓐ	50-80	5.0-8.0
Ⓑ	22-35	2.2-3.5
Ⓒ	22-35	2.2-3.5

REMOVAL AND DISASSEMBLY

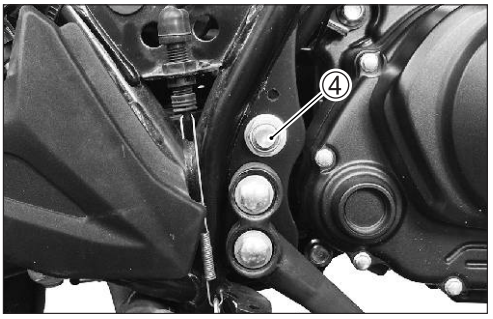
- Remove the rear wheel.(Refer to page 5-25)
- Remove the rear right absorber upper nut ①, take off the upper end of rear right absorber ② from the chassis.



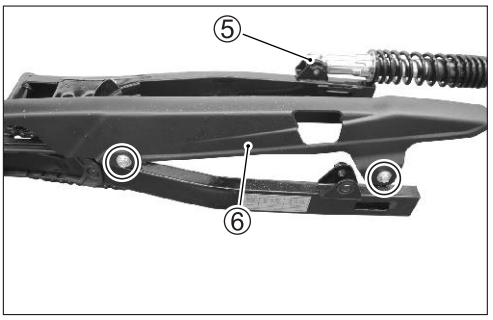
- Remove the rear left and right absorbers ③.



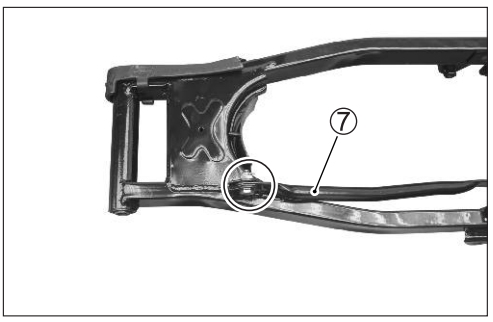
- Remove the rear swing arm shaft nut④, draw out the rear swing arm shaft, take off the rear swing arm.



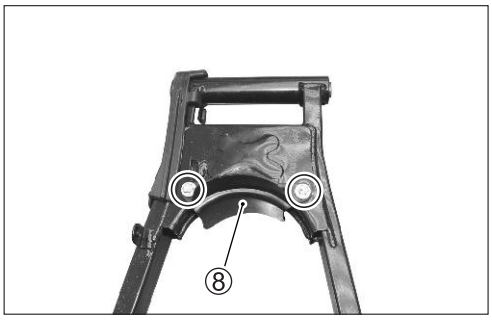
- Remove the rear right absorber ⑤ and chain case ⑥.



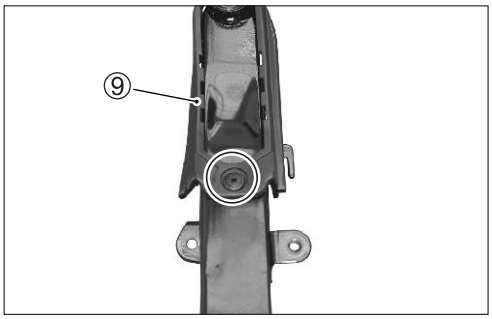
- Remove the rear torque link ⑦.



- Remove the rear fender plate cover⑧.



- Remove the chain buffer⑨.



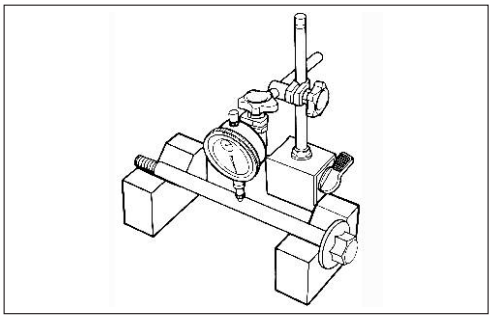
INSPECTION

ROCKER ARM SHAFT

- Using a dial gauge, check the pivot shaft for runout. Replace it if the runout exceeds the limit.

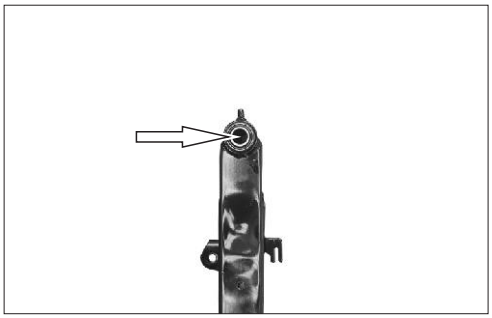
DATA Pivot shaft runout limit: 0.6 mm

TOOL 09900-20606: Dial gauge (1/100)
09900-20701: Magnetic stand
09900-21304: V-block



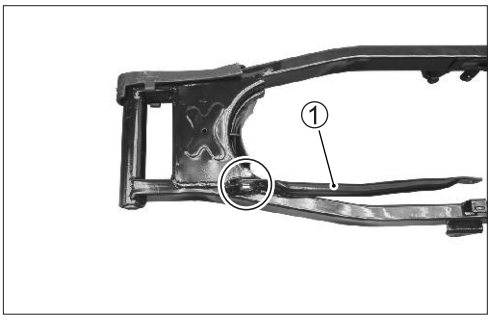
BUSHING

Inspect the swing arm shaft bushing. Replace the bushing if any unusual is found.



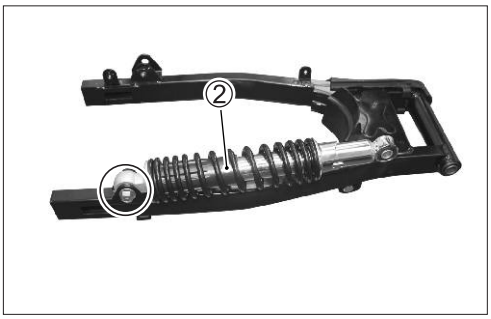
REASSEMBLY

Reassemble and remount the swing arm in the reverse order of disassembly and removal, and also carry out the following steps.



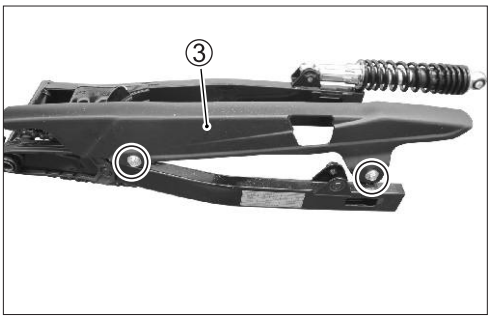
- Install the rear torque link①, tighten the rear torque link nut to the specified torque.

Rear torque link nut: 10-16 N•m

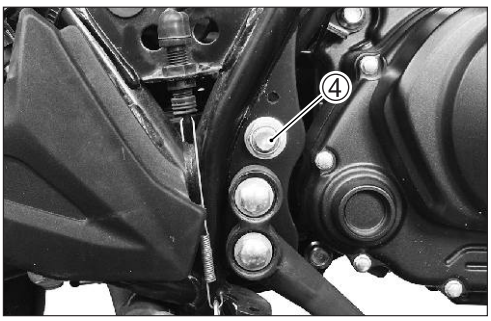


- Install the rear right absorber②to the rear swing arm, tighten the lower mounting bolt to specified torque.

Rear absorber lower mounting bolt: 22-35 N•m

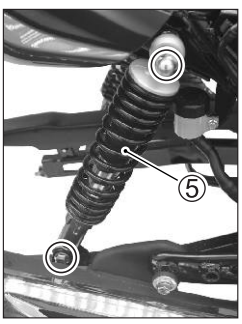


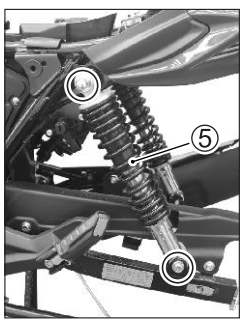
- Pass the swing arm into the drive chain, install the chain case③to the rear swing arm.



- Install the rear swing arm, tighten the rear swing arm shaft nut④to the specified torque.

Rear swing shaft nut: 50-80 N•m





- Install the rear absorbers⑤, tighten the bolts and nuts to the specified torque.

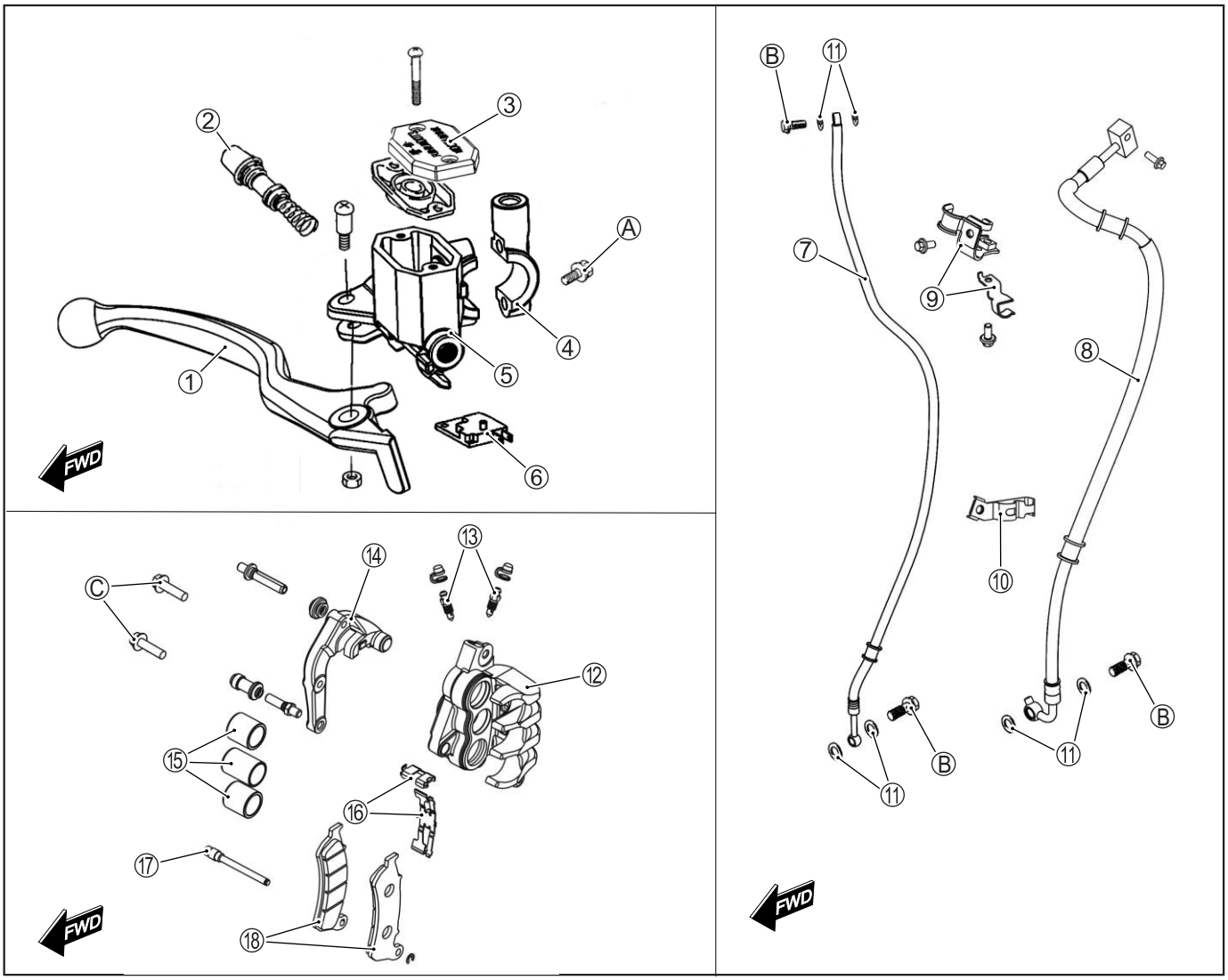
Rear absorber mounting bolt/nut: 22-35 N•m

After install the rear swing arm and rear wheel, adjust the following items.

- Chain sag. (Refer to page 2-18)
- Tire pressure. (Refer to page 2-23)

FRONT BRAKE

CONSTRUCTION



①	Front brake lever	⑫	Front brake caliper
②	Piston	⑬	Air bleeder valve
③	Reservoir tank cover	⑭	Brake caliper bracket
④	Holder	⑮	Caliper piston
⑤	Front brake master cylinder	⑯	Spring pad
⑥	Front brake light switch	⑰	Pin
⑦	Front brake hose	⑱	Brake pads
⑧	Rear brake hose	Ⓐ	Bolt
⑨	Brake hose clamp	Ⓑ	Bolt
⑩	Brake hose clamp	Ⓒ	Bolt
⑪	Brake hose washer		

ITEM	N•m	kgf•m
Ⓐ	8-12	0.8-1.2
Ⓑ	20-25	2.0-2.5
Ⓒ	18-28	1.8-2.8

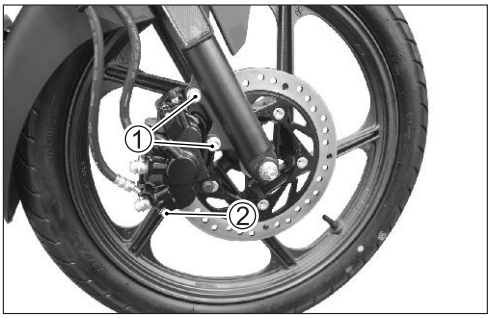
⚠ WARNING

The brake system of this motorcycle is filled with a DOT3 brake fluid. Do not use or mix different types of fluid such as silicone-based and petroleum-based fluid for refilling the system, otherwise serious damage will be caused.
Do not use any brake fluid taken from old or used or unsealed containers. Never re-use the brake fluid left over from the last servicing and stored for long periods.
When storing the brake fluid, seal the container completely and keep away from children.
When replenishing brake fluid, take care not to get dust into fluid.
When washing brake components, use new brake fluid. Never use cleaning solvent.
A contaminated brake disc or brake pad reduces braking performance. Discard contaminated pads and clean the disc with high quality brake cleaner or neutral detergent.

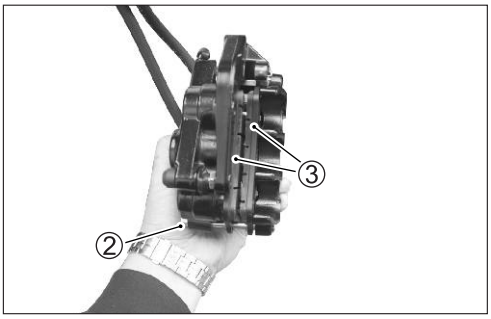
⚠ CAUTION

Immediately and completely wipe off any brake fluid contacting any part of the motorcycle. The brake fluid reacts chemically with paint, plastics and rubber materials, etc., and will damage them severely.

Brake Pad Replacement



- Loosen brake caliper mounting bolts ①.
- Loosen brake pads mounting bolts ②.



- Remove the brake pads ③ by removing the brake pad mounting bolt ②.

⚠ CAUTION

Replace brake pads as a set otherwise braking performance will be adversely affected.

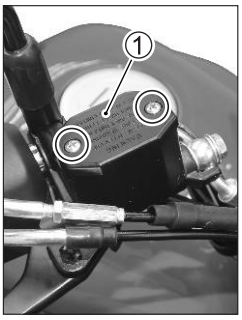
- Reinstall the brake pads and brake caliper in the reverse order of removal.
- Tighten the brake caliper mounting bolt to the specified torque.

Brake caliper mounting bolts: 18~28 N•m

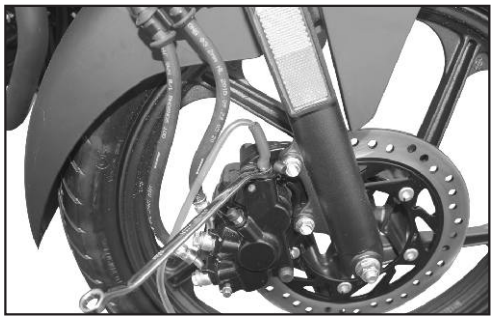
⚠ CAUTION

While installing the brake caliper, push the pistons all the way into the brake caliper.

BRAKE FLUID REPLACEMENT

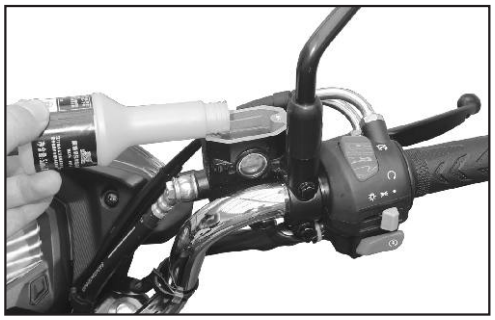


- Support the motorcycle with the main stand, keep the steering straight.
- Remove the master cylinder cover①and diaphragm.
- Draw the brake fluid as mush as possible.
- Add new brake fluid to the master cylinder.



- Attach the special tool hose to the air bleeder valve.
- Loosen the brake caliper bleed valve, start the special tool, draw out the brake fluid in the brake system, and add new brake fluid in the master cylinder as necessary.

TOOL **11F14-035: Pneumatic fluid drawer**



- Tighten the brake caliper bleed valve, take off the special tool.
- Add the new brake fluid in the master cylinder to the upper level.
- Check the brake performance, if the brake force is insufficient, apply air bleeding to the brake system (Refer to page 2-14).

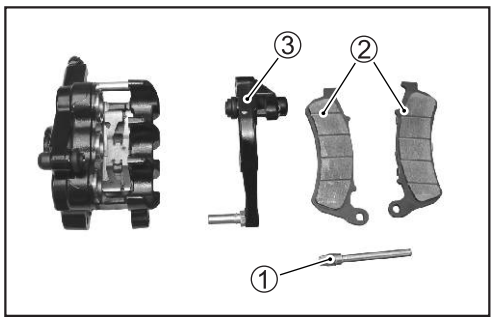
CALIPER REMOVAL AND DISASSEMBLY

- Place a clean rag underneath the union bolt on the brake caliper to catch any split brake fluid.
- Remove the brake hose from the caliper by removing the union bolt ①and catch the brake fluid in a suitable receptacle.
- Remove the brake caliper mounting bolt ②, take off the brake caliper.

⚠ CAUTION

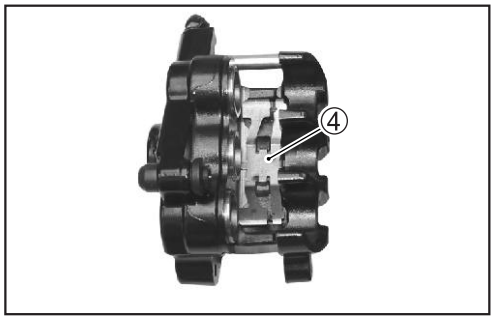
Never re-use the brake fluid left over from the last servicing and stored for long periods.

BRAKE CALIPER DISASSEMBLY



- Remove the sliding pin ①, brake pads ②and bracket③.

- Remove the pad spring ④.



- Place a clean rag over the pistons to prevent it from popping out and then force out the pistons using compress air.

⚠ CAUTION

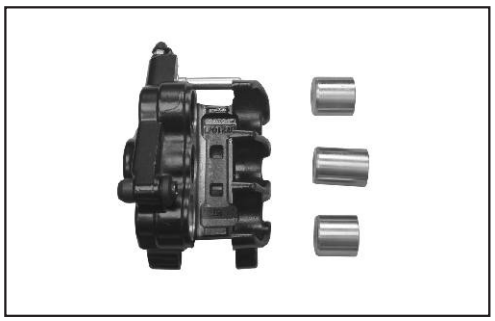
Do not use high pressure air to prevent piston damage.

- Remove the piston, piston boot and piston seal.

INSPECTION

CALIPER CYLINDER

- Inspect the brake caliper cylinder wall for nicks, scratches and other damage. If any damage is found, replace the caliper with a new one.

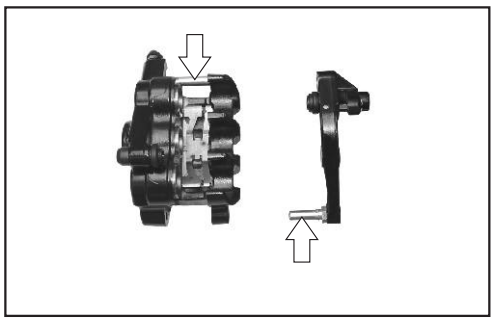


PISTON

- Inspect the brake caliper piston surface for any scratches and other damage. If any damage is found, replace the piston with a new one.

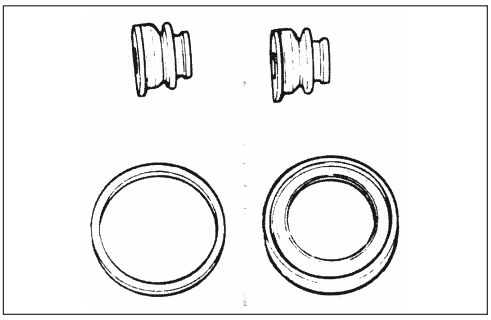
BRAKE CALIPER SLIDING PIN

- Inspect the brake caliper sliding pin for wear and other damage. If any damage is found, replace it with a new one.



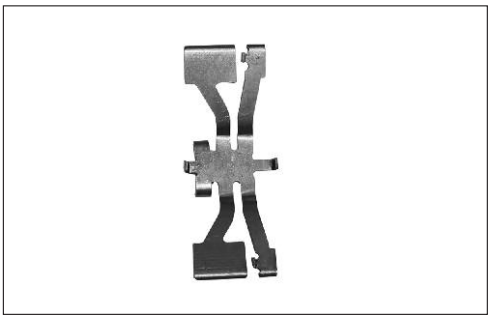
RUBBER PARTS

- Inspect the rubber parts for cracks and damage. If any defects are found, replace the affected parts.



PAD SPRING

- Inspect the pad spring for damage and excessive bend. If any defects are found, replace it with a new one.

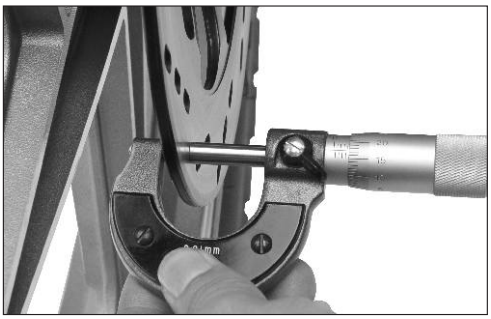


BRAKE DISC

- Visually inspect brake disc for cracks or damage, and measure the brake disc thickness with a micrometer.

TOOL **09900-20205: Micrometer (0-25 mm)**

DATA **Brake disc thickness limit: 3.5 mm**



- Measure runout of the brake disc with a dial gauge.

TOOL **09900-20606: Dial gauge (1/100 mm)**
09900-20701: Magnetic stand

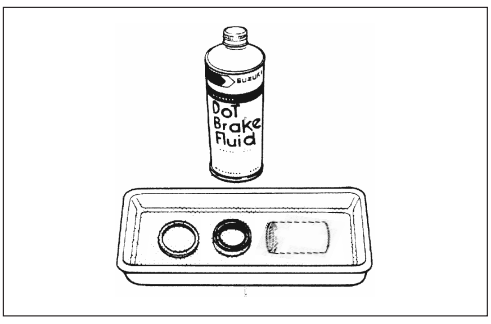
DATA **Brake disc runout limit: 0.3 mm**

REASSEMBLY

Reassemble and remount the brake caliper in the reverse orders of disassembly and removal, and also carry out the following steps.

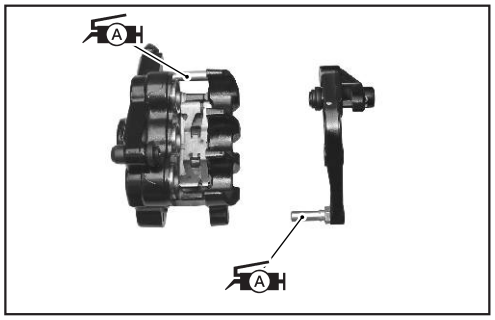
⚠ WARNING

Wash the master cylinder components with fresh brake fluid before reassembly.
Never use cleaning solvent or gasoline to wash them.
Apply brake fluid to the cylinder bore and all internal parts before inserting into the bore.



- Apply the super grease to sliding pin.

99000-25010: **SUPER GREASE "A"**



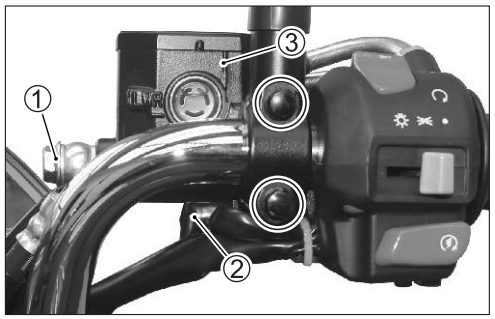
- Tighten the brake caliper mounting bolt ①and brake hose union bolt ②to the specific torque.

Brake hose union bolt: 20~25 N•m
Brake caliper mounting bolt:18~28 N•m

- Refill the brake fluid after reassembling brake caliper. (Refer to page 5-38)

⚠ WARNING

Brake fluid, if it leaks, will interfere with safe running and immediately discolor painted surfaces.
Check the brake hoses for cracks and hose joint for leak-age before riding.



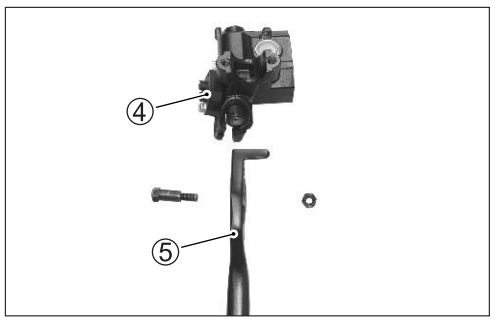
MASTER CYLINDER REMOVAL AND DISASSEMBLY

- Drain the brake fluid from the master cylinder.
- Place a clean rag underneath the union bolt on the master cylinder to catch any split brake fluid, unscrew the union ① bolt and disconnect the brake hose from the master cylinder joint.
- Disconnect the front brake switch coupler ②.
- Remove the two clamp bolts and take off the master cylinder.

⚠ CAUTION

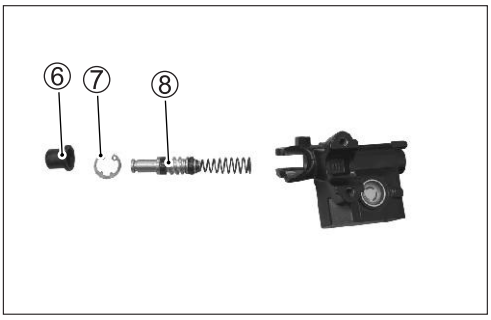
Immediately and completely wipe off any brake fluid contacting any part of the motorcycle. The fluid reacts chemically with paint, plastics, rubber materials, etc. and will damage them severely.

- Remove the brake lever ④and front brake switch ⑤.



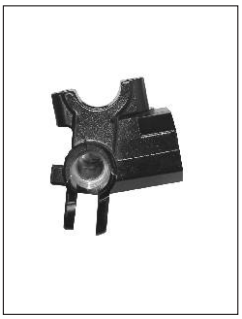
- Remove the dust-proof seal boot.
- Remove the circlip, piston and spring using special tool.

TOOL **09900-06108: Snap ring pliers**



INSPECTION

- Inspect the master cylinder bore for any scratches or other damage.
- Inspect the piston surface for scratches or other damage.
- Inspect the primary cup, secondary cup and dust seal boot for wear or damage.

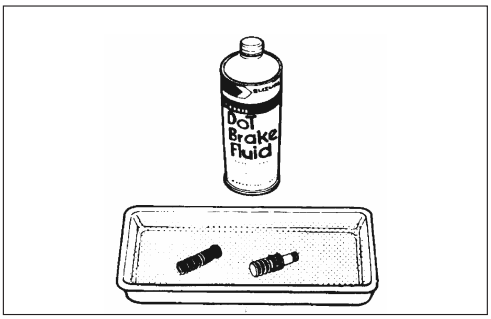


REASSEMBLY

Reassemble and remount the master cylinder in the reverse orders of disassembly and removal, and also carry out the following steps:

- Wash the master cylinder components with fresh brake fluid before reassembly.

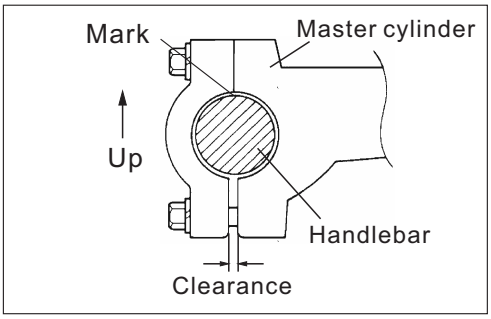
BF **Brake fluid specification: DOT3**



⚠ CAUTION

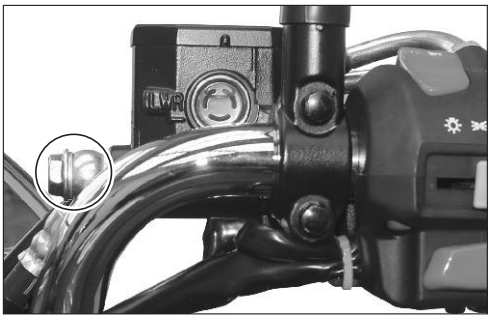
When washing the components, use the specified brake fluid. Never use different types of fluid or cleaning solvents such as gasoline, kerosine, etc.
Do not wipe the brake fluid off after washing the components.
Apply brake fluid to the cylinder bore and all internal parts before inserting into the bore.

- When remounting the master cylinder on the handlebar, tighten the clamp bolt for upside as shown first. Align the mark on the handlebar with the mating slit.



- Connect the brake hose, tighten the union bolt to the specified torque.

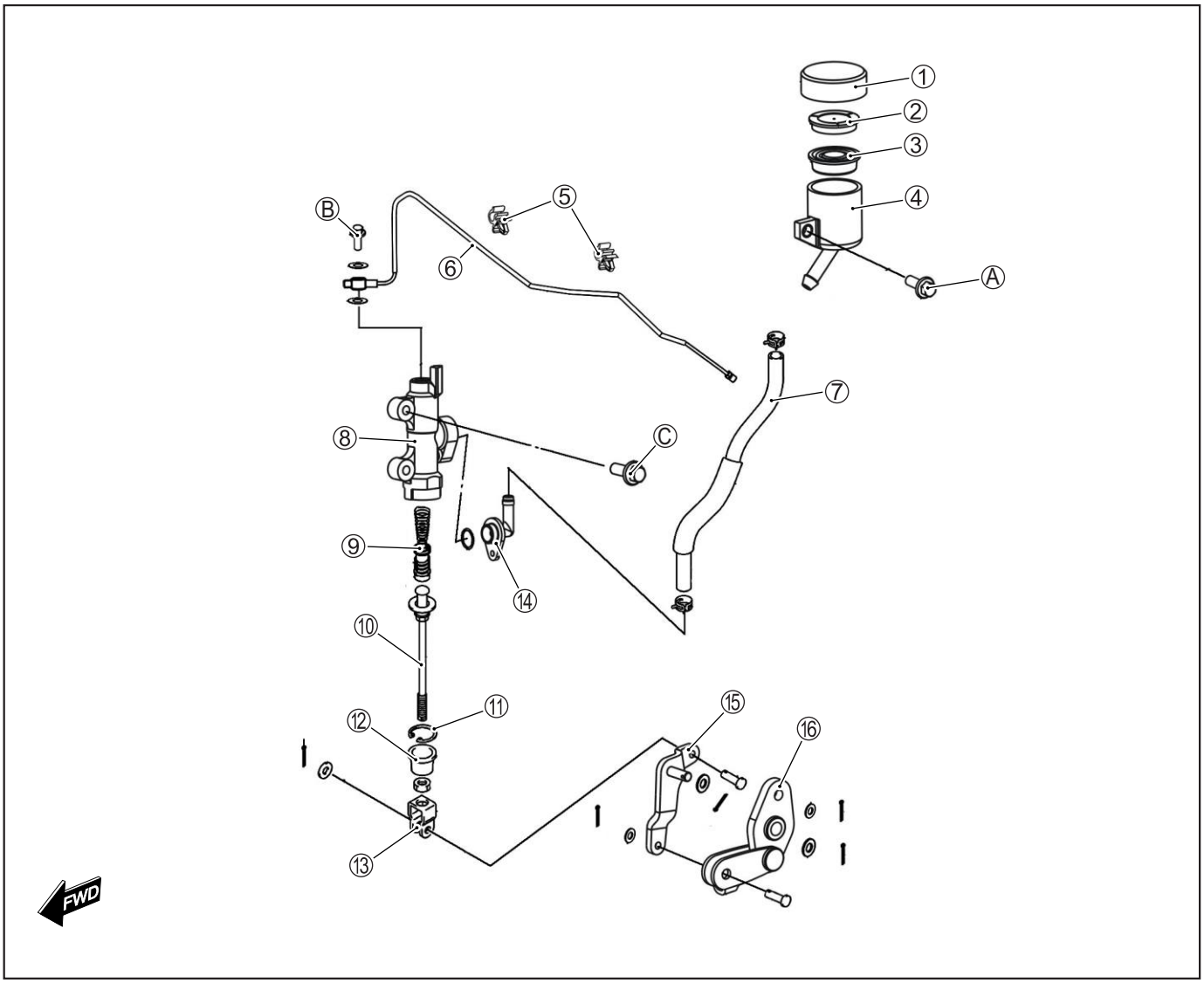
Brake hose union bolt: 20~25 N•m



- Refill the brake fluid after installing the master cylinder. (Refer to page 6-42)

COMBINED BRAKE SYSTEM

CONSTRUCTION

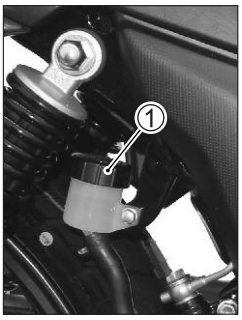


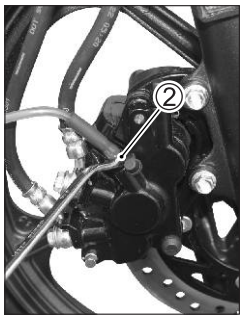
①	Rear master cylinder cap	⑪	Circlip
②	Reservoir tank cap	⑫	Dust-proof cap
③	Diaphragm	⑬	Rear master cylinder yoke
④	Reservoir tank	⑭	Rear master cylinder connector
⑤	Bundy tube clamp	⑮	CBS transmission bar
⑥	Bundy tube	⑯	CBS conrod
⑦	Rear brake fluid hose	Ⓐ	Flanged bolt
⑧	Rear master cylinder	Ⓑ	Brake fluid bolt
⑨	Spring	Ⓒ	Master cylinder mounting bolt
⑩	Piston		

ITEM	N•m	kgf•m
Ⓐ	8-12	0.8-1.2
Ⓑ	20-25	2.0-2.5
Ⓒ	18-28	1.8-2.8

REAR BRAKE MASTER CYLINDER AND FLUID RESERVOIR TANK REMOVAL AND DISASSEMBLY

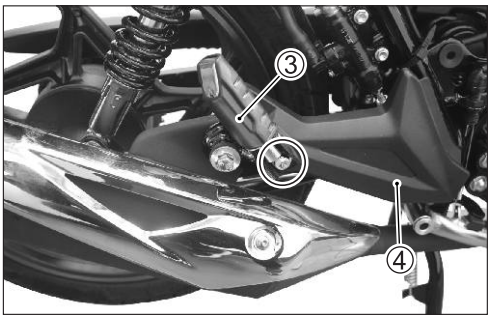
REMOVAL



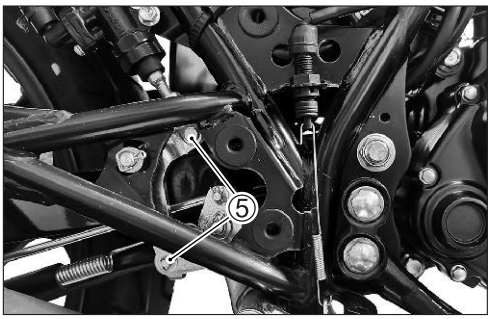


- Remove the rear brake fluid reservoir tank cap ①, seal plate and diaphragm.
- Attach the special tool hose to the air bleeder valve ②, loosen the brake caliper bleed valve nut, start the special tool, draw out the brake fluid in the brake system.

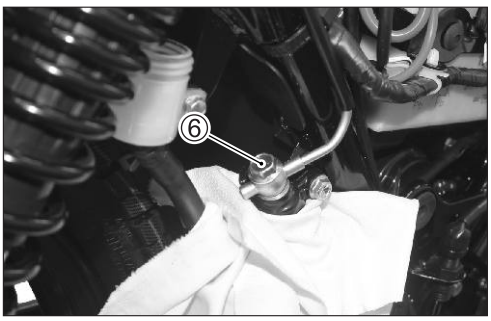
TOOL **11F14-035: Pneumatic fluid drawer**



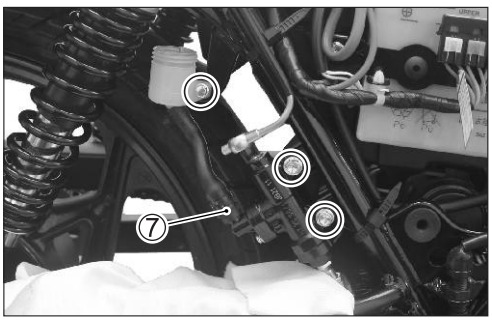
- Remove the rear footrest ③, take off the rear footrest protection plate ④.



- Remove the rear brake linked rod cotter pin ⑤, remove the rear brake linked rod.



- Place a clean cloth around the brake hose, remove the rear brake fluid mounting bolt ⑥.



- Remove the rear brake fluid reservoir tank and rear brake master cylinder⑦.

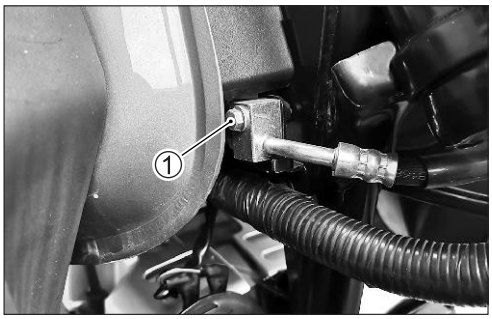
- Remove the Buddy hose ⑧.

REASSEMBLY

Reassemble and remount the rear brake system in the reverse orders of disassembly and removal, and also carry out the following steps:

- When installing the Bundy hose, tighten the bundy hose nut ①with hand first, then tighten the Bundy hose nut to the specified torque.

Bundy hose nut: 12-17 N•m

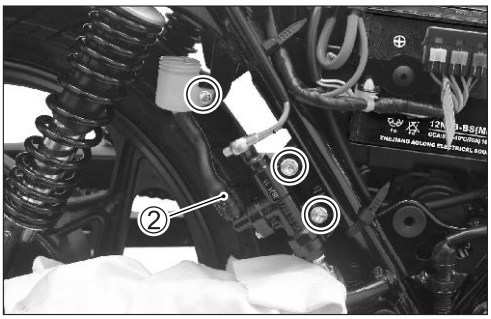


⚠ CAUTION

Refer to page 9-21 for Bundy hose routing.

- Install the rear brake master cylinder ②and reservoir tank tighten the mounting bolt to the specified torque.

Rear brake master cylinder mounting bolt: 8-12 N•m
Rear brake reservoir tank mounting bolt: 20-26 N•m

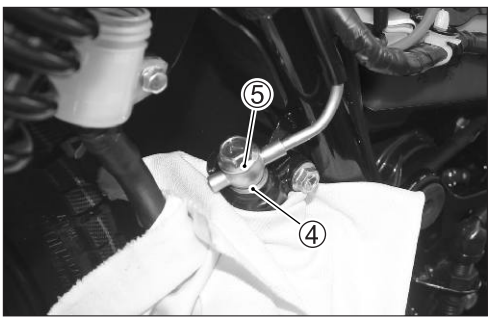


- Install the rear footrest ③, tighten the bolt/nut to the specified torque.

Rear footrest mounting bolt/nut: 40-60 N•m

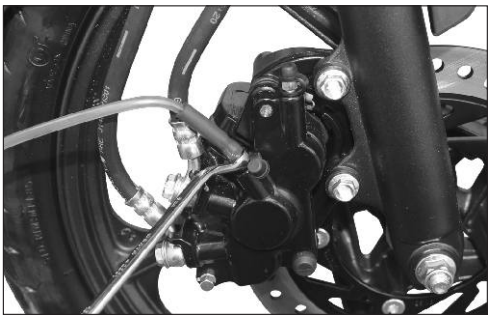
- When install the rear brake hose union bolt, pass the bolt from right to left, the position mark ④ should be fitted at left side of position point ⑤as shown.
- When install the washer, the coarse side faces to brake hose side.
- Tighten the union bolt to the specified torque.

Brake hose union bolt: 20-25 N•m



REFILL THE BRAKE FLUID

- Fill the rear brake master cylinder with brake fluid.



- Attach the special tool hose to the air bleeder valve ④, start the special tool draw out the brake fluid slowly, add the brake fluid into the reservoir tank as the same time until the brake fluid flows out of hose.

11F14-035: Pneumatic fluid drawer

- After filling the brake fluid to the rear brake master cylinder, apply the air bleeding to the brake fluid circuit.(Refer to page 2-14)

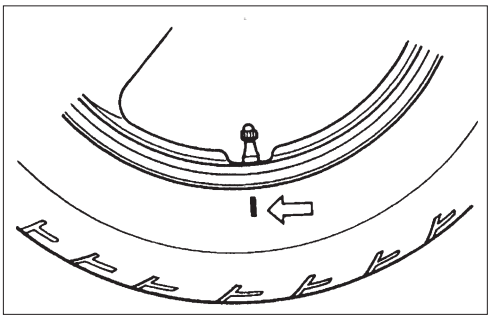
TIRE AND WHEEL

TIRE REMOVAL

- The most critical factor of tubeless tire is the seal between the wheel rim and the tire bead. For this reason, it is recommended to use a tire changer that can satisfy this sealing requirement and can make the operation efficient as well as functional.

⚠ CAUTION

For operating procedures, refer to the instructions supplied by the tire changer manufacturer.

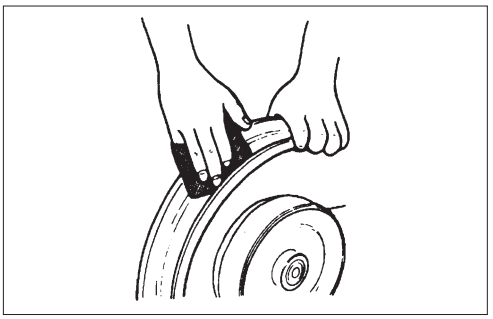


NOTE:
When removing the tire in case of repair or inspection, mark the tire with a chalk to indicate the tire position relative to the valve position.
Even though the tire is refitted to the original position after repairing puncture, the tire may have to be balanced again since such repair can cause imbalance.

INSPECTION

WHEEL

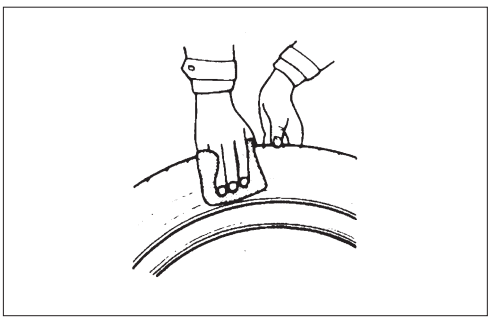
Wipe the wheel clean and check for the following point.



- Distortion and crack.
- Any flaws and scratches at the bead seating area.
- Wheel runout. (Refer to page 6-4)

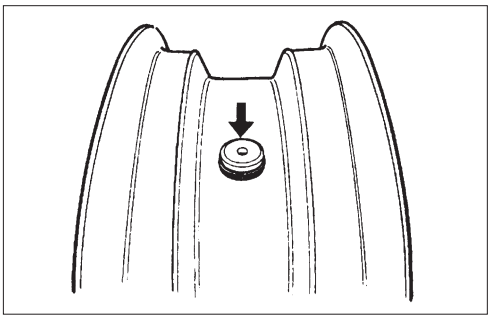
TIRE

Tire must be checked for the following points:



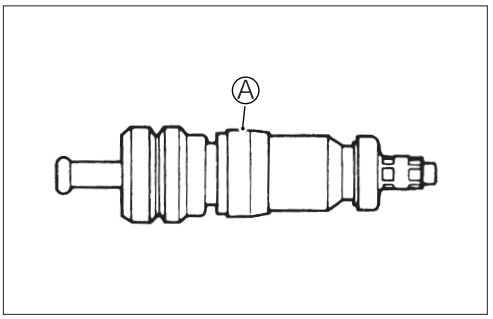
- Nick and rupture on side wall
- Tire tread depth
- Tread separation
- Abnormal, uneven wear on tread
- Surface damage on bead
- Localized tread wear due to skidding (flat spot)
- Abnormal condition of inner liner.

VALVE INSTALLATION



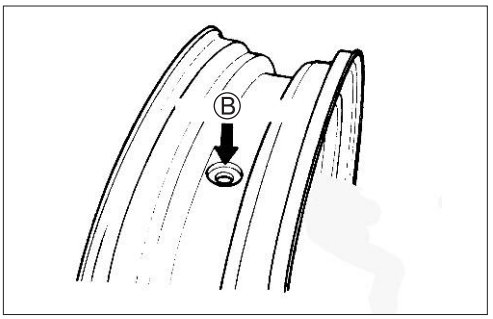
- Inspect the valve after the tire is removed from the rim.
- Replace the valve with a new one if the seal Ⓐ rubber is peeling or has damage.

NOTE:
If the external appearance of the valve shows no abnormal condition, removing of the valve is not necessary.
If the seal has abnormal deformation, replace the valve with a new one.



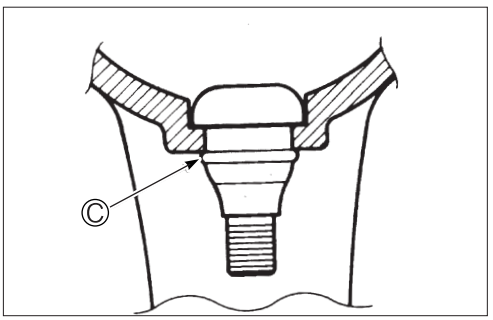
- Cleaned off any dust or rust around the valve hole Ⓑ.
- Install the new valve in the rim.

NOTE:
To properly install the valve into the valve hole, apply a special tire lubricant or neutral soapy liquid to the valve.

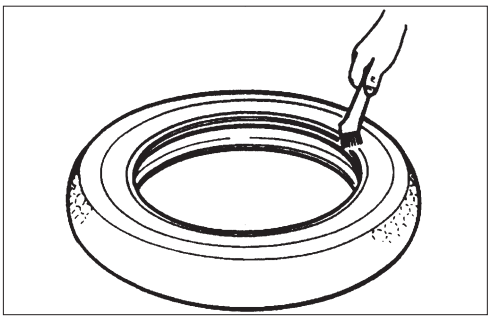


⚠ CAUTION

Be careful not to damage the lip Ⓒ of valve.



TIRE INSTALLATION

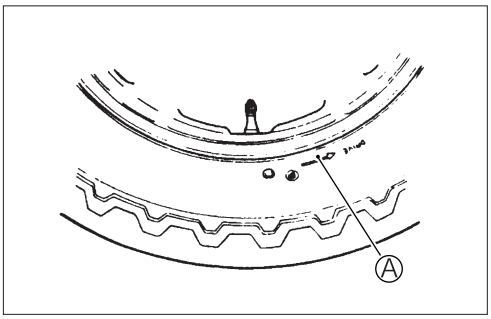


- Apply tire lubricant or neutral soapy to the tire bead.

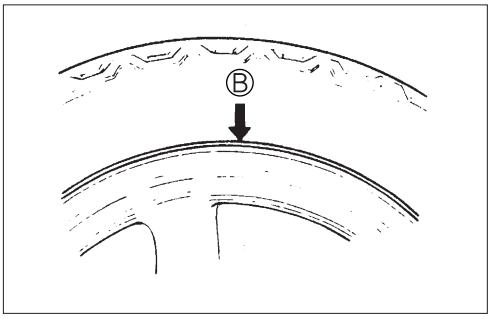
⚠ CAUTION

Never use oil, grease or gasoline.

When installing the tire onto the wheel, observe the following points:



- When installing the tire, the arrow on the side wall should point to the direction of wheel rotation.
- Align the chalk mark put on the tire at the time of removal with the valve position.
- For installation procedure of tire onto the wheel, follow the instructions given by the tire changer manufacturer.
- Bounce the tire several times while rotating. This makes the tire bead expand outward to contact the wheel, thereby facilitating air inflation.
- Inflate the tire.



⚠ WARNING

Do not inflate the tire more than 400 kPa (4.0 kgf/cm²). If inflated beyond this limit, the tire can burst and possibly cause injury. Do not stand directly over the tire while inflating. In the case of preset pressure air inflate, pay special care for the set pressure adjustment.

- In this condition, check the rim line Ⓑ cast on the tire side walls. The line must be equidistant from the wheel rim all around. If the distance between the rim line and wheel rim varies, this indicates that the bead is not properly seated. If this is the case, deflate the tire completely and unseat the bead for both sides. Coat the bead with lubricant and fit the tire again.
- When the bead has been fitted properly, adjust the pressure to specification.
- As necessary, adjust the tire balance.

⚠ CAUTION

Do not run with a repaired tire at a high speed.

DATA Cold inflation tire pressure

	Solo riding		Dual riding	
	kPa	kg/cm²	kPa	Kg/cm²
Front wheel	250	2.50	250	2.50
Rear wheel	250	2.50	250	2.50

ELECTRICAL

CONTENTS

ELECTRIC PARTS LOCATION 7-1
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FUEL LEVEL GAUGE 7-13
COMBINATION METER 7-14
LIGHTS 7-18
SWITCHES 7-21
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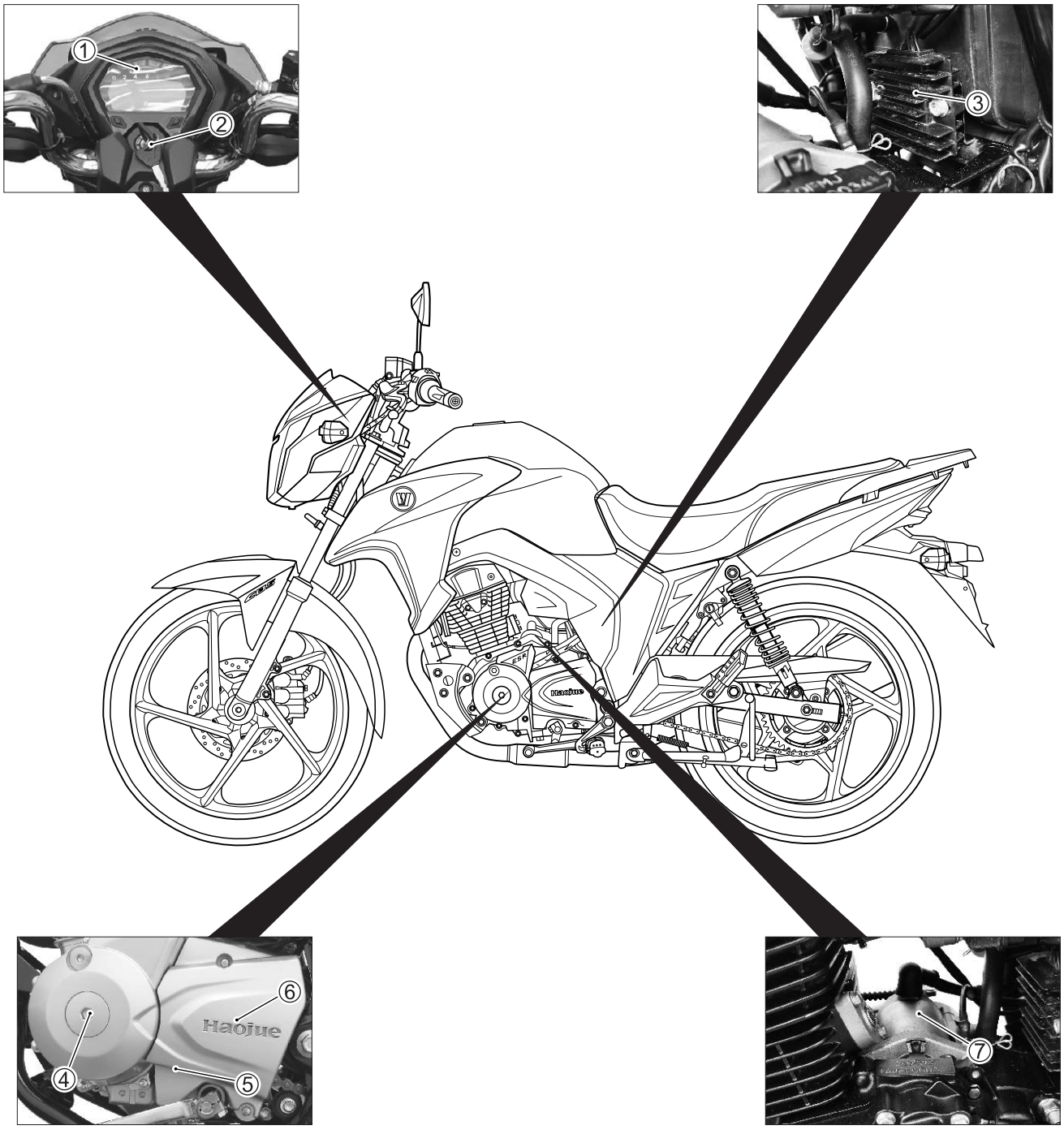
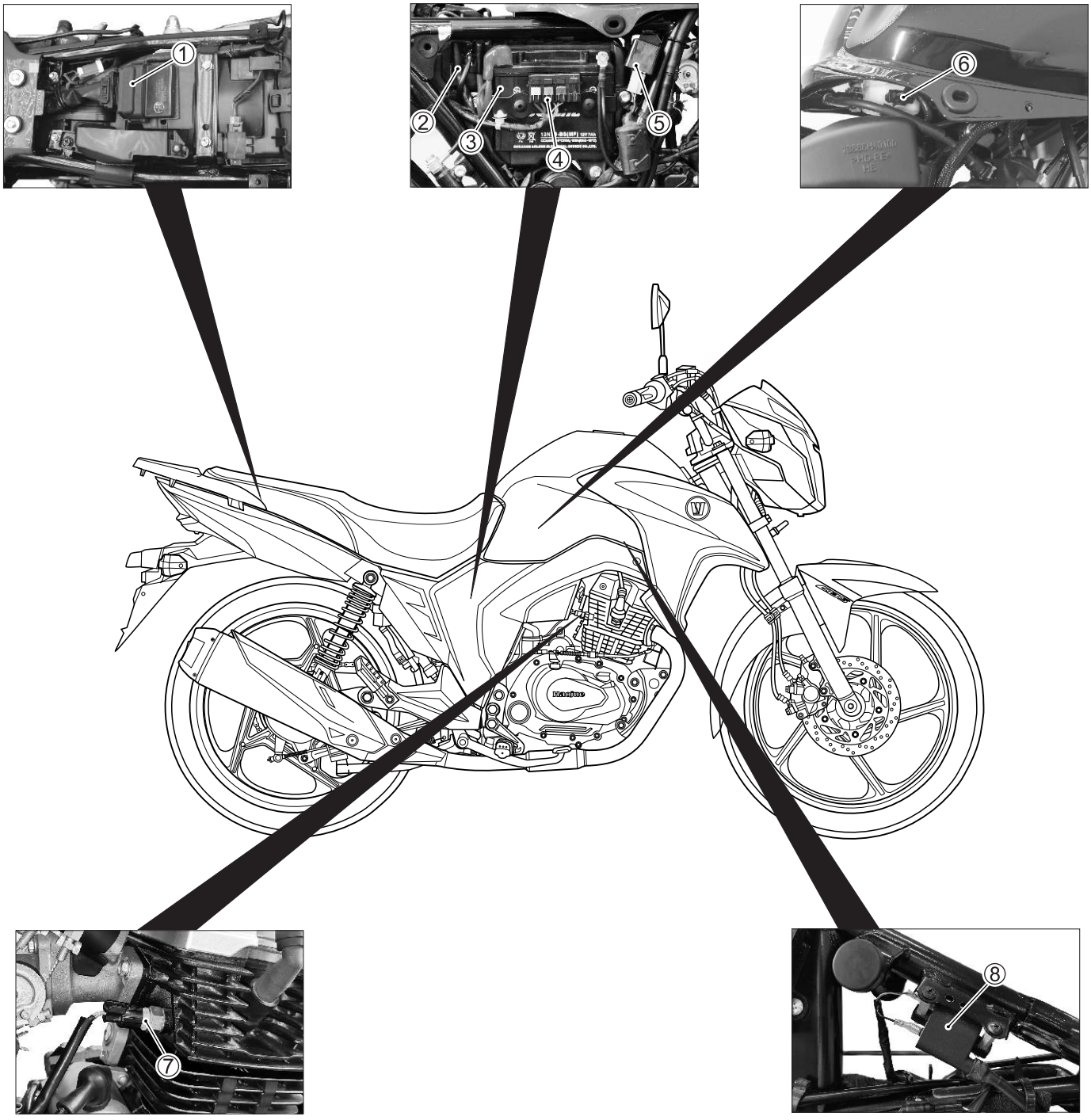
ELECTRIC PARTS LOCATION

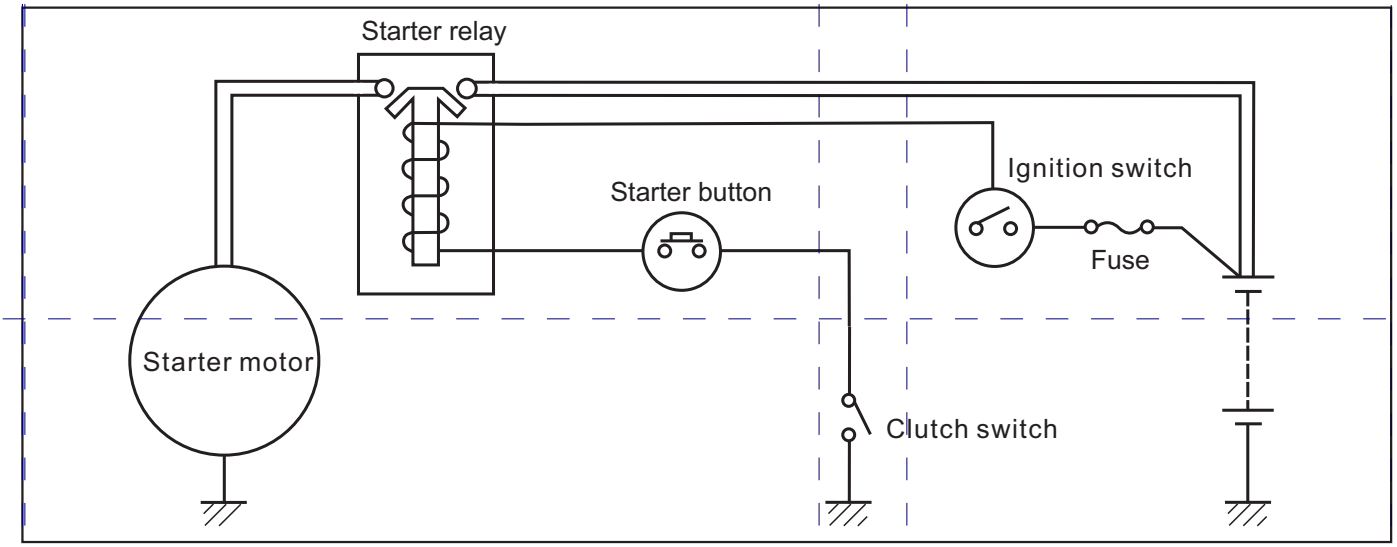


①	Combination meter	④	Generator and CKPS	⑦	Starter motor
②	Ignition switch	⑤	Gear position switch		
③	Rectifier	⑥	Speed sensor		



①	ECM	④	Fuse box	⑦	Engine temperature sensor
②	Starter replay	⑤	Turn signal relay	⑧	Ignition coil
③	Battery	⑥	Fuel level sensor		

STARTER SYSTEM



TROUBLESHOOTING

Inspection: One of the front and rear brake light switches is on.
Battery is full-charged.
Fuse is not blown before the diagnosis.

Starter motor will not run.

Check whether to hear the click noise from the starter relay when the starter button is pushed.

Clicks →

Check whether run the starter motor when connect the starter motor ⊕ terminal to the battery terminal directly. (Do not use thin wire, because a large amount of current flows.)

- **Run** →
  - Faulty starter relay
  - Loose or disconnected starter motor wire
- **Not run** →
  - Faulty starter motor

No click

Measure the starter relay voltage at the starter relay connectors (between O ⊕ and Y/G ⊖) when the starter button is pushed.

No voltage →
- Faulty ignition switch
- Faulty front/rear brake light switch
- Faulty starter button
- Poor contact of connector
- Open circuit in wiring harness

Voltage measured

Check the starter relay (Refer to page 7-7)

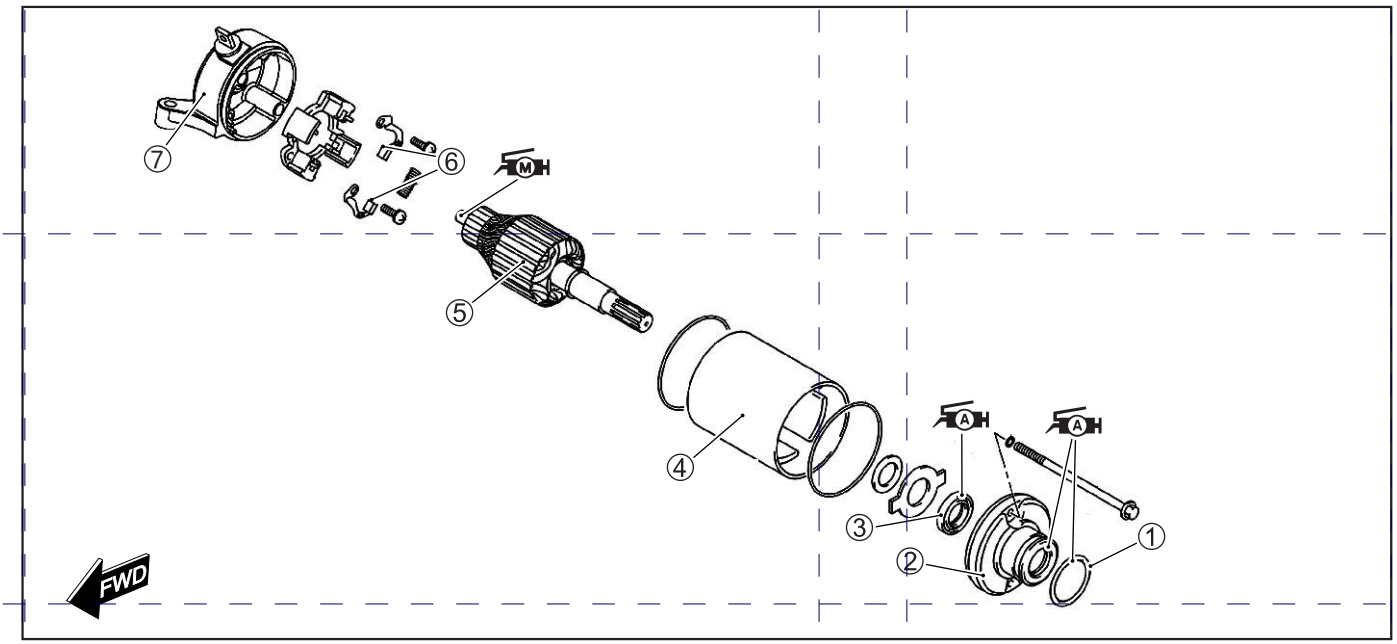
Incorrect →
- Faulty starter relay

Starter motor runs, but does not start engine.

- Poor contact of connector or open circuit in wiring harness
- Fault ECM
- Fault ignition coil
- Fault CKP sensor
- Fault starter clutch

STARTER MOTOR REMOVAL AND DISASSEBMLY

- Remove the starter motor. (Refer to page 3-7)
- Disassemble the starter motor as shown.

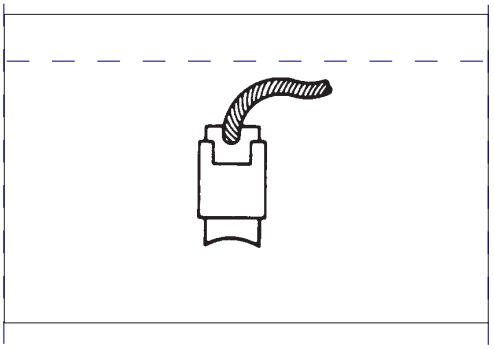


①	O-ring	④	Motor shell	⑦	Rear cover
②	Front cover	⑤	Armature shaft		
③	Oil seal	⑥	Carbon brush		

STARTER MOTOR INSPECTION

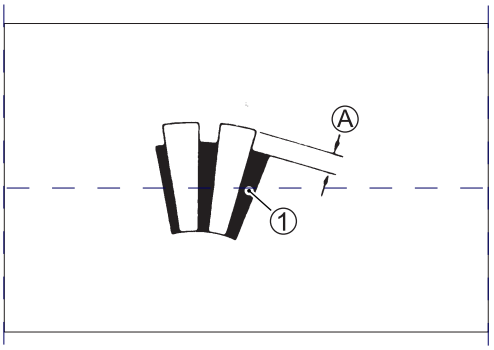
CARBON BRUSHES

Inspect carbon brush for cracks or abnormal wear. If any defects are found, replace with a new carbon brush assembly.



COMMUTATOR

Inspect the commutator surface for abnormal wear or undercut Ⓐ.
If the commutator surface shows abnormal wear, replace with a commutator.
If the commutator surface is worn unevenly, polish with No. 400 sand paper and wiped with a clean dry cloth.
If there is no undercut, scrape out insulator ①with a saw blade.

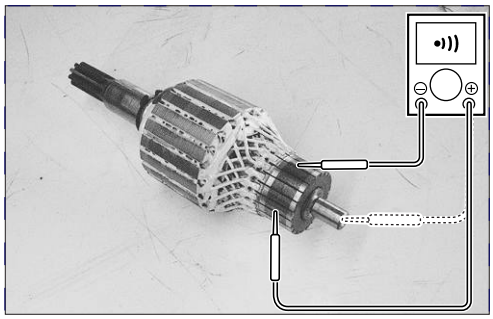


ARMATURE COIL

- Inspect for continuity between each segment and between each segment and the armature shaft using the multi-circuit tester.
- If there is no continuity between the segments or there is continuity between the segments and shaft, replace the armature with a new one.

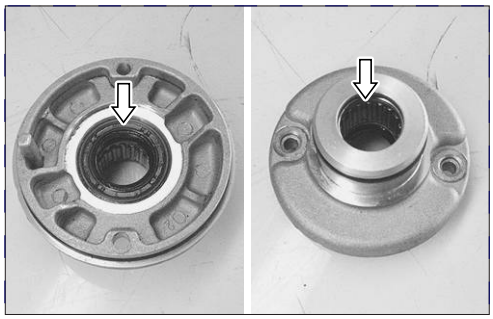
09900-25008: Multi-circuit tester set

Tester knob indication: Continuity test



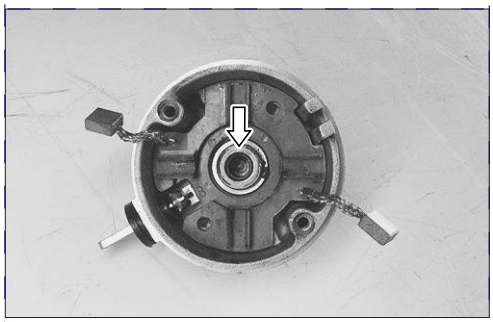
OIL SEAL AND BEARING

Inspect the oil seal lip for damage or leakage.
Inspect the bearing for abnormal noise and smooth movement.
If any defects are found, replace the failed part.



BUSH

Inspect the bush for wear and damage.
If any defects are found, replace the failed part.



STARTER MOTOR REASSEMBLY

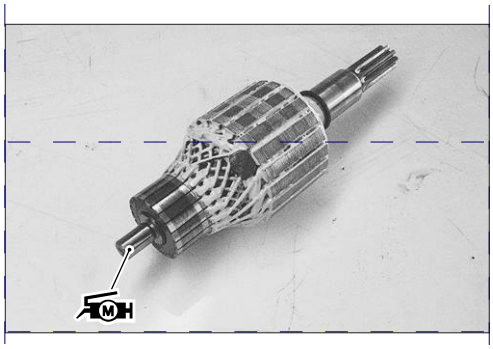
Reassemble the starter motor in the reverse order of disassembly.

- Apply grease to the lip of the oil seal and bearing.

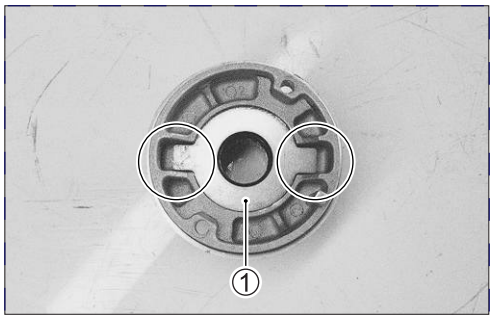
99000-25010: SUPER GREASE "A"

- Apply a small quantity of Moly paste to the armature shaft.

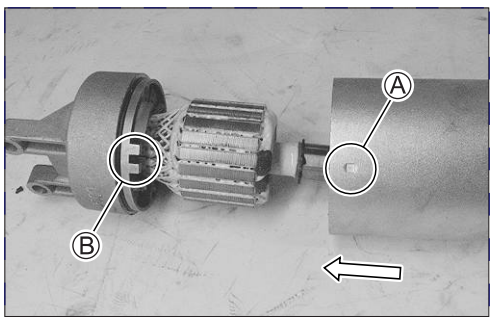
99000-25140: Moly paste



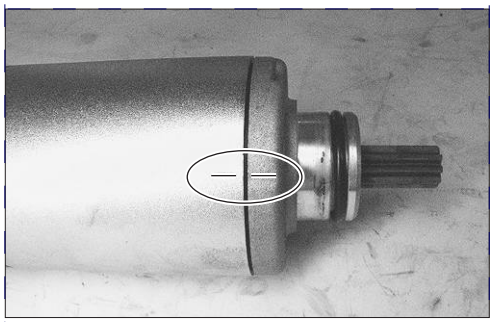
- Install the washer ① to the front cover as shown.



- When fixing the rear cover to the starter motor shell, align the protrusion Ⓐ of the starter motor shell with the notch Ⓑ of the rear cover.

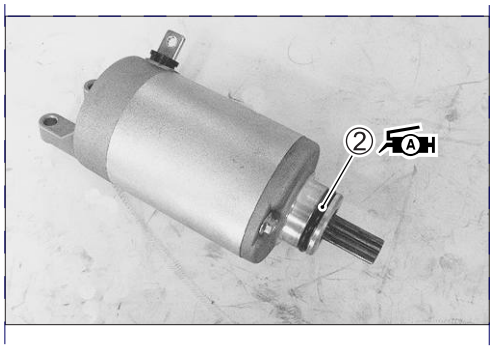


- When installing the front cover, align the match mark on the starting motor shell with the match mark on the front cover.

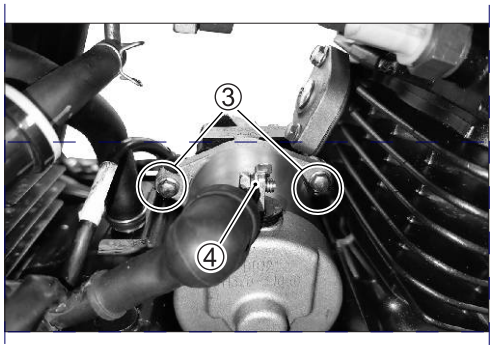


- Apply a few quantity of grease to O-ring ②.

99000-25010: SUPER GREASE "A"

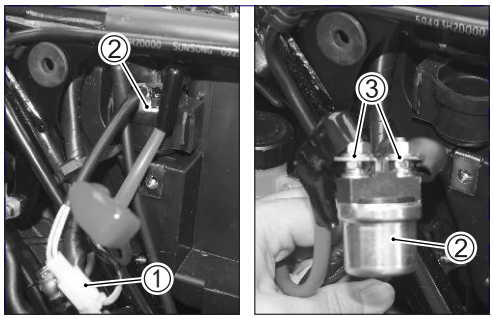


- Tighten the starter motor mounting bolts ③.
- Tighten the starter motor wire mounting bolt ④.

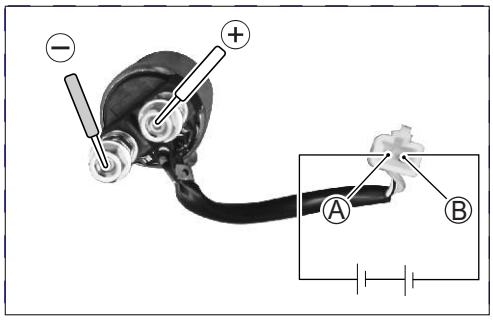


STARTER RELAY INSPECTION

- Remove the battery.(Refer to page 2-3)
- Disconnect the starter relay coupler ①, take off the starter relay ② from the jacket, disconnect the starter relay wire, remove the starter relay.



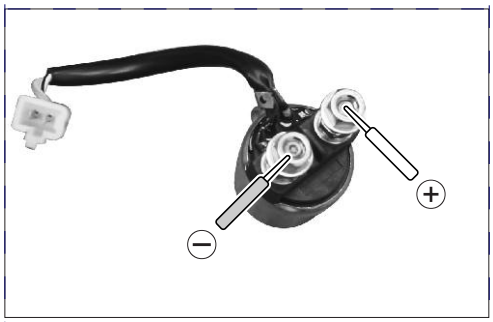
- Apply 12V to the terminal Ⓐ and terminal Ⓑ, check for continuity between the terminals with multi-circuit tester. If a “click” is not heard, the starter relay is not working.



- Measure the relay coil resistance between the terminals using the multi-circuit tester. If the resistance is not within the specified value, replace the starter relay with a new one.

TOOL **09900-25008: Multi-circuit tester set**

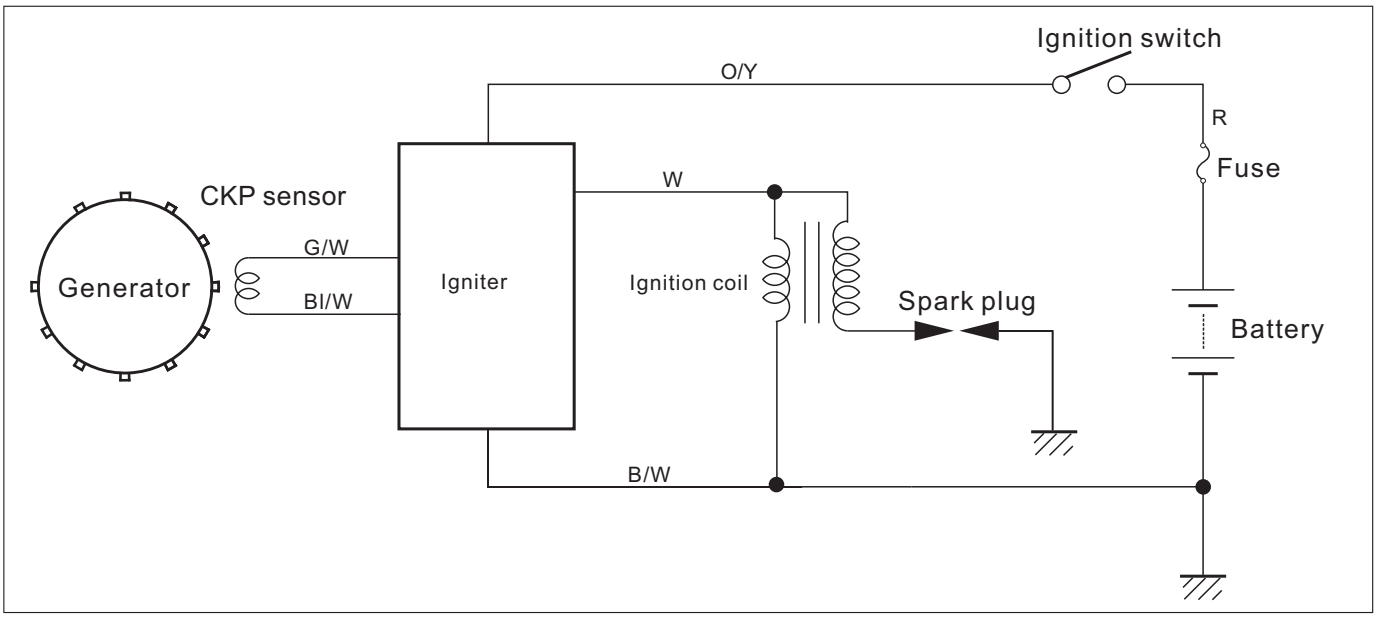
DATA **Starter relay resistence: 4-6Ω**



IGNITION SYSTEM

DESCRIPTION

In the capacitor discharged ignition system, the electrical energy generated by the generator charges the capacitor. The energy is released in a signal surge at the specified ignition timing, and current flows through the primary side of the ignition coil. A high voltage current is induced in the secondary windings of the ignition coil resulting in strong spark between the spark plug gap.



TROUBLESHOOTING

No spark or poor spark

* Make sure the ignition switch is at "ON" position, the fuse is not blown and grasp the clutch lever before check.

Inspect the battery voltage
- **Less than 12.4 V** → ● Recharge the battery or replace the battery

12.4 V

Check the ECM couplers for poor contact
- **Loose** → ● Poor contact of couplers

Correct

Inspect the ignition system components and couplers.
- **Incorrect** → ● Faulty ignition switch
  ● Broken wire harness or poor contact of related connectors

Correct

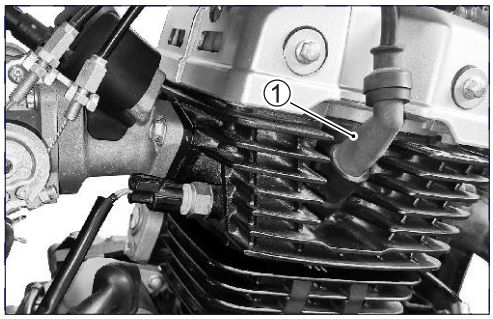
Inspect the resistance of CKP sensor (Bl/W and G/W)
- **Incorrect** → ● Faulty CKP sensor
  ● Metal particle stuck on the CKP sensor or rotor tip

Correct

- ● Faulty ECM
- ● Faulty ignition coil or its connector

IGNITION COIL

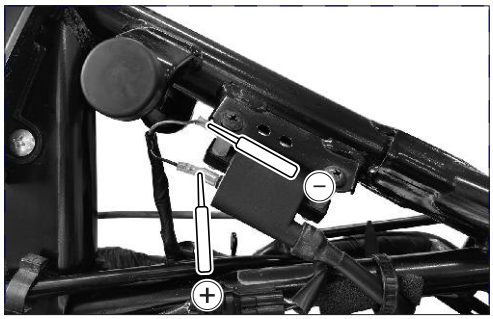
- Remove the spark plug cap ①.

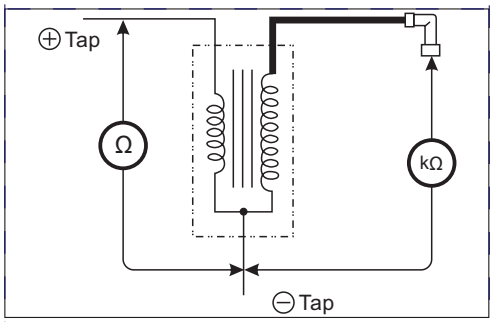


- Remove the fuel tank.(Refer to page 5-1)
- Measure the ignition coil resistence with the multi-circuit tester.

09900-25008: Multi-circuit tester set

Primary: 3.1~4.3 Ω (tap-tap)
Secondary: 28.5~38.5 KΩ (tap - Plug cap)



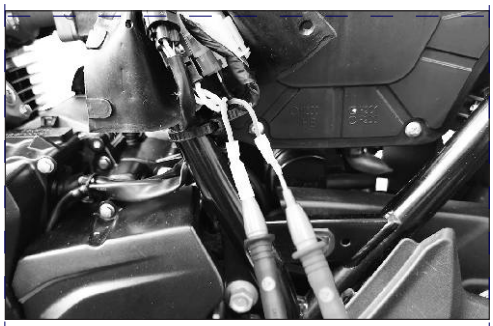


CRANKSHAFT POSITION SENSOR RESISTANCE

- Disconnect the CKPS coupler.
- Measure the resistance between CKPS terminals.

09900-25008: Multi-circuit tester set

CKP sensor resistance: 115-155 Ω (G/W-Bl/W)



SPARK PLUG

(Refer to page 2-7)

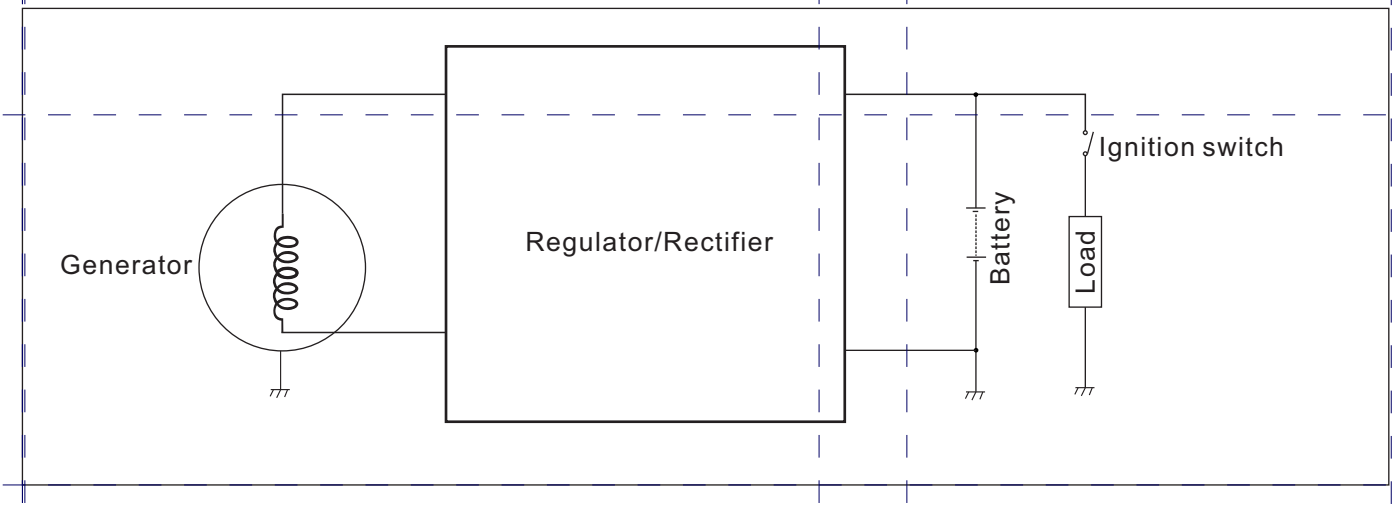
ECM

If the ignition system is defective and no problem is found in the inspection above, replace the ECM.

CHARGING SYSTEM

DESCRIPTION

The circuit of the charging system is indicated in the figure, which is composed of an AC generator, regulator/rectifier unit and battery.
The AC current generated from AC generator is converted by rectifier and is turned into DC current, then is charges the battery.



TROUBLESHOOTING

Battery runs down quickly

Check accessories which waste electric power excessively — **Installed** → ● **Remove accessories**

Not installed

Inspect battery leak current — **Leak** → ● **Short circuit of wiring harness.** ● **Loose or disconnected wires.** ● **Faulty battery.**

No leak

Inspect charging voltage between battery terminals. — **Correct** → ● **Faulty battery.** ● **Abnormal driving status.**

Incorrect

Inspect continuity of generator coils — **No continuity** → ● **Faulty generator coils or disconnected lead wires.**

Correct

Inspect generator no-load voltage — **Incorrect** → ● **Faulty generator rotor.**

Correct

(to next page)

(Continue the last page)

Inspect regulator/rectifier. — Incorrect → • Faulty regulator/rectifier

Correct

Inspect wirings — Incorrect → • Short circuit of wiring harness.
• Poor contact of couplers

Correct → • Fault battery

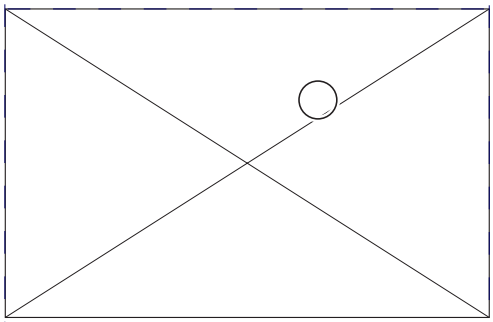
Others

Battery overcharge → • Faulty regulator/rectifier
• Faulty battery
• Poor contact of generator wire coupler.

INSPECTION

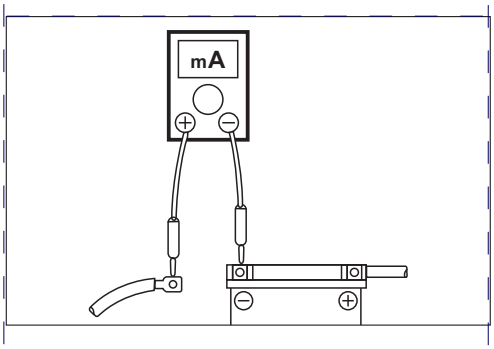
BATTERY LEAK CURRENT INSPECTION

- Remove the left side cover. (Refer to page 6-1)
- Turn off the ignition switch.
- Disconnect the battery (-) lead wire.
- Measure the current between the battery (-) terminal and the specified ground cable terminal. If the reading exceeds the specified value, then leakage is present.



TOOL **09900-25008: Multi-circuit tester set**

DATA **Battery leakage: less than 0.5 mA**

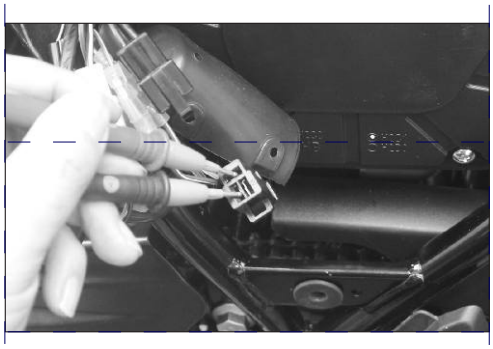


⚠ CAUTION

*** The leak current might be large, turn the tester to high range first when connecting the meter.**
*** Do not turn on the ignition switch when measuring current.**

GENERATOR COIL INSPECTION

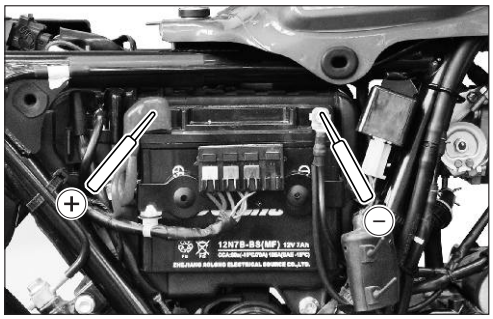
- Remove the right side cover.
- Disconnect generator coupler .
- Measure the resistence between the lead wires, if the resistence is incorrect, replace the generator coil.



TOOL **09900-25008: Multi-circuit tester set**

DATA **Generator coil resistance: 0.5-1.3 Ω (Black-Black)**

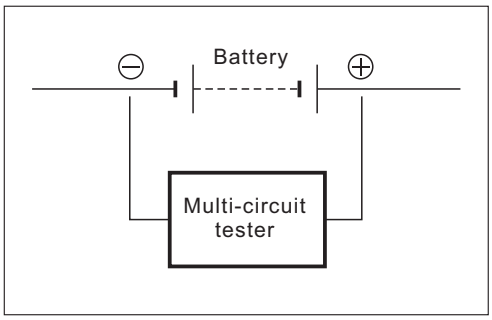
CHARGING OUTPUT INSPECTION



- Remove the left side cover.
- Start the engine, keep it running at 5 000 r/min, with the lighting switch turned on and dimmer switch turned to "≡O" position.
- Measure the DC voltage between the battery terminal and with a multi-circuit tester. If the tester reads under the specified value, inspect the generator coil and regulator/ rectifier.

⚠ CAUTION

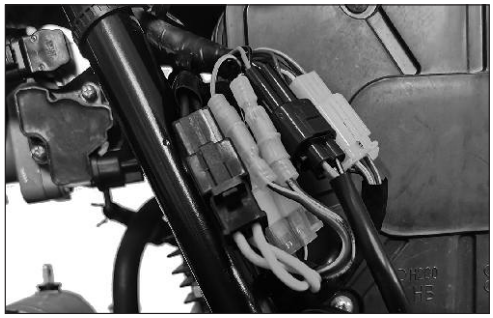
When making this test, be sure that the battery is fully-charged status.



TOOL 09900-25008: Multi-circuit tester set

DATA Charging output: 14.0-15.0V at 5 000 r/min

AC GENERATOR NO-LOAD PERFORMANCE



- Disconnect the generator coupler.
- Start the engine and keep it running at 5 000r/min.
- Measure voltage between generator terminal and ground by using the multi-circuit tester. If the reading is below standard values, replace with a new generator coil. (Refer to page 3-33)

NOTE:

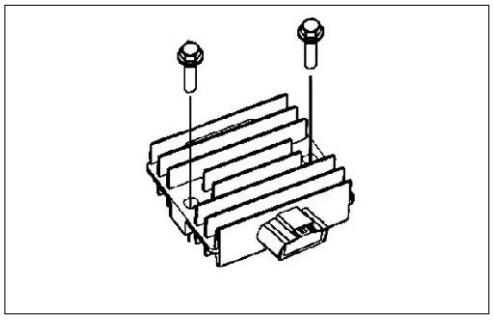
It is not necessary to remove the engine from the chassis during removing the generator coil.

TOOL 09900-25008: Multi-circuit tester set

DATA Generator no-load voltage:
Over 100 V (AC), at 5 000r/min

REGULATOR / RECTIFIER INSPECTION

If the inspections above are normal, replace the rectifier with a new one.

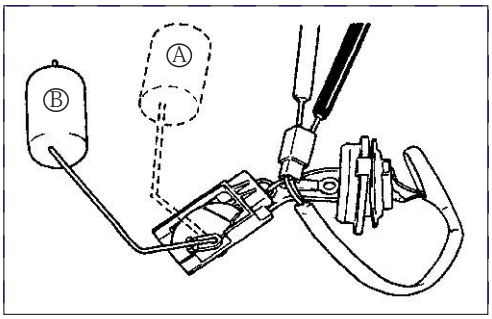


FUEL LEVEL

FUEL LEVEL SENSOR

- Remove the fuel lever sensor. (Refer to page 5-3)
- Check the resistance of each float position with a multi-circuit tester.
- If the resistance measured is incorrect, replace the fuel level sensor with a new one.
- The relation between the position of the fuel level sensor float and resistance is shown in the following table.

Float position	Resistance(Ω)
Ⓐ Full	5-9
Ⓑ Empty	98-102



TOOL **09900-25008: Multi-circuit tester set**

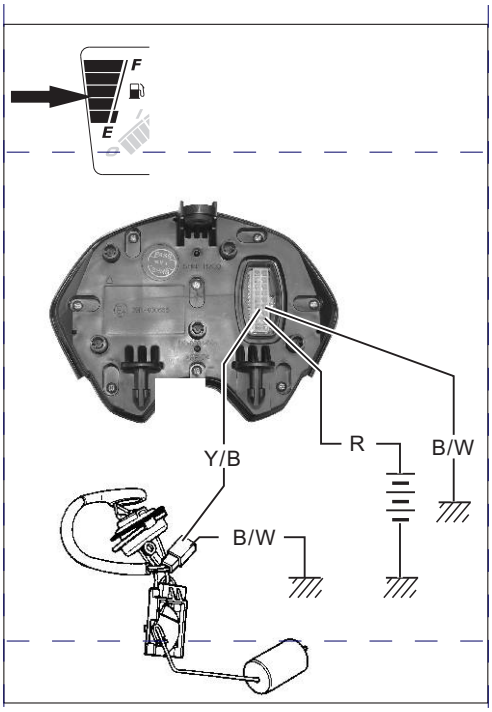
⚠ CAUTION

When inspecting the gauge resistance, be sure to disconnect the battery lead wire, or a mutil-circuit tester may be damaged.

FUEL LEVEL INDICATOR

- Disconnect fuel level gauge coupler.
- To test the fuel meter two different checks may be used.
- The first, connect a jumper wire between B/W and Y/B wires coming from the main wiring harness. With turning on the ignition switch, the fuel meter should indicted “F”.
- The second test will check the accuracy of the meter in the full and empty positions.
- Fuel meter is normal if its pointer indicates the “E” (empty) position when the specified resistance is applied to the circuit and if its pointer indicates the “F” (full) position when the resistor is changed to 5-9 ohms. If either one or both indications are abnormal, replace the fuel meter with a new one.

Resistance	5-9Ω	98-102Ω
Float position	F	E



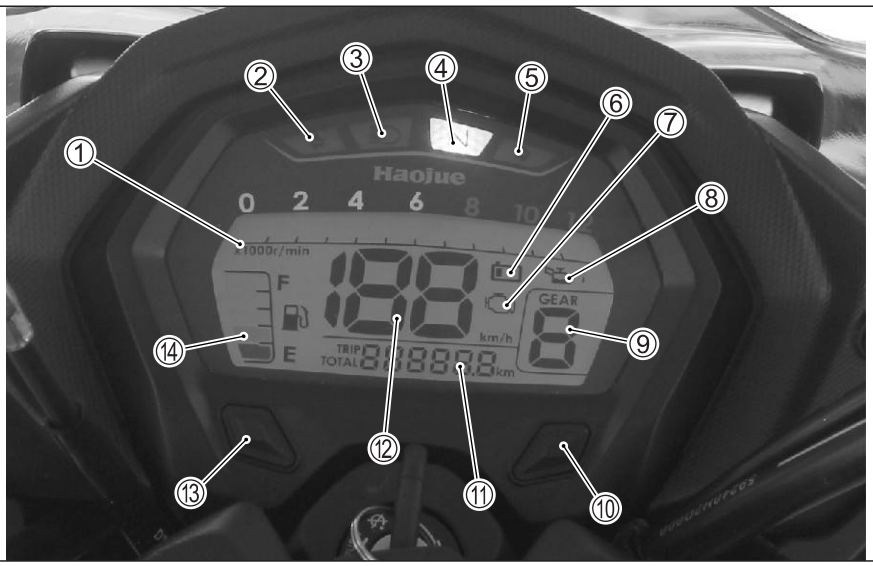
COMBINATION METER

Combination Meter System Description

This combination meter mainly consists of LCD (Liquid Crystal Display) and LED (Light Emitting Diode). The LCD indicates speedometer, tachometer, ODO / Trip A/ Trip B / Clock / Gear position indicator / oil change indicator / low battery voltage indicator and fuel level indicator respectively.

LED (Light Emitting Diode)

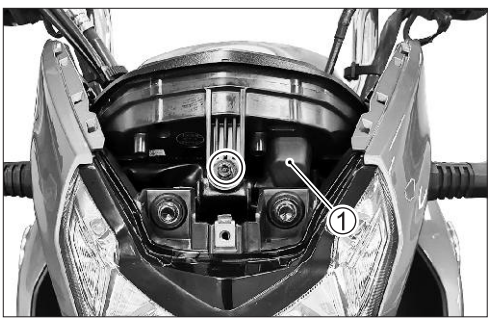
LED is used for the instrument panel light and each indicator light. LED is maintenance free.
LED is less power consuming and more resistant to vibration compared to the bulb.



①	LCD (Tachometer)	⑧	LED (Oil change indicator light)
②	LED (Left turn signal light)	⑨	LCD (Gear position indicator light)
③	LED (High beam indicator light)	⑩	ADJ button
④	LED (Neutral indicator light)	⑪	LCD (Odometer, Trip meter A, Trip meter B)
⑤	LED (Right turn signal light)	⑫	LCD (Speedometer)
⑥	LCD (Low battery voltage indicator light)	⑬	SEL button
⑦	LED (Malfunction indication light)	⑭	LCD (Fuel level indicator)

REMOVAL AND DISASSEMBLY

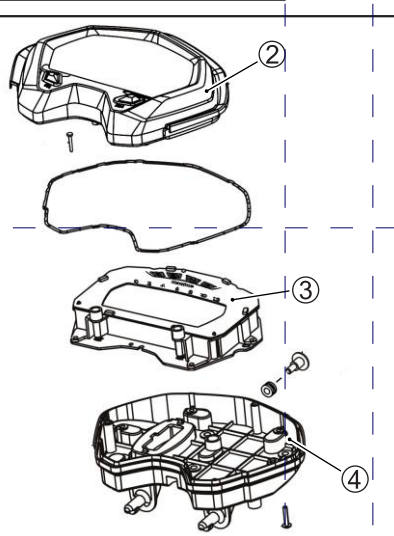
- Remove the windshield and headlamp upper cover. (Refer to page 6-4).
- Remove the bolt, disconnect the meter coupler ①, take off the combination meter.



- Disassemble the meter assembly as follows.

⚠ CAUTION

Do not disassemble the meter panel ③.



②	Meter upper case	③	Meter panel	④	Meter lower case

INSPECTION

LED (Light Emitting Diode)

Check that the LEDs (Lower fuel indicator light and oil change indicator light) immediately light up when the ignition switch is turned on. Check that other LEDs (Neutral indicator light, turn signal indicator lights and Hi beam indicator light) light up/go off by operating the gearshift lever, turn signal and dimmer switch. If abnormal condition is found, replace the combination meter unit with a new one after checking its wire harness/coupler.

TACHOMETER SELF-DIAGNOSE

The tachometer indicator displays maximum r/min then back “zero” when the ignition switch is turned on. If abnormal condition is found, replace the combination meter unit with a new one after checking its wire harness/coupler.

OIL CHANGE INDICATOR LIGHT

The oil change indicator ① comes on to remind you to change the engine oil. The indicator comes on at initial 1000 km and preset intervals thereafter. The preset interval is adjustable between 2 000 and 3 000 km. Reset the indicator after changing the engine oil to turn off the indicator.

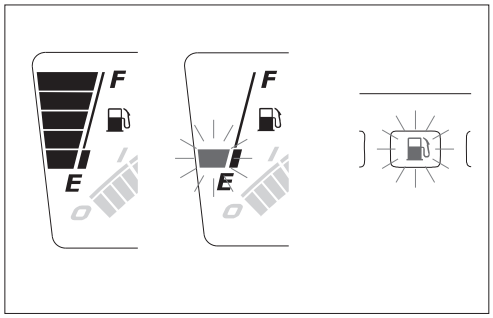
TO RESET THE OIL CHANGE INTERVAL

Press and hold the “ADJ” button ② and “SEL” button ③ for 2 seconds until the “3 000” blinks at the ODO meter. Press the “ADJ” button or “SEL” button to change the interval between “3 000” and “2 000”, press the “ADJ” button or “SEL” button for 2 seconds, oil change indicator change light goes off.

⚠ CAUTION

The preset interval can be adjusted after odometer reaches 1000 km.
Reset the indicator after initial engine oil replacement.
Reset the indicator after oil replacement even if the indicator is not displayed.

FUEL LEVEL INDICATOR INSPECTION



The fuel level indicator displays 5 portions, when 5 portions are solid when means the fuel is full, when only 1 portion or low fuel level indicator light blinks together, it means the remaining fuel is about 2.5L, please add the fuel as soon as possible.

⚠ CAUTION

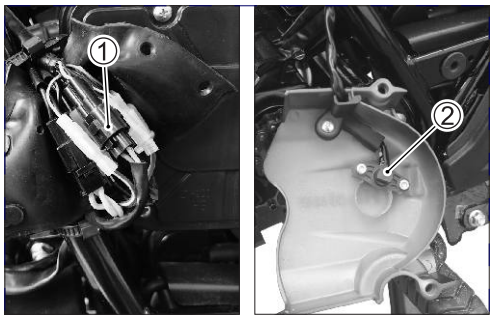
When the fuel level indicator display portions come on and go off one by one. it means fuel level sensor wire open circuit or short circuit.
Turn on and off the ignition switch repeatedly, accelerate or decelerate quickly, lean motorcycle, or go uphill or downhill will lead to the change of fuel level indicator which is normal.

SPEEDOMETER AND ODOMETER

If the speedometer, ODO meter and trip meter don't work properly, check the speed sensor and coupler connection, if no defects are found, replace the combination meter.

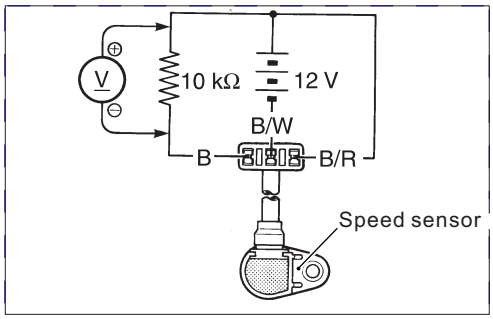
SPEED SENSOR

- Remove the left side cover. (Refer to page 6-1)
- Remove the engine sprocket cover.(Refer to page 3-3)
- Disconnect the speed sensor coupler ①.
- Take off the speed sensor ②.

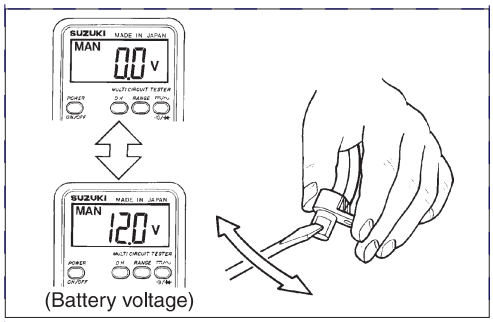


- Connect a battery, a 10kΩ resistance and a multi-circuit tester to the speed sensor as shown in the illustration.

09900-25008: Multi-circuit tester set



- Move the blade screw driver on the speed sensor surface as shown in the illustration, the voltage reading on the tester should be changed accordingly (0V to 12V).
  If the reading doesn't change, replace the speed sensor with a new one.



CAUTION

The highest voltage reading should be the same as the battery voltage during the test.

- Install the speed sensor.

ODOMETER

To change the display, push the "SEL" button ①.The display changes in the order ODO, Trip A and Trip B. When the Trip A/B is displayed, press and hold the "SEL" button for 2 s to reset the trip meter.

LOW BATTERY VOLTAGE INDICATOR LIGHT

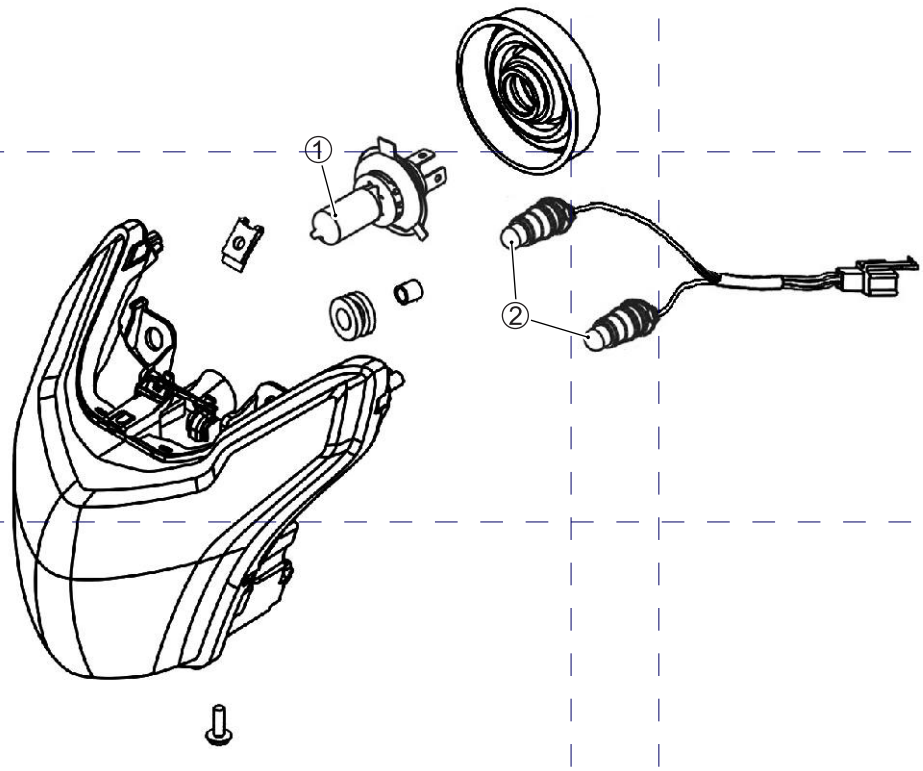
The low battery voltage indicator light blinks when the battery voltage is low. Shut down the high power appliance to prevent the battery over discharge. Re-charge the battery in time.

CAUTION

The battery indication light blinks and the meter can not display normally when the battery power wire is disconnected or fuse blows, check the power supply circuit.

LIGHTS

HEADLAMP AND POSITION LIGHT



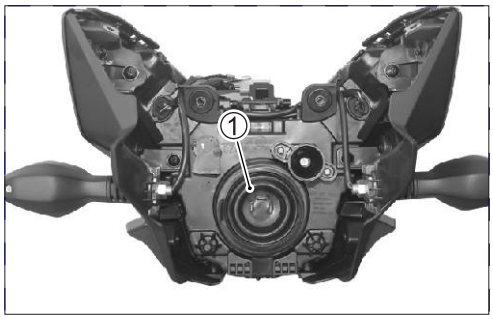
Headlamp①: 12V 35W/35W
Position light②: 12V 5W

⚠ CAUTION

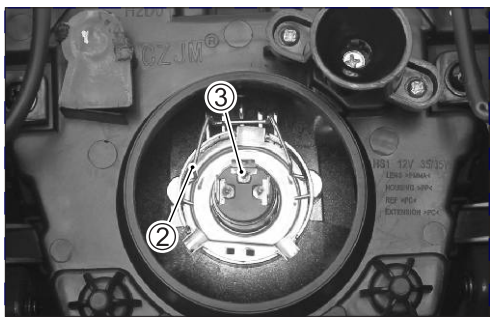
Remove the bulb when the bulb is cool.
If you have touched the bulb with your bare hands, clean it with a cloth moistened with alcohol or soapy water to maintain lens clarity.
Do not use bulb of other specification.

HEADLAMP BULB

- Remove the cowling and headlamp. (Refer to page 6-2)
- Remove the rubber cover ①.

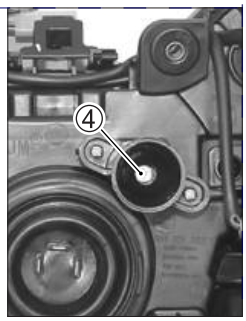


- Loosen the headlamp mounting spring ②, remove the headlamp bulb③.



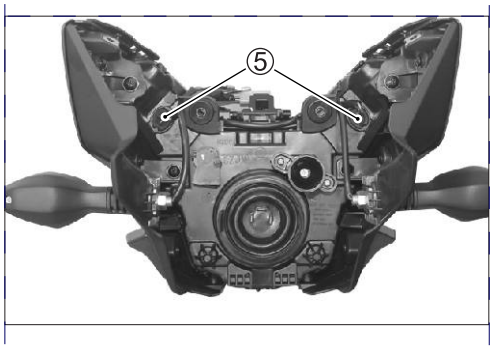
HEADLAMP LIGHT ADJUSTMENT

- Turn the headlamp adjust screw ④ to adjust the headlamp light beam vertically.

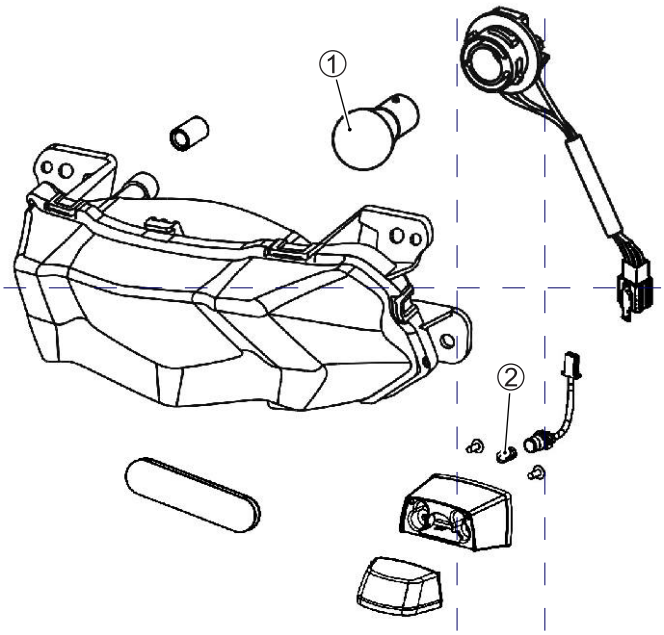


POSITION LIGHT

- Remove the cowling and headlamp assembly.
  (Refer to page 6-2)
- Remove the position light ⑤, replace the position light bulb.



TAIL LIGHT/BRAKE LIGHT AND LICENSE LIGHGT



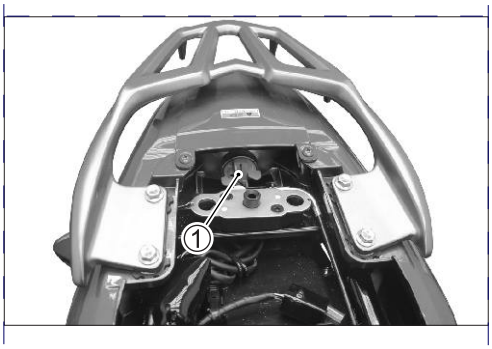
Tail light/Brake light ①: 12V 5W/21W
License light②: 12V 5W

⚠ CAUTION

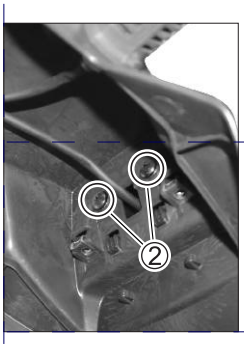
Remove the bulb when the bulb is cool.
If you have touched the bulb with your bare hands, clean it with a cloth moistened with alcohol or soapy water to maintain lens clarity.
Do not use bulb of other specification.

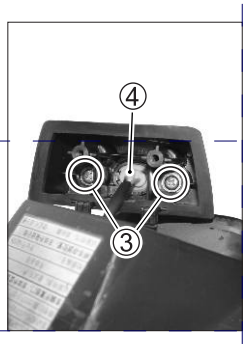
BULB REPLACEMENT

- Remove the saddle. (Refer to page 6-1)
- Remove the tail light/brake light ①, replace the tail light / brake light bulb.



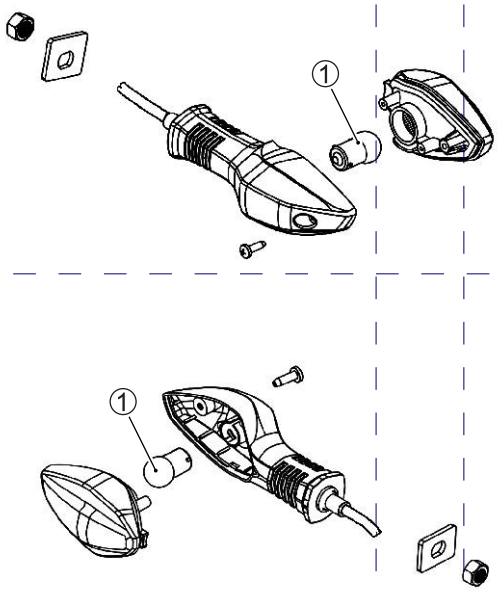
- Remove the rear fender inner plate. (Refer to page 6-5)
- Remove the license light cover mounting screw ②and ③.
- Remove the rear license light socket from rear license light cover.
- Remove the rear license bulb.





- Reassemble the lights in the reverse order of removal.

FRONT AND REAR TURN SIGNAL LIGHTS



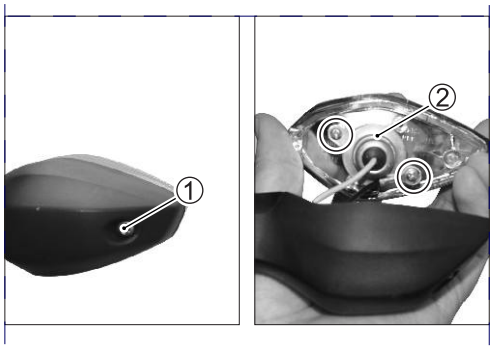
Turn signal light ①: 12V 10W

⚠ CAUTION

If you have touched the bulb with your bare hands, clean it with a cloth moistened with alcohol or soapy water to maintain lens clarity.
Do not use bulb of other specifications.

REAR SIGNAL LIGHT BULB REPLACEMENT

- Remove the turn signal light cover mounting screw ①.
- Remove the turn signal light socket mounting screws, take off the turn signal light socket ②.
- Replace the turn signal light bulb.



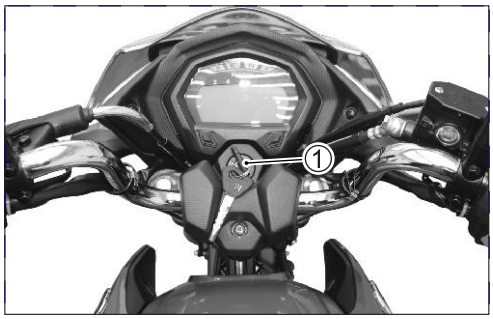
SWITCHES

- Inspect each switch for continuity with the pocket tester. If any abnormality is found, replace the respective switch with a new one.

09900-25008: Multi-circuit tester set

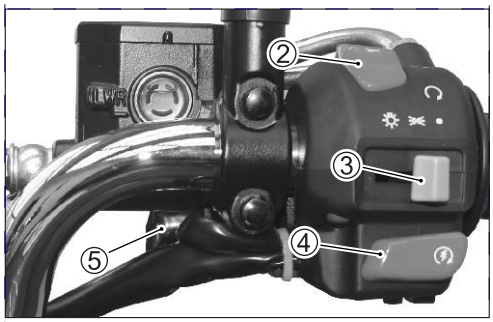
IGNITION SWITCH ①

	R	O	B/W	BI/W
	○	○		
			○	○
			○	○



ENGINE STOP SWITCH ②

	O/B	O/W
	○	○



CLUTCH SWITCH ③

	Gr	O	Y/W
●			
	○	○	
	○	○	○

STARTER SWITCH ④

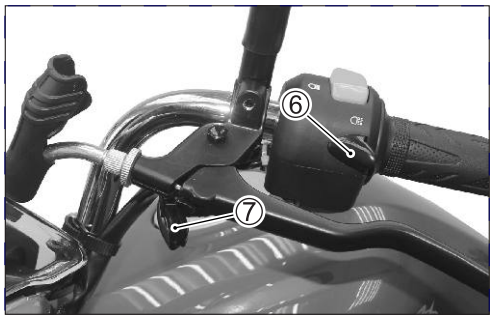
	B/R	Y/G
	○	○

FRONT BRAKE LIGHT SWITCH ⑤

	B	B
Off		
On	○	○

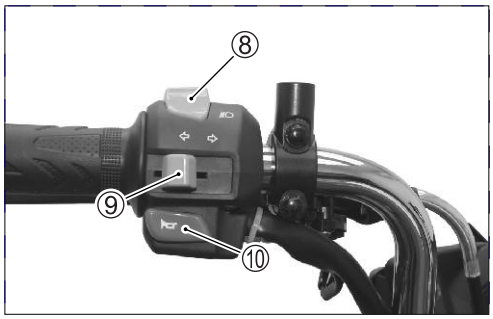
PASS SWITCH ⑥

	Y	O
≡O	O	O



CLUTCH SWITCH ⑦

	B/R	B/Y
OFF		
ON	O	O



DIMMER SWITCH ⑧

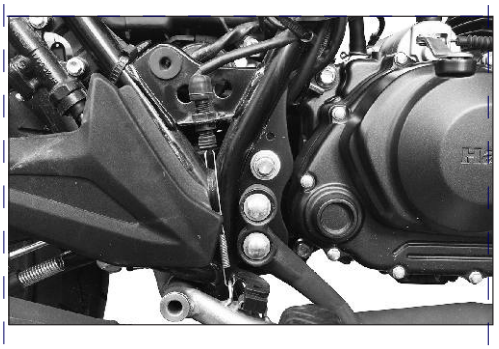
	Y/W	W	Y
≡O	O		O
≡O	O	O	

TURN SIGNAL LIGHT SWITCH ⑨

	Lg	LBl	B
⇨		O	O
•			
⇦	O	O	

HORN SWITCH ⑩

	B/Bl	B/W
▶⊂	O	O

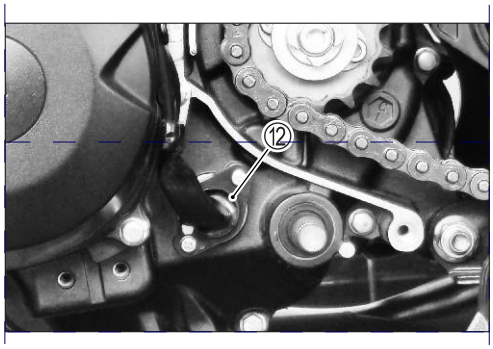


REAR BREAK LIGHT SWITCH ⑪

	O	W/B
Off		
On	O	O

GEAR POSITION INDICATOR LIGHT SWITCH ⑫

	Ground	Bl	W/Y	R/B	G/Bl	Y/Bl	Br/R
Neutral	O	O					
1st	O		O				
2nd	O			O			
3rd	O				O		
4th	O					O	
5th	O						O



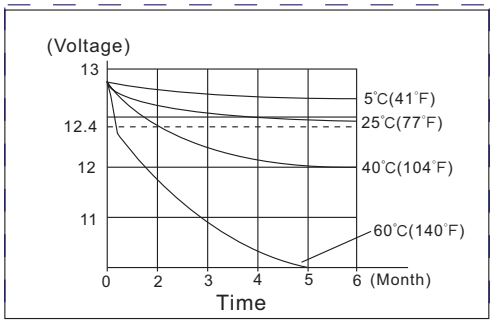
BATTERY

- Remove the left side cover. (Refer to page 6-1)
- Remove the battery. (Refer to page 2-3)
- Measure the battery voltage using the multi circuit tester. If the voltage reading is less than 12.4V, charge the battery with a battery charger.

NOTE:

For charging the battery, make sure to use the charger specially designed for MF battery. Otherwise, the battery may be overcharged resulting in shortened service life.

Please charge complementary regularly.
(In the ambient temperature 25°C every three months.)



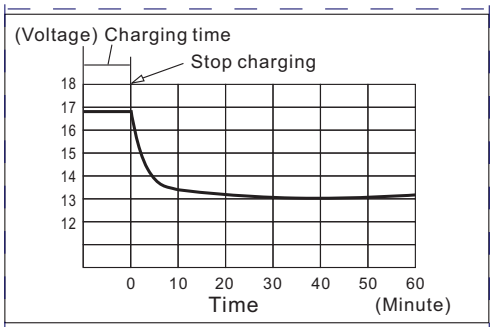
NOTE:

Different according to a storage state.

- Charge fully at the time of a voltage check regularly no to be less than 12.4V during a storage.

Please charge a battery for the following steps.

1. Supplementary charging the battery and wait 30 minutes.
2. Measure the battery voltage. If the voltage reading is less than 12.4V, recharge the battery.
3. Wait 30 minutes again and measure the battery voltage. If the voltage is less than 12.4V, change to a new battery.



Charging time: Current 0.2CA, for 2-5 hours

- After charging the battery, install the battery in the reverse order of removal.

⚠ CAUTION

Reversing the battery lead wires can damage the charging system and the battery.
Always attach the red lead to the ⊕ positive terminal first, then the black lead to the ⊖ negative terminal.

SERVICING INFORMATION

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TROUBLESHOOTING

FI SYSTEM MALFUNCTION CODE AND DEFECTIVE CONDITION

(Refer to page 4-14 to 4-17)

ENGINE

Complaint	Symptom and possible causes	Remedy
Engine will not start or is hard to start.	**Compression too low**	
	1. Valve clearance out of adjustment.	Adjust.
	2. Worn valve guides or poor seating.	Repair or replace.
	3. Mistiming valves.	Adjust.
	4. Excessively worn piston rings.	Replace.
	5. Worn-down cylinder bore.	Replace.
	6. Too slowly starter motor cranks.	See electrical section.
	7. Poor seating of spark plug.	Tighten.
	Plug not sparking	
	1. Fouled spark plug.	Clean or replace.
	2. Wet spark plug.	Clean or dry.
	3. Defective ignition coil.	Replace.
	4. Defective CKP sensor.	Replace.
	5. Defective ECM.	Replace.
	6. Open-circuited wiring connections.	Replace.
	7. Open or short in ignition coil cord.	Repair or replace.
	No fuel reaching the injector	
	1. Clogged fuel filter or fuel hose.	Clean or replace.
	2. Defective fuel pump.	Replace.
	3. Defective fuel pressure regulator.	Replace.
	4. Defective fuel injector.	Replace.
	5. Defective fuel pump relay.	Replace.
	6. Defective ECM.	Replace.
	7. Open-circuited wiring connection.	Inspect and repair.
	Incorrect fuel/air mixture	
	1. TP sensor out of adjustment.	Adjust.
	2. Defective fuel pump.	Replace.
	3. Defective fuel pressure regulator.	Replace.
	4. Defective TP sensor.	Replace.
	5. Defective CKP sensor.	Replace.
	6. Defective IAP sensor.	Replace.
	7. Defective ECM.	Replace.
	8. Defective ECT sensor.	Replace.
	9. Defective IAT sensor.	Replace.
	10. Clogged ISC valve air passage way.	Repair or replace.

Complaint	Symptom and possible causes	Remedy
Engine idles poorly.	1. Valve clearance out of adjustment. 2. Poor seating of valves. 3. Defective valve guides. 4. Worn down camshaft. 5. Too wide spark plug gap. 6. Defective ignition coil. 7. Defective CKP sensor. 8. Defective ECM. 9. Defective TP sensor. 10. Defective fuel pump. 11. Damaged or cracked vacuum hose. 12. Damaged or clogged ISC valve. 13. ISC bad learning.	Adjust. Replace or repair. Replace. Replace. Adjust or replace. Replace. Replace. Replace. Replace. Replace. Replace. Replace or repair. Reset learned value.
Engine stalls often.	**Incorrect fuel/air mixture** 1. Defective IAP sensor or circuit. 2. Clogged fuel filter. 3. Defective fuel pump. 4. Defective fuel pressure regulator. 5. Defective ET sensor. 6. Defective thermostat. 7. Defective IAT sensor. 8. Damaged or cracked vacuum hose. 9. Damaged or clogged ISC valve. **Fuel injector improperly operating** 1. Defective fuel injector. 2. No injection signal from ECM. 3. Open or short circuited wiring connection. 4. Defective battery or low battery voltage. **Control circuit or sensor improperly operating** 1. Defective ECM. 2. Defective fuel pressure regulator. 3. Defective TP sensor. 4. Defective IAT sensor. 5. Defective CKP sensor. 6. Defective ECT sensor. 7. Defective fuel pump relay. 8. Defective ISC valve. 9. ISC bad learning. **Engine internal parts improperly operating** 1. Fouled spark plug. 2. Defective CKP sensor or ECM. 3. Clogged fuel hose. 4. Out of adjustment valve clearance.	 Repair or replace. Clean or replace. Replace. Replace. Replace. Replace. Replace. Replace or repair. Clean or replace. Replace. Repair or replace. Repair or replace. Replace or recharge. Replace. Replace. Replace. Replace. Replace. Replace. Replace. Replace. Reset the learn value. Clean. Replace. Clean. Adjust.

Complaint	Symptom and possible causes	Remedy
Noisy engine.	**Excessive valve chatter**	
	1. Too large valve clearance.	Adjust.
	2. Weakened or broken valve springs.	Replace.
	3. Worn rocker arm or cam surface.	Replace.
	4. Worn and burnt camshaft journal.	Replace.
	Noise seems to come from piston	
	1. Worn down piston or cylinder.	Replace.
	2. Carbon combustion chamber fouled with carbon.	Clean.
	3. Worn piston pin or piston pin bore.	Replace.
	4. Worn piston rings or ring grooves.	Replace.
	Noise seems to come from timing chain	
	1. Stretched chain.	Replace.
	2. Worn sprocket.	Replace.
	3. Tension adjuster not working.	Repair or replace.
	Noise seems to come from clutch	
	1. Worn splines of counter shaft or hub	Replace.
	2. Worn teeth of clutch plates	Replace.
	3. Distorted clutch plates, driven and drive.	Replace.
	4. Worn clutch release bearing.	Replace.
	Noise seems to come from crankshaft	
	1. Worn bearing.	Replace bearing.
	2. Worn and burnt big-end bearing.	Replace bearing.
	3. Worn and burnt journal of bearing.	Replace bearing.
	4. Too large clearance.	Replace.
	Noise seems to come from balancer	
	1. Worn and burnt journal of bearing.	Replace.
	Noise seems to come from transmission	
	1. Worn or rubbing gears.	Replace.
	2. Worn splines.	Replace.
	3. Worn or rubbing primary gears.	Replace.
	4. Worn bearings.	Replace.

Complaint	Symptom and possible causes	Remedy
Engine runs poorly in high speed range.	**Defective engine internal/electrical parts**	
	1. Weakened valve springs.	Replace.
	2. Worn camshaft.	Replace.
	3. Valve timing out of adjustment.	Adjust.
	4. Too narrow spark plug gaps.	Adjust.
	5. Ignition not advanced sufficiently due to poorly working timing advance circuit.	Replace ECM.
	6. Defective ignition coil.	Replace.
	7. Defective CKP sensor.	Replace.
	8. Defective ECM.	Replace.
	9. Clogged air cleaner element.	Clean.
	10. Clogged fuel hose, resulting in inadequate fuel supply to injector.	Clean.
	11. Defective fuel pump.	Replace.
	12. Defective TP sensor.	Replace.
	Defective air flow system	
	1. Clogged air cleaner element.	Replace.
	2. Defective throttle body.	Adjust or replace.
	3. Sucking air from throttle body joint.	Repair or replace.
	4. Defective ECM.	Replace.
	Defective control circuit or sensor	
	1. Low fuel pressure.	Repair or replace.
	2. Defective TP sensor.	Replace.
	3. Defective IAT sensor.	Replace.
	4. Defective CKP sensor.	Replace.
	5. Defective GP switch.	Replace.
	6. Defective IAP sensor.	Replace.
	7. Defective ECM.	Replace.
	8. TP sensor out of adjustment.	Adjust.
	9. Defective ISC valve.	Replace.

Complaint	Symptom and possible causes	Remedy
Engine lacks power.	**Defective engine internal/electrical parts**	
	1. Loss of valve clearance.	Adjust.
	2. Weakened valve springs.	Replace.
	3. Valve timing out of adjustment.	Adjust.
	4. Worn piston rings or cylinder.	Replace.
	5. Poor seating of valves.	Repair.
	6. Fouled spark plug.	Clean or replace.
	7. Incorrect spark plug.	Adjust or replace.
	8. Clogged fuel injector.	Replace.
	9. Clogged air cleaner element.	Clean.
	10. TP sensor out of adjustment.	Adjust.
	11. Sucking air from throttle body or vacuum hose.	Repair or replace.
	12. Too much engine oil.	Drain excess oil.
	13. Defective fuel pump or ECM.	Replace.
	14. Defective CKP sensor and ignition coil.	Replace.
	Defective control circuit or sensor	
	1. Low fuel pressure.	Repair or replace.
	2. Defective TP sensor.	Replace.
	3. Defective IAT sensor.	Replace.
	4. Defective CKP switch.	Replace.
	5. Defective IAP sensor.	Replace.
	6. Defective ECM.	Replace.
	7. TP sensor out of adjustment.	Adjust.
	8. Defective ISC valve.	Replace.
Engine overheats.	**Defective engine internal parts**	
	1. Heavy carbon deposit on piston crowns.	Clean.
	2. Not enough oil in the engine.	Add oil.
	3. Defective oil pump or clogged oil circuit.	Replace or clean.
	4. Sucking air from intake pipe.	Tighten or replace.
	5. Use incorrect engine oil.	Replace.
	Lean fuel/air mixture	
	1. Short-circuited IAP sensor/lead wire.	Repair or replace.
	2. Short-circuited IAT sensor/lead wire.	Repair or replace.
	3. Sucking air from intake pipe joint.	Repair or replace.
	4. Defective fuel injector.	Replace.
	5. Defective ECT sensor.	Replace.
	Other factors	
	1. Ignition timing is too advanced due to defective timing advance system (ECT sensor, GP switch, CKP sensor and ECM)	Replace.
	2. Drive chain is too tight.	Adjust.
	3. ISC bad learning.	Reset the learned value.

Complaint	Symptom and possible causes	Remedy
Dirty or heavy exhaust smoke.	1. Too much engine oil in the engine. 2. Worn piston rings or cylinders. 3. Worn valve guides. 4. Scored or scuffed cylinder walls. 5. Worn valves stems. 6. Defective stem oil seal.	Drain excess oil. Replace. Replace. Replace. Replace. Replace.
Slipping clutch.	1. Weakened clutch springs. 2. Worn or distorted pressure plate. 3. Distorted clutch plates. 4. Clutch release screw out of adjustment.	Replace. Replace. Replace. Adjust.
Dragging clutch.	1. Some clutch spring weakened while others are not. 2. Distorted pressure plate or clutch plates. 3. Clutch release screw out of adjustment.	Replace. Replace. Adjust.
Transmission will not shift.	1. Broken gearshift cam. 2. Distorted gearshift forks. 3. Worn gearshift pawl.	Replace. Replace. Replace.
Transmission will not shift back.	1. Broken return spring on shift shaft. 2. Rubbing or stickily shift shaft. 3. Distorted or worn gearshift forks.	Replace. Repair. Replace.
Transmission jumps out of gear.	1. Worn shifting gears on drive-shaft or counter-shaft. 2. Distorted or worn gearshift forks. 3. Weakened stopper spring on gearshift stopper. 4. Worn gearshift cam stopper plate.	Replace. Replace. Replace. Replace.

CHASSIS

Complaint	Symptom and possible causes	Remedy
Steering feels too heavy or stiff.	1. Steering stem nut over tightened. 2. Worn bearing or race in steering stem. 3. Distorted steering stem. 4. Not enough pressure in tires.	Adjust. Replace. Replace. Adjust.
Steering oscillation.	1. Loss of balance between right and left front suspensions. 2. Distorted front absorber. 3. Distorted front axle or crooked tire.	Replace. Repair or replace. Replace.
Wobbly front wheel.	1.Distorted wheel rim. 2.Warn-down front wheel bearings. 3.Defective or incorrect tire. 4.Loose nut on axle.	Replace. Replace. Replace. Re-tighten.
Front suspension too soft.	1. Weakened springs. 2. Not enough fork oil.	Replace. Refill.
Front suspension too stiff.	1. Fork oil too viscous. 2. Too much fork oil.	Replace. Remove excess oil.
Noisy front suspension.	1. Not enough fork oil. 2. Loose nuts on suspension.	Refill. Re-tighten.
Wobbly rear wheel.	1. Distorted wheel rim. 2. Worn-down rear wheel bearings. 3. Defective or incorrect tire. 4. Loose nut of axle. 5. Worn swing arm bushings. 6. Loosen bolt on the rear absorber.	Replace. Replace. Replace. Re-tighten. Replace. Re-tighten.
Rear suspension too soft.	1. Weakened springs. 2. Rear suspension adjuster improperly set.	Replace. Adjust.
Rear suspension too stiff.	1. Rear suspension adjuster improperly set. 2. Worn swing arm bushings or bearing.	Adjust. Replace.
Noisy rear suspension.	1. Loose bolts of suspension. 2. Worn swing arm bushings or bearing.	Re-tighten. Replace.

BRAKE

Complaint	Symptom and possible causes	Remedy
Insufficient brake power.	1. Leakage of brake fluid from hydraulic system. 2. Worn pads. 3. Friction surfaces of pad are dirty oil or dust. 4. Worn disc. 5. Air in hydraulic system. 6. Not enough brake fluid in the reservoir.	Repair or replace. Replace. Clean disc/pads or replace. Replace. Bleed air. Replenish.
Brake squeaking.	1. Carbon adhesion on pad surface. 2. Tilted pad. 3. Damaged wheel bearing. 4. Loosen front wheel axle nut or rear wheel axle nut. 5. Worn pads. 6. Foreign material in brake fluid. 7. Clogged return port of master cylinder.	Repair surface with sandpaper. Correct pad fitting or replace. Replace. Tighten to specified torque. Replace. Replace brake fluid. Disassemble and clean master cylinder.
Excessive brake lever stroke.	1. Air in hydraulic system. 2. Insufficient brake fluid. 3. Improper quality of brake fluid.	Bleed air. Replenish fluid to specified level; bleed air. Replace with correct fluid.
Leakage of brake fluid.	1. Insufficient tightening of connection joints 2. Cracked hose. 3. Worn piston and/or cup.	Tighten to specified torque. Replace. Replace piston and/or cup.
Brake drags.	1. Rusty part. 2. Insufficient brake lever or brake pedal pivot lubrication.	Clean and lubricate. Lubricate.

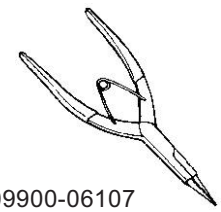
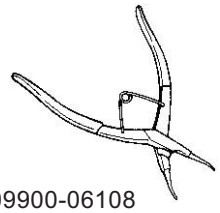
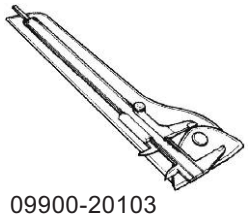
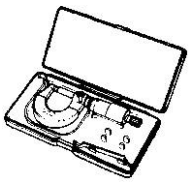
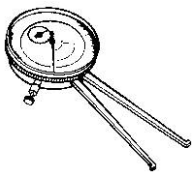
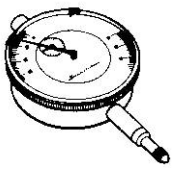
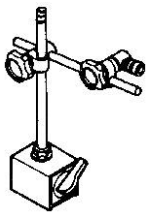
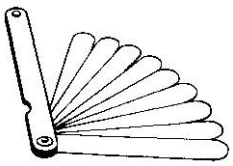
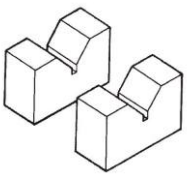
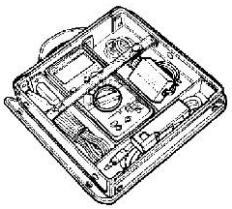
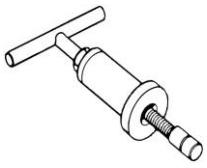
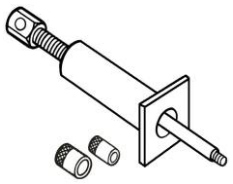
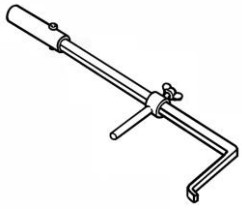
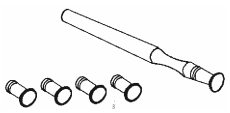
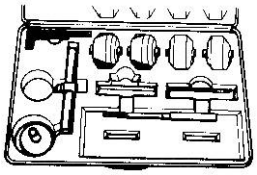
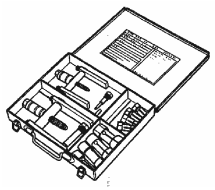
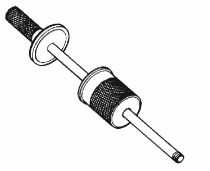
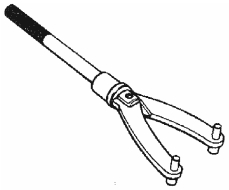
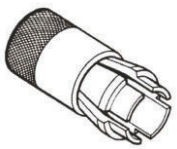
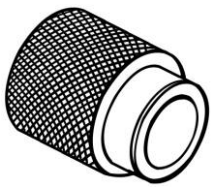
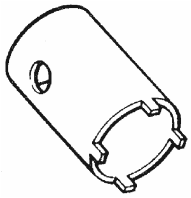
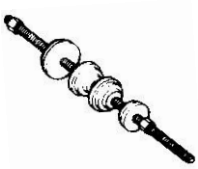
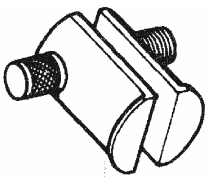
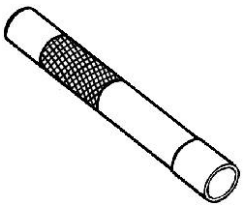
ELECTRICAL

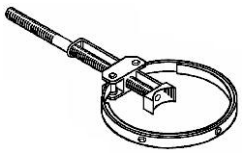
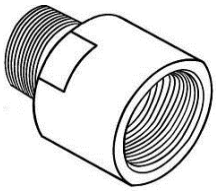
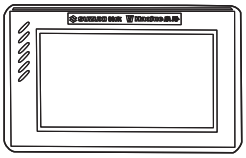
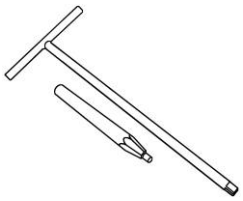
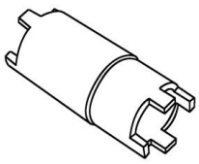
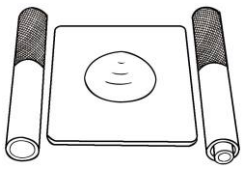
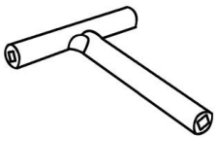
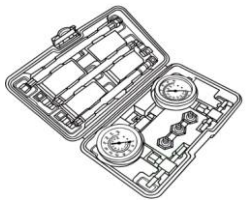
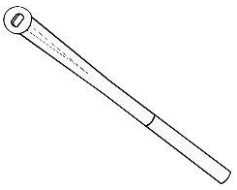
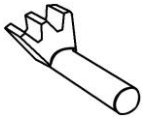
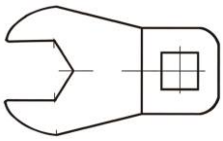
Complaint	Symptom and possible causes	Remedy
No sparking or poor sparking.	1. Defective ignition coil. 2. Defective spark plug. 3. Defective CKP sensor. 4. Defective ECM. 5. Defective TO sensor. 6. Open-circuited wiring connections.	Replace. Replace. Replace. Replace. Replace. Check and repair.
Premature carbon buildup on spark plug.	1. Mixture too rich. 2. Excessively high idling speed. 3. Incorrect gasoline. 4. Dirty air cleaner element. 5. Too cold spark plug.	Inspect FI system. Inspect FI system. Change. Replace. Replace with hot type plug.
Spark plug becomes fouled too soon.	1. Worn piston rings. 2. Worn piston or cylinder. 3. Excessive clearance of valve stems in valve guides. 4. Worn stem oil seal.	Replace. Replace. Replace. Replace.
Spark plug electrodes overheat or burn.	1. Too hot spark plug. 2. Overheated the engine. 3. Loose spark plug. 4. Too lean mixture.	Replace with cold type plug. Tune up. Re-tighten. Inspect FI system.
Generator does not charge.	1. Open or short circuited lead wires, or loose lead connections. 2. Short-circuited, grounded or open generator coil. 3. Short-circuited or punctured rectifier.	Repair or replace or tighten. Replace. Replace.
Generator does charge, but charging rate is below the specification.	1. Lead wires tend to get shorted or open-circuited or loosely connected at terminals 2. Grounded or open-circuited generator coil. 3. Defective regulator/rectifier. 4. Defective cell plates in the battery.	Repair or re-tighten. Replace. Replace. Replace the battery.
Generator overcharges.	1. Internal short-circuit in the battery. 2. Damaged or defective regulator/rectifier. 3. Poorly grounded regulator/rectifier.	Replace the battery. Replace. Clean and tighten the ground wire.
Unstable charging.	1. Lead wire insulation frayed due to vibration, resulting in intermittent short-circuiting. 2. Internally shorted generator. 3. Defective regulator/rectifier.	Repair or replace. Replace. Replace.
Starter button is not effective.	1. Run down battery. 2. Defective switch contacts. 3. Brushes not seating properly on starter motor commutator. 4. Defective starter relay. 5. Defective main fuse.	Repair or replace. Replace. Repair or replace. Replace. Replace.

BATTERY

Complaint	Symptom and possible causes	Remedy
Battery runs down quickly.	1. The charging method is not correct.	Check the generator, regulator/rectifier and circuit connections, make necessary adjustments to obtain specified charging.
	2. Cell plates have lost much of their active material as result of over-charging.	Replace the battery and correct the charging system.
	3. A shorted circuit condition exists within the battery.	Replace the battery.
	4. Poor connection between couplers.	Check and correct.
	5. Battery is too old.	Replace the battery.
Battery overcharge	1. Defective rectifier.	Replace the battery and the regulator / rectifier.
	2. Defective battery.	Replace the battery.
	3. Poor generator wire coupler connection.	Inspection and repair.
"Sulfation", acidic white powdery substance or spots on cell plates.	1. Cracked battery.	Replace the battery.
	2. The battery was left unused in a cold climate for too long.	Replace the battery.

Special tool

 09900-06107 Snap ring pliers (open type)	 09900-06108 Snap ring pliers (closing type)	 09900-20103 Vernier calipers (1/50mm, 300mm)	 09900-20202 Micrometer (1/100mm, 25~50mm)	 09900-20203 Micrometer (1/100mm, 50~75mm)
 09900-20205 Micrometer (1/100mm,0~25mm)	 09900-20508 Cylinder bore gauge kit (1/100mm,40~80mm)	 09900-20605 Dial calipers (1/100mm,10~34mm)	 09900-20606 Dial meter (1/100mm,10mm)	 09900-20701 Magnetic bracket
 09900-20803 Thickness gauge	 09900-21304 V-block (set) (100mm)	 09900-25008 Multi-circuit tester	 09910-32812 Crankshaft installer	 09910-34510 Piston pin puller
 09913-50121 Oil seal remover	 09913-70210 Bearing installer set (Φ10-75)	 09916-10911 Valve seat cutter set	 09916-21110 Valve cutter set	 (Crankcase separating tool
 09921-20240 Bearing remover set	 09924-84510 Bearing installer set	 09930-30104 Slide shaft	 09930-40113 Sprocket holder	 09940-50112 Fork oil seal installer
 09940-53311 Bearing installer	 09940-14911 Steering stem nut remover and insaller	 09941-34513 Steering racer installer	 09941-54911 Bearing remover	 09941-74910 Steering bearing installer

 11F14-001 Rotor holder	 11F14-004 Generator rotor remover	 11F14-009 Diagnosis device	 11F14-010 Front absorber tools	 11F14-011 4 claws nut socket
 11F14-018 Valve remover and installer	 11F14-019 Tappet adjust driver	 11F14-023 Pressure test tool	 11F14-030 Balancer shaft holder 8	 11F14-033 6 pins test switch
 11F14-035 Pneumatic fluid drawer	 11F14-036 Gear stopper	 11J14-001 Balancer shaft nut wrench 19		

TIGHTENING TORQUE

ENGINE

ITEM		N•m	Kg•m
Crankcase bolt		10-15	1.0-1.5
Oil filter cover		20-28	2.0-2.8
Crankshaft bearing pin retainer bolt		8-12	0.8-1.2
Engine sprocket bolt		10-15	1.0-1.5
Locating pin mounting bolt		10-15	1.0-1.5
Stopper cam mounting bolt		10-15	1.0-1.5
Balancer shaft nut		25-35	2.5-3.5
Oil pump mounting screw		8-12	0.8-1.2
Oil drain bolt		25-30	2.5-3.0
Rocker arm shaft retainer bolt		8-12	0.8-1.2
Generator stator bolt		5-8	0.5-0.8
CKPS coil bolt		3-6	0.3-0.6
Clutch lock nut		40-45	4.0-4.5
Oil filter rotor lock nut		40-45	4.0-5.0
Rotor cover screw		3-6	0.3-0.6
Clutch cover bolt		10-15	1.0-1.5
Cylinder head nut		25-35	2.5-3.5
Crankshaft sprocket bolt		10-15	1.0-1.5
Generator rotor bolt		45-50	4.5-5.0
Generator cover bolt		10-15	1.0-1.5
Spark plug		15-20	1.5-2.0
Engine mounting nut	Front	33-39	3.3-3.9
	Rear	50-60	5.0-6.0
Engine hanging plate nut		22-33	2.2-3.3
ET sensor		13-16	1.3-1.6

CHASSIS

ITEM	Kg•m	N•m
Front axle nut	3.6-5.2	36-52
Front fork upper clamp bolt	2.2-3.5	22-35
Steering stem lock nut	3.5-5.5	35-55
Lower bracket clamp bolt	2.5-3.5	25-35
Handlebar clamp bolt	1.2-2.0	12-20
Front footrest mounting bolt	4.0-6.0	40-60
Front brake disc mounting bolt	1.8-2.8	18-28
Front brake caliper mounting bolt	1.8-2.8	18-28
Brake hose union bolt	2.0-2.5	20-25
Bleeder valve	0.7-1.0	7-10
Rear absorber bolt/nut (Upper&lower)	2.2-3.5	22-35
Rear axle nut	5.0-8.0	50-80
Rear footrest mounting bolt	4.0-6.0	40-60
Rear sprocket nut	3.0-3.7	30-37
Rear swing arm bolt/nut	5.0-8.0	50-80
Rear torque link	1.0-1.6	10-16
Brake cam lever	0.8-1.2	8-12
Bundy hose mounting nut	1.2-1.7	12-17
Exhaust pipe bolt	1.1-1.5	11-15
Muffler mounting bolt	3.5-4.5	35-45
Oxygen sensor	2.0-3.0	20-30

TIGHTENING TORQUE CHART

Diameter(mm)	Conventional or “4” marked		“7” marked	
	N•m	Kg•m	N•m	Kg•m
4	1.0-2.0	0.1-0.2	15-3.0	0.15-0.3
5	2.0-4.0	0.2-0.4	3.0-6.0	0.3-0.6
6	4.0-7.0	0.4-0.7	8.0-12	0.8-1.2
8	10-16	1.0-1.6	18-28	1.8-2.8
10	22-35	2.2-3.5	40-60	4-6
12	35-55	3.5-5.5	70-100	7-10
14	50-80	5.0-8.0	110-160	11-16
16	80-130	8.0-13	170-250	17-25
18	130-190	13-19	200-280	20-28

SERVICE DATA

CYLINDER HEAD + VALVE

Unit: mm

ITEM	STANDARD		LIMIT
Valve diam.	IN.	29.64	——
	EX.	25.64	——
Valve clearance (when cold)	IN.	0.04-0.08	——
	EX.	0.09-0.13	——
Valve guide I.D.	IN.	5.00-5.01	——
	EX.	5.00-5.01	——
Valve stem O.D.	IN.	4.975-4.990	——
	EX.	4.955-4.970	——
Valve deflection	IN.& EX.	——	0.35
Valve stem runout	IN.& EX.	——	0.05
Valve head thickness	IN.& EX.	——	0.5
Valve seat width	IN.	0.9-1.1	——
	EX.	0.9-1.1	——
Valve head radial runout	IN.& EX.	——	0.03
Valve spring free length	IN.	——	34.8
	EX.	——	35.4
Cylinder head distortion	——		0.05
Cylinder head cover distortion	——		0.05

CAM + ROCKER

Unit: mm

ITEM	STANDARD		LIMIT
Cam height	IN.	33.05-33.29	32.1
	EX.	32.94-33.18	32.0
Rocker arm I.D	10.00-10.02		——
Rocker arm shaft O.D	9.97-9.99		——

CYLINDER + PISTON + PISTON RING

Unit: mm

ITEM	STANDARD		LIMIT
Compression pressure	1 000-1 400 kPa		800 kPa
Piston to cylinder clearance	0.03-0.05		0.12
Cylinder bore	60.00-60.01		60.13
Piston diam.	Measure at 10 mm from the skirt end 59.98-59.99		59.855
Cylinder distortion	——		0.05
Piston ring free end gap	1st	Approx 6.5	5.2
	2nd	Approx 6.6	5.3
Piston ring end gap	1st	0.10-0.22	0.50
	2nd	0.20-0.32	0.50
Piston ring to groove clearance	1st	——	0.18
	2nd	——	0.15
Piston ring groove width	1st	0.81-0.82	——
	2nd	0.81-0.82	——
	Oil ring	1.51-1.52	——
Piston ring thickness	1st	0.76-0.78	——
	2nd	0.76-0.78	——
Piston pin bore	14.00-14.01		14.05
Piston pin O.D.	13.99-14.00		13.95

CONROD + CRANKSHAFT

Unit: mm

ITEM	STANDARD	LIMIT
Conrod small end I.D.	14.00-14. 02	14.04
Conrod big end side clearance	0.10-0.35	0.6
Conrod big end width	15.95-16.00	——
Conrod deflection	——	3.0
Crankshaft web to web width	$48^{+0.1}$	——
Crankshaft runout	——	0.05

CLUTCH

Unit: mm

ITEM	STANDARD	LIMIT
Clutch cable play	4	——
Drive plate thickness	2.92-3.08	2.62
Drive plate claw width	15.75-15.85	15.25
Driven plate distortion	——	0.10
Clutch spring free length	38.1	36.2

TRANSMISSION + DRIVE CHAIN

ITEM	STANDARD		LIMIT
Primary reduction ratio	3.091		——
Final reduction ratio	2.933		——
Gear ratios	Low	3.077	——
	2nd	1.875	——
	3rd	1.409	——
	4th	1.12	——
	Top	0.938	——
Shift fork to groove clearance	0.10-0.24 mm		0.44 mm
Shift fork groove width	5.10-5.17 mm		——
Shift fork thickness	4.93-5.00 mm		——
Drive chain	Type	KMC428HD	——
	Links	120 links	——
	20 pitch length	——	259.4 mm
Drive chain slack	10-20 mm		——

FI SENSORS

ITEM	STANDARD		NOTE
CKP sensor resistence	115-155 Ω		
IAP sensor power supply voltage	4.5-5.5 V		
TP sensor power supply voltage	4.5-5.5 V		
TP sensor output voltage	Close	Approx. 1.1 V	
	Open	Approx. 4.4 V	
ET sensor input voltage	4.5-5.5 V		
ET sensor resistence	Approx 2.645 KΩ at 20°C		
IAT sensor input voltage	4.5-5.5 V		
TO sensor input voltage	4.5-5.5 V		
TO sensor voltage	Normal	0.4-1.4 V	
	Tilt	3.7-4.4 V	Tilt 65º
Injector voltage	Battery voltage		
Oxygen sensor power supply voltage	4.5-5.5 V		
Fuel pump power supply voltage	Battery voltage		
ISC valve resistence	20 Ω at 20°C		

THROTTLE BODY

ITEM	STANDARD
Bore size	MTB28
I.D. NO	D28
Throttle cable play	2.0-4.0mm

INJECTOR, FUEL PUMP AND FUEL PRESSURE REGULATOR

ITEM	STANDARD	NOTE
Injector resistence	12 Ω at 20°C	
Fuel pressure adjuster open pressure	Approx 300 kPa	
Fuel pump power supply voltage	Battery voltage	

ELECTRICAL

ITEM		SPECIFICATION		NOTE
Spark plug		Type	NGK:DR8EA	
		Gap	0.8-0.9 mm	
Spark plug performance		Over 8mm at 1 atm.		
CKP sensor resistance		115-155 Ω		
Ignition coil resistance		Primary	3.1-4.3 Ω	Terminal-Terminal
		Secondary	28.5-38.5 kΩ	Plug cap-Terminal
Generator coil resistance		0.5-1.3 Ω		
Generator no-load voltage (When cold)		100 V or higher, at 5 000r/min		
Starter relay resistance		4-6 Ω		
Battery	Model	12N7B-4B		
	Capacity	12V(7Ah)		
Fuse	Lights	15A		
	Signal	10A		
	Main fuse	20A		

LIGHT

ITEM		SPECIFICATION	ITEM	SPECIFICATION
Headlamp	HI	35W	Neutral indicator light	LED
	LO	35W	Malfunction indicator light	LED
Tail/Brake light		5W/21W	High beam indicator light	LED
Turn signal light		10W	Turn signal indicator light	LED
Position light		5W	Gear position indicator light	LCD
License light		5W		

TIRE

ITEM		SPECIFICATION	LIMIT
Cold inflation tire pressure (Solo riding)	Front	175 kPa (1.75 kgf/cm²)	——
	Rear	225 kPa (2.25 kgf/cm²)	——
Cold inflation tire pressure (Dual riding)	Front	175 kPa (1.75 kgf/cm²)	——
	Rear	225 kPa (2.25 kgf/cm²)	——
Tire size	Front	80/100-18 47P	——
	Rear	100/80-18 59P	——

BRAKE + WHEEL

ITEM	STANDARD		LIMIT
Rear brake pedal height	10-20		——
Brake disc thickness	Front	3.8-4.2	3.5
Brake disc runout	Front	——	0.30
Brake drum I.D	Rear	——	130.7
Brake fluid type	DOT3		——
Wheel rim runout	Front and rear	——	2.0
Wheel axle runout	Front and rear	——	0.25
Dynamic wheel balance	Front and rear	8 g	——

SUSPENSION

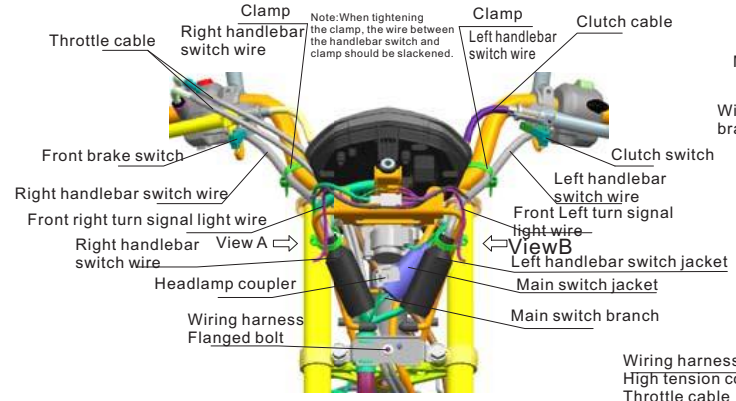
Unit: mm

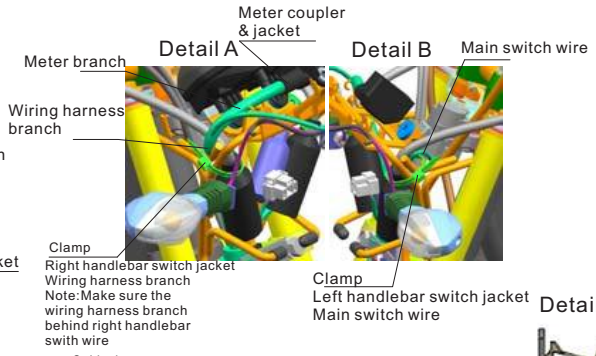
ITEM	STANDARD	LIMIT
Front fork stroke	110	——
Front fork spring free length	422	413.6
Fork fork oil type	MX4#	——
Front fork oil capacity(each leg)	152±1ml	——
Rear shock absorber spring	The 2nd position	——
Rear swing arm shaft runout	——	0.6

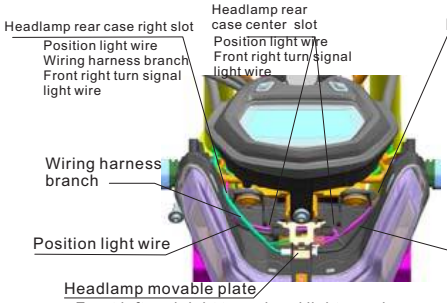
FUEL + OIL

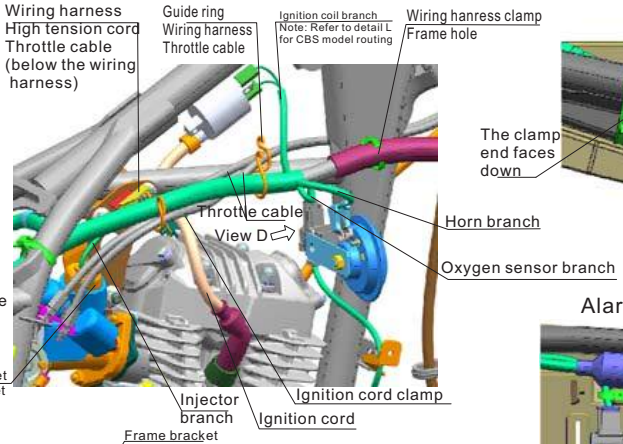
ITEM	STD/SPEC.		NOTE
Fuel type	Use unleaded gasoline of grade 89 octane (Research Method) or higher. Ethanol gasoline E10.		
Fuel tank capacity	16.5 L		
Engine oil type and grade	SAE 10W-40, API SL or higher		
Engine oil capacity	Oil change	1 100 ml	
	Overhaul	1 200 ml	

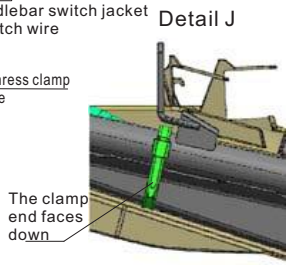
WIRE ROUTING

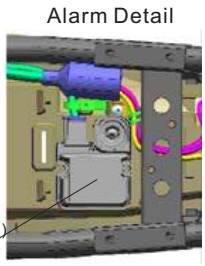


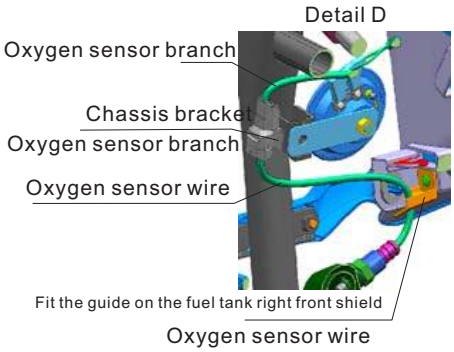


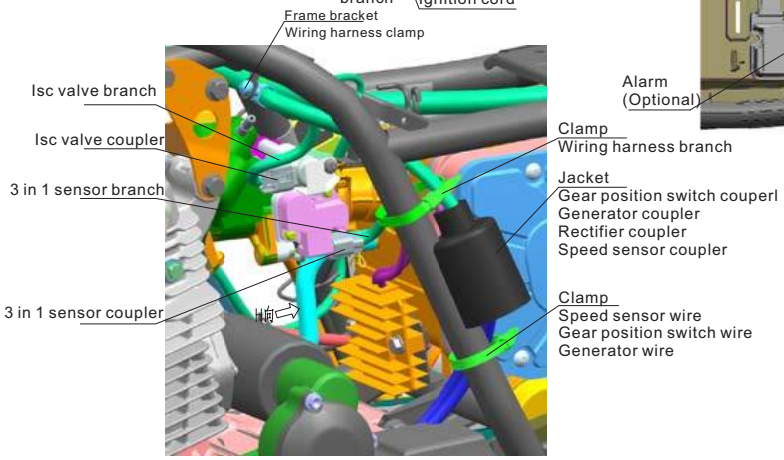


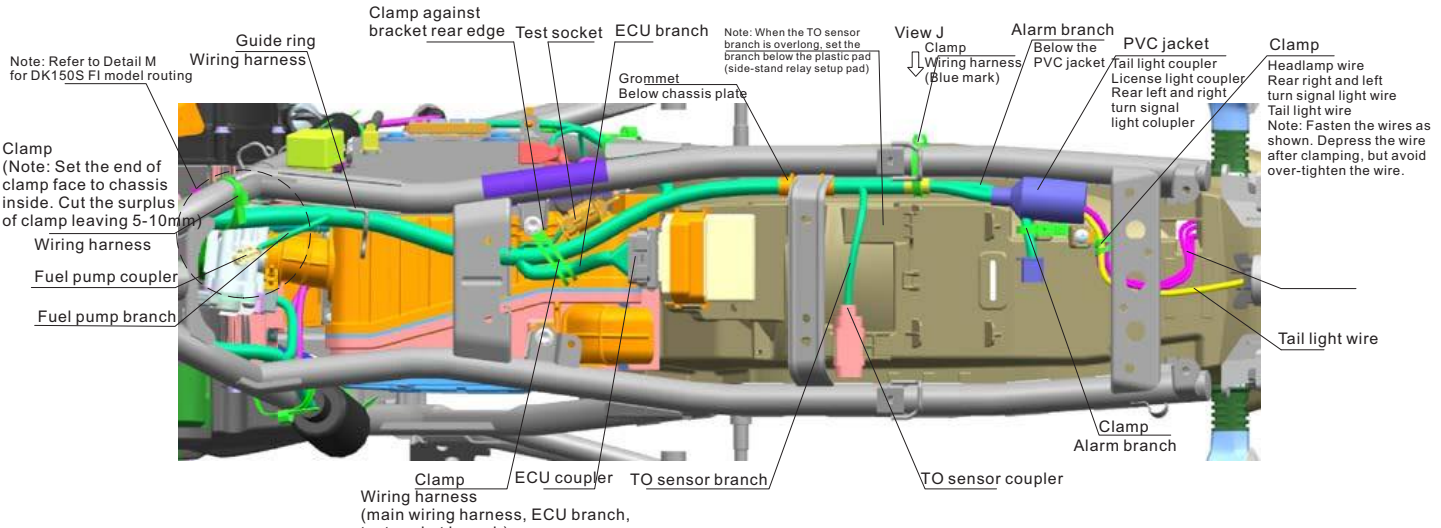


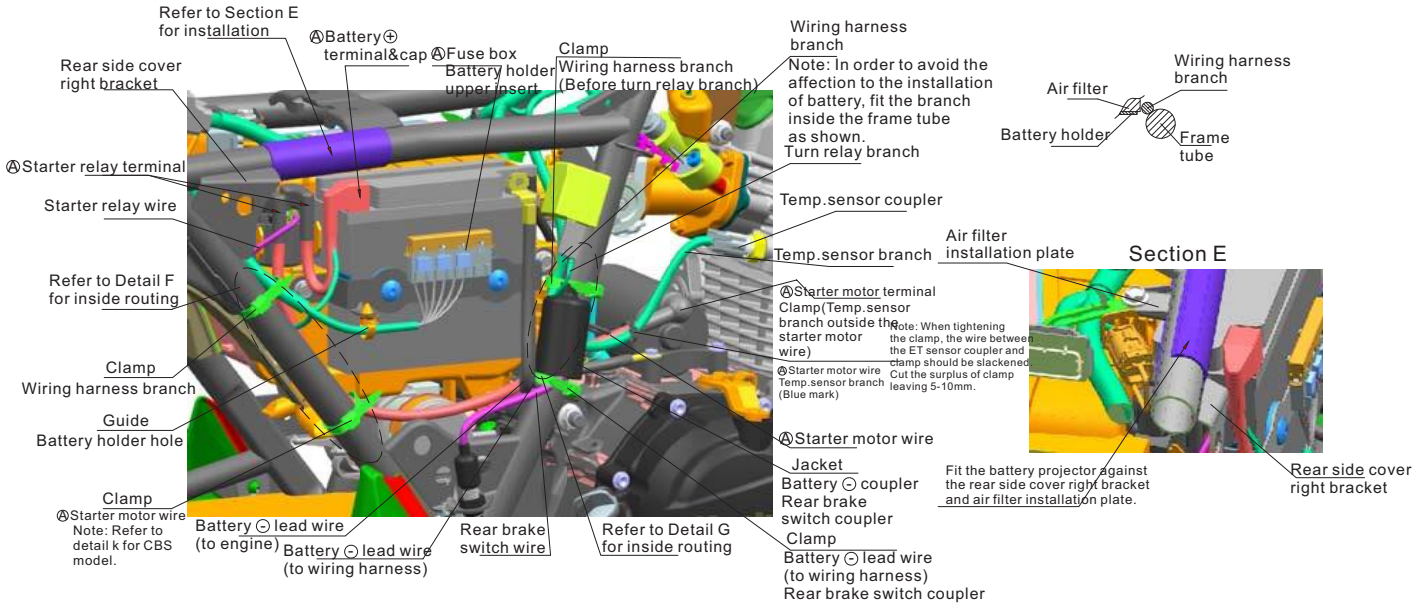


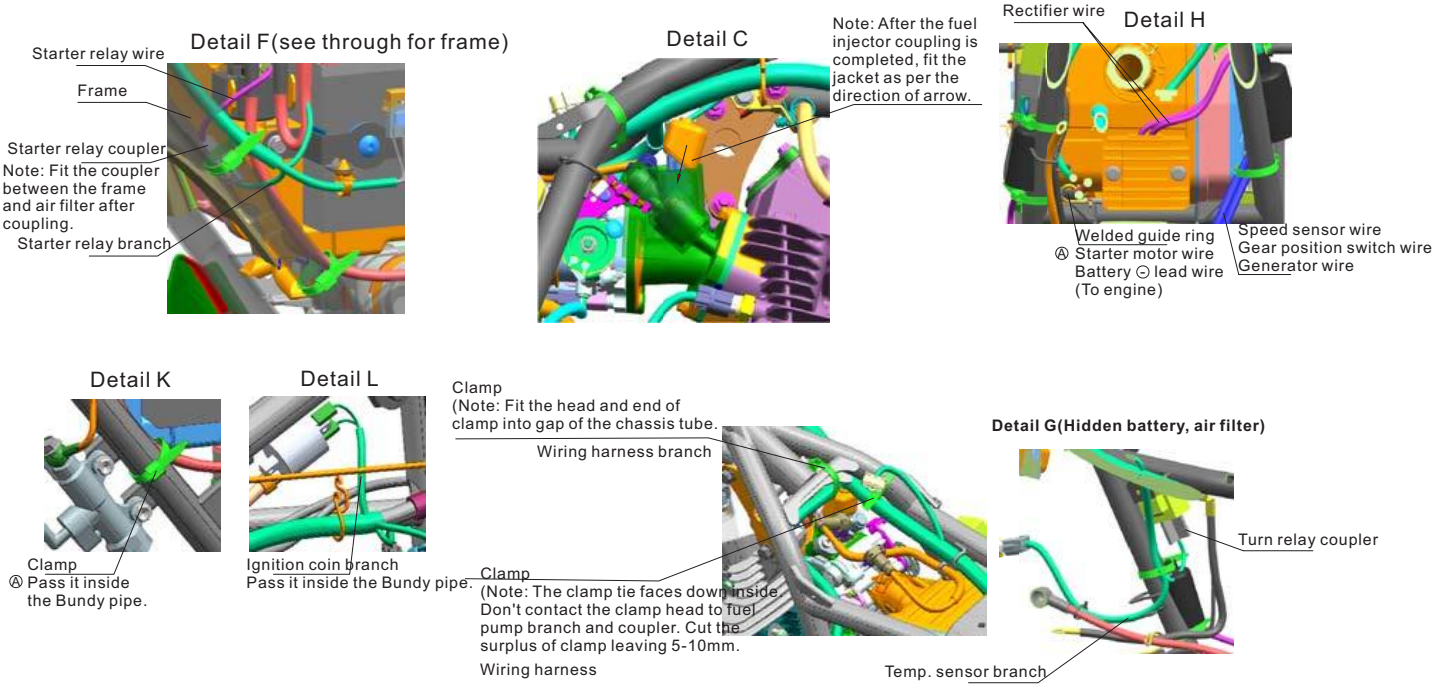






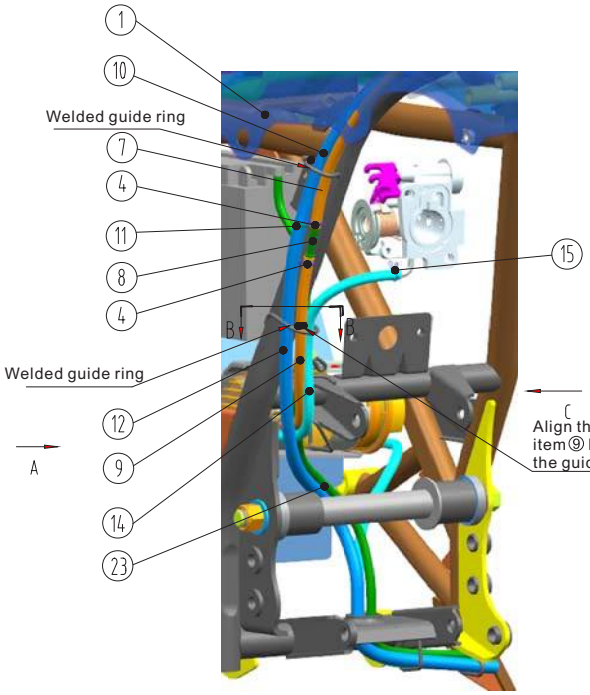


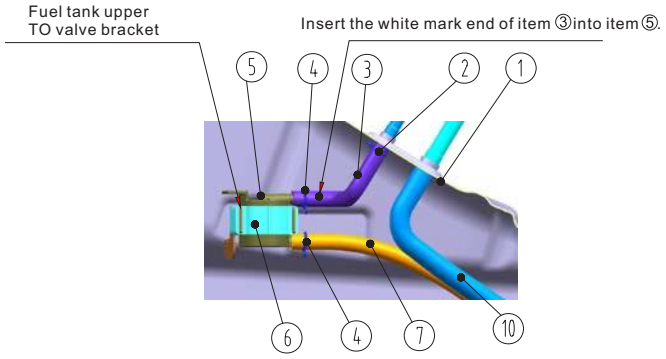


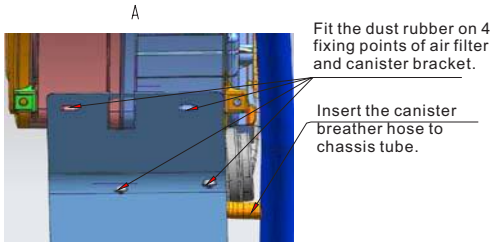


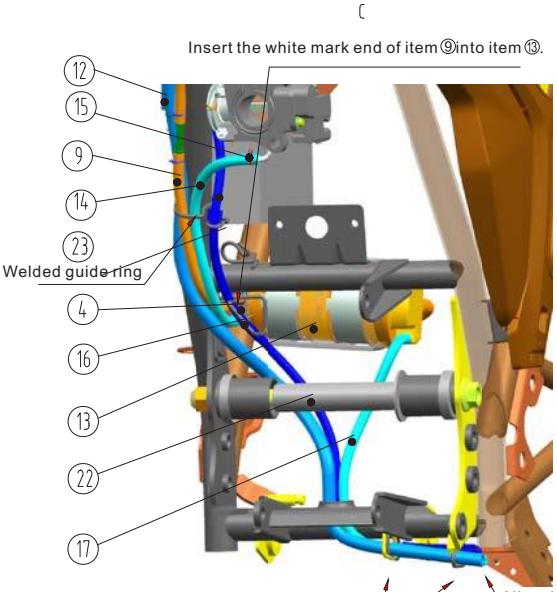
	Ⓐ Routing part	Control	Requirement
1	Battery ⊕ terminal Starter relay terminal Starer motor terminal	The clearance between the terminal and metal part.	Above 8mm
2	Battery ⊕ terminal to fuse	Routing area.	Free of burr and sharp edge. Do not pass the wire at movable portion. Use clamp to fix the wire at specified position. Wire should not interfere with other part.
3	Battery ⊕ terminal to starter relay		
4	Starter relay to starter motor		
5	CKPS signal terminal	Coupling method.	Couple the male and female terminal as per wire color. Install the jacket after coupling.

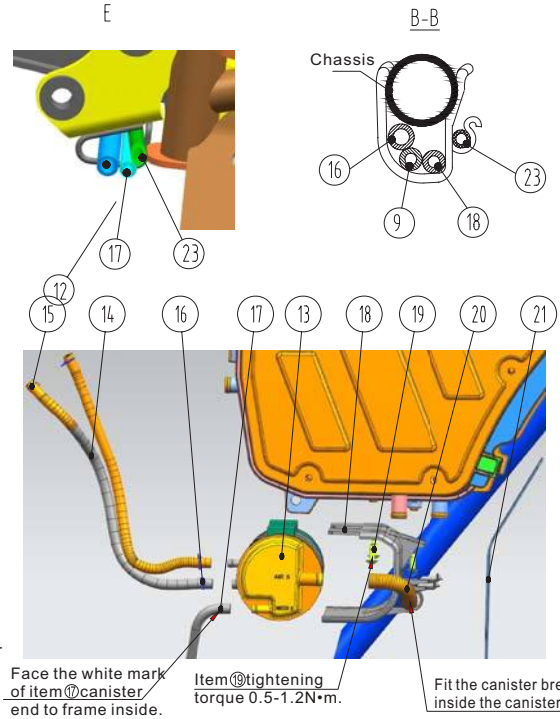
EMISSION EVAPORATION CONTROL SYSTEM







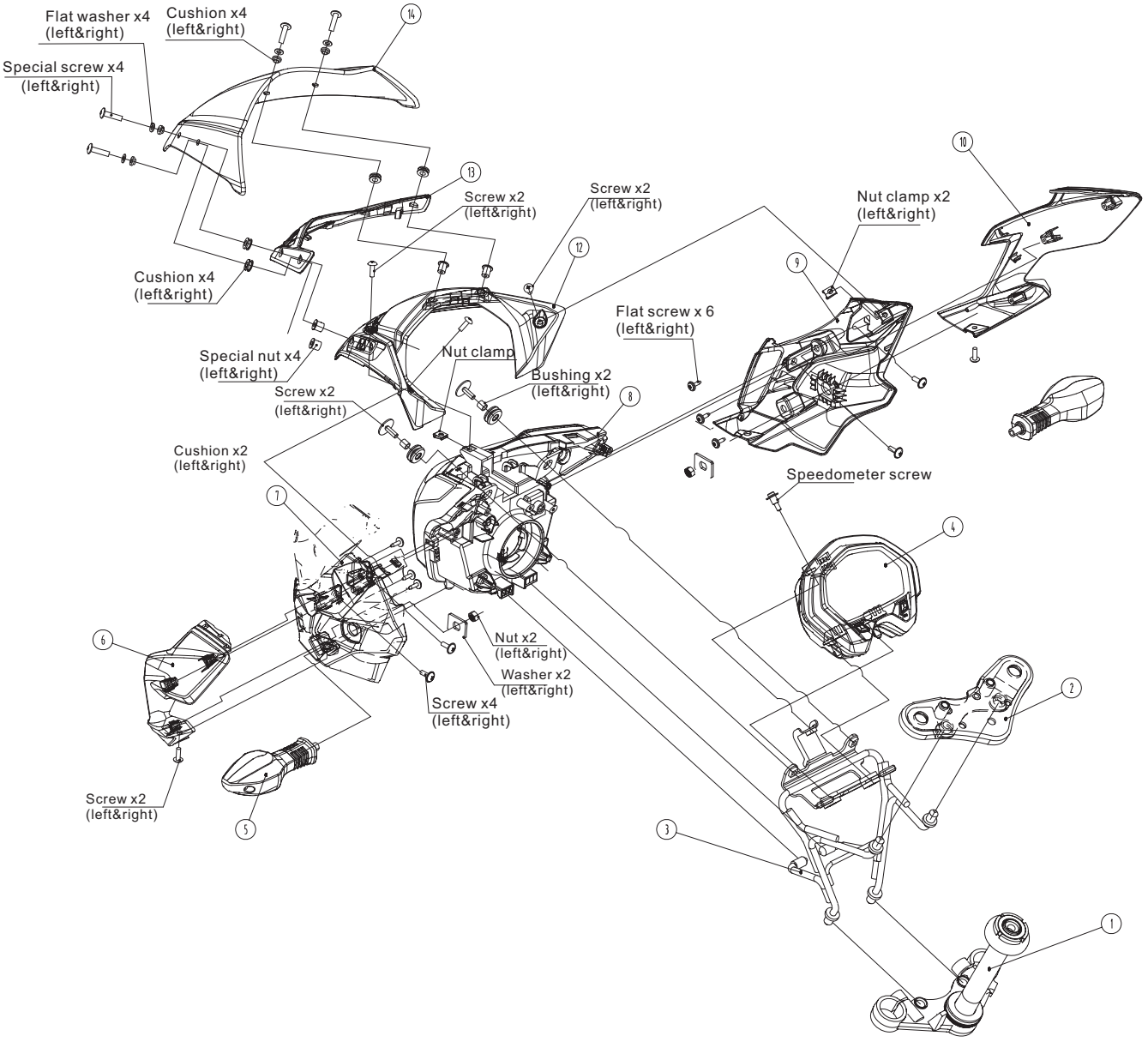




Align the item⑫fuel tank over-flow hose II white band with chassis guide ring outside.

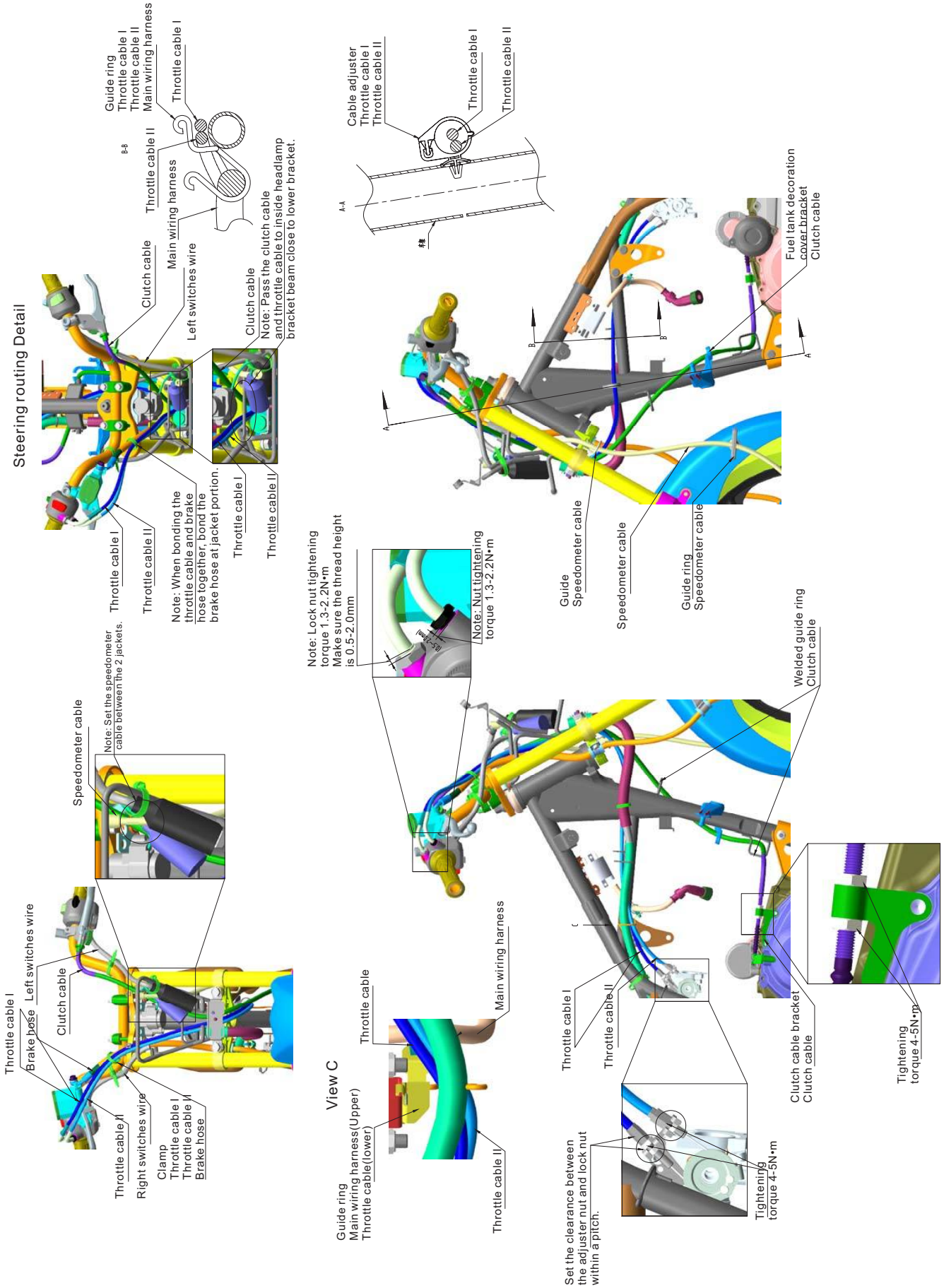
①	Fuel tank	⑨	Hose between TO valve & canister hose II	⑰	Canister drain hose
②	Clamp	⑩	Fuel tank overflow hose I	⑱	Canister bracket
③	Hose between fuel tank and TO valve	⑪	Connector	⑲	Screw
④	Clamp	⑫	Fuel tank overflow hose II	⑳	Canister breather hose
⑤	TO valve	⑬	Canister	㉑	Dust rubber
⑥	TO valve jacket	⑭	Absorber hose	㉒	Rear swing arm
⑦	Hose between TO valve & connector	⑮	Clamp	㉓	Battery breather hose
⑧	Connector	⑯	Clamp		

COWLING INSTALLION

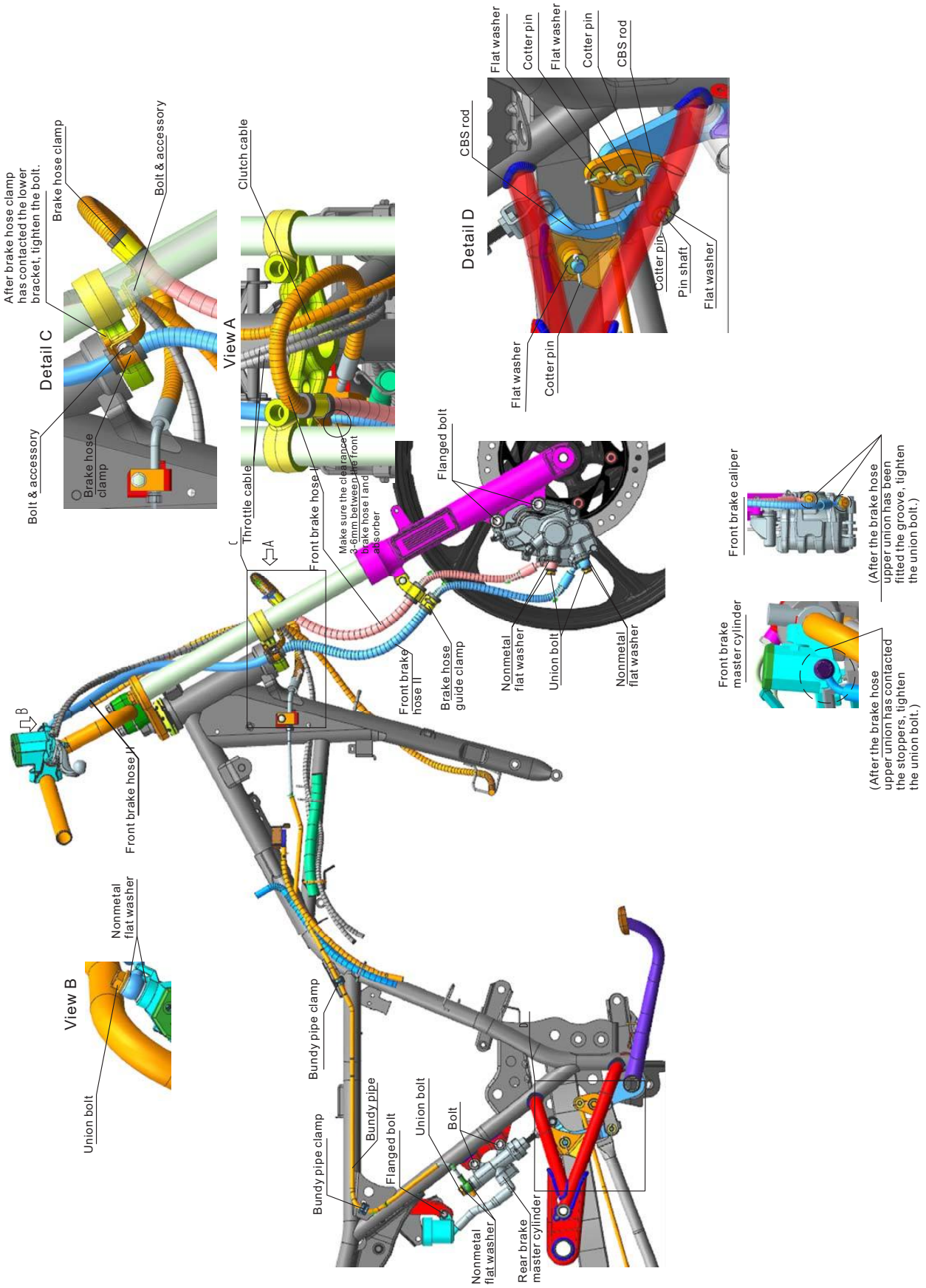


①	Lower bracket	⑧	Headlamp
②	Upper bracket	⑨	Headlamp case right cover
③	Headlamp bracket	⑩	Headlamp right decoration cover
④	Meter	⑪	Front right turn signal light
⑤	Front left turn signal light	⑫	Headlamp upper cover
⑥	Front left turn signal light decoration cover	⑬	Windshield bracket
⑦	Headlamp case left cover	⑭	Windshield

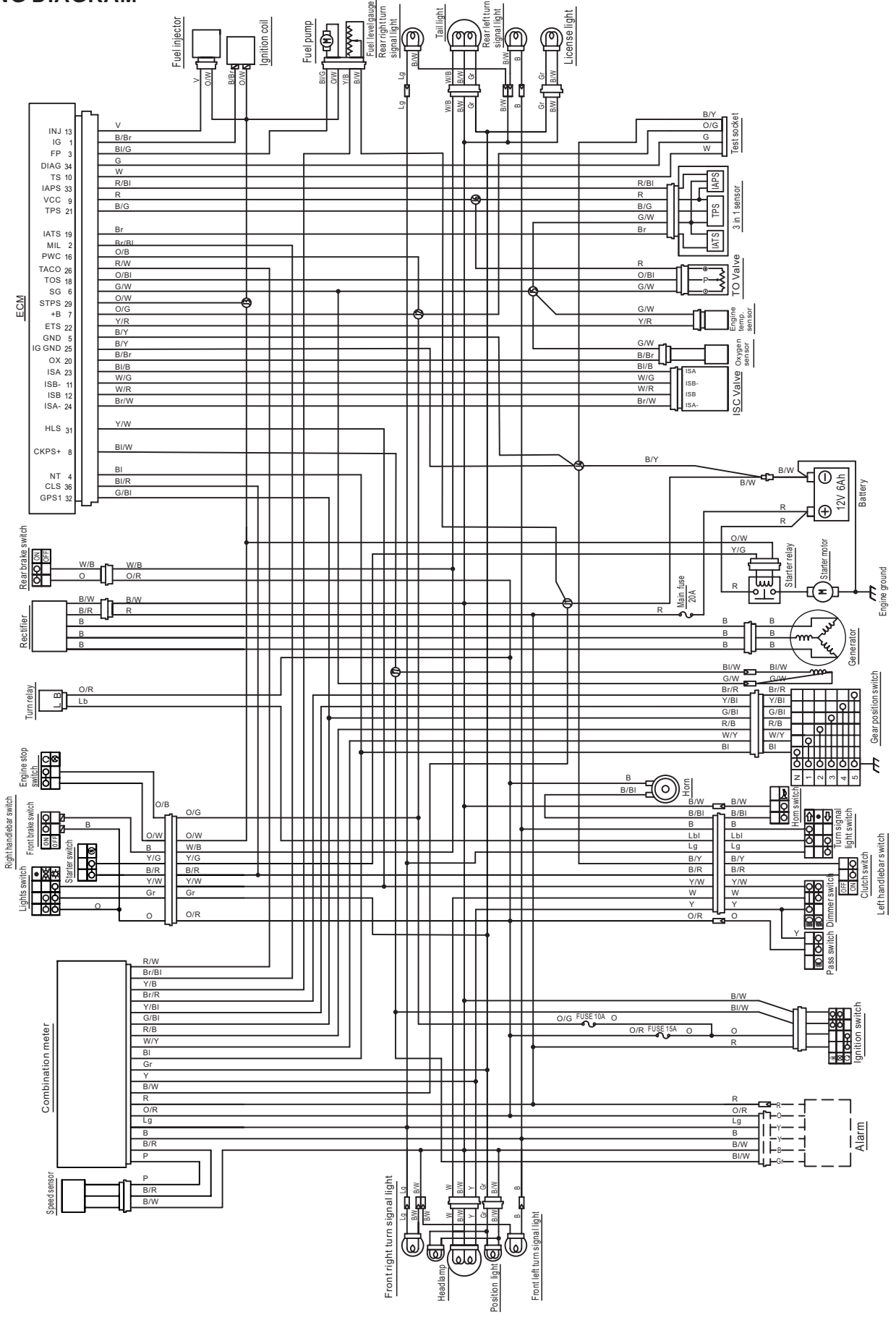
CABLE ROUTING



FRONT BRAKE HOSE ROUTING



WIRING DIAGRAM



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